

Singular supports, characteristic cycles, and microlocalization of constructible sheaves

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Abstract

For a constructible sheaf $\mathcal{F}$ on a smooth scheme X over a perfect field, the singular support $SS(\mathcal{F})$ and the characteristic cycle $CC(\mathcal{F})$ are defined on the cotangent bundle. In analogy with the construction of Kashiwara–Schapira via microlocalization in the transcendental setting, we construct the microlocal versions $SS_\mu(\mathcal{F})$ and $CC_\mu(\mathcal{F})$ of $SS(\mathcal{F})$ and $CC(\mathcal{F})$ respectively, and prove

$$SS_\mu(\mathcal{F}) = SS(\mathcal{F}), \quad CC_\mu(\mathcal{F}) = \text{cl}CC(\mathcal{F}),$$

where cl is the cycle class map.

As a consequence of the second equality, we show that the cohomological characteristic classes are the cycle classes associated with the corresponding characteristic cycles, as conjectured in (*Invent. Math.* 207: 597–695, 2017). As further applications, we give new proofs of the index formula and Bloch’s conductor formula in the proper case, independently of Abe’s homotopical approach via ∞ -categories.

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1 Introduction

For a constructible sheaf $\mathcal{F}$ on a smooth scheme X over a field k of characteristic $p > 0$, the singular support $SS(\mathcal{F})$ and the characteristic cycle $CC(\mathcal{F})$ are defined on the cotangent bundle T^*X [Bei16], [Sai17a]. In a transcendental context, the singular supports and the characteristic cycles allow a construction using the microlocalization $\mu\mathcal{H}om(\mathcal{F}, \mathcal{F})$ defined on the cotangent bundle [KS90]. We study a similar construction in the algebraic setting.

Let η_0 be the generic point of the henselization of $\mathbf{A}_k^1$ at the origin. Choose a geometric point $\bar{\eta}_0$ above η_0 and a compatible separable closure $\bar{k}$ of k , and write $G_{\eta_0} = \text{Gal}(\bar{\eta}_0/\eta_0)$ and $G_k = \text{Gal}(\bar{k}/k)$. Let Λ be a finite local ring of residue characteristic $\ell \neq p$. We fix a non-trivial character $\psi : \mathbf{F}_p \rightarrow \Lambda^\times$. For $\mathcal{F} \in D_{\text{ctf}}(X, \Lambda)$, the microlocalization $\mu\mathcal{H}om(\mathcal{F}, \mathcal{F})$ is an object of the derived category $D_{\text{ctf}}(T^*X \times_k \eta_0, \Lambda)$ of bounded constructible complexes of finite tor-dimension on $T^*X_{\bar{k}}$ equipped with a continuous semilinear G_{η_0} -action, where the action on the base is induced by the quotient $G_{\eta_0} \rightarrow G_k$. It is defined as the Fourier transform of the specialization $\nu\mathcal{H}om(\mathcal{F}, \mathcal{F}) \in D_{\text{ctf}}(TX_{\bar{k}}, \Lambda)$ of

$$(1.0.1) \quad \mathcal{H}_{\mathcal{F}} := \mathcal{H}om(\text{pr}_1^* \mathcal{F}, \text{pr}_2^! \mathcal{F})$$

along the diagonal $\delta : X \hookrightarrow X \times X$. Here and in what follows, we omit R from the notation for derived functors. The support of $\mu\mathcal{H}om(\mathcal{F}, \mathcal{F})$ descends to a unique closed conical subset $SS_{\mu}(\mathcal{F}) \subseteq T^*X$, which we define in Definition 3.1.4. The construction of the cohomology class $CC_{\mu}(\mathcal{F})$ supported on $SS_{\mu}(\mathcal{F})$, given in Definition 4.1.2, refines the construction of the characteristic class $C_{X/k}(\mathcal{F})$ in [AS07]. Recall that $C_{X/k}(\mathcal{F}) \in H^0(X, K_X)$ is defined as the composition

$$(1.0.2) \quad \Lambda \xrightarrow{1_{\mathcal{F}}} \delta^! \mathcal{H}_{\mathcal{F}} \rightarrow \delta^* \mathcal{H}_{\mathcal{F}} \xrightarrow{\text{ev}} K_X,$$

where $K_X = a^! \Lambda$ for the structure morphism $a : X \rightarrow \text{Spec } k$. Taking adjoints and applying the microlocalization functor, we obtain

$$(1.0.3) \quad \Lambda \rightarrow \mu\mathcal{H}om(\mathcal{F}, \mathcal{F}) \rightarrow (e^{\vee*} K_X)|_{T^*X_{\bar{k}}},$$

where $e^\vee: T^*X \rightarrow X$ is the projection. The composition defines a G_k -invariant cohomology class supported on $SS_\mu(\mathcal{F})_{\bar{k}}$, which descends uniquely to a class $CC_\mu(\mathcal{F}) \in H_{SS_\mu(\mathcal{F})}^0(T^*X, e^{\vee*}K_X)$.

We prove the following equality between $SS_\mu(\mathcal{F})$ and $SS(\mathcal{F})$.

Theorem 1.0.1 (Theorem 3.3.6 and Theorem 4.9.12). *Let X be a smooth k -scheme and $\mathcal{F} \in D_{\text{ctf}}(X, \Lambda)$. We have*

$$(1.0.4) \quad SS_\mu(\mathcal{F}) = SS(\mathcal{F}).$$

The assertion also holds for constructible complexes with coefficients in a finite extension $E/\mathbf{Q}_\ell$, interpreted on the pro-étale site.

Following a suggestion of Beilinson, one can use Wang's Fourier transform [Wang] to construct SS_μ and CC_μ uniformly over fields of arbitrary characteristic, including characteristic zero; see Definition 3.1.9, Proposition 3.1.10, and Remark 4.1.3. This approach simplifies some arguments and avoids both the choice of an additive character and the associated coefficient extension in Lemma 2.1.1.

We outline our proof of (1.0.4). The first step is to prove the following inclusion (cf. Theorem 3.2.5):

$$(1.0.5) \quad SS_\mu(\mathcal{F}) \subseteq SS(\mathcal{F}).$$

By a standard reduction, we may assume that $X = P = \mathbf{P}^n$ is a projective space of dimension $n > 1$. Let $P^\vee$ be the dual projective space and let

$$(1.0.6) \quad Q = \{(x, [H]) : x \in H\} \subseteq P \times P^\vee, \quad p : Q \rightarrow P, \quad p^\vee : Q \rightarrow P^\vee,$$

be the incidence correspondence and its projections respectively. Let U be the largest open subset of Q such that $p|_U$ is universally locally acyclic relatively to $\mathcal{G} = p^{\vee*}Rp_*^\vee p^*\mathcal{F}[2n-2](n-1)$. Beilinson's results on Radon transforms [Bei16, Theorem 3.2] identify the projectivization of $SS(\mathcal{F})$ with $Q \setminus U$. It therefore suffices to show that $\mu\mathcal{H}om(\mathcal{F}, \mathcal{F})$ vanishes on the cotangent lines parametrized by U .

We consider the base change of the diagonal $\delta_P : P \hookrightarrow P \times P$ along $p \times p$

$$(1.0.7) \quad \begin{array}{ccc} Q \times_P Q & \xrightarrow{i_p} & Q \times Q \\ \downarrow & & \downarrow p \times p \\ P & \xrightarrow{\delta_P} & P \times P. \end{array}$$

It induces natural morphisms $\Pi : T_{Q \times_P Q}(Q \times Q) \simeq Q \times_P Q \times_P TP \rightarrow T_P(P \times P) \simeq TP$ and $\Pi^\vee : Q \times_P Q \times_P T^*P \rightarrow T^*P$, which are the projections. A key step is to construct the following retraction (see Corollary 2.4.3):

$$(1.0.8) \quad \nu_{\delta_P}(\mathcal{H}_{\mathcal{F}}) \longrightarrow \Pi_*\nu_{i_p}(\mathcal{H}_{\mathcal{G}}) \longrightarrow \nu_{\delta_P}(\mathcal{H}_{\mathcal{F}}).$$

Applying the Fourier transform to (1.0.8), we obtain $\mu\mathcal{H}om(\mathcal{F}, \mathcal{F})$ as a retract of

$$(1.0.9) \quad F_\psi \Pi_*\nu_{i_p}(\mathcal{H}_{\mathcal{G}}) \simeq \Pi_*^\vee F_\psi \nu_{i_p}(\mathcal{H}_{\mathcal{G}}).$$

We then estimate the support

$$(1.0.10) \quad \Sigma := \text{supp}_k(F_\psi \nu_{i_p}(\mathcal{H}_\mathcal{G})) \subseteq Q \times_P Q \times_P T^*P,$$

where $\text{supp}_k(F_\psi \nu_{i_p}(\mathcal{H}_\mathcal{G}))$ is the unique closed subset of $Q \times_P Q \times_P T^*P$ such that

$$(1.0.11) \quad (\text{supp}_k(F_\psi \nu_{i_p}(\mathcal{H}_\mathcal{G})))_{\bar{k}} = \text{supp}(F_\psi \nu_{i_p}(\mathcal{H}_\mathcal{G})).$$

We use a deformation method to construct a linear action of $T(Q \times Q/P^\vee \times P^\vee)$ on $T_{Q \times_P Q}(Q \times Q)$ under which $\nu_{i_p}(\mathcal{H}_\mathcal{G})$ is invariant (cf. Lemma 2.4.4). We use this linear action to deduce that Σ is contained in the union of the zero section and the tautological line bundle on the diagonal $Q \subseteq Q \times_P Q$. We use the universal local acyclicity of $p|_U$ with respect to $\mathcal{G}$ to deduce that $\Sigma|_{U \times_P U}$ is contained in the zero section 0_{T^*P} . Combining the two estimations, we obtain (1.0.5).

The next step is to prove a support estimate for microlocalization along a smooth closed subscheme.

Theorem 1.0.2 (Theorem 3.2.6). *Let $Z \hookrightarrow X$ be a closed immersion of smooth k -schemes and $\mathcal{F} \in D_{\text{ctf}}(X, \Lambda)$. Let $T_Z X$ be the normal bundle of $Z \hookrightarrow X$ and $F_{\psi, T_Z X/Z}$ the Fourier transform on it. We define*

$$(1.0.12) \quad \mu_{Z/X}(\mathcal{F}) = F_{\psi, T_Z X/Z}(\nu_{Z/X} \mathcal{F}) \in D_{\text{ctf}}(T_Z^* X \times_k \eta_0, \Lambda),$$

where $\nu_{Z/X}(\mathcal{F})$ is the specialization of $\mathcal{F}$ along $Z \hookrightarrow X$, and η_0 is the generic point of the henselization of $\mathbf{A}^1$ at 0. We have:

$$(1.0.13) \quad \text{supp}_k(\mu_{Z/X}(\mathcal{F})) \subseteq SS_\mu(\mathcal{F}) \cap T_Z^* X \subseteq SS(\mathcal{F}) \cap T_Z^* X,$$

where $\text{supp}_k(\mu_{Z/X}(\mathcal{F})) \subseteq T_Z^* X$ is the unique closed subset of $T_Z^* X$ such that

$$(1.0.14) \quad (\text{supp}_k(\mu_{Z/X}(\mathcal{F})))_{\bar{k}} = \text{supp}(\mu_{Z/X}(\mathcal{F})) \quad \text{in} \quad T_{Z_{\bar{k}}} X_{\bar{k}}.$$

This is proved by constructing a retraction

$$(1.0.15) \quad \mu_{Z/X}(\mathcal{F}) \rightarrow (q^\vee)^* \mu \mathcal{H}om(\mathcal{F}, \mathcal{F}) \otimes \mu_{Z/X}(\mathcal{F}) \longrightarrow \mu_{Z/X}(\mathcal{F}),$$

where $q^\vee : T_Z^* X \hookrightarrow T^* X \times_X Z$ is dual to the quotient bundle map $q : TX \times_X Z \rightarrow T_Z X$. Then (1.0.5) implies (1.0.13).

We establish a Künneth formula for $\mu \mathcal{H}om$ in Lemma 3.3.1. Let X and Y be smooth k -schemes. For constructible complexes $\mathcal{F} \in D_{\text{ctf}}(X, \Lambda)$ and $\mathcal{G} \in D_{\text{ctf}}(Y, \Lambda)$, we construct a canonical isomorphism

$$(1.0.16) \quad \mu \mathcal{H}om(\mathcal{F} \boxtimes \mathcal{G}, \mathcal{F} \boxtimes \mathcal{G}) \simeq \mu \mathcal{H}om(\mathcal{F}, \mathcal{F}) \boxtimes \mu \mathcal{H}om(\mathcal{G}, \mathcal{G}).$$

This gives the product formulas

$$(1.0.17) \quad SS_\mu(\mathcal{F} \boxtimes \mathcal{G}) = SS_\mu(\mathcal{F}) \times SS_\mu(\mathcal{G}), \quad CC_\mu(\mathcal{F} \boxtimes \mathcal{G}) = CC_\mu(\mathcal{F}) \boxtimes CC_\mu(\mathcal{G}).$$

Finally, we prove the equality (1.0.4) using (1.0.5), Theorem 1.0.2 together with the Künneth formula (1.0.17) for SS_μ .

Theorem 1.0.3 (Theorem 4.6.2 and Theorem 4.9.2). *Let k be a perfect field and let X be a smooth k -scheme of pure dimension d . For any $\mathcal{F} \in D_{\text{ctf}}(X, \Lambda)$, we have*

$$(1.0.18) \quad CC_\mu(\mathcal{F}) = \text{cl} CC(\mathcal{F}),$$

where $\text{cl} : Z_d(SS(\mathcal{F})) \rightarrow H_{SS(\mathcal{F})}^0(T^*X, e_X^{\vee*} K_{X/k})$ is the cycle class map.

The assertion also holds for constructible complexes with coefficients in a finite extension $E/\mathbf{Q}_\ell$, interpreted on the pro-étale site.

We sketch the proof of (1.0.18). First, we prove the equality in Corollary 4.2.5 for a tamely ramified sheaf $\mathcal{F}$ on a curve by computing $\nu\mathcal{H}om(\mathcal{F}, \mathcal{F})$. Then, using a global argument based on a theorem of Katz–Gabber [Kat86], we prove Proposition 4.6.1 for curves. For the proof, we use an index formula for CC_μ proved in Corollary 4.3.3 and compare it with the Grothendieck–Ogg–Shafarevich formula. The index formula in Corollary 4.3.3 is a special case of the compatibility of CC_μ with proper push-forward in Proposition 4.3.2.

To prove Theorem 4.6.2 in higher dimension, we prove that $CC_\mu(\mathcal{F})$ also satisfies the characterization of $CC(\mathcal{F})$ by the Milnor formula. By the equality $SS_\mu(\mathcal{F}) = SS(\mathcal{F})$, the theorem of Beilinson $\dim SS(\mathcal{F}) = \dim X$ implies $\dim SS_\mu(\mathcal{F}) \leq \dim X$. By the weak cohomological purity, this makes it suffice to compare the coefficients in $CC_\mu(\mathcal{F})$ and in $CC(\mathcal{F})$ of irreducible components of dimension $\dim X$. By the compatibility with push-forward for a closed immersion, it is reduced to the case where X is projective. Then, by taking a Lefschetz pencil and by the compatibility with proper push-forward in Proposition 4.3.2, the Milnor formula is reduced to the equality $CC_\mu = \text{cl} CC$ for the direct image on $\mathbf{P}^1$ proved in Proposition 4.6.1.

As an application of Theorem 1.0.3, we prove a conjecture on cohomological characteristic classes formulated by the first named author.

Conjecture 1.0.4 (Saito, [Sai17a, Conjecture 6.8]). *Assume k is a perfect field. Let Z be a k -scheme, and let $i : Z \rightarrow X$ be a closed immersion into a smooth k -scheme X . Let Λ be a finite local ring whose residue characteristic is invertible in k . For every $\mathcal{F} \in D_{\text{ctf}}(Z, \Lambda)$, we have*

$$(1.0.19) \quad C_{Z/k}(\mathcal{F}) = \text{cl}_Z(cc_{Z,0}(\mathcal{F})) \quad \text{in } H^0(Z, K_{Z/k}),$$

where $cc_{Z,0}(\mathcal{F}) = 0_X^! CC(i_* \mathcal{F}) \in \text{CH}_0(Z)$ is the refined Gysin pull-back of $CC(i_* \mathcal{F})$ by the zero section $0_X : X \rightarrow T^*X$ (see [Sai17a, Definition 6.7]).

In [YZ25, Theorem 1.2], the third named author and Zhao prove this conjecture in the case where Z is smooth and quasi-projective by using non-acyclicity classes and the fibration method. Using microlocalization, we prove Conjecture 1.0.4 in full generality.

Theorem 1.0.5 (Theorem 4.8.2). *Conjecture 1.0.4 holds.*

We apply the results for ℓ -adic coefficients to establish two results for finite coefficients. The first is the index formula for the Euler–Poincaré characteristic, which is proved in [Sai17a, Theorem 7.13] under the assumption that the variety is projective.

Corollary 1.0.6 (Corollary 4.9.5). *Let X be a smooth scheme over a perfect field k of characteristic $p > 0$, and let Λ be a finite local ring over $\mathbf{Z}/\ell^n \mathbf{Z}$ for a prime number $\ell \neq p$. Let $i : Z \hookrightarrow X$ be a closed immersion and $\mathcal{F} \in D_{\text{ctf}}(Z, \Lambda)$. Assume Z is proper over k . Then we have an equality in $\mathbf{Z}$*

$$(1.0.20) \quad \chi(Z_{\bar{k}}, \mathcal{F}) = \deg cc_{Z,0}(\mathcal{F}).$$

The second is the geometric case of the conductor formula conjectured by Bloch in [Blo87, Conjecture], which is proved in [Sai21, Corollary 2.2.4] under the assumption that the morphism f is projective.

Corollary 1.0.7 (Corollary 4.9.8). *Let $f : X \rightarrow Y$ be a proper flat morphism of smooth schemes over a perfect field k of characteristic $p > 0$, and let Λ be a finite local ring over $\mathbf{Z}/\ell^m\mathbf{Z}$ for a prime number $\ell \neq p$. Let $y \in Y$ be a closed point. Assume that $\dim X = n$, $\dim Y = 1$ and that there is a dense open subset $V \subseteq Y$ such that the base change $f_V : X \times_Y V \rightarrow V$ is smooth. Then we have an equality in $\mathbf{Z}$*

$$(1.0.21) \quad -a_y(f_*\Lambda) = (-1)^n \deg_{k(y)} \left(c_{nX_y}^X(\Omega_{X/Y}^1) \cap [X] \right),$$

where $c_{nX_y}^X(\Omega_{X/Y}^1)$ is the localized Chern class defined in [Blo87, Section 1].

We also note that Abe [Abe] has studied the index formula and the conductor formula from a homotopical perspective using ∞ -categories.

We briefly sketch the contents of each section. We recall basic properties of Fourier transforms and prove some complements in Section 2.1. In Section 2.2, we introduce deformation to normal bundle. The specialization functor is introduced in Section 2.3 following Verdier [Ver83]. The specialization is defined by applying the nearby cycles functor to the deformation to normal bundle. The microlocalization is defined as the Fourier transform of the specialization. In Section 2.4, we prove some results on retraction.

In Section 3.1, we define $SS_\mu(\mathcal{F})$ using the microlocalization $\mu\mathcal{H}om(\mathcal{F}, \mathcal{F})$. We prove (1.0.5) and Theorem 1.0.2 in Section 3.2, and we establish the equality (1.0.4) in Section 3.3.

In Section 4.1, we define $CC_\mu(\mathcal{F})$ using the microlocalization $\mu\mathcal{H}om(\mathcal{F}, \mathcal{F})$. We compute CC_μ in Section 4.2 for a tamely ramified sheaf on a curve; this computation is used in the proof of the curve case of (1.0.18). In Sections 4.3 and 4.4, we study the compatibility with proper push-forward and with smooth pull-back respectively. We prove the Künneth formula (1.0.17) in Section 4.5. We prove (1.0.18) in Section 4.6. In Section 4.7, we give an explicit and independent computation of CC_μ for a wildly ramified Artin–Schreier sheaf on a curve. As an application of Theorem 1.0.3, we prove Theorem 1.0.5 in Section 4.8. In Section 4.9, we extend the results to ℓ -adic sheaves, and prove the index formula (1.0.20) and Bloch’s conductor formula (1.0.21).

When the first named author studied the microlocalization of étale sheaves with Ahmed Abbes in 2004, they thought that this approach would not work since an alternative approach using the blow-up taken in [AS07] worked well. The first named author recently computed the example in Section 4.7 and realized that they had reached the negative conclusion too quickly. The same question was also considered independently by D.-C. Cisinski and A. Khan.

During 2016–2018, when the third named author was a postdoctoral researcher at the University of Regensburg, Denis-Charles Cisinski suggested developing a theory of singular support and microlocalization in the motivic setting. While a notion of singular support for motives was successfully constructed, the difficulties arising from wild ramification led the third named author to remain skeptical about the existence of a satisfactory theory of microlocalization. After completing the joint work with Zhao [YZ25], however, he came to believe that an appropriate form of microlocalization might exist, although no substantial

progress was made at that stage.¹ It was only after the first named author had taken the first step in this work that the key idea emerged. More specifically, the idea of deforming after the base change (1.0.7) was inspired by the construction of non-acyclicity classes in [YZ25, 4.2–4.3].

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Notation and Conventions

- (1) Throughout this article, k denotes a field of characteristic $p > 0$, and $\bar{k}$ denotes a separable closure of k . Unless otherwise stated, all schemes are separated and of finite type over k .
- (2) Let Λ be a finite local ring over $\mathbf{Z}/\ell^n\mathbf{Z}$ for a prime number $\ell \neq p$ and an integer $n \geq 1$. For a noetherian scheme X over k , $D_c^b(X, \Lambda) \supset D_{\text{ctf}}(X, \Lambda)$ denote the derived category of bounded constructible complexes of Λ -modules on X and its full subcategory of complexes of finite Tor-dimension.
- (3) We fix a non-trivial character $\psi : \mathbf{F}_p \rightarrow \Lambda^\times$ whenever such a character exists; otherwise use the coefficient extension of Lemma 2.1.1. Let $\mathcal{L}_\psi$ be the Artin–Schreier sheaf on $\mathbf{A}_k^1$ corresponding to the character ψ . For a morphism $f : X \rightarrow \mathbf{A}_k^1$, we put $\mathcal{L}_\psi(f) = f^*\mathcal{L}_\psi$.
- (4) For a smooth k -scheme X , we write TX and T^*X for its tangent and cotangent bundles, respectively. If $f : X \rightarrow Y$ is a smooth morphism, we write $T(X/Y)$ and $T^*(X/Y)$ for the relative tangent and cotangent bundles, respectively. For a regular immersion $i : Z \hookrightarrow X$, we write $T_Z X$ and $T_Z^* X$ for the normal and conormal bundles of Z in X , respectively.
- (5) Unless explicitly stated otherwise, we omit R and L from the notation for derived functors.
- (6) For a separated morphism $a : X \rightarrow S$, write $K_{X/S} = a^!\Lambda$ and $D_{X/S}(-) = \mathcal{H}om(-, K_{X/S})$. We write D_X when the base is $\text{Spec } k$.

¹There is a link between $\mu\mathcal{H}om(\mathcal{F}, \mathcal{F})$ and the object $\delta_1^* \delta^\Delta \mathcal{H}_{\mathcal{F}}$; this link will be investigated in a later paper.

2 Fourier transforms and the specialization

2.1 Complements on Fourier transforms

Let k be a field of characteristic $p > 0$. Let Λ be a finite local ring over $\mathbf{Z}/\ell^n\mathbf{Z}$ for a prime number $\ell \neq p$ and an integer $n \geq 1$. If such a character exists, we fix a non-trivial character $\psi: \mathbf{F}_p \rightarrow \Lambda^\times$. Let $\psi^\vee: \mathbf{F}_p \rightarrow \Lambda^\times$ be the character defined by $\psi^\vee(x) = \psi(-x) = \psi(x)^{-1}$.

In general, the following lemma shows that such a ψ exists after a finite étale faithfully flat extension of the coefficient ring Λ .

Lemma 2.1.1. *There exist a finite étale faithfully flat local Λ -algebra Λ' and a non-trivial character $\psi: \mathbf{F}_p \rightarrow (\Lambda')^\times$.*

Proof. Since p is invertible in Λ , the algebra $B = \Lambda[T]/(1 + T + \cdots + T^{p-1})$ is finite étale over Λ . Since B is Artinian, it is a finite product of nonzero local rings. Choose one local factor Λ' . It is finite étale over Λ , hence finite free of positive rank because Λ is local. In particular, Λ' is faithfully flat over Λ .

The image ζ of T in Λ' satisfies $\zeta^p = 1$, and $\zeta - 1$ is invertible. Hence ζ has order p , and

$$(2.1.1) \quad \psi: \mathbf{F}_p \longrightarrow (\Lambda')^\times, \quad \bar{a} \longmapsto \zeta^a,$$

is a non-trivial character. □

Definition 2.1.2. Let k be a field of characteristic $p > 0$ and $\psi: \mathbf{F}_p \rightarrow \Lambda^\times$ be a non-trivial character.

1. Let $\mathbf{A}^1 = \text{Spec } k[t] \rightarrow \mathbf{A}^1 = \text{Spec } k[x]$ be the Artin-Schreier covering defined by $t^p - t = x$ and identify the Galois group with $\mathbf{F}_p$ by the action of $a \in \mathbf{F}_p$ defined by $t \mapsto t + a$. Let $\mathcal{L}_\psi$ be the locally constant constructible sheaf of free Λ -modules of rank 1 on $\mathbf{A}^1 = \text{Spec } k[x]$ trivialized by the Artin-Schreier covering $\mathbf{A}^1 \rightarrow \mathbf{A}^1$ corresponding to the character $\psi: \mathbf{F}_p \rightarrow \Lambda^\times$.

2. Let X be a noetherian scheme over k and let E be a vector bundle over X . Let $E^\vee$ be the dual vector bundle and

$$(2.1.2) \quad \begin{array}{ccc} E & \xleftarrow{\text{pr}_1} & E \times_X E^\vee \xrightarrow{\text{pr}_2} E^\vee \\ & \mu = \langle, \rangle \downarrow & \\ & \mathbf{A}^1 & \end{array}$$

be the projections and the canonical pairing. We define the (naive) Fourier transform

$$(2.1.3) \quad F = F_\psi = F_{\psi, E/X}: D_c^b(E, \Lambda) \rightarrow D_c^b(E^\vee, \Lambda)$$

by $F_\psi \mathcal{F} = \text{pr}_{2!}(\text{pr}_1^* \mathcal{F} \otimes \mu^* \mathcal{L}_\psi)$ and the (naive) dual Fourier transform

$$(2.1.4) \quad F^\vee = F_{\psi^\vee} = F_{\psi^\vee, E^\vee/X}: D_c^b(E^\vee, \Lambda) \rightarrow D_c^b(E, \Lambda)$$

by $F_{\psi^\vee} \mathcal{G} = \text{pr}_{1!}(\text{pr}_2^* \mathcal{G} \otimes \mu^* \mathcal{L}_{\psi^\vee})$.

The functor $F_\psi: D_c^b(E, \Lambda) \rightarrow D_c^b(E^\vee, \Lambda)$ induces $F_\psi: D_{\text{ctf}}^b(E, \Lambda) \rightarrow D_{\text{ctf}}^b(E^\vee, \Lambda)$. Let $0: X \rightarrow E$ and $0^\vee: X \rightarrow E^\vee$ be the 0-sections and let $e: E \rightarrow X$ and $e^\vee: E^\vee \rightarrow X$ denote the canonical morphisms. Fourier transforms satisfy the following elementary properties.

Lemma 2.1.3. *Let $e: E \rightarrow X$ be a vector bundle of rank n over a noetherian scheme X over k and let $e^\vee: E^\vee \rightarrow X$ be the dual vector bundle.*

1. *We have a canonical isomorphism*

$$(2.1.5) \quad 0^{\vee*} F_\psi \rightarrow e_!$$

of functors $D_c^b(E, \Lambda) \rightarrow D_c^b(X, \Lambda)$.

2. *We have a canonical isomorphism*

$$(2.1.6) \quad F_\psi 0_* \rightarrow e^{\vee*},$$

of functors $D_c^b(X, \Lambda) \rightarrow D_c^b(E^\vee, \Lambda)$.

3. *The adjoint*

$$(2.1.7) \quad F_\psi e^! \rightarrow 0_*^\vee$$

of the composition $0^{\vee} F_\psi e^! \rightarrow e_! e^! \rightarrow 1$ of the morphism induced by (2.1.5) and the adjunction is an isomorphism of functors $D_c^b(X, \Lambda) \rightarrow D_c^b(E^\vee, \Lambda)$.*

The isomorphism (2.1.7) is equivalently formulated as an isomorphism

$$(2.1.8) \quad F_\psi e^* \rightarrow 0_*^\vee(- \otimes e_! \Lambda).$$

In particular, for any $\mathcal{M} \in D_c^b(X, \Lambda)$, we have

$$(2.1.9) \quad \text{supp} F_\psi(e^* \mathcal{M}) \subseteq 0^\vee(X) \subseteq E^\vee.$$

Proof. 1. By proper base change theorem, we obtain an isomorphism (2.1.5).

2. Let $\mathcal{F} \in D_c^b(X, \Lambda)$. Since the restriction $\mu^* \mathcal{L}_\psi|_{X \times_X E^\vee}$ is canonically identified with Λ , we have an isomorphism $F_\psi 0_* \mathcal{F} = \text{pr}_{2!}(\text{pr}_1^* 0_* \mathcal{F} \otimes \mu^* \mathcal{L}_\psi) \rightarrow e^{\vee*} \mathcal{F}$ (2.1.6).

3. Let $\mathcal{F} \in D_c^b(X, \Lambda)$. We have an isomorphism $F_\psi(e^! \mathcal{F}) = \text{pr}_{2!}(\text{pr}_2^* e^{\vee!} \mathcal{F} \otimes \mu^* \mathcal{L}_\psi) \rightarrow e^{\vee!} \mathcal{F} \otimes F_\psi \Lambda$ by the projection formula. By the proper base change theorem, the restriction of $F_\psi \Lambda = \text{pr}_{2!} \mu^* \mathcal{L}_\psi$ on the complement $E^\vee - 0^\vee(X)$ of the 0-section vanishes. Hence the isomorphism (2.1.5) induces an isomorphism (2.1.7).

By the canonical isomorphism $e^*(-) \otimes e^! \Lambda \rightarrow e^!$ of Poincaré duality, by identifying $e^{\vee*} - \otimes 0_*^\vee e_! \Lambda = 0_*^\vee(- \otimes e_! \Lambda)$, we obtain (2.1.8). $\square$

The following is a fundamental result of Katz and Laumon.

Proposition 2.1.4. 1. ([KL85, Théorème (2.4.4)]). *Let $j: \mathbf{G}_m \times \mathbf{A}^1 \rightarrow \mathbf{A}^1 \times \mathbf{P}^1$ be the open immersion over k and let $\mathcal{L}_\psi(y/x)$ denote the pull-back of $\mathcal{L}_\psi$ by the morphism $y/x: \mathbf{G}_m \times \mathbf{A}^1 = \text{Spec } k[x^{\pm 1}, y] \rightarrow \mathbf{A}^1$. Then, the projection $\text{pr}_1: \mathbf{A}^1 \times \mathbf{P}^1 \rightarrow \mathbf{A}^1$ is universally locally acyclic relatively to $j_! \mathcal{L}_\psi(y/x)$.*

2. ([KL85, (a) in the proof of Théorème (2.4.1)]). *Let X be a noetherian scheme and consider the cartesian diagram*

$$(2.1.10) \quad \begin{array}{ccc} \mathbf{A}_X^2 & \xrightarrow{j_2} & \mathbf{P}_X^1 \times_X \mathbf{A}_X^1 \\ \text{pr}_1 \downarrow & & \downarrow \overline{\text{pr}}_1 \\ \mathbf{A}_X^1 & \xrightarrow{j} & \mathbf{P}_X^1. \end{array}$$

Then, the morphism $j_{2!}(\text{pr}_1^(-) \otimes \mu^* \mathcal{L}_\psi) \rightarrow j_{2*}(\text{pr}_1^*(-) \otimes \mu^* \mathcal{L}_\psi)$ is an isomorphism.*

Proof. 1. First, we show the universal local acyclicity on $\mathbf{A}^2 = \operatorname{Spec} k[x, y] \subseteq \mathbf{A}^1 \times \mathbf{P}^1$. It suffices to show the universal local acyclicity after the base change by the purely inseparable morphism $\mathbf{A}^1 \rightarrow \mathbf{A}^1$ sending x to x^p . The normalization of the covering of $\mathbf{A}^2$ defined by $t^p - t = y/x^p$ is $\mathbf{A}^2 = \operatorname{Spec} k[x, s] \rightarrow \mathbf{A}^2 = \operatorname{Spec} k[x, y]$ given by $s^p - x^{p-1}s = y$ for $s = xt$ and the projection $\mathbf{A}^2 = \operatorname{Spec} k[x, s] \rightarrow \mathbf{A}^1 = \operatorname{Spec} k[x]$ is smooth. Hence the assertion follows from the local acyclicity of smooth morphisms.

Next, we show the universal local acyclicity on $\mathbf{A}^1 \times (\mathbf{P}^1 - \{0\})$. Changing the coordinates and notation, it suffices to show that the projection $\mathbf{A}^2 = \operatorname{Spec} k[x, y] \rightarrow \mathbf{A}^1 = \operatorname{Spec} k[x]$ is universally locally acyclic with respect to $j_! \mathcal{L}_\psi(1/xy)$ for the open immersion $j: \mathbf{G}_m^2 \rightarrow \mathbf{A}^2$. Let $g: \mathbf{A}^1 \times \mathbf{G}_m = \operatorname{Spec} k[x, u^{\pm 1}] \rightarrow \mathbf{A}^2 = \operatorname{Spec} k[x, u]$ be the open immersion. By the generic universal local acyclicity [SGA 4 $\frac{1}{2}$ -TF, Corollaire 2.16], the projection $p_1: \mathbf{A}^2 = \operatorname{Spec} k[x, u] \rightarrow \mathbf{A}^1 = \operatorname{Spec} k[x]$ is universally locally acyclic relatively to $g_! \mathcal{L}_\psi(1/u)$. The blow-up $q: P \rightarrow \mathbf{A}^2$ at the origin is the union of two affine planes $A = \mathbf{A}^2 = \operatorname{Spec} k[x, y]$ and $B = \mathbf{A}^2 = \operatorname{Spec} k[u, v]$ where $u = xy$ and $x = uv$ respectively. Since q is an isomorphism outside the exceptional divisor E , the composition $p_1 \circ q: P \rightarrow \mathbf{A}^1$ is locally acyclic on the complement $P - E$. Further, by the first step above, the composition $p_1 \circ q: P \rightarrow \mathbf{A}^1$ is locally acyclic relatively to $q^* g_! \mathcal{L}_\psi(1/u)$ except possibly at the origins of A and B . Since $q: P \rightarrow \mathbf{A}^2$ is proper, by proper base change theorem, the composition $p_1 \circ q: P \rightarrow \mathbf{A}^1$ is everywhere locally acyclic relatively to $q^* g_! \mathcal{L}_\psi(1/u)$. Since the restriction of $q^* g_! \mathcal{L}_\psi(1/u)$ on A is the same as $j_! \mathcal{L}_\psi(1/xy)$, the assertion follows.

2. Let $\mathcal{F} \in D_c^b(\mathbf{A}_X^1, \Lambda)$. By 1, the projection $\overline{\operatorname{pr}}_1: \mathbf{P}_X^1 \times_X \mathbf{A}_X^1 \rightarrow \mathbf{P}_X^1$ is locally acyclic relatively to $j_{2!} \mu^* \mathcal{L}_\psi$. Hence by [Ill77, Proposition 2.10], the morphism $j_{2!}(\operatorname{pr}_1^* \mathcal{F} \otimes \mu^* \mathcal{L}_\psi) = \overline{\operatorname{pr}}_1^* j_* \mathcal{F} \otimes j_{2!} \mu^* \mathcal{L}_\psi \rightarrow j_{2*}(\operatorname{pr}_1^* \mathcal{F} \otimes \mu^* \mathcal{L}_\psi)$ is an isomorphism. $\square$

Proposition 2.1.4.2 implies that the canonical morphism

$$(2.1.11) \quad F_\psi \mathcal{F} = \operatorname{pr}_{2!}(\operatorname{pr}_1^* \mathcal{F} \otimes \mu^* \mathcal{L}_\psi) \rightarrow \operatorname{pr}_{2*}(\operatorname{pr}_1^* \mathcal{F} \otimes \mu^* \mathcal{L}_\psi)$$

is an isomorphism [KL85, Théorème (2.4.1)].

Corollary 2.1.5. *Let E be a line bundle on X and $E^\vee$ be the dual. Let $g: E^\times = E - X \rightarrow E$ and $g^\vee: E^{\vee \times} = E^\vee - X \rightarrow E^\vee$ be the open immersions of the complements of the 0-sections. Let $0: X \rightarrow E$ be the 0-section and define a morphism $0_* \Lambda_X \rightarrow g_! \Lambda_{E^\times}[1]$ by the exact sequence $0 \rightarrow g_! \Lambda_{E^\times} \rightarrow \Lambda_E \rightarrow 0_* \Lambda_X \rightarrow 0$. Then, there exists a unique isomorphism*

$$(2.1.12) \quad F_\psi g_! \Lambda_{E^\times}[1] \rightarrow g_*^\vee \Lambda_{E^{\vee \times}}$$

fitting in the commutative diagram

$$(2.1.13) \quad \begin{array}{ccc} F_\psi 0_* \Lambda_X & \xrightarrow{(2.1.6)} & \Lambda_{E^\vee} \\ \downarrow & & \downarrow \\ F_\psi g_! \Lambda_{E^\times}[1] & \xrightarrow{(2.1.12)} & g_*^\vee \Lambda_{E^{\vee \times}}. \end{array}$$

Proof. By the isomorphism $F_\psi \Lambda_E \rightarrow 0_*^\vee \Lambda_X(-1)[-2]$, the exact sequence $0 \rightarrow g_! \Lambda_{E^\times} \rightarrow \Lambda_E \rightarrow \Lambda_X \rightarrow 0$ induces an isomorphism $(F_\psi 0_* \Lambda_X)|_{E^{\vee \times}} = \Lambda_{E^{\vee \times}} \rightarrow F_\psi g_! \Lambda_{E^\times}[1]|_{E^{\vee \times}}$. By adjunction, there exists a unique morphism $F_\psi g_! \Lambda_{E^\times}[1] \rightarrow g_*^\vee \Lambda_{E^{\vee \times}}$ making (2.1.13) commutative. Its restriction on $E^{\vee \times}$ is an isomorphism.

It suffices to show that the morphism $F_\psi g_! \Lambda \rightarrow g_*^\vee g^{\vee*} F_\psi g_! \Lambda$ is an isomorphism. We consider the commutative diagram

$$(2.1.14) \quad \begin{array}{ccc} F_\psi g_! \Lambda & \longrightarrow & \mathrm{pr}_{2*}(\mathrm{pr}_1^* g_! \Lambda \otimes \mu^* \mathcal{L}) \\ \downarrow & & \downarrow \\ g_*^\vee g^{\vee*} F_\psi g_! \Lambda & \longrightarrow & g_*^\vee \mathrm{pr}_{2*}((\mathrm{pr}_1^* g_! \Lambda \otimes \mu^* \mathcal{L})|_{E \times_X E^{\vee \times}}). \end{array}$$

By the isomorphism (2.1.11), the horizontal arrows are isomorphisms. It suffices to show that the right vertical arrow is an isomorphism. Since $e_* g_! \Lambda = 0$, the morphism $\mathrm{pr}_{2*}(\mathrm{pr}_1^* g_! \Lambda \otimes \mu^* \mathcal{L}) \rightarrow \mathrm{pr}_{2*}((g_! \Lambda \boxtimes g_*^\vee \Lambda) \otimes \mu^* \mathcal{L})$ is an isomorphism. Let $g_2^\vee: E \times_X E^{\vee \times} \rightarrow E \times_X E^\vee$ be the open immersion. Then $\mathrm{pr}_{2*}((g_! \Lambda \boxtimes g_*^\vee \Lambda) \otimes \mu^* \mathcal{L}) \rightarrow \mathrm{pr}_{2*} g_{2*}^\vee((\mathrm{pr}_1^* g_! \Lambda \otimes \mu^* \mathcal{L})|_{E \times_X E^{\vee \times}})$ is an isomorphism and the assertion follows. $\square$

Proposition 2.1.6 ([Lau87, Théorème (1.2.2.1)]). *Let $e: E \rightarrow X$ be a vector bundle over a noetherian scheme X over k and let $e^\vee: E^\vee \rightarrow X$ be the dual vector bundle. There exists a canonical isomorphism*

$$(2.1.15) \quad i: F_\psi F_{\psi^\vee} \rightarrow - \otimes e^{\vee*} e_! \Lambda$$

of functors $D_c^b(E^\vee, \Lambda) \rightarrow D_c^b(E^\vee, \Lambda)$.

Proof. Let $\mathcal{L}_\psi(x(y^\vee - x^\vee))$ denote the pull-back of $\mathcal{L}_\psi$ by the morphism $E^\vee \times_X E \times_X E^\vee \rightarrow \mathbf{A}^1$ sending $(x^\vee, x, y^\vee)$ to $\mu(x, y^\vee - x^\vee)$. We consider the cartesian diagram

$$(2.1.16) \quad \begin{array}{ccccc} E & \xleftarrow{\mathrm{pr}_1} & E \times_X E^\vee & \xrightarrow{\mathrm{pr}_2} & E^\vee \\ \mathrm{pr}_2 \uparrow & & \mathrm{pr}_{23} \uparrow & \nearrow \mathrm{pr}_3 & \uparrow \mathrm{pr}_2 \\ E^\vee \times_X E & \xleftarrow{\mathrm{pr}_{12}} & E^\vee \times_X E \times_X E^\vee & \xrightarrow{\mathrm{pr}_{13}} & E^\vee \times_X E^\vee \\ \mathrm{pr}_1 \downarrow & \nwarrow \mathrm{pr}_1 & \delta \times 1 \uparrow & & \uparrow \delta \\ E^\vee & & E^\vee \times_X E & \xrightarrow{\mathrm{pr}_1} & E^\vee. \end{array}$$

By the proper base change theorem applied to the lower right square, the pull-back by the lower middle vertical arrow induces isomorphisms

$$(2.1.17) \quad \mathrm{pr}_{13!} \mathcal{L}_\psi(x(y^\vee - x^\vee)) \rightarrow \mathrm{pr}_{13!}(\delta \times 1)_*(\delta \times 1)^* \mathcal{L}_\psi(x(y^\vee - x^\vee)) \rightarrow \delta_* \mathrm{pr}_{1!} \Lambda \rightarrow \delta_* e^{\vee*} e_! \Lambda.$$

By proper base change theorem, the projection formula and by (2.1.17), we obtain isomorphisms

$$(2.1.18) \quad \begin{aligned} F_\psi F_{\psi^\vee} &\rightarrow \mathrm{pr}_{3!}(\mathrm{pr}_1^* - \otimes \mathcal{L}_\psi(x(y^\vee - x^\vee))) \rightarrow \mathrm{pr}_{2!}(\mathrm{pr}_1^* - \otimes \mathrm{pr}_{13!} \mathcal{L}_\psi(x(y^\vee - x^\vee))) \\ &\rightarrow \mathrm{pr}_{2!}(\delta_*(- \otimes e^{\vee*} e_! \Lambda)) \rightarrow - \otimes e^{\vee*} e_! \Lambda. \end{aligned}$$

$\square$

In the following, we fix ψ and drop it from the notation. The isomorphism (2.1.15) is equivalently reformulated as an isomorphism

$$(2.1.19) \quad i: F(F^\vee(-) \otimes e^! \Lambda) \rightarrow 1.$$

Proposition 2.1.7. *Let $e: E \rightarrow X$ be a vector bundle of rank n over a noetherian scheme X over k and let $e^\vee: E^\vee \rightarrow X$ be the dual vector bundle. The diagram*

$$(2.1.20) \quad \begin{array}{ccc} F(F^\vee e^{\vee!}(-) \otimes e^! \Lambda) & \xrightarrow{i} & e^{\vee!} \\ \textcolor{red}{(2.1.8)} \downarrow & & \uparrow (-1)^n \text{can} \\ F(0_*(-) \otimes e^! \Lambda) & \xrightarrow{\textcolor{red}{(2.1.6)}} & e^{\vee*}(-) \otimes e^{\vee*} 0^* e^! \Lambda \end{array}$$

of functors $D_c^b(X, \Lambda) \rightarrow D_c^b(E^\vee, \Lambda)$ is commutative. The upper horizontal arrow is defined by (2.1.19) and the right vertical arrow is $(-1)^n$ -times the morphism induced by the canonical isomorphism $e^{\vee*} 0^* e^! \Lambda \rightarrow e^{\vee!} \Lambda$.

Proof. We consider closed immersions

$$(2.1.21) \quad E^\vee \times_X E^\vee \xrightarrow{(1,0,1)} E^\vee \times_X E \times_X E^\vee \xleftarrow{\delta \times 1} E^\vee \times_X E$$

defined by $(\delta \times 1)(x^\vee, x) = (x^\vee, x, x^\vee)$. Let $p_1, p_3: E^\vee \times_X E^\vee \rightarrow E^\vee$ and $q: E^\vee \times_X E \rightarrow E^\vee$ be the restrictions of the projections $\text{pr}_1, \text{pr}_3: E^\vee \times_X E \times_X E^\vee \rightarrow E^\vee$. By the proof of Proposition 2.1.6, the upper horizontal morphism $i: F(F^\vee e^{\vee!} \mathcal{F} \otimes e^! \Lambda) \rightarrow e^{\vee!} \mathcal{F}$ is induced by the composition via upper right in the diagram

$$(2.1.22) \quad \begin{array}{ccc} \text{pr}_{3!}(\text{pr}_1^* e^{\vee!} \mathcal{F} \otimes \mathcal{L}((y^\vee - x^\vee)x) \otimes \text{pr}_2^* e^! \Lambda) & \longrightarrow & q_!(e^{\vee!} \mathcal{F} \boxtimes e^! \Lambda) \\ \downarrow & & \downarrow \\ p_{3!}(\text{pr}_1^* e^{\vee!} \mathcal{F} \otimes (e^\vee \times e^\vee)^* 0^* e^! \Lambda) & \longrightarrow & e^{\vee!} \mathcal{F} \end{array}$$

for $\mathcal{F} \in D_c^b(X, \Lambda)$. The upper horizontal arrow and the left vertical arrow are induced by the pull-back by the immersions in (2.1.21) respectively. The right vertical arrow is induced by the isomorphism $\text{Tr}_{E/X}: e_! e^! \rightarrow 1$. The lower horizontal arrow is induced by the isomorphism $\text{Tr}_{E^\vee/X}: e_!^\vee e^{\vee!} \rightarrow 1$ for the first factor $E^\vee$ and the canonical isomorphism $e^{\vee*} 0^* e^! \Lambda \rightarrow e^{\vee!} \Lambda$. The arrows in (2.1.20) via lower left are induced by the corresponding arrows in (2.1.22). Hence it suffices to show that (2.1.22) is $(-1)^n$ -commutative. By the projection formula, we may assume $\mathcal{F} = \Lambda$.

After changing the coordinate $(x^\vee, x, y^\vee)$ by $(y^\vee - x^\vee, x, y^\vee)$, the immersions in (2.1.21) are the base change by $E^\vee \rightarrow X$ of the immersions $E^\vee \rightarrow E \times_X E^\vee \leftarrow E$ and $\mathcal{L}((y^\vee - x^\vee)x)$ is the pull-back of $\mathcal{L}(z^\vee x)$. Hence, it suffices to show Lemma 2.1.8.1 below. $\square$

Lemma 2.1.8. 1. *Let $e: E \rightarrow X$ be a vector bundle of rank n and $e^\vee: E^\vee \rightarrow X$ be the dual. Then, the diagram*

$$(2.1.23) \quad \begin{array}{ccccc} (e \times e^\vee)_! \mu^* \mathcal{L}_\psi(n)[2n] & \xlongequal{\quad} & e_!^\vee F_\psi e^! \Lambda & \xrightarrow{\textcolor{red}{(2.1.7)}} & e_!^\vee 0_*^\vee \Lambda \\ \parallel & & & & \downarrow \\ e_! F_\psi^\vee e^{\vee!} \Lambda & \xrightarrow{\textcolor{red}{(2.1.7)}} & e_! 0_* \Lambda & \longrightarrow & \Lambda \end{array}$$

of isomorphisms of sheaves on X is $(-1)^n$ -commutative.

2. Let $\mu: \mathbf{A}^2 = \text{Spec } k[x, y] \rightarrow \mathbf{A}^1 = \text{Spec } k[t]$ be the morphism defined by $t = xy$ and let $\Phi_0 \Lambda$ denote the vanishing cycles complex at $0 \in \mathbf{A}^2$ with respect to μ . Then, we have

an isomorphism $R^1\mu_!\Lambda \rightarrow \Lambda$, an exact sequence $0 \rightarrow \Phi_0^1\Lambda \rightarrow R^2\mu_!\Lambda \rightarrow \Lambda(-1) \rightarrow 0$ and $R^q\mu_!\Lambda = 0$ for $q \neq 1, 2$.

3. Let k be an algebraically closed field and let σ be the involution of $\mathbf{A}^2 = \text{Spec } k[x, y]$ switching x and y . Then σ acts on $H_c^2(\mathbf{A}^2, \mathcal{L}_\psi(xy))$ by -1 .

Proof. 1. Since the assertion is local on X , we may assume that $E = \mathbf{A}_X^n$. By the Künneth formula, we may assume that $n = 1$. Further, we may assume that $X = \text{Spec } k$ for an algebraically closed field k . Hence, the assertion is reduced to 3.

2. We construct a relative compactification $\bar{\mu}: P \rightarrow \mathbf{A}^1$. The second projection $\mathbf{P}^1 \times \mathbf{A}^1 \rightarrow \mathbf{A}^1 = \text{Spec } k[t]$ is proper. Let $P \rightarrow \mathbf{P}^1 \times \mathbf{A}^1$ be the blow-up at the origin of $\mathbf{A}^2 = \text{Spec } k[y, t] \subseteq \mathbf{P}^1 \times \mathbf{A}^1$. Let L_0 be the proper transform of $(y = 0) \subseteq \mathbf{A}^2$ and L_∞ be the inverse image of $(\mathbf{P}^1 \times \mathbf{A}^1) - \mathbf{A}^2$. Then, the complement $P - (L_0 \cup L_\infty)$ is identified with $\mathbf{A}^2 = \text{Spec } k[x, y]$ by $xy = t$ and hence $P \rightarrow \mathbf{P}^1 \times \mathbf{A}^1 \rightarrow \mathbf{A}^1 = \text{Spec } k[t]$ is a relative compactification of $\mu: \mathbf{A}^2 = \text{Spec } k[x, y] \rightarrow \mathbf{A}^1$.

Let $j: \mathbf{A}^2 = \text{Spec } k[x, y] = P - (L_0 \cup L_\infty) \rightarrow P$ be the open immersion. The morphism $\bar{\mu}: P \rightarrow \mathbf{A}^1$ is smooth except at $0 \in \mathbf{A}^2$ and L_0 and L_∞ are sections of $\bar{\mu}: P \rightarrow \mathbf{A}^1$. Hence the morphism $P \rightarrow \mathbf{A}^1$ is locally acyclic relatively to $j_!\Lambda$ except at the origin $0 \in \mathbf{A}^2$. Let $\bar{\eta}_0$ be the generic geometric point of the strict henselization at $0 \in \mathbf{A}^1$ and let $\Phi_0\Lambda$ denote the vanishing cycles complex at 0 with respect to μ . By the Picard–Lefschetz formula, the complex $\Phi_0\Lambda$ is acyclic except at degree 1 and the Λ -module $\Phi_0^1\Lambda$ is free of rank 1. Let $+ = \mathbf{A}^2 \times_{\mathbf{A}^1} 0$ and $\mathbf{G}_{m, \bar{\eta}_0} = \mathbf{A}^2 \times_{\mathbf{A}^1} \bar{\eta}_0$ be the fibers of $\mu: \mathbf{A}^2 \rightarrow \mathbf{A}^1$. Then, the assertion follows from the distinguished triangle $\Gamma_c(+, \Lambda) \rightarrow \Gamma_c(\mathbf{G}_{m, \bar{\eta}_0}, \Lambda) \rightarrow \Phi_0\Lambda \rightarrow$.

3. By the projection formula, we have a canonical isomorphism $\mu_!\mathcal{L}_\psi(xy) \rightarrow \mathcal{L}_\psi \otimes \mu_!\Lambda$. Hence by 2. and $\Gamma_c(\mathbf{A}^1, \mathcal{L}_\psi) = 0$, we have a canonical isomorphism $\Phi_0^1\Lambda \rightarrow H_c^0(\mathbf{A}^1, \mathcal{L}_\psi \otimes R^2\mu_!\Lambda) \rightarrow H_c^2(\mathbf{A}^2, \mathcal{L}_\psi(xy))$. Since σ acts on $\Phi_0^1\Lambda$ by -1 by the Picard–Lefschetz formula, the assertion follows. $\square$

We study relations of Fourier transforms with pull-back and push-forward by linear morphisms of vector bundles.

Proposition 2.1.9 ([Lau87, Théorème (1.2.2.4)]). *Let $a: E \rightarrow E'$ be a linear morphism of vector bundles on a noetherian scheme X over k and $a^\vee: E'^\vee \rightarrow E^\vee$ be the dual. Let F and F' denote the Fourier transforms for E and E' respectively. We have a canonical isomorphism*

$$(2.1.24) \quad d: F'a_! \rightarrow a^{\vee*}F$$

of functors $D_c^b(E, \Lambda) \rightarrow D_c^b(E'^\vee, \Lambda)$.

The morphism (2.1.6) is the case where $a: X \rightarrow E$ is the 0-section.

Proof. By applying the proper base change theorem to the cartesian squares in the commutative diagram

$$(2.1.25) \quad \begin{array}{ccccc} & & E \times_X E^\vee & \longrightarrow & E^\vee \\ & \swarrow & \uparrow & & \uparrow a^\vee \\ E & \longleftarrow & E \times_X E'^\vee & \longrightarrow & E'^\vee \\ \downarrow a & & \downarrow & \nearrow & \\ E' & \longleftarrow & E' \times_X E'^\vee & & \end{array}$$

and by the projection formula, we obtain isomorphisms

$$\begin{aligned}
(2.1.26) \quad F' a_! &= \mathrm{pr}'_{2!}(\mathrm{pr}_1'^*(a_! -) \otimes \mathcal{L}_\psi(x' x'^\vee)) \rightarrow \mathrm{pr}'_{2!}(a \times 1_{E'^\vee})_!(\mathrm{pr}_1'^*(-) \otimes \mathcal{L}_\psi(a(x) x'^\vee)) \\
&\rightarrow \mathrm{pr}'_{2!}(1_E \times a^\vee)^*(\mathrm{pr}_1^*(-) \otimes \mathcal{L}_\psi(x x^\vee)) \\
&\rightarrow a^{\vee*} \mathrm{pr}_{2!}(\mathrm{pr}_1^*(-) \otimes \mathcal{L}_\psi(x x^\vee)) = a^{\vee*} F.
\end{aligned}$$

□

Corollary 2.1.10. *Let the notation be as in Proposition 2.1.9 and let $F^\vee$ and $F'^\vee$ denote the dual Fourier transforms for E and E' respectively (cf. (2.1.4)).*

1. *There exists a unique isomorphism*

$$(2.1.27) \quad d^\vee: F(a^*(-) \otimes e^! \Lambda) \rightarrow a_!^\vee F'(- \otimes e'^! \Lambda)$$

of functors $D_c^b(E', \Lambda) \rightarrow D_c^b(E^\vee, \Lambda)$ such that the diagram

$$\begin{array}{ccc}
F(a^* F'^\vee(-) \otimes e^! \Lambda) & \xrightarrow{d^\vee} & a_!^\vee F'(F'^\vee(-) \otimes e'^! \Lambda) \\
d \uparrow & & \downarrow i_{E'} \\
F(F^\vee a_!^\vee(-) \otimes e^! \Lambda) & \xrightarrow{i_E} & a_!^\vee
\end{array}
\quad (2.1.28)$$

is commutative. The left vertical arrow is an isomorphism induced by the isomorphism d for $a^\vee: E'^\vee \rightarrow E^\vee$ under the canonical identifications $E^{\vee\vee} = E$ and $E'^{\vee\vee} = E'$.

If $a: E \rightarrow E'$ is smooth, (2.1.27) is equivalently reformulated as an isomorphism

$$(2.1.29) \quad d^\vee: F a^! \rightarrow a_*^\vee F'.$$

2. *The isomorphism $F e^! \rightarrow 0_!^\vee$ (2.1.7) is the special case of $d^\vee$ (2.1.29) where $a: E \rightarrow X$ is the structure morphism. Namely, the diagram*

$$\begin{array}{ccc}
F(e^*(-) \otimes e^! \Lambda) & = & F e^! \xrightarrow{(2.1.7)} 0_!^\vee \\
d \uparrow & \nearrow i_E & \\
F(F^\vee 0_!^\vee(-) \otimes e^! \Lambda) & &
\end{array}
\quad (2.1.30)$$

is commutative.

Proof. 1. The functor $F'^\vee: D_c^b(E'^\vee, \Lambda) \rightarrow D_c^b(E', \Lambda)$ is an equivalence of categories by Proposition 2.1.6. Hence the commutative diagram (2.1.28) uniquely defines a natural isomorphism $d^\vee$.

2. By the characterization of $d^\vee$, it suffices to show that the diagram (2.1.30) is commutative. By the proof of Proposition 2.1.7, $i_E: F(F^\vee 0_!^\vee(-) \otimes e^! \Lambda) \rightarrow 0_!^\vee$ equals the composition of the isomorphism $F(F^\vee 0_!^\vee(-) \otimes e^! \Lambda) \rightarrow \mathrm{pr}_{1!}(0_!^\vee(-) \boxtimes e^! \Lambda)$ induced by the pull-back by $\delta \times 1: E^\vee \times E \rightarrow E^\vee \times E \times E^\vee$ and the isomorphism $\mathrm{pr}_{1!}(0_!^\vee(-) \boxtimes e^! \Lambda) \rightarrow 0_!^\vee(-) \otimes e_! e^! \Lambda \rightarrow 0_!^\vee$ induced by $\mathrm{Tr}_{E/X}$. Since $F(e^*(-) \otimes e^! \Lambda) \rightarrow F e^! \rightarrow 0_!^\vee$ (2.1.7) is also induced by $\mathrm{Tr}_{E/X}$, we have the equality as required. □

Let $a: E \rightarrow E'$ be a linear morphism of vector bundles on a noetherian scheme X . Canonical morphisms

$$(2.1.31) \quad \mathrm{adj}_a: a_! e^! \rightarrow e'^!,$$

$$(2.1.32) \quad \mathrm{adj}_a: 0_! \rightarrow a^! 0_!'$$

are defined to be the adjoints of the canonical isomorphisms $e^! \rightarrow a^!e'^!$ and $a_!0_! \rightarrow 0_!'$. In particular, (2.1.31) for the 0-section $a = 0': X \rightarrow E'$ defines $\text{adj}_{e'}: 0_! \rightarrow e'^!$. Similarly,

$$(2.1.33) \quad \text{res}_a: a^*0'_* \rightarrow 0_*,$$

$$(2.1.34) \quad \text{res}_a: e'^* \rightarrow a_*e^*$$

are defined as the adjoint of $0'_* \rightarrow a_*0_*$ and $a^*e'^* \rightarrow e^*$. In particular, (2.1.33) for the canonical morphism $a: E \rightarrow E' = X$ defines $\text{res}_e: e^* \rightarrow 0_*$. We show that the Fourier transforms of adjunction morphisms are identified with the restriction morphisms. We also show that dual statements involve signs.

Proposition 2.1.11. *Let $a: E \rightarrow E'$ be a linear morphism of vector bundles on a noetherian scheme X over k and $a^\vee: E^\vee \rightarrow E'^\vee$ be the dual. Let F and F' denote the Fourier transforms for E and E' respectively.*

1. *The diagram*

$$(2.1.35) \quad \begin{array}{ccc} F'a_!e^! & \xrightarrow{d} & a^{\vee*}Fe^! \xrightarrow{(2.1.7)_E} a^{\vee*}0_*^\vee \\ \text{adj}_a \downarrow & & \swarrow \text{res}_{a^\vee} \\ F'e^! & \xrightarrow{(2.1.7)_{E'}} & 0_*'^\vee \end{array}$$

is commutative. In particular, the diagram

$$(2.1.36) \quad \begin{array}{ccc} F_\psi e^! & \xrightarrow{(2.1.7)} & 0_*^\vee \\ \text{adj}_e \uparrow & & \uparrow \text{res}_{e^\vee} \\ F_\psi 0_! & \xrightarrow{(2.1.6)} & e^{\vee*} \end{array}$$

is commutative.

2. *Assume that $a: E \rightarrow E'$ is an immersion of codimension $r = n' - n = \text{rank } E' - \text{rank } E$. Then, the diagram*

$$(2.1.37) \quad \begin{array}{ccc} F'a_*e^* & \xrightarrow{d} & a^{\vee*}Fe^* \xrightarrow{(2.1.8)_E} a^{\vee*}0_*^\vee(-\otimes e_!\Lambda) \\ \text{res}_a \uparrow & & \nearrow \text{adj}_{a^\vee} \\ F'e'^* & \xrightarrow{(2.1.8)_{E'}} & 0_*'^\vee(-\otimes e'_!\Lambda) \end{array}$$

is $(-1)^r$ -commutative. The slant arrow is defined by canonically identifying the upper right corner $a^{\vee*}0_*^\vee(-\otimes e_!\Lambda)$ with $a^{\vee!}0_!^\vee(-\otimes e'_!\Lambda)$. In particular, the diagram

$$(2.1.38) \quad \begin{array}{ccc} F_\psi 0_*' & \xrightarrow{(2.1.6)} & e'^{\vee*} \\ \text{res} \uparrow & & \uparrow \text{adj} \\ F_\psi e'^* & \xrightarrow{(2.1.8)} & 0_*'^\vee(-\otimes e'_!\Lambda) \end{array}$$

is $(-1)^{n'}$ -commutative.

Proof. 1. By taking the adjoint, it suffices to show that the diagram

$$(2.1.39) \quad \begin{array}{ccc} 0'^{\vee*} F' a_! e^! & \xrightarrow{d} & 0'^{\vee*} a^{\vee*} F e^! \xrightarrow{(2.1.7)_E} 0'^{\vee*} a^{\vee*} 0_*^{\vee} \\ \text{adj}_a \downarrow & & \nearrow \text{res}_{a^{\vee}} \\ 0'^{\vee*} F' e^! & \xrightarrow{(2.1.7)_{E'}} & 1 \end{array}$$

is commutative. By the definition of (2.1.7) as the adjoint and by the isomorphism (2.1.5), it is reduced to the commutative diagram

$$(2.1.40) \quad \begin{array}{ccc} e'_! a_! e^! & \longrightarrow & e_! e^! \\ \text{adj}_a \downarrow & & \downarrow \\ e'_! e^! & \longrightarrow & 1 \end{array}$$

where the arrows without labels are canonical isomorphisms.

For (2.1.36), it is enough to consider the case $a = 0: X \rightarrow E$.

2. By taking the adjoint, it suffices to show that the diagram

$$(2.1.41) \quad \begin{array}{ccc} a_!^{\vee} F' a_* e^* & \xrightarrow{d} & a_!^{\vee} a^{\vee*} F e^* \xrightarrow{(2.1.8)_E} 0_!^{\vee} (- \otimes e'_! \Lambda) \\ \text{res}_a \uparrow & & \nearrow \text{can} \\ a_!^{\vee} F' e'^* & \xrightarrow{(2.1.8)_{E'}} & a_!^{\vee} 0_!^{\vee} (- \otimes e'_! \Lambda) \end{array}$$

is $(-1)^r$ -commutative. Since every term is supported on the 0-section of $E^{\vee}$, it suffices to show the $(-1)^r$ -commutativity after applying $e_!^{\vee}$. Then by the projection formula, it suffices to show the $(-1)^r$ -commutativity after evaluating the functors at Λ . Namely, it is reduced to showing that the diagram

$$(2.1.42) \quad \begin{array}{ccc} e_!^{\vee} F' a_* \Lambda_E & \xrightarrow{d} & e_!^{\vee} a^{\vee*} F \Lambda_E \\ \text{res}_a \uparrow & & \downarrow (2.1.8)_E \\ e_!^{\vee} F' \Lambda_{E'} & \xrightarrow{(2.1.8)_{E'}} & e_!^{\vee} \Lambda_{E'} \end{array}$$

is $(-1)^r$ -commutative. Since the assertion is local on X , by the Künneth formula, we may further assume that $a: X = E \rightarrow E'$ is the 0-section. In this case, the $(-1)^r = (-1)^{n'}$ -commutativity follows from Lemma 2.1.8.1.

For (2.1.38), it is enough to consider the case $a = 0: X \rightarrow E'$. □

We deduce an analogue of Proposition 2.1.11 for $d^{\vee}$.

Proposition 2.1.12. *Let $a: E \rightarrow E'$ be a linear morphism of vector bundles on a noetherian scheme X over k and $a^{\vee}: E'^{\vee} \rightarrow E^{\vee}$ be the dual. Let $n = \text{rank } E$, $n' = \text{rank } E'$ and $r = n - n'$.*

1. *The diagram*

$$(2.1.43) \quad \begin{array}{ccc} F(a^* 0'_*(-) \otimes e^! \Lambda) & \xrightarrow{d^{\vee}} & a_!^{\vee} F'(0'_*(-) \otimes e'^! \Lambda) \xrightarrow{(2.1.6)_{E'}} a_!^{\vee} e'^{\vee!} \\ \text{res}_a \downarrow & & \nearrow (-1)^r \text{adj}_{a^{\vee}} \\ F(0_*(-) \otimes e^! \Lambda) & \xrightarrow{(2.1.6)_E} & e^{\vee!} \end{array}$$

is commutative. The bottom horizontal arrow $(2.1.6)_E$ is the composition $F(0_*(-) \otimes e^! \Lambda) \xrightarrow{(2.1.6)} e^{\vee*}(-) \otimes e^{\vee*} 0^* e^! \Lambda \rightarrow e^{\vee*}(-) \otimes e^{\vee!} \Lambda \rightarrow e^{\vee!}$ and $(2.1.6)_{E'}$ in the upper line is defined similarly.

2. Assume that a is smooth. Then the diagram

$$(2.1.44) \quad \begin{array}{ccc} F a^! 0'_! & \xrightarrow{d^\vee} & a_*^\vee F' 0'_! \xrightarrow{(2.1.6)_{E'}} a_*^\vee e'^{\vee*} \\ \text{adj}_a \uparrow & & \nearrow \text{res}_{a^\vee} \\ F 0_! & \xrightarrow{(2.1.6)_E} & e^{\vee*} \end{array}$$

is commutative.

In [KW00, Lemma 13.5], the diagram (2.1.43) is studied with an unspecified isomorphism.

Proof. 1. Applying the functor $F(- \otimes e^! \Lambda)$ to the diagram (2.1.35) for $a^\vee: E'^\vee \rightarrow E^\vee$, we obtain a commutative diagram

$$(2.1.45) \quad \begin{array}{ccc} F(F^\vee a_!^\vee e'^{\vee!}(-) \otimes e^! \Lambda) & \xrightarrow{d} & F(a^* F'^\vee e'^{\vee!}(-) \otimes e^! \Lambda) \xrightarrow{(2.1.7)_{E'^\vee}} F(a^* 0'_*(-) \otimes e^! \Lambda) \\ \text{adj}_{a^\vee} \downarrow & & \nwarrow \text{res}_a \\ F(F^\vee e^{\vee!}(-) \otimes e^! \Lambda) & \xrightarrow{(2.1.7)_{E^\vee}} & F(0_*(-) \otimes e^! \Lambda). \end{array}$$

We consider the composition of this with (2.1.43) by identifying the slant arrow in (2.1.45) and the left vertical arrow in (2.1.43). By Proposition 2.1.7, the composition $F(F^\vee e^{\vee!}(-) \otimes e^! \Lambda) \rightarrow F(0_*(-) \otimes e^! \Lambda) \rightarrow e^{\vee!}$ of the lower line is $(-1)^n i_E$. Hence by Proposition 2.1.11.1, it suffices to show that the composition of the upper line $F(F^\vee a_!^\vee e'^{\vee!}(-) \otimes e^! \Lambda) \rightarrow a_!^\vee e'^{\vee!}$ is $(-1)^{n'} i_E$. To prove this, we consider the diagram

$$(2.1.46) \quad \begin{array}{ccccc} F(F^\vee a_!^\vee e'^{\vee!}(-) \otimes e^! \Lambda) & \xrightarrow{d} & F(a^* F'^\vee e'^{\vee!}(-) \otimes e^! \Lambda) & \xrightarrow{(2.1.7)_{E'^\vee}} & F(a^* 0'_*(-) \otimes e^! \Lambda) \\ i_E \downarrow & & \downarrow d^\vee & & \downarrow d^\vee \\ a_!^\vee e'^{\vee!} & \xleftarrow{i_{E'}} & a_!^\vee F'(F'^\vee e'^{\vee!}(-) \otimes e'^! \Lambda) & \xrightarrow{(2.1.7)_{E'^\vee}} & a_!^\vee F'(0'_*(-) \otimes e'^! \Lambda). \end{array}$$

The upper line is the same as that of (2.1.45) and the right vertical arrow is the upper left horizontal arrow in (2.1.43). By Proposition 2.1.7, the composition of the lower line is $(-1)^{n'}$ -times the upper right horizontal arrow in (2.1.43) labelled $(2.1.6)_{E'}$. The left square is commutative by Corollary 2.1.10 and the right square is commutative by the functoriality of $d^\vee$. Hence the composition of the upper lines in (2.1.45) and (2.1.43) is $(-1)^{n'} i_E$ and the diagram (2.1.43) is commutative.

2. Applying the functor $F(- \otimes e^! \Lambda)$ to the diagram (2.1.37) for $a^\vee: E'^\vee \rightarrow E^\vee$, we obtain a $(-1)^r$ -commutative diagram

$$(2.1.47) \quad \begin{array}{ccc} F(F^\vee a_*^\vee e'^{\vee*}(-) \otimes e^! \Lambda) & \xrightarrow{d} & F(a^* F'^\vee e'^{\vee*}(-) \otimes e^! \Lambda) \xrightarrow{(2.1.8)_{E'^\vee}} F(a^* 0'_*(-) \otimes a^! \Lambda) \\ \text{res}_{a^\vee} \uparrow & & \nearrow \text{adj}_a \\ F(F^\vee e^{\vee*}(-) \otimes e^! \Lambda) & \xrightarrow{(2.1.8)_{E^\vee}} & F 0_{*}. \end{array}$$

At the upper right corner and at the lower right corner, we identify $a^*(e'^*e_!^\vee\Lambda) \otimes e^!\Lambda$ with $a^!\Lambda$ and $e^*(e_!^\vee\Lambda) \otimes e^!\Lambda$ with Λ respectively by the canonical isomorphisms. We consider the composition of (2.1.47) with (2.1.44) by identifying the slant arrow in (2.1.47) and the left vertical arrow in (2.1.44), similarly as in the proof of 1. By Proposition 2.1.7, the composition $F(F^\vee e^{\vee*}(-) \otimes e^!\Lambda) \rightarrow F0_* \rightarrow e^{\vee*}$ of the lower line is $(-1)^n i_E$. Hence by Proposition 2.1.11.2, it suffices to show that the composition of the upper line $F(F^\vee a_*^\vee e'^{\vee*}(-) \otimes e^!\Lambda) \rightarrow a_*^\vee e'^{\vee*}$ is $(-1)^{n'} i_E$. To prove this, we consider the diagram

$$(2.1.48) \quad \begin{array}{ccccc} F(F^\vee a_*^\vee e'^{\vee*}(-) \otimes e^!\Lambda) & \xrightarrow{d} & F(a^* F'^\vee e'^{\vee*}(-) \otimes e^!\Lambda) & \xrightarrow{(2.1.8)_{E'^\vee}} & F(a^* 0'_*(-) \otimes e^!\Lambda) \\ i_E \downarrow & & \downarrow d^\vee & & \downarrow d^\vee \\ a_*^\vee e'^{\vee*} & \xleftarrow{i_{E'}} & a_*^\vee F'(F'^\vee e'^{\vee*}(-) \otimes e^!\Lambda) & \xrightarrow{(2.1.8)_{E'^\vee}} & a_!^\vee F'0'_* \end{array}$$

The left square is commutative by Corollary 2.1.10 and the right square is commutative by the functoriality of $d^\vee$. By Proposition 2.1.7, the composition of the lower line is $(-1)^{n'}$ -times the upper right horizontal arrow in (2.1.44) labelled $(2.1.6)_{E'}$. Hence the composition of the upper lines in (2.1.47) and (2.1.44) is $(-1)^{n'} i_E$ and the diagram (2.1.44) is commutative. $\square$

Let $X \rightarrow Y$ be a separated morphism of finite type of noetherian schemes over k and E be a vector bundle on Y . Let $b: E \times_Y X \rightarrow E$ and $b^\vee: E^\vee \times_Y X \rightarrow E^\vee$ be the base change morphisms for E and for the dual vector bundle. Let F_X and F_Y denote the Fourier transforms for $E \times_Y X$ and E respectively. Then we have canonical isomorphisms

$$(2.1.49) \quad b_!^\vee F_X \rightarrow F_Y b_!,$$

$$(2.1.50) \quad b^{\vee*} F_Y \rightarrow F_X b^*$$

by the proper base change theorem.

Let X be a smooth curve over a field k and $0 \in X$ be a k -rational point. Let $j: U = X - \{0\} \rightarrow X$ be the open immersion. We identify $R^1 j_* \Lambda(1)$ with Λ_0 by the isomorphism

$$(2.1.51) \quad \Lambda_0 \rightarrow R^1 j_* \Lambda(1)$$

defined by the image of a uniformizer by the boundary morphism $j_* \mathbf{G}_m \rightarrow R^1 j_* \Lambda(1)$. Let $e: E \rightarrow X$ be a line bundle over X and $e^\vee: E^\vee \rightarrow X$ be the dual. Let $E_0 = E \times_X 0$ and $E_0^\vee = E^\vee \times_X 0$ be the fibers. Let $j_E: E - E_0 \rightarrow E$ and $j_{E^\vee}: E^\vee - E_0^\vee \rightarrow E^\vee$ be the base change. Let $E_0^\times = E_0 - \{0\}$ and $E_0^{\vee\times} = E_0^\vee - \{0\}$ be the complement of the origin and let $g_0: E_0^\times \rightarrow E_0$ and $g_0^\vee: E_0^{\vee\times} \rightarrow E_0^\vee$ be the open immersions. We identify the non-trivial cohomology sheaf $\mathcal{H}^{-1}(e^* j_* K_U) = e^* R^1 j_* \Lambda(1)$ at the top degree with Λ_{E_0} by the isomorphism (2.1.51). Define a subcomplex $\mathcal{N} = \mathcal{N}_\Lambda \in D_{\text{ctf}}(E, \Lambda)$ of $e^* j_* K_U$ on E by replacing $\mathcal{H}^{-1}(e^* j_* K_U) = e^* R^1 j_* \Lambda(1) = \Lambda_{E_0}$ by the subsheaf $g_{0!} \Lambda_{E_0^\times}$. Here and in the following, by abuse of notation, we identify a sheaf on a closed subscheme of E with its direct image on E and similarly for $E^\vee$. By the definition, we have a canonical isomorphism

$$(2.1.52) \quad 0^* \mathcal{N}_\Lambda = 0^* e^* \mathcal{H}^{-2}(j_* K_U)[2] \rightarrow K_X.$$

Lemma 2.1.13. 1. *We have a distinguished triangle*

$$(2.1.53) \quad \mathcal{N}_\Lambda \rightarrow e^* j_* K_U \oplus g_{0!} \Lambda_{E_0^\times}[1] \rightarrow \Lambda_{E_0}[1] \rightarrow .$$

2. Let $+: 0^\vee(X) \cup E_0^\vee \rightarrow E^\vee$ be the closed immersion of the union of the 0-section and the fiber $E_0^\vee$. Then, there exists a unique morphism

$$(2.1.54) \quad F_\psi \mathcal{N}_\Lambda \rightarrow +! +^! e^{\vee*} K_X$$

fitting in the commutative diagram

$$(2.1.55) \quad \begin{array}{ccc} F_\psi \mathcal{N}_\Lambda & \xrightarrow{(2.1.54)} & +! +^! e^{\vee*} K_X \\ \downarrow & & \downarrow \text{adj} \\ F_\psi 0_* K_X & \xrightarrow{(2.1.6)} & e^{\vee*} K_X. \end{array}$$

The left vertical arrow is induced by the adjoint of $0^* \mathcal{N}_\Lambda \rightarrow K_X$ (2.1.52) and the right vertical arrow is the adjunction morphism.

Proof. 1. We have a canonical morphism $\mathcal{N}_\Lambda \rightarrow e^* j_* K_U \oplus g_{0!} \Lambda_{E_0^\vee}[1]$ by construction. The morphisms of cohomology sheaves are isomorphisms except for degree -1 . For degree -1 , the morphism is an injection and the cokernel is Λ_{E_0} . Hence, we obtain (2.1.53).

2. By abuse of notation, we identify a sheaf on a closed subscheme of $E^\vee$ with its direct image on $E^\vee$. Applying the Fourier transform to (2.1.53), we obtain a distinguished triangle

$$(2.1.56) \quad F_\psi \mathcal{N}_\Lambda \rightarrow 0_*^\vee j_* \Lambda \oplus g_{0*}^\vee \Lambda \rightarrow \Lambda_0(-1)[-1] \rightarrow .$$

Hence $F_\psi \mathcal{N}_\Lambda$ is supported on the union $+= 0^\vee(X) \cup E_0^\vee$ and the composition $F_\psi \mathcal{N}_\Lambda \rightarrow F_\psi 0_* K_X \rightarrow e^{\vee*} K_X$ of the arrows in (2.1.55) via lower left induces $F_\psi \mathcal{N}_\Lambda \rightarrow +! +^! e^{\vee*} K_X$ (2.1.54) making (2.1.55) commutative. $\square$

Proposition 2.1.14. *Let $i: E_0^\vee \rightarrow E^\vee$ be the closed immersion and let $\Lambda_{0^\vee(X)} \rightarrow 0^{\vee!} e^{\vee*} K_X$ and $\Lambda_{E_0^\vee} \rightarrow i^! e^{\vee*} K_X$ be the canonical isomorphisms defined by the cycle classes $[0^\vee(X)]$ and $[E_0^\vee]$. Define $+! +^! e^{\vee*} K_X \rightarrow 0_*^\vee j_* \Lambda \oplus g_{0*}^\vee \Lambda$ to be that induced by the inverses of these isomorphisms. Then, the diagram*

$$(2.1.57) \quad \begin{array}{ccc} F_\psi \mathcal{N}_\Lambda & \xrightarrow{(2.1.54)} & +! +^! e^{\vee*} K_X \\ (2.1.53) \downarrow & & \downarrow \\ F_\psi(e^* j_* K_U \oplus g_{0!} \Lambda_{E_0^\vee}[1]) & \xrightarrow{(2.1.8) \oplus (2.1.12)} & 0_*^\vee j_* \Lambda \oplus g_{0*}^\vee \Lambda \end{array}$$

is (-1) -commutative and the morphism (2.1.54) is an isomorphism.

Proof. First, we show the (-1) -commutativity for the first component. It suffices to consider the restriction on $U = X - \{0\}$, by adjunction. The restriction of $0_! \Lambda \rightarrow \mathcal{N}_\Lambda \rightarrow 0_* K_X$ on U equals that of the composition of the adjoint $0_! \Lambda \rightarrow e^* K_X$ and the restriction $e^* K_X \rightarrow 0_* K_X$. By Proposition 2.1.12.1, its Fourier transform is (-1) -times the composition of the restriction $\Lambda \rightarrow 0_*^\vee \Lambda$ and the adjoint $0_!^\vee \Lambda \rightarrow e^{\vee*} K_X$. Since the adjoint $0_!^\vee \Lambda \rightarrow e^{\vee*} K_X$ is defined by the cycle class $[0^\vee(X)]$, the (-1) -commutativity for the first component follows.

We show the (-1) -commutativity for the second component. Since the identification $\Lambda \rightarrow R^1 j_* \Lambda(1)$ (2.1.51) is defined by sending 1 to the class of a uniformizer t , its image

by the boundary map $R^1 j_* \Lambda(1) \rightarrow H_0^2(X, \Lambda(1))$ is the minus $-[0]$ of the cycle class by [SGA 4 $\frac{1}{2}$ -Cycle, 2.1.3]. Thus, the (-1) -commutativity for the second component follows.

We show that (2.1.54) is an isomorphism. By cohomological purity, $+! +^! e^{\vee*} K_X$ fits in a distinguished triangle $+! +^! e^{\vee*} K_X \rightarrow 0_*^{\vee} j_* \Lambda \oplus g_{0*}^{\vee} \Lambda \rightarrow \Lambda_0(-1)[-1] \rightarrow$. The lower horizontal arrow is an isomorphism by Corollary 2.1.5. Since the diagram (2.1.57) is (-1) -commutative, by comparing this with (2.1.56), we see that (2.1.54) is an isomorphism. $\square$

By purity, the Λ -module $H_{0^{\vee}(X) \cup E_0^{\vee}}^0(E^{\vee}, e^{\vee*} K_X)$ is free of rank 2 generated by the cycle classes $[0^{\vee}(X)]$ and $[E_0^{\vee}]$.

Corollary 2.1.15. *Let I denote the inertia group at $0 \in E_0$.*

1. *We have a distinguished triangle*

$$(2.1.58) \quad 0^! \mathcal{N}_{\Lambda} \rightarrow j_* \Lambda \oplus \Gamma(I, \Lambda)_0 \rightarrow H^1(I, \Lambda)_0[-1] \rightarrow .$$

The last two terms are placed at $0 \in E_0 \cap 0(X) \subseteq E$.

2. *Let $\Lambda \rightarrow 0^! \mathcal{N}_{\Lambda}$ be a morphism and let $\Lambda \rightarrow F_{\psi} \mathcal{N}_{\Lambda} \rightarrow +^! e^{\vee*} K_X$ be the composition of the Fourier transform of its adjoint $0_! \Lambda \rightarrow \mathcal{N}_{\Lambda}$ and (2.1.54). If the composition $\Lambda \rightarrow 0^! \mathcal{N}_{\Lambda} \rightarrow j_* \Lambda \oplus \Gamma(I, \Lambda)_0$ with the first arrow in (2.1.58) maps $1 \in \Lambda$ to $(a, b) \in \Lambda \oplus H^0(I, \Lambda)$, then the composition $\Lambda \rightarrow +^! e^{\vee*} K_X$ defines an element $-a[0^{\vee}(X)] - b[E_0^{\vee}] \in H_{0^{\vee}(X) \cup E_0^{\vee}}^0(E^{\vee}, e^{\vee*} K_X)$.*

Proof. 1. For terms in (2.1.53), we have canonical isomorphisms $0^! e^* j_* K_U \rightarrow j_* \Lambda$, $0^! g_{0!} \Lambda_{E_0^{\times}} \rightarrow \Gamma(I, \Lambda)_0[-1]$ and $0^! \Lambda_{E_0} \rightarrow H^1(I, \Lambda)_0[-2]$. Hence (2.1.53) induces (2.1.58).

2. Let $g_0^{\vee}: E_0^{\vee} - \{0\} \rightarrow E_0^{\vee}$ be the open immersion to the fiber at 0. Then the Fourier transform of the first arrow in (2.1.58) defines the upper line in the commutative diagram

$$(2.1.59) \quad \begin{array}{ccccc} F_{\psi} 0^! \mathcal{N}_{\Lambda} & \longrightarrow & F_{\psi} (j_* \Lambda \oplus \Gamma(I, \Lambda)_0) & \xrightarrow{(2.1.6)} & e^{\vee*} j_* \Lambda \oplus e^{\vee*} \Gamma(I, \Lambda)_0 \\ \downarrow & & \downarrow & & \downarrow \text{res} \\ F_{\psi} \mathcal{N}_{\Lambda} & \longrightarrow & F_{\psi} (e^* j_* K_U \oplus g_{0!} \Lambda_{E_0^{\times}}[1]) & \longrightarrow & 0_*^{\vee} j_* \Lambda \oplus g_{0*}^{\vee} \Lambda. \end{array}$$

The lower line is (2.1.57) via lower left. Thus the assertion follows from the (-1) -commutativity of (2.1.57). $\square$

Proposition 2.1.16. *Let X and X' be noetherian schemes over k . Let $e: E \rightarrow X$ and $e': E' \rightarrow X'$ be vector bundles. For $\mathcal{F} \in D_{\text{ctf}}(E, \Lambda)$ and $\mathcal{G} \in D_{\text{ctf}}(E', \Lambda)$, there is a natural isomorphism*

$$(2.1.60) \quad F_{\psi, E \times E' / X \times X'}(\mathcal{F} \boxtimes \mathcal{G}) \simeq F_{\psi, E / X}(\mathcal{F}) \boxtimes F_{\psi, E' / X'}(\mathcal{G}).$$

Proof. We have natural isomorphisms

$$(2.1.61) \quad (E \times E') \times_{X \times X'} (E \times E')^{\vee} \simeq (E \times E') \times_{X \times X'} (E^{\vee} \times E'^{\vee}) \simeq (E \times_X E^{\vee}) \times (E' \times_{X'} E'^{\vee}).$$

Consider the following commutative diagram

$$(2.1.62) \quad \begin{array}{ccc} (E \times E') \times_{X \times X'} (E \times E')^{\vee} & \xrightarrow{\simeq} & (E \times_X E^{\vee}) \times (E' \times_{X'} E'^{\vee}) \\ \downarrow \langle, \rangle_{E \times E'} & & \downarrow \langle, \rangle_{E, <, \rangle_{E'}} \\ \mathbf{A}^1 & \xleftarrow{\text{add}} & \mathbf{A}^1 \times \mathbf{A}^1. \end{array}$$

Since $\text{add}^* \mathcal{L}_\psi = \mathcal{L}_\psi \boxtimes \mathcal{L}_\psi$, we obtain an isomorphism

$$(2.1.63) \quad \mathcal{L}_\psi(\langle \cdot, \cdot \rangle_{E \times E'}) \simeq \mathcal{L}_\psi(\langle \cdot, \cdot \rangle_E) \boxtimes \mathcal{L}_\psi(\langle \cdot, \cdot \rangle_{E'}).$$

Now (2.1.60) is the following composition of isomorphisms

$$(2.1.64) \quad \begin{aligned} & F_{\psi, E \times E' / X \times X'}(\mathcal{F} \boxtimes \mathcal{G}) \\ & \simeq (\text{pr}_{E^\vee \times E'^\vee})_! (\text{pr}_{E \times E'}^* (\mathcal{F} \boxtimes \mathcal{G}) \otimes \mathcal{L}_\psi(\langle \cdot, \cdot \rangle_{E \times E'})) \\ & \stackrel{(2.1.63)}{\simeq} (\text{pr}_{E^\vee} \times \text{pr}_{E'^\vee})_! ((\text{pr}_E^* \mathcal{F} \boxtimes \text{pr}_{E'}^* \mathcal{G}) \otimes (\mathcal{L}_\psi(\langle \cdot, \cdot \rangle_E) \boxtimes \mathcal{L}_\psi(\langle \cdot, \cdot \rangle_{E'}))) \\ & \stackrel{(1)}{\simeq} \text{pr}_{E^\vee}^! (\text{pr}_E^* \mathcal{F} \otimes \mathcal{L}_\psi(\langle \cdot, \cdot \rangle_E)) \boxtimes \text{pr}_{E'^\vee}^! (\text{pr}_{E'}^* \mathcal{G} \otimes \mathcal{L}_\psi(\langle \cdot, \cdot \rangle_{E'})) \\ & \simeq F_{\psi, E/X}(\mathcal{F}) \boxtimes F_{\psi, E'/X'}(\mathcal{G}), \end{aligned}$$

where (1) is an isomorphism by the Künneth formula. $\square$

Corollary 2.1.17 ([Lau87, proof of Proposition 1.2.2.7]). *Let X be a noetherian scheme over k . Let $e : E \rightarrow X$ and $e' : E' \rightarrow X$ be vector bundles. For $\mathcal{F} \in D_{\text{ctf}}(E, \Lambda)$ and $\mathcal{G} \in D_{\text{ctf}}(E', \Lambda)$, there is a natural isomorphism*

$$(2.1.65) \quad F_{\psi, E \times_X E' / X}(\mathcal{F} \boxtimes_X \mathcal{G}) \simeq F_{\psi, E/X}(\mathcal{F}) \boxtimes_X F_{\psi, E'/X}(\mathcal{G}).$$

Proof. Let $i : E \times_X E' \hookrightarrow E \times E'$ and $i^\vee : E^\vee \times_X E'^\vee \hookrightarrow E^\vee \times E'^\vee$ be inclusions. By the isomorphism (2.1.50) and Proposition 2.1.16, we have natural isomorphisms

$$(2.1.66) \quad \begin{aligned} F_{\psi, E \times_X E' / X}(\mathcal{F} \boxtimes_X \mathcal{G}) & \simeq F_{\psi, E \times_X E' / X}(i^*(\mathcal{F} \boxtimes \mathcal{G})) \\ & \stackrel{(2.1.50)}{\simeq} i^{\vee*} F_{\psi, E \times E' / X \times X}(\mathcal{F} \boxtimes \mathcal{G}) \\ & \stackrel{(2.1.60)}{\simeq} i^{\vee*} (F_{\psi, E/X}(\mathcal{F}) \boxtimes F_{\psi, E'/X}(\mathcal{G})) \\ & \simeq F_{\psi, E/X}(\mathcal{F}) \boxtimes_X F_{\psi, E'/X}(\mathcal{G}). \end{aligned}$$

$\square$

Corollary 2.1.18 ([Lau87, Proposition 1.2.2.7]). *Let $e : E \rightarrow X$ be a vector bundle, and let*

$$(2.1.67) \quad \text{add} : E \times_X E \rightarrow E$$

be the addition morphism. For $\mathcal{F}, \mathcal{G} \in D_{\text{ctf}}(E, \Lambda)$, we have a natural isomorphism

$$(2.1.68) \quad F_{\psi, E/X}(\text{add}_!(\mathcal{F} \boxtimes_X \mathcal{G})) \simeq F_{\psi, E/X}(\mathcal{F}) \otimes F_{\psi, E/X}(\mathcal{G}).$$

Proof. The dual morphism

$$(2.1.69) \quad \text{add}^\vee = \delta_{e^\vee} : E^\vee \rightarrow E^\vee \times_X E^\vee$$

is the diagonal morphism of $e^\vee : E^\vee \rightarrow X$. By Proposition 2.1.9 and Corollary 2.1.17, we have the following natural isomorphisms

$$(2.1.70) \quad \begin{aligned} F_{\psi, E/X}(\text{add}_!(\mathcal{F} \boxtimes_X \mathcal{G})) & \simeq \delta_{e^\vee}^* F_{\psi, E \times_X E/X}(\mathcal{F} \boxtimes_X \mathcal{G}) \\ & \simeq \delta_{e^\vee}^* (F_{\psi, E/X}(\mathcal{F}) \boxtimes_X F_{\psi, E/X}(\mathcal{G})) \\ & \simeq F_{\psi, E/X}(\mathcal{F}) \otimes F_{\psi, E/X}(\mathcal{G}). \end{aligned}$$

$\square$

Definition 2.1.19. Let X be a noetherian scheme over k . Let $e : E \rightarrow X$ and $w : W \rightarrow X$ be vector bundles. Let $\beta : W \rightarrow E$ be a linear morphism of vector bundles over X . Let add_E be the addition map

$$(2.1.71) \quad \text{add}_E : E \times_X E \rightarrow E, (v_1, v_2) \mapsto v_1 + v_2.$$

The morphism β induces an action α of W on E :

$$(2.1.72) \quad \alpha = \text{add}_E \circ (\beta \times \text{id}_E) : W \times_X E \rightarrow E, (\ell, v) \mapsto \beta(\ell) + v.$$

Write $\text{pr}_E = w \times \text{id}_E : W \times_X E \rightarrow E$ for the projection. For $\mathcal{K} \in D_{\text{ctf}}(E, \Lambda)$, a W -invariant structure on $\mathcal{K}$ is an isomorphism

$$(2.1.73) \quad \alpha^* \mathcal{K} \simeq \text{pr}_E^* \mathcal{K}.$$

If $\mathcal{K}$ admits a W -invariant structure, then we say that $\mathcal{K}$ is W -invariant.

Remark 2.1.20. If the isomorphism (2.1.73) satisfies the usual cocycle condition, then $\mathcal{K}$ is W -equivariant. We do not need this stronger condition.

Proposition 2.1.21. *We use the notation of Definition 2.1.19. Assume $\mathcal{K} \in D_{\text{ctf}}(E, \Lambda)$ is W -invariant. Consider the dual morphism*

$$(2.1.74) \quad \beta^\vee : E^\vee \rightarrow W^\vee,$$

and its kernel $\ker(\beta^\vee) \subseteq E^\vee$. Then we have

$$(2.1.75) \quad \text{supp}(F_{\psi, E/X} \mathcal{K}) \subseteq \ker(\beta^\vee).$$

Proof. Consider the dual morphisms

$$(2.1.76) \quad \alpha^\vee = (\beta^\vee \times \text{id}_{E^\vee}) \circ \delta_{e^\vee} = (\beta^\vee, \text{id}_{E^\vee}) : E^\vee \rightarrow W^\vee \times_X E^\vee,$$

$$(2.1.77) \quad \text{pr}_E^\vee = (0_{W^\vee}, \text{id}_{E^\vee}) : E^\vee \rightarrow W^\vee \times_X E^\vee,$$

where $0_{W^\vee} : E^\vee \rightarrow W^\vee$ is the zero morphism. Note that α and pr_E are smooth of the same relative dimension. By Poincaré duality and (2.1.73), we have an isomorphism

$$(2.1.78) \quad \alpha^! \mathcal{K} \simeq \text{pr}_E^! \mathcal{K}.$$

Then by (2.1.29), we obtain a natural isomorphism

$$(2.1.79) \quad (\beta^\vee, \text{id})_* F_{\psi, E/X}(\mathcal{K}) \simeq (0_{W^\vee}, \text{id})_* F_{\psi, E/X}(\mathcal{K}).$$

Taking the supports of both sides of (2.1.79), we obtain

$$(2.1.80) \quad (\beta^\vee, \text{id})(\text{supp } F_{\psi, E/X}(\mathcal{K})) = (0_{W^\vee}, \text{id})(\text{supp } F_{\psi, E/X}(\mathcal{K})).$$

By taking the first projection, we obtain $\beta^\vee(\text{supp } F_{\psi, E/X}(\mathcal{K})) \subseteq 0_{W^\vee}$. This implies (2.1.75). $\square$

Lemma 2.1.22. *With the notation of Definition 2.1.19, for $\mathcal{F} \in D_{\text{ctf}}(W, \Lambda)$ and $\mathcal{G} \in D_{\text{ctf}}(E, \Lambda)$, we have a natural isomorphism*

$$(2.1.81) \quad F_{\psi, E/X}(R\alpha_!(\text{pr}_1^* \mathcal{F} \otimes \text{pr}_2^* \mathcal{G})) \simeq \beta^{\vee*} F_{\psi, W/X}(\mathcal{F}) \otimes F_{\psi, E/X}(\mathcal{G}).$$

Proof. Consider the commutative diagram

$$(2.1.82) \quad \begin{array}{ccccc} & & W \times_X E & \xrightarrow{\text{pr}_1} & W \\ & \swarrow \text{pr}_2 & \downarrow \beta \times \text{id} & & \downarrow \beta \\ E & \xleftarrow{\text{pr}_2} & E \times_X E & \xrightarrow{\text{pr}_1} & E, \end{array}$$

where the right square is cartesian. By Corollary 2.1.18 and Proposition 2.1.9, we have

$$(2.1.83) \quad \begin{aligned} F_{\psi, E/X}(\alpha_!(\mathcal{F} \boxtimes \mathcal{G})) &\simeq F_{\psi, E/X}(\text{add}_{E!}(\beta \times \text{id})_!(\mathcal{F} \boxtimes \mathcal{G})) \\ &\simeq F_{\psi, E/X}(\text{add}_{E!}(\beta_! \mathcal{F} \boxtimes \mathcal{G})) \\ &\stackrel{(2.1.68)}{\simeq} F_{\psi, E/X}(\beta_! \mathcal{F}) \otimes F_{\psi, E/X}(\mathcal{G}) \\ &\stackrel{(2.1.24)}{\simeq} \beta^{\vee*} F_{\psi, W/X}(\mathcal{F}) \otimes F_{\psi, E/X}(\mathcal{G}). \end{aligned}$$

□

2.2 Deformation to normal bundle

First, we study dilatations of schemes smooth over an affine line $\mathbf{A}^1 = \text{Spec } k[u]$.

Definition 2.2.1. Let k be a field and let $Z \rightarrow X$ be a closed immersion of schemes smooth over the affine line $\mathbf{A}^1$. Let $\{0\} \subseteq \mathbf{A}^1 = \text{Spec } k[u]$ be the closed subscheme defined by the ideal (u) and define closed subschemes $Z_0 = Z \times_{\mathbf{A}^1} \{0\} \subseteq X_0 = X \times_{\mathbf{A}^1} \{0\} \subseteq X$. Let $\overline{D}_Z X \rightarrow X$ be the blow-up at $Z_0 \subseteq X$ and define the dilatation $D_Z X \subseteq \overline{D}_Z X$ to be the open subscheme obtained by removing the proper transform of $X_0 \subseteq X$.

If $X = \text{Spec } A$ is affine and $Z \subseteq X$ is defined by an ideal $I \subseteq A$, then u is a non-zero divisor in A and $D_Z X = \text{Spec } A[I/u]$ for the subring $A[I/u] \subseteq A[1/u]$.

Lemma 2.2.2. 1. *The immersion $Z \rightarrow X$ is lifted uniquely to $Z \rightarrow D_Z X$.*

2. *The closed fiber $D_Z X \times_{\mathbf{A}^1} \{0\}$ is canonically identified with the normal bundle $T_{Z_0} X_0$.*

3. *The scheme $D_Z X$ is smooth over $\mathbf{A}^1$.*

Proof. 1. Since $Z_0 \subseteq Z$ is a Cartier divisor, the assertion follows from the universality of dilatations.

2. The exceptional divisor of the blow-up $\overline{D}_Z X \rightarrow X$ is canonically identified with the projective space bundle $\mathbf{P}(T_{Z_0} X)$. Since $T_{Z_0} X = T_{Z_0} X_0 \oplus \mathbf{A}_{Z_0}^1$, the intersection $\mathbf{P}(T_{Z_0} X) \cap D_Z X$ is identified with $T_{Z_0} X_0$.

3. We may assume that $X = \text{Spec } A$ and that $Z \subseteq X$ is defined by an ideal $I \subseteq A$. Then $D_Z X = \text{Spec } A[I/u]$ for the subring $A[I/u] \subseteq A[1/u]$ and is flat over $\mathbf{A}^1$. Since $D_Z X \times_{\mathbf{A}^1} 0 = T_{Z_0} X_0$ is smooth over k , $D_Z X$ is smooth over $\mathbf{A}^1$. □

Let

$$(2.2.1) \quad \begin{array}{ccc} Z & \longrightarrow & X \\ \downarrow & & \downarrow \\ W & \longrightarrow & Y \end{array}$$

be a commutative diagram of smooth schemes over $\mathbf{A}^1$ such that the horizontal arrows are closed immersions. Then, we have a canonical morphism $D_Z X \rightarrow D_W Y$. We say that the diagram (2.2.1) is transversal if it is cartesian and if the morphism $\mathcal{O}_X \otimes_{\mathcal{O}_Y}^L \mathcal{O}_W \rightarrow \mathcal{O}_Z$ is an isomorphism. Further if the diagram (2.2.1) is transversal, then the diagram

$$(2.2.2) \quad \begin{array}{ccc} D_Z X & \longrightarrow & X \\ \downarrow & & \downarrow \\ D_W Y & \longrightarrow & Y \end{array}$$

is cartesian.

Lemma 2.2.3. *Let*

$$(2.2.3) \quad \begin{array}{ccc} Z & \longrightarrow & X \\ \downarrow & & \downarrow \\ W & \longrightarrow & Y \end{array}$$

be a commutative diagram of smooth schemes over $\mathbf{A}^1$ such that the horizontal arrows are closed immersions.

1. Assume that $X \rightarrow Y$ is a closed immersion and that the diagram (2.2.3) is cartesian. Then the morphism $D_Z X \rightarrow D_W Y$ is a closed immersion.
2. Assume that $X \rightarrow Y$ and $Z \rightarrow W$ are smooth. Then the morphism $D_Z X \rightarrow D_W Y$ is smooth.

Proof. 1. We may assume that $Y = \text{Spec } B$ and $X = \text{Spec } A$ are affine and $Z \subseteq X$ and $W \subseteq Y$ are defined by ideals $I \subseteq A$ and $J \subseteq B$. By the assumption that the diagram (2.2.3) is cartesian, the surjection $B \rightarrow A$ induces a surjection $J \rightarrow I$. Hence the induced morphism $B[J/u] \rightarrow A[I/u]$ is a surjection.

2. By the assumption that $X \rightarrow Y$ is smooth, the base change of the morphism $D_Z X \rightarrow D_W Y$ by $\mathbf{G}_m \rightarrow \mathbf{A}^1$ is smooth. Further by the assumption that $Z \rightarrow W$ is smooth, the morphism $T_{Z_0} X_0 \rightarrow T_{W_0} Y_0$ is smooth. Hence the morphism $D_Z X \rightarrow D_W Y$ of smooth schemes over $\mathbf{A}^1$ is smooth. $\square$

We apply the construction above to define the deformation to normal bundle.

Definition 2.2.4. Let k be a field and let $i: Z \rightarrow X$ be a closed immersion of smooth schemes over k . We define the deformation $A_Z X$ to the normal bundle $T_Z X$ to be the dilatation $D_{Z \times \mathbf{A}^1}(X \times \mathbf{A}^1)$ for the closed immersion $Z \times \mathbf{A}^1 \rightarrow X \times \mathbf{A}^1$ of smooth schemes over the affine line $\mathbf{A}^1 = \text{Spec } k[u]$.

More explicitly, if the closed immersion i is defined by the coherent ideal sheaf $I \subseteq \mathcal{O}_X$, then

$$(2.2.4) \quad A_Z X = \text{Spec}_X(\mathcal{R}_{I/\mathcal{O}_X}),$$

where $\mathcal{R}_{I/\mathcal{O}_X}$ is the extended Rees algebra

$$(2.2.5) \quad \mathcal{R}_{I/\mathcal{O}_X} = \mathcal{O}_X[u, Iu^{-1}] = \bigoplus_{m \in \mathbf{Z}} I^m u^{-m} \subseteq \mathcal{O}_X[u, u^{-1}],$$

with the convention that $I^m = \mathcal{O}_X$ for $m \leq 0$.

By Lemma 2.2.2.3, the element u defines a smooth morphism

$$(2.2.6) \quad A_Z X \rightarrow \mathbf{A}^1,$$

which fits in a cartesian diagram

$$(2.2.7) \quad \begin{array}{ccccc} T_Z X & \longrightarrow & A_Z X & \longleftarrow & X \times \mathbf{G}_m \\ \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \mathbf{A}^1 & \longleftarrow & \mathbf{G}_m. \end{array}$$

Together with the natural projection $A_Z X \rightarrow X$, the morphism (2.2.6) gives

$$(2.2.8) \quad \pi_{Z/X} : A_Z X \rightarrow X \times \mathbf{A}^1.$$

We consider the diagonal immersion $\delta : \Delta_X = X \rightarrow X \times X$. Then, the normal bundle $T_X(X \times X)$ is the tangent bundle TX . Hence, we have a cartesian diagram

$$(2.2.9) \quad \begin{array}{ccccc} TX & \longrightarrow & A_X(X \times X) & \longleftarrow & X \times X \times \mathbf{G}_m \\ \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \mathbf{A}^1 & \longleftarrow & \mathbf{G}_m. \end{array}$$

Lemma 2.2.5. *Let*

$$(2.2.10) \quad \begin{array}{ccc} Z & \xhookrightarrow{i} & X \\ \downarrow g & & \downarrow f \\ W & \xhookrightarrow{j} & Y \end{array}$$

be a commutative diagram of smooth schemes such that the horizontal arrows are closed immersions.

1. *There is a natural morphism*

$$(2.2.11) \quad A_g(f) : A_Z X \rightarrow A_W Y,$$

which fits into a commutative diagram

$$(2.2.12) \quad \begin{array}{ccc} A_Z X & \xrightarrow{A_g(f)} & A_W Y \\ \downarrow \pi_{Z/X} & & \downarrow \pi_{W/Y} \\ X \times \mathbf{A}^1 & \xrightarrow{f \times \text{id}} & Y \times \mathbf{A}^1 \end{array}$$

2. *If f and g are smooth, then $A_g(f)$ is smooth. In particular, if f is smooth, then the morphism $A_X(X \times X) \rightarrow A_Y(Y \times Y)$ is smooth.*

3. *If f and g are closed immersions and if (2.2.10) is cartesian, then $A_g(f)$ is a closed immersion. In particular, if f is a closed immersion, then the morphism $A_X(X \times X) \rightarrow A_Y(Y \times Y)$ is a closed immersion.*

4. *If (2.2.10) is transversal, i.e., it is cartesian and $\mathcal{O}_Z \simeq \mathcal{O}_X \otimes_{\mathcal{O}_Y}^L \mathcal{O}_W$, then (2.2.12) is cartesian.*

Proof. Consider the diagram

$$(2.2.13) \quad \begin{array}{ccc} Z \times \mathbf{A}^1 & \hookrightarrow & X \times \mathbf{A}^1 \\ \downarrow & & \downarrow \\ W \times \mathbf{A}^1 & \hookrightarrow & Y \times \mathbf{A}^1. \end{array}$$

The assertions follow by applying (2.2.2) and Lemma 2.2.3 to this diagram. $\square$

Consider the diagram (2.2.10). Its induced morphism $A_g(f)$ in (2.2.11) is defined over $\mathbf{A}^1$. Over $\mathbf{G}_m$, the morphism (2.2.11) is identified with $f \times \text{id} : X \times \mathbf{G}_m \rightarrow Y \times \mathbf{G}_m$:

$$(2.2.14) \quad \begin{array}{ccc} A_Z(X) \times_{\mathbf{A}^1} \mathbf{G}_m & \xrightarrow{A_g(f) \times \text{id}} & A_W(Y) \times_{\mathbf{A}^1} \mathbf{G}_m \\ \parallel & & \parallel \\ X \times \mathbf{G}_m & \xrightarrow{f \times \text{id}} & Y \times \mathbf{G}_m. \end{array}$$

On the special fiber, the morphism (2.2.11) is the natural map $T_Z X \rightarrow T_W Y$ between normal bundles:

$$(2.2.15) \quad \begin{array}{ccc} A_Z(X) \times_{\mathbf{A}^1} \{0\} & \xrightarrow{A_g(f) \times \text{id}} & A_W(Y) \times_{\mathbf{A}^1} \{0\} \\ \parallel & & \parallel \\ T_Z X & \xrightarrow{T_g f} & T_W Y. \end{array}$$

Remark 2.2.6. Let $i : Z \hookrightarrow X$ be a closed immersion between smooth schemes. The structure morphism $\pi_{Z/X} : A_Z(X) \rightarrow X \times \mathbf{A}^1$ can be described by the following diagram:

$$(2.2.16) \quad \begin{array}{ccc} A_Z(X) & \xrightarrow{A_i(\text{id})} & A_X(X) \\ \pi_{Z/X} \downarrow & & \simeq \downarrow \pi_{X/X} \\ X \times \mathbf{A}^1 & \xlongequal{\quad} & X \times \mathbf{A}^1. \end{array}$$

Proposition 2.2.7. *Let $i : Z \hookrightarrow X$ and $j : W \hookrightarrow Y$ be closed immersions between smooth schemes over k . We have a cartesian diagram*

$$(2.2.17) \quad \begin{array}{ccc} A_{Z \times W}(X \times Y) & \longrightarrow & A_Z(X) \\ \downarrow & & \downarrow \\ A_W(Y) & \longrightarrow & \mathbf{A}^1. \end{array}$$

Proof. We work locally on X and on Y , and then we may assume that $X = \text{Spec}(A)$ and $Y = \text{Spec}(B)$ are affine. Let $I \subseteq A$ be the ideal defining Z in X , and let $J \subseteq B$ be the ideal defining W in Y . Then

$$(2.2.18) \quad K = I \otimes_k B + A \otimes_k J$$

is the ideal defining $Z \times W$ in $X \times Y$. It suffices to prove that

$$(2.2.19) \quad \mathcal{R}_{K/A \otimes_k B} \simeq \mathcal{R}_{I/A} \otimes_{k[u]} \mathcal{R}_{J/B}.$$

By definition, both sides of (2.2.19) are subrings of

$$(2.2.20) \quad A \otimes_k B[u, u^{-1}] \simeq A[u, u^{-1}] \otimes_{k[u]} B[u, u^{-1}]$$

generated by $A \otimes_k B[u]$, $(I \otimes_k B)u^{-1}$ and $(A \otimes_k J)u^{-1}$. $\square$

Corollary 2.2.8. *Let $j : Z \hookrightarrow X$ and $i : X \hookrightarrow Y$ be closed immersions between smooth schemes, and let $f : Y \rightarrow X$ be a smooth morphism such that $f \circ i = \text{id}$. Then the following diagram is cartesian:*

$$(2.2.21) \quad \begin{array}{ccc} A_Z(Y) & \xrightarrow{A_{\text{id}}(f)} & A_Z(X) \\ \downarrow A_j(\text{id}) & & \downarrow \pi_{Z/X} \\ A_X(Y) & \xrightarrow{(f \times \text{id}) \circ \pi_{X/Y}} & X \times \mathbf{A}^1. \end{array}$$

Proof. Since the formation of deformation spaces is compatible with étale base change, we may work étale locally on X and Y . Using the smoothness of f together with the section i , we may therefore reduce to the following case by [Fu11, Theorem 2.5.8]

$$(2.2.22) \quad X = X \times \{w\} \hookrightarrow Y = X \times W \xrightarrow{f=\text{pr}_X} X$$

where W is a smooth scheme over k and $w \in W$ is a k -rational point. By Proposition 2.2.7, we obtain cartesian diagrams

$$(2.2.23) \quad \begin{array}{ccccc} A_Z(Y) = A_{Z \times \{w\}}(X \times W) & \longrightarrow & A_{\{w\}}(W) & \longleftarrow & A_{X \times \{w\}}(X \times W) = A_X(Y) \\ \downarrow & & \downarrow & & \downarrow \\ A_Z(X) & \longrightarrow & \mathbf{A}^1 & \longleftarrow & A_X(X) = X \times \mathbf{A}^1 \end{array}$$

and the assertion follows. $\square$

The following construction will be used to define an invariant structure on specialization.

Construction 2.2.9. Let $f : X \rightarrow Y$ be a smooth morphism, and let $i : Z \hookrightarrow X$ be a regular immersion between smooth k -schemes. Let $\delta_f : X \hookrightarrow X \times_Y X$ be the diagonal morphism. Applying Corollary 2.2.8 to the following diagram

$$(2.2.24) \quad \begin{array}{ccccc} Z \hookrightarrow X & \xrightarrow{\delta_f} & X \times_Y X & \xrightarrow{\text{pr}_{f,2}} & X \\ & \searrow & \downarrow \text{pr}_{f,1} & & \downarrow f \\ & & X & \xrightarrow{f} & Y, \end{array}$$

we obtain a cartesian diagram

$$(2.2.25) \quad \begin{array}{ccc} A_Z(X \times_Y X) & \xrightarrow{A_{\text{id}}(\text{pr}_{f,1})} & A_Z(X) \\ \downarrow A_i(\text{id}) & & \downarrow \pi_{Z/X} \\ A_X(X \times_Y X) & \xrightarrow{\pi_{\delta_f}} & X \times \mathbf{A}^1, \end{array}$$

where $\pi_{\delta_f} = (\text{pr}_{f,1} \times \text{id}) \circ \pi_{X/X \times_Y X}$. The second projection $\text{pr}_{f,2} : X \times_Y X \rightarrow X$ induces the following action:

$$(2.2.26) \quad \tilde{\alpha}_{Z/X/Y} : A_X(X \times_Y X) \times_{X \times \mathbf{A}^1} A_Z(X) \xleftarrow[\simeq]{(A_i(\text{id}), A_{\text{id}}(\text{pr}_{f,1}))} A_Z(X \times_Y X) \xrightarrow{A_{\text{id}}(\text{pr}_{f,2})} A_Z(X).$$

Unwinding the definitions, on the generic fiber, (2.2.26) is given by

$$(2.2.27) \quad \text{pr}_{f,2} : X \times_Y X \rightarrow X,$$

whereas on the special fiber, (2.2.26) induces an action

$$(2.2.28) \quad \alpha_{Z/X/Y} : T(X/Y)|_Z \times_Z T_Z X = T(X/Y) \times_X T_Z X \rightarrow T_Z X.$$

Similarly, the morphism induced by $\text{pr}_{f,1}$ is the projection onto the $A_Z(X)$ -factor:

$$(2.2.29) \quad \text{pr}_{A_Z(X)} : A_X(X \times_Y X) \times_{X \times \mathbf{A}^1} A_Z(X) \xleftarrow[\simeq]{(A_i(\text{id}), A_{\text{id}}(\text{pr}_{f,1}))} A_Z(X \times_Y X) \xrightarrow{A_{\text{id}}(\text{pr}_{f,1})} A_Z(X).$$

On the generic fiber, the morphism (2.2.29) is given by

$$(2.2.30) \quad \text{pr}_{f,1} : X \times_Y X \rightarrow X.$$

On the special fiber, (2.2.29) is the projection

$$(2.2.31) \quad \text{pr}_{T_Z X} : T(X/Y)|_Z \times_Z T_Z X = T(X/Y) \times_X T_Z X \rightarrow T_Z X,$$

where $T(X/Y)$ is the relative tangent bundle of $X \rightarrow Y$ and is isomorphic to the normal bundle of $\delta_f : X \rightarrow X \times_Y X$.

2.3 The specialization functor

We briefly recall the definition and functorialities of the nearby cycles functor. Let $\mathcal{O}_K$ be a henselian discrete valuation ring. Let $S^{\text{ur}} = \text{Spec } \mathcal{O}_{K^{\text{ur}}}$ be the maximal unramified extension of $S = \text{Spec } \mathcal{O}_K$. Let $s, \eta \in S$ be the closed point and the generic point. Let $\bar{s} \in S^{\text{ur}}$ be the closed point and $\bar{\eta} = \text{Spec } K_{\text{sep}} \rightarrow S^{\text{ur}}$ be a geometric point above η . Let Λ be a finite ring and assume that the cardinality is invertible in $\mathcal{O}_K$. For a scheme X of finite type over S , let

$$(2.3.1) \quad \begin{array}{ccccccc} X_{\bar{s}} & \xrightarrow{\bar{i}} & X \times_S S^{\text{ur}} & \xleftarrow{\bar{j}} & X_{\bar{\eta}} & \xrightarrow{j} & X_{\eta} \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ \bar{s} & \longrightarrow & S^{\text{ur}} & \longleftarrow & \bar{\eta} & \longrightarrow & \eta \end{array}$$

be a cartesian diagram. Let $D_c^b(X_s \times_s \eta, \Lambda)$ be the derived category of constructible sheaves on $X_{\bar{s}}$ with continuous action of $\text{Gal}(\bar{\eta}/\eta)$ compatible with the action on $X_{\bar{s}}$ as in [Del73, Exposé XIII, Construction 1.2.4 (b)]. Then the nearby cycles functor

$$(2.3.2) \quad \Psi : D_c^b(X_{\eta}, \Lambda) \rightarrow D_c^b(X_s \times_s \eta, \Lambda)$$

is defined by $\Psi = \bar{i}^* \bar{j}_* j^*$. Its composition $D_c^b(X, \Lambda) \rightarrow D_c^b(X_s \times_s \eta, \Lambda)$ with the pull-back is also denoted by Ψ .

Let $f: X \rightarrow Y$ be a morphism of schemes separated of finite type over S . Then the base change morphisms define morphisms of functors

$$(2.3.3) \quad \alpha: f_s^* \Psi_Y \rightarrow \Psi_X f_\eta^*,$$

$$(2.3.4) \quad \Psi_Y f_{\eta*} \rightarrow f_{s*} \Psi_X,$$

$$(2.3.5) \quad \gamma: f_{s!} \Psi_X \rightarrow \Psi_Y f_{\eta!}.$$

The morphisms (2.3.3) and (2.3.4) are adjoint to each other. If f is proper, (2.3.4) and (2.3.5) are isomorphisms inverse to each other by the proper base change theorem. If f is smooth, (2.3.3) is an isomorphism by the smooth base change theorem. A canonical morphism

$$(2.3.6) \quad \beta: \Psi_X f_\eta^! \rightarrow f_s^! \Psi_Y$$

is defined as the adjoint of (2.3.5).

Let us describe α and β more explicitly. Consider the cartesian diagram

$$(2.3.7) \quad \begin{array}{ccccccc} X_{\bar{s}} & \xrightarrow{\bar{i}_X} & X \times_S S^{\text{ur}} & \xleftarrow{\bar{j}_X} & X_{\bar{\eta}} & \xrightarrow{j_X} & X_\eta \\ f_s \downarrow & & \downarrow \tilde{f} & & \downarrow f_{\bar{\eta}} & & \downarrow f_\eta \\ Y_{\bar{s}} & \xrightarrow{\bar{i}_Y} & Y \times_S S^{\text{ur}} & \xleftarrow{\bar{j}_Y} & Y_{\bar{\eta}} & \xrightarrow{j_Y} & Y_\eta. \end{array}$$

Let $\mathcal{G} \in D_{\text{ctf}}(Y_\eta, \Lambda)$. The map $\alpha(\mathcal{G}): f_s^* \Psi_Y(\mathcal{G}) \rightarrow \Psi_X(f_\eta^* \mathcal{G})$ is obtained by restricting the base change map

$$(2.3.8) \quad \tilde{f}^* \bar{j}_Y^* j_Y^* \mathcal{G} \rightarrow \bar{j}_X^* f_{\bar{\eta}}^* j_Y^* \mathcal{G}$$

to the special fiber. The map $\beta(\mathcal{G}): \Psi_X(f_\eta^! \mathcal{G}) \rightarrow f_s^! \Psi_Y(\mathcal{G})$ is adjoint to

$$(2.3.9) \quad f_{s!} \Psi_X(f_\eta^! \mathcal{G}) \xrightarrow{\gamma(f_\eta^! \mathcal{G})} \Psi_Y(f_{\eta!} f_\eta^! \mathcal{G}) \xrightarrow{\Psi_Y(\varepsilon_{f_\eta, \mathcal{G}})} \Psi_Y(\mathcal{G}),$$

where $\varepsilon_{f_\eta} := \varepsilon_{f_\eta, \mathcal{G}}: f_{\eta!} f_\eta^! \mathcal{G} \rightarrow \mathcal{G}$ is the counit.

The morphisms α (2.3.3) and γ (2.3.5) are compatible with the isomorphism of projection formula as follows.

Lemma 2.3.1. *Let $f: X \rightarrow Y$ be a morphism of separated schemes of finite type over $S = \text{Spec } \mathcal{O}_K$. Let $\mathcal{F} \in D_{\text{ctf}}(X_\eta, \Lambda)$ and $\mathcal{G} \in D_{\text{ctf}}(Y_\eta, \Lambda)$. Then the diagram*

$$(2.3.10) \quad \begin{array}{ccc} f_{s!} \Psi_X(\mathcal{F}) \otimes \Psi_Y(\mathcal{G}) & \xrightarrow[\simeq]{\text{PF}_{f_s}} & f_{s!}(\Psi_X(\mathcal{F}) \otimes f_s^* \Psi_Y(\mathcal{G})) \\ \gamma(\mathcal{F}) \otimes 1 \downarrow (2.3.5) & & (2.3.3) \downarrow f_{s!}(1 \otimes \alpha(\mathcal{G})) \\ \Psi_Y(f_{\eta!} \mathcal{F}) \otimes \Psi_Y(\mathcal{G}) & & f_{s!}(\Psi_X(\mathcal{F}) \otimes \Psi_X(f_\eta^* \mathcal{G})) \\ \downarrow m & & \downarrow m \\ \Psi_Y(f_{\eta!} \mathcal{F} \otimes \mathcal{G}) & & f_{s!} \Psi_X(\mathcal{F} \otimes f_\eta^* \mathcal{G}) \\ \Psi_Y(\text{PF}_{f_\eta}) \downarrow \simeq & & (2.3.5) \downarrow \gamma(\mathcal{F} \otimes f_\eta^* \mathcal{G}) \\ \Psi_Y(f_{\eta!}(\mathcal{F} \otimes f_\eta^* \mathcal{G})) & \xlongequal{\quad} & \Psi_Y(f_{\eta!}(\mathcal{F} \otimes f_\eta^* \mathcal{G})) \end{array}$$

is commutative, where m is the canonical tensor map for the nearby cycles functor, and PF_{f_s} and PF_{f_η} are the isomorphisms defined by the projection formula for the morphisms f_s and f_η respectively.

Proof. We denote the counits $\bar{j}_X^* \bar{j}_{X*} \xrightarrow{\varepsilon} \text{id}$ and $\bar{j}_Y^* \bar{j}_{Y*} \xrightarrow{\varepsilon} \text{id}$ both by ε . Let $a: \tilde{f}^* \bar{j}_{Y*} \rightarrow \bar{j}_{X*} f_{\bar{\eta}}^*$ be the base change morphism, which is adjoint to

$$(2.3.11) \quad \bar{j}_X^* \tilde{f}^* \bar{j}_{Y*} \simeq f_{\bar{\eta}}^* \bar{j}_Y^* \bar{j}_{Y*} \xrightarrow{f_{\bar{\eta}}^* \varepsilon} f_{\bar{\eta}}^*.$$

Let $c: \tilde{f}_! \bar{j}_{X*} \rightarrow \bar{j}_{Y*} f_{\bar{\eta}}^!$ be the morphism adjoint to

$$(2.3.12) \quad b: \bar{j}_Y^* \tilde{f}_! \bar{j}_{X*} \xrightarrow[\simeq]{\text{BC}} f_{\bar{\eta}}^! \bar{j}_X^* \bar{j}_{X*} \xrightarrow{f_{\bar{\eta}}^! \varepsilon} f_{\bar{\eta}}^!.$$

For $\bar{j} = \bar{j}_X$ or $\bar{j}_Y$, let $\mu: \bar{j}_* A \otimes \bar{j}_* B \rightarrow \bar{j}_*(A \otimes B)$ be the canonical tensor morphism, which is adjoint to

$$(2.3.13) \quad \bar{j}^*(\bar{j}_* A \otimes \bar{j}_* B) \simeq \bar{j}^* \bar{j}_* A \otimes \bar{j}^* \bar{j}_* B \xrightarrow{\varepsilon \otimes \varepsilon} A \otimes B.$$

By the definitions of a and μ , we have commutative diagrams

$$(2.3.14) \quad \begin{array}{ccc} f_{\bar{\eta}}^* \bar{j}_Y^* \bar{j}_{Y*} & \xrightarrow{f_{\bar{\eta}}^* \varepsilon} & f_{\bar{\eta}}^* \\ \simeq \downarrow & & \uparrow \varepsilon \\ \bar{j}_X^* \tilde{f}^* \bar{j}_{Y*} & \xrightarrow{\bar{j}_X^* a} & \bar{j}_X^* \bar{j}_{X*} f_{\bar{\eta}}^* \end{array} \quad \begin{array}{ccc} \bar{j}_X^*(\bar{j}_{X*} A \otimes \bar{j}_{X*} B) & \xrightarrow{\bar{j}_X^* \mu} & \bar{j}_X^* \bar{j}_{X*}(A \otimes B) \\ \simeq \downarrow & & \downarrow \varepsilon \\ \bar{j}_X^* \bar{j}_{X*} A \otimes \bar{j}_X^* \bar{j}_{X*} B & \xrightarrow{\varepsilon \otimes \varepsilon} & A \otimes B. \end{array}$$

Let $\mathcal{F}' = \mathcal{F}|_{X_{\bar{\eta}}}$ and $\mathcal{G}' = \mathcal{G}|_{Y_{\bar{\eta}}}$. Recall that $\Psi_X(\mathcal{F}) = \bar{i}_X^* \bar{j}_{X*} \mathcal{F}'$, $\Psi_Y(\mathcal{G}) = \bar{i}_Y^* \bar{j}_{Y*} \mathcal{G}'$ and $\bar{i}_Y^* \tilde{f}_! \simeq f_{s!} \bar{i}_X^*$. Under these identifications, (2.3.10) is obtained by applying $\bar{i}_Y^*$ to the diagram (2.3.15) below, using the compatibility of the projection formula with restriction to closed subschemes. Thus, it suffices to prove that the following diagram commutes.

$$(2.3.15) \quad \begin{array}{ccc} \tilde{f}_! \bar{j}_{X*} \mathcal{F}' \otimes \bar{j}_{Y*} \mathcal{G}' & \xrightarrow{\text{PF}} & \tilde{f}_!(\bar{j}_{X*} \mathcal{F}' \otimes \tilde{f}^* \bar{j}_{Y*} \mathcal{G}') \\ \downarrow c \otimes 1 & & \downarrow \tilde{f}_!(1 \otimes a) \\ \bar{j}_{Y*} f_{\bar{\eta}}^! \mathcal{F}' \otimes \bar{j}_{Y*} \mathcal{G}' & & \tilde{f}_!(\bar{j}_{X*} \mathcal{F}' \otimes \bar{j}_{X*} f_{\bar{\eta}}^* \mathcal{G}') \\ \downarrow \mu & & \downarrow \tilde{f}_! \mu \\ \bar{j}_{Y*}(f_{\bar{\eta}}^! \mathcal{F}' \otimes \mathcal{G}') & & \tilde{f}_! \bar{j}_{X*}(\mathcal{F}' \otimes f_{\bar{\eta}}^* \mathcal{G}') \\ \downarrow \bar{j}_{Y*}(\text{PF}) & & \downarrow c \\ \bar{j}_{Y*} f_{\bar{\eta}}^!(\mathcal{F}' \otimes f_{\bar{\eta}}^* \mathcal{G}') & \xlongequal{\quad} & \bar{j}_{Y*} f_{\bar{\eta}}^!(\mathcal{F}' \otimes f_{\bar{\eta}}^* \mathcal{G}'). \end{array}$$

By the adjunction $\bar{j}_Y^* \dashv \bar{j}_{Y*}$, it suffices to show that the two composites in (2.3.15) have the same adjoint.

By the definitions of c and μ , the adjoint of the composite along the left-hand path is

$$(2.3.16) \quad \begin{aligned} \bar{j}_Y^*(\tilde{f}_! \bar{j}_{X*} \mathcal{F}' \otimes \bar{j}_{Y*} \mathcal{G}') &\simeq \bar{j}_Y^* \tilde{f}_! \bar{j}_{X*} \mathcal{F}' \otimes \bar{j}_Y^* \bar{j}_{Y*} \mathcal{G}' \\ &\xrightarrow{b \otimes \varepsilon} f_{\bar{\eta}}^! \mathcal{F}' \otimes \mathcal{G}' \xrightarrow{\text{PF}} f_{\bar{\eta}}^!(\mathcal{F}' \otimes f_{\bar{\eta}}^* \mathcal{G}'). \end{aligned}$$

By (2.3.14), the adjoint of the composite along the top and right-hand edges is

$$\begin{aligned}
(2.3.17) \quad & \bar{j}_Y^* (\bar{f}_! \bar{j}_{X*} \mathcal{F}' \otimes \bar{j}_{Y*} \mathcal{G}') \xrightarrow{\text{PF}} \bar{j}_Y^* \bar{f}_! (\bar{j}_{X*} \mathcal{F}' \otimes \bar{f}^* \bar{j}_{Y*} \mathcal{G}') \\
& \simeq \bar{f}_{\bar{\eta}}^! (\bar{j}_X^* \bar{j}_{X*} \mathcal{F}' \otimes \bar{f}_{\bar{\eta}}^* \bar{j}_Y^* \bar{j}_{Y*} \mathcal{G}') \\
& \xrightarrow{\bar{f}_{\bar{\eta}}^! (\varepsilon \otimes \bar{f}_{\bar{\eta}}^* \varepsilon)} \bar{f}_{\bar{\eta}}^! (\mathcal{F}' \otimes \bar{f}_{\bar{\eta}}^* \mathcal{G}').
\end{aligned}$$

By the compatibility of the projection formula with restriction to open subschemes, this composite coincides with the composite in (2.3.16). $\square$

We recall a canonical construction (cf. [Sai17a, (8.3)]). Let $f : X \rightarrow Y$ be a separated morphism of schemes, and let $\mathcal{F}, \mathcal{G} \in D_{\text{ctf}}(Y, \Lambda)$. We have a relative purity morphism

$$(2.3.18) \quad c_f = c_{f, \mathcal{F}, \mathcal{G}} : f^* \mathcal{F} \otimes f^! \mathcal{G} \rightarrow f^! (\mathcal{F} \otimes \mathcal{G}),$$

which is obtained by adjunction from the following composition

$$(2.3.19) \quad f_! (f^* \mathcal{F} \otimes f^! \mathcal{G}) \simeq \mathcal{F} \otimes f_! f^! \mathcal{G} \rightarrow \mathcal{F} \otimes \mathcal{G}.$$

By the definition of $c_{f, \mathcal{F}, \mathcal{G}}$, we have a commutative diagram

$$\begin{array}{ccc}
f_! (f^* \mathcal{F} \otimes f^! \mathcal{G}) & \xrightarrow{f_! c_{f, \mathcal{F}, \mathcal{G}}} & f_! f^! (\mathcal{F} \otimes \mathcal{G}) \\
\text{PF}_f \downarrow \simeq & & \downarrow \varepsilon_{f, \mathcal{F} \otimes \mathcal{G}} \\
\mathcal{F} \otimes f_! f^! \mathcal{G} & \xrightarrow{1 \otimes \varepsilon_{f, \mathcal{G}}} & \mathcal{F} \otimes \mathcal{G},
\end{array}$$

where PF_f is the isomorphism defined by the projection formula for the morphism f . We put $c_{f, \mathcal{F}} := c_{f, \mathcal{F}, \Lambda}$. We say that f is $\mathcal{F}$ -transversal if $c_{f, \mathcal{F}}$ is an isomorphism (cf. [Sai17a, Definition 8.5]).

Lemma 2.3.2. *Let $f : X \rightarrow Y$ be a morphism of separated schemes of finite type over $S = \text{Spec } \mathcal{O}_K$. For any $\mathcal{F} \in D_{\text{ctf}}(Y_\eta, \Lambda)$ and $\mathcal{G} \in D_{\text{ctf}}(Y_\eta, \Lambda)$, we have a commutative diagram in $D_{\text{ctf}}(X_s \times_s \eta, \Lambda)$*

$$\begin{array}{ccccc}
\Psi_X(f_\eta^* \mathcal{F}) \otimes \Psi_X(f_\eta^! \mathcal{G}) & \xrightarrow{m} & \Psi_X(f_\eta^* \mathcal{F} \otimes f_\eta^! \mathcal{G}) & \xrightarrow{\Psi_X(c_{f_\eta})} & \Psi_X(f_\eta^! (\mathcal{F} \otimes \mathcal{G})) \\
\alpha(\mathcal{F}) \otimes 1 \uparrow (2.3.3) & & & & \downarrow \beta(\mathcal{F} \otimes \mathcal{G}) (2.3.6) \\
(2.3.21) \quad f_s^* \Psi_Y(\mathcal{F}) \otimes \Psi_X(f_\eta^! \mathcal{G}) & & & & \\
1 \otimes \beta(\mathcal{G}) \downarrow (2.3.6) & & & & \\
f_s^* \Psi_Y(\mathcal{F}) \otimes f_s^! \Psi_Y(\mathcal{G}) & \xrightarrow{c_{f_s}} & f_s^! (\Psi_Y(\mathcal{F}) \otimes \Psi_Y(\mathcal{G})) & \xrightarrow{f_s^! m} & f_s^! \Psi_Y(\mathcal{F} \otimes \mathcal{G})
\end{array}$$

where m is the canonical tensor map for the nearby cycles functor.

Proof. We show the following equality

$$(2.3.22) \quad \beta(\mathcal{F} \otimes \mathcal{G}) \circ \Psi_X(c_{f_\eta}) \circ m \circ (\alpha(\mathcal{F}) \otimes 1) = f_s^! m \circ c_{f_s} \circ (1 \otimes \beta(\mathcal{G}))$$

by comparing the adjoints under $f_{s!} \dashv f_s^!$.

To compute the adjoint of the left hand side, we consider the diagram

(2.3.23)

$$\begin{array}{ccccc}
f_{s!}(f_s^* \Psi_Y(\mathcal{F}) \otimes \Psi_X(f_\eta^! \mathcal{G})) & \xrightarrow{f_{s!}(\alpha(\mathcal{F}) \otimes 1)} & f_{s!}(\Psi_X(f_\eta^* \mathcal{F}) \otimes \Psi_X(f_\eta^! \mathcal{G})) \\
\downarrow \text{PF}_{f_s} \simeq & & \downarrow f_{s!} m \\
\Psi_Y(\mathcal{F}) \otimes f_{s!} \Psi_X(f_\eta^! \mathcal{G}) & & f_{s!} \Psi_X(f_\eta^* \mathcal{F} \otimes f_\eta^! \mathcal{G}) \xrightarrow{f_{s!} \Psi_X(c_{f_\eta})} f_{s!} \Psi_X(f_\eta^! (\mathcal{F} \otimes \mathcal{G})) \\
\downarrow 1 \otimes \gamma_{f_\eta^! \mathcal{G}} & (3) \quad \downarrow \gamma_{f_\eta^* \mathcal{F} \otimes f_\eta^! \mathcal{G}} & \downarrow \gamma_{f_\eta^! (\mathcal{F} \otimes \mathcal{G})} \\
\Psi_Y(\mathcal{F}) \otimes \Psi_Y(f_\eta^! f_\eta^! \mathcal{G}) & \xrightarrow{m} & \Psi_Y(\mathcal{F} \otimes f_\eta^! f_\eta^! \mathcal{G}) \xrightarrow{\Psi_Y(1 \otimes \varepsilon_{f_\eta})} \Psi_Y(\mathcal{F} \otimes \mathcal{G}). \\
& & \downarrow \Psi_Y(\varepsilon_{f_\eta}) \\
& & \Psi_Y(f_\eta^! (f_\eta^* \mathcal{F} \otimes f_\eta^! \mathcal{G})) \xrightarrow{\Psi_Y(f_\eta^! c_{f_\eta})} \Psi_Y(f_\eta^! f_\eta^! (\mathcal{F} \otimes \mathcal{G})) \\
& & \downarrow \Psi_Y(\text{PF}_{f_\eta}) \simeq \\
& & \Psi_Y(\mathcal{F} \otimes f_\eta^! f_\eta^! \mathcal{G}) \xrightarrow{\Psi_Y(1 \otimes \varepsilon_{f_\eta})} \Psi_Y(\mathcal{F} \otimes \mathcal{G}).
\end{array}$$

The composition of the right column is the adjoint of $\beta(\mathcal{F} \otimes \mathcal{G})$ by (2.3.9). Hence the adjoint of the left hand side $\beta(\mathcal{F} \otimes \mathcal{G}) \circ \Psi_X(c_{f_\eta}) \circ m \circ (\alpha(\mathcal{F}) \otimes 1)$ of the equality (2.3.22) equals the composition via upper right. By naturality of the morphism γ (2.3.5) and the commutative diagram (2.3.20), the squares (1) and (2) are commutative respectively. By Lemma 2.3.1 applied to the morphism f , $f_\eta^! \mathcal{G} \in D_{\text{ctf}}(X_\eta, \Lambda)$ and $\mathcal{F} \in D_{\text{ctf}}(Y_\eta, \Lambda)$, the left rectangle (3) is commutative. Thus the diagram (2.3.23) is commutative and the adjoint of the left side of the equality (2.3.22) equals the composition via the lower left.

To compute the adjoint of the right hand side, we consider the diagram

(2.3.24)

$$\begin{array}{ccccc}
f_{s!}(f_s^* \Psi_Y(\mathcal{F}) \otimes \Psi_X(f_\eta^! \mathcal{G})) & \xrightarrow{f_{s!}(1 \otimes \beta(\mathcal{G}))} & f_{s!}(f_s^* \Psi_Y(\mathcal{F}) \otimes f_s^! \Psi_Y(\mathcal{G})) & \xrightarrow{f_{s!} c_{f_s}} & f_{s!} f_s^! (\Psi_Y(\mathcal{F}) \otimes \Psi_Y(\mathcal{G})) \\
\downarrow \text{PF}_{f_s} & & \downarrow \text{PF}_{f_s} & & \downarrow \varepsilon_{f_s} \\
\Psi_Y(\mathcal{F}) \otimes f_{s!} \Psi_X(f_\eta^! \mathcal{G}) & \xrightarrow{1 \otimes f_{s!} \beta(\mathcal{G})} & \Psi_Y(\mathcal{F}) \otimes f_{s!} f_s^! \Psi_Y(\mathcal{G}) & \xrightarrow{1 \otimes \varepsilon_{f_s}} & \Psi_Y(\mathcal{F}) \otimes \Psi_Y(\mathcal{G}) \\
\downarrow 1 \otimes \gamma_{f_\eta^! \mathcal{G}} & & \downarrow 1 \otimes \Psi_Y(\varepsilon_{f_\eta}) & & \downarrow m \\
\Psi_Y(\mathcal{F}) \otimes \Psi_Y(f_\eta^! f_\eta^! \mathcal{G}) & \xrightarrow{m} & \Psi_Y(\mathcal{F} \otimes f_\eta^! f_\eta^! \mathcal{G}) & \xrightarrow{\Psi_Y(1 \otimes \varepsilon_{f_\eta})} & \Psi_Y(\mathcal{F} \otimes \mathcal{G}).
\end{array}$$

The adjoint of the right hand side $f_s^! m \circ c_{f_s} \circ (1 \otimes \beta(\mathcal{G}))$ of the equality (2.3.22) equals the composition through the upper right. By (2.3.20), the upper right square is commutative. By the functoriality of the isomorphism of projection formula, the upper left square is commutative. By naturality of the tensor map and its compatibility with the unit, the lower triangle is commutative. The definition of $\beta(\mathcal{G})$ as the adjoint of (2.3.9) implies that the upper triangle (without $\Psi_Y(\mathcal{F})$) is commutative. Thus the diagram (2.3.24) is commutative and the adjoint of the right side of the equality (2.3.22) equals the composition via the lower left. Thus, we get the equality (2.3.22). $\square$

We recall the definition of monodromic sheaves from [Ver83, Définition 3.1].

Definition 2.3.3. Let X be a noetherian scheme over a field k and E be a vector bundle over X . For a geometric point x of E , define a morphism $w_x: \mathbf{G}_{m,x} \rightarrow E$ by $w_x(t) = t \cdot x$.

We say that $\mathcal{F} \in D_c^b(E, \Lambda)$ is monodromic if for every geometric point x of E and for every integer $q \in \mathbf{Z}$, the cohomology sheaf $\mathcal{H}^q(w_x^* \mathcal{F})$ on $\mathbf{G}_{m,x}$ is locally constant and tamely ramified along $0, \infty$.

We study a property of a locally constant sheaf on $\mathbf{G}_m$ tamely ramified along $0, \infty$. Recall that k is a field of characteristic $p > 0$ and Λ is a finite ring such that the cardinality is invertible in k .

Lemma 2.3.4. *Let $\mathcal{M}$ be a locally constant constructible sheaf on $\mathbf{G}_m = \text{Spec } k[x^{\pm 1}]$ tamely ramified along $0, \infty \in \mathbf{P}^1$ and let $j: \mathbf{G}_m \rightarrow \mathbf{A}^1$ be the open immersion. Let $\bar{k}$ be a separable closure of k .*

1. *The canonical morphism $\Gamma(\mathbf{G}_{m, \bar{k}}, \mathcal{M}) \rightarrow (j_* \mathcal{M})_{\bar{0}}$ is an isomorphism.*
2. $\Gamma_c(\mathbf{A}_{\bar{k}}^1, j_* \mathcal{M}) = 0$.

Proof. 1. Let I_0^t be the tame inertia group at 0 and let M be the stalk of $\mathcal{M}$ at the generic geometric point of $\mathbf{G}_m$. Then, the canonical morphism $I_0^t \rightarrow \pi_1(\mathbf{G}_{m, \bar{k}})^t$ to the tame fundamental group is an isomorphism and we obtain a commutative diagram

$$(2.3.25) \quad \begin{array}{ccc} \Gamma(\mathbf{G}_{m, \bar{k}}, \mathcal{M}) & \longrightarrow & j_* \mathcal{M}_{\bar{0}} \\ \uparrow & & \uparrow \\ \Gamma(\pi_1(\mathbf{G}_{m, \bar{k}})^t, M) & \longrightarrow & \Gamma(I_0^t, M) \end{array}$$

of isomorphisms.

2. Let $j_\infty: \mathbf{A}^1 \rightarrow \mathbf{P}^1$ be the open immersion. Then, we have a distinguished triangle $\Gamma_c(\mathbf{A}_{\bar{k}}^1, j_* \mathcal{M}) \rightarrow \Gamma(\mathbf{P}_{\bar{k}}^1, j_{\infty*} j_* \mathcal{M}) \rightarrow j_{\infty*} j_* \mathcal{M}|_{\infty} \rightarrow$. The second morphism is identified with $\Gamma(\mathbf{G}_{m, \bar{k}}, \mathcal{M}) \rightarrow j_{\infty*} j_* \mathcal{M}|_{\infty}$ and is an isomorphism similarly as in 1. Hence we have $\Gamma_c(\mathbf{A}_{\bar{k}}^1, j_* \mathcal{M}) = 0$. $\square$

We recall the definition of the specialization functor from [Ver83]. Let $i: Z \rightarrow X$ be a closed immersion of smooth schemes over a field k . We consider the cartesian diagram

$$(2.3.26) \quad \begin{array}{ccccc} T_Z X & \longrightarrow & A_Z X & \longleftarrow & X \times \mathbf{G}_m \\ \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \mathbf{A}^1 & \longleftarrow & \mathbf{G}_m. \end{array}$$

Let η_0 be the generic point of the henselization of $\mathbf{A}^1$ at 0, and let $\bar{k}$ be a separable closure of k . Let Ψ be the nearby cycles functor with respect to the middle vertical arrow in (2.3.26) at $0 \in \mathbf{A}^1$ and let $p_1: X \times \mathbf{G}_m \rightarrow X$ denote the projection. We define the specialization functor

$$(2.3.27) \quad \nu_i = \nu_{Z/X}: D_c^b(X, \Lambda) \rightarrow D_c^b(T_Z X \times_k \eta_0, \Lambda)$$

by $\nu_{Z/X} = \Psi \circ p_1^*$. We adopt the following convention throughout the paper.

Convention 2.3.5. For a morphism $r: U \rightarrow V$ of schemes over k , we also write r for the induced morphism of topoi

$$(2.3.28) \quad r: U \times_k \eta_0 \rightarrow V \times_k \eta_0.$$

For $\mathcal{F} \in D_c^b(U, \Lambda)$, we also denote by $\mathcal{F}$ its pull-back along the canonical projection $U \times_k \eta_0 \rightarrow U$. Its underlying object in $D_c^b(U_{\bar{k}}, \Lambda)$ is $\mathcal{F}|_{U_{\bar{k}}}$, equipped with the semilinear G_{η_0} -action induced by its natural G_k -action via the quotient $G_{\eta_0} \rightarrow G_k$.

Lemma 2.3.6. *Consider a cartesian diagram of smooth k -schemes*

$$(2.3.29) \quad \begin{array}{ccc} Z & \xhookrightarrow{i} & X \\ \downarrow g & & \downarrow f \\ W & \xhookrightarrow{j} & Y, \end{array}$$

where i and j are closed immersions. Let $p : T_Z X \rightarrow Z$ be the projection map, and let $\mathcal{F} \in D_c^b(X, \Lambda)$. Assume that f and g are smooth and that f is universally locally acyclic relatively to $\mathcal{F}$. Then there is a canonical isomorphism

$$(2.3.30) \quad \nu_{Z/X}(\mathcal{F}) \simeq p^* i^* \mathcal{F} \quad \text{in} \quad D_c^b(T_Z X \times_k \eta_0, \Lambda),$$

where the right-hand side is interpreted according to Convention 2.3.5.

Proof. By Lemma 2.2.5, we have a cartesian diagram

$$(2.3.31) \quad \begin{array}{ccc} A_Z(X) & \xrightarrow{A_g(f)} & A_W(Y) \\ \downarrow & & \downarrow \\ X & \xrightarrow{f} & Y. \end{array}$$

Then $A_g(f)$ is universally locally acyclic with respect to $\mathcal{F}|_{A_Z(X)}$. Since $A_W(Y) \rightarrow \mathbf{A}^1$ is smooth, it follows from [SGA 4 $\frac{1}{2}$ -TF, Th. finitude, Lemme 2.14] (see also [Ill77, Corollaire 2.7]) that the morphism

$$(2.3.32) \quad A_Z(X) \xrightarrow{A_g(f)} A_W(Y) \rightarrow \mathbf{A}^1$$

is universally locally acyclic with respect to $\mathcal{F}|_{A_Z(X)}$. Applying the nearby cycles functor yields a natural isomorphism $\nu_{Z/X}(\mathcal{F}) \simeq \mathcal{F}|_{T_Z X} \simeq p^* i^* \mathcal{F}$. $\square$

Let

$$(2.3.33) \quad \begin{array}{ccc} Z & \longrightarrow & X \\ g \downarrow & & \downarrow f \\ W & \longrightarrow & Y \end{array}$$

be a commutative diagram of smooth schemes over k such that the horizontal arrows are closed immersions. Then, we have a commutative diagram

$$(2.3.34) \quad \begin{array}{ccccc} T_Z X & \longrightarrow & A_Z X & \longleftarrow & X \times \mathbf{G}_m \\ T_f \downarrow & & A_f = A_g(f) \downarrow & & \downarrow \\ T_W Y & \longrightarrow & A_W Y & \longleftarrow & Y \times \mathbf{G}_m \end{array}$$

and the morphisms (2.3.3), (2.3.4), (2.3.5), (2.3.6) define morphisms

$$(2.3.35) \quad \alpha : T_f^* \nu_{W/Y} \rightarrow \nu_{Z/X} f^*,$$

$$(2.3.36) \quad \nu_{W/Y} f_* \rightarrow T_{f*} \nu_{Z/X},$$

$$(2.3.37) \quad \gamma : T_{f!} \nu_{Z/X} \rightarrow \nu_{W/Y} f_!,$$

$$(2.3.38) \quad \beta : \nu_{Z/X} f^! \rightarrow T_f^! \nu_{W/Y}.$$

Proposition 2.3.7. *Let $i: Z \rightarrow X$ be a closed immersion of smooth schemes over k and $\mathcal{F} \in D_c^b(X, \Lambda)$.*

1 ([Ver83, Section 8 (SP1)]). *The specialization $\nu_{Z/X}\mathcal{F}$ on T_ZX is monodromic.*

2 (cf. [Ver83, Section 8 (SP5)]). *The morphisms $0^*\nu_{Z/X}\mathcal{F} \rightarrow i^*\mathcal{F}$ (2.3.35) and $i^!\mathcal{F} \rightarrow 0^!\nu_{Z/X}\mathcal{F}$ (2.3.38) for the immersion $A_ZZ = Z \times \mathbf{A}^1 \rightarrow A_ZX$ are isomorphisms.*

Proof. 2. The first isomorphism is [Ver83, Section 8 (SP5)]. The following proof of the second isomorphism is similar to the proof for the first isomorphism loc. cit. It suffices to show that $0^!\nu_{Z/X}\mathcal{F} = 0$ assuming that $i^!\mathcal{F} = 0$. Let $j: U = X - Z \rightarrow X$ be the open immersion of the complement. Then, the assumption $i^!\mathcal{F} = 0$ means that the canonical morphism $\mathcal{F} \rightarrow j_*j^*\mathcal{F}$ is an isomorphism. By blow-up, we may assume that $Z \subseteq X$ is a Cartier divisor of X and that there exists a cartesian diagram

$$\begin{array}{ccc} Z & \longrightarrow & X \\ \downarrow & & \downarrow \\ 0 & \longrightarrow & \mathbf{A}^1 \end{array}$$

where the vertical arrows are smooth. We prove the assertion by induction on the dimension d of the support of $\mathcal{F}$. If $d = 0$, then $\mathcal{F} = 0$ on a neighborhood of Z and the assertion holds. Assume $d \geq 1$.

First, we show $0^!\nu_{Z/X}\mathcal{F} = 0$ generically on Z . By replacing X by an open neighborhood of the generic point of Z , we may assume that $j^*\mathcal{F}$ is locally constant. By further replacing X by a finite scheme over X and shrinking X and using the isomorphism (2.3.36) for finite morphisms, we may assume that the cohomology sheaves of $j^*\mathcal{F}$ are constant and further $j^*\mathcal{F} = \mathcal{H}^0 j^*\mathcal{F}[0]$. Let $j_1: U \times \mathbf{G}_m \rightarrow A_ZX - A_ZZ$ be the open immersion. Then, since $A_ZX - A_ZZ \rightarrow \mathbf{A}^1$ is smooth, the direct image $R^0 j_{1*} \text{pr}_1^* j^*\mathcal{F}$ is a constant sheaf. Let $j_2: A_ZX - A_ZZ \rightarrow A_ZX$ be the open immersion and let $j_{2,0}: T_ZX - Z \rightarrow T_ZX$ be its restriction on the closed fiber. Further, since A_ZX and A_ZZ are smooth over $\mathbf{A}^1$, we have $\nu_{Z/X}\mathcal{F} = (j_{2*} R^0 j_{1*} \text{pr}_1^* j^*\mathcal{F})|_{T_ZX} = j_{2,0*}((R^0 j_{1*} \text{pr}_1^* j^*\mathcal{F})|_{T_ZX - Z})$. Thus we have $0^!\nu_{Z/X}\mathcal{F} = 0$ generically on Z .

We show $0^!\nu_{Z/X}\mathcal{F} = 0$ on the whole Z . Since the assertion is local on X , we may take an étale morphism $X \rightarrow \mathbf{A}^n$ such that $Z \subseteq X$ is the inverse image of $\mathbf{A}^{n-1} \subseteq \mathbf{A}^n$. By considering the normalization of $\mathbf{A}^n$ in X and the compatibility with finite direct image, we may assume $Z = \mathbf{A}^{n-1} \subseteq X = \mathbf{A}^n$. Since the support of $0^!\nu_{Z/X}\mathcal{F}$ is of dimension $< n - 1$, there exists a linear projection $X = \mathbf{A}^n \rightarrow X' = \mathbf{A}^{n-1}$ such that $Z = \mathbf{A}^{n-1} \rightarrow Z' = \mathbf{A}^{n-2}$ is finite on the support S of $0^!\nu_{Z/X}\mathcal{F}$. Let $\bar{X} = X' \times \mathbf{P}^1 \rightarrow X'$ and $\bar{q}: \bar{Z} = Z' \times \mathbf{P}^1 \rightarrow Z'$ be projective completions and let $\bar{\mathcal{F}} = j_{X*}\mathcal{F}$ for the open immersion $j_X: X \rightarrow \bar{X}$. Then, we have $0^!\nu_{Z/X}\mathcal{F} = (0^!\nu_{\bar{Z}/\bar{X}}\bar{\mathcal{F}})|_Z$ and $\bar{q}: \bar{Z} \rightarrow Z'$ is finite on the union $S \cup (\bar{Z} - Z)$ containing the support of $0^!\nu_{\bar{Z}/\bar{X}}\bar{\mathcal{F}}$. By the induction hypothesis and the compatibility with finite direct images, we have $\bar{q}_* 0^!\nu_{\bar{Z}/\bar{X}}\bar{\mathcal{F}} = 0$ and hence $0^!\nu_{\bar{Z}/\bar{X}}\bar{\mathcal{F}} = 0$. Thus, we obtain $0^!\nu_{Z/X}\mathcal{F} = 0$. $\square$

Let $i: Z \hookrightarrow X$ be a closed immersion of smooth schemes. We denote by

$$(2.3.39) \quad 0: Z \hookrightarrow T_ZX$$

the zero section. Consider the map

$$(2.3.40) \quad \tilde{i} = A_{\text{id}}(i): A_Z(Z) = Z \times \mathbf{A}^1 \hookrightarrow A_Z(X).$$

Let S be the strict henselization of $\mathbf{A}^1$ at 0 and let $\bar{s}$ and $\bar{\eta}$ be the closed point and a geometric generic point of S respectively. The morphism $\tilde{i} = A_{\text{id}}(i)$ gives the diagram

$$(2.3.41) \quad \begin{array}{ccccc} Z_{\bar{s}} & \xrightarrow{s_Z} & Z \times S & \xleftarrow{j_Z} & Z_{\bar{\eta}} \\ 0 \downarrow & & \downarrow \tilde{i} & & \downarrow i_{\bar{\eta}} \\ T_{Z_s} X_{\bar{s}} & \xrightarrow{s_X} & A_Z(X)_S & \xleftarrow{j_X} & X_{\bar{\eta}}. \end{array}$$

Here j_X and j_Z are the geometric generic fiber maps.

Let $\mathcal{F} \in D_{\text{ctf}}(X, \Lambda)$ and $\mathcal{G} \in D_{\text{ctf}}(Z, \Lambda)$. Applying (2.3.35), and (2.3.38) to the map $\tilde{i} = A_{\text{id}}(i)$, we get natural morphisms

$$(2.3.42) \quad \alpha(\mathcal{F}) : 0^* \nu_{Z/X}(\mathcal{F}) \xrightarrow{(2.3.35)} i^* \mathcal{F}, \quad \beta(\mathcal{F}) : i^! \mathcal{F} \xrightarrow{(2.3.38)} 0^! \nu_{Z/X}(\mathcal{F}),$$

which are isomorphisms by Proposition 2.3.7. Also by applying (2.3.37), we get an isomorphism

$$(2.3.43) \quad \gamma(\mathcal{G}) : 0_! \mathcal{G} \rightarrow \nu_{Z/X}(i_! \mathcal{G}).$$

Taking $\mathcal{F} = \Lambda$, we get isomorphisms

$$(2.3.44) \quad i^! \Lambda \xrightarrow[\simeq]{\beta(\Lambda)} 0^! \nu_{Z/X}(\Lambda) \simeq 0^! \Lambda.$$

Proposition 2.3.8. *Let $i : Z \hookrightarrow X$ be a closed immersion of smooth k -schemes, and let $\mathcal{F} \in D_{\text{ctf}}(X, \Lambda)$ and $\mathcal{G} \in D_{\text{ctf}}(X, \Lambda)$. Let η_0 be the generic point of the henselization of $\mathbf{A}^1$ at 0, and write $0 : Z \hookrightarrow T_Z X$ for the zero section. Using the notation of Convention 2.3.5, we have a commutative diagram in $D_{\text{ctf}}(Z \times_k \eta_0, \Lambda)$*

$$(2.3.45) \quad \begin{array}{ccc} i^* \mathcal{F} \otimes i^! \mathcal{G} & \xrightarrow{c_{i, \mathcal{F}, \mathcal{G}}} & i^! (\mathcal{F} \otimes \mathcal{G}) \\ \alpha_{\mathcal{F}} \otimes 1 \uparrow \simeq & & \downarrow \simeq \beta_{\mathcal{F} \otimes \mathcal{G}} \\ 0^* \nu_{Z/X}(\mathcal{F}) \otimes i^! \mathcal{G} & & \\ \simeq \downarrow 1 \otimes \beta_{\mathcal{G}} & & \\ 0^* \nu_{Z/X}(\mathcal{F}) \otimes 0^! \nu_{Z/X}(\mathcal{G}) & \xrightarrow{c_0} 0^! (\nu_{Z/X}(\mathcal{F}) \otimes \nu_{Z/X}(\mathcal{G})) \xrightarrow{0^! m} 0^! \nu_{Z/X}(\mathcal{F} \otimes \mathcal{G}), \end{array}$$

where m is the canonical tensor map for the nearby cycles functor.

In particular, setting $\mathcal{G} = \Lambda$, we obtain that the morphism i is $\mathcal{F}$ -transversal if and only if 0 is $\nu_{Z/X}(\mathcal{F})$ -transversal.

Proof. This follows from Lemma 2.3.2, applied to the morphism $\tilde{i} = A_{\text{id}}(i) : A_Z(Z) = Z \times \mathbf{A}^1 \hookrightarrow A_Z(X)$. $\square$

Now we construct an invariant structure on specializations.

Lemma 2.3.9. *Let $f : X \rightarrow Y$ be a smooth morphism, and let $i : Z \hookrightarrow X$ be a regular immersion between smooth k -schemes. For $\mathcal{K} \in D_{\text{ctf}}(Y, \Lambda)$, the complex $\nu_{Z/X}(f^* \mathcal{K}) \in D_{\text{ctf}}(T_Z X \times_k \eta_0, \Lambda)$ is $T(X/Y)|_Z$ -invariant for the action of $T(X/Y)|_Z$ on $T_Z X$ defined by (2.2.28).*

Proof. Since $f \circ \text{pr}_{f,1} = f \circ \text{pr}_{f,2}$, there is an isomorphism

$$(2.3.46) \quad \text{pr}_{f,1}^* f^* \mathcal{K} \simeq \text{pr}_{f,2}^* f^* \mathcal{K}.$$

By Lemma 2.2.5.2, the morphisms (2.2.26) and (2.2.29) are smooth over $\mathbf{A}^1$. Applying (2.3.3) to these two morphisms yields natural isomorphisms

$$(2.3.47) \quad \nu_{Z/X \times_Y X} \text{pr}_{f,2}^* f^* \mathcal{K} \simeq \alpha_{Z/X/Y}^* \nu_{Z/X} f^* \mathcal{K},$$

$$(2.3.48) \quad \nu_{Z/X \times_Y X} \text{pr}_{f,1}^* f^* \mathcal{K} \simeq \text{pr}_{T_Z X}^* \nu_{Z/X} f^* \mathcal{K}.$$

Combining (2.3.46), (2.3.47), and (2.3.48), we obtain an isomorphism

$$(2.3.49) \quad \alpha_{Z/X/Y}^* \nu_{Z/X} f^* \mathcal{K} \simeq \text{pr}_{T_Z X}^* \nu_{Z/X} f^* \mathcal{K} \quad \text{in } D_{\text{ctf}}((T(X/Y) \times_X T_Z X) \times_k \eta_0, \Lambda).$$

This proves the assertion. $\square$

We recall the internal Hom object $\mathcal{H} = \mathcal{H}om$ for cohomological correspondences (cf. [LZ22]) and define its specialization version.

Definition 2.3.10. Let X be a smooth scheme over a field k . We identify the normal bundle $T_X(X \times X)$ for the diagonal $\delta = \delta_X: X \rightarrow X \times X$ with the tangent bundle TX . Let η_0 denote the generic point of the henselization of $\mathbf{A}^1$ at 0. We define bifunctors

$$(2.3.50) \quad \mathcal{H} = \mathcal{H}_X: D_{\text{ctf}}(X, \Lambda)^{\text{op}} \times D_{\text{ctf}}(X, \Lambda) \rightarrow D_{\text{ctf}}(X \times X, \Lambda),$$

$$(2.3.51) \quad \nu \mathcal{H}om: D_{\text{ctf}}(X, \Lambda)^{\text{op}} \times D_{\text{ctf}}(X, \Lambda) \rightarrow D_{\text{ctf}}(TX \times_k \eta_0, \Lambda),$$

by $\mathcal{H}_X(\mathcal{F}_1, \mathcal{F}_2) = \mathcal{H}om_{X \times X}(\text{pr}_1^* \mathcal{F}_1, \text{pr}_2^* \mathcal{F}_2)$ and $\nu \mathcal{H}om = \nu_{X/X \times X}(\mathcal{H}_X)$ (2.3.27). If $\mathcal{F}_1 = \mathcal{F}_2 = \mathcal{F}$, we also write

$$(2.3.52) \quad \mathcal{H}_{\mathcal{F}} = \mathcal{H}_X(\mathcal{F}, \mathcal{F}).$$

Applying the construction of 2.2.9 to $i: Z \hookrightarrow X$ and $X \rightarrow \text{Spec}(k)$, we obtain

$$(2.3.53) \quad \tilde{\alpha}: A_X(X \times X) \times_{X \times \mathbf{A}^1} A_Z(X) \rightarrow A_Z(X),$$

$$(2.3.54) \quad \alpha: TX|_Z \times_Z T_Z X = TX \times_X T_Z X \rightarrow T_Z X.$$

We denote by $q: TX|_Z \rightarrow T_Z X$ the quotient bundle morphism. Unwinding the definitions, we see that for a geometric point $z \in Z$, the action

$$(2.3.55) \quad \alpha_z: T_z X \times (T_Z X)_z \rightarrow (T_Z X)_z$$

is given as follows:

$$(2.3.56) \quad \alpha_z = \text{add} \circ (q_z \times \text{id}): T_z X \times T_z X / T_z Z \rightarrow T_z X / T_z Z.$$

Consider the following commutative diagram

$$(2.3.57) \quad \begin{array}{ccccc} & A_Z(X) & \xrightarrow{A_i(\text{id})} & A_X(X) & \\ & \downarrow A_{\text{id}}(\delta_X) & & \downarrow A_{\text{id}}(\delta_X) & \\ A_Z(X) & \xleftarrow{A_{\text{id}}(\text{pr}_2)} A_Z(X \times X) & \xrightarrow{A_i(\text{id})} & A_X(X \times X) & \\ & \downarrow A_{\text{id}}(\text{pr}_1) & & \downarrow & \\ & A_Z(X) & \longrightarrow & A_X(X) = X \times \mathbf{A}^1. & \end{array}$$

The bottom square of (2.3.57) is cartesian by Corollary 2.2.8. Hence the upper square of (2.3.57) is also cartesian, which gives the following commutative diagram over the special fiber (note that $T_Z(X \times X) \simeq TX \times_X T_ZX$ by (2.2.25))

$$(2.3.58) \quad \begin{array}{ccccc} & & T_ZX & \longrightarrow & X \\ & & \downarrow (0_{TX}, \text{id}) & & \downarrow 0_{TX} \\ T_ZX & \xleftarrow{\alpha} & TX \times_X T_ZX & \xrightarrow{\text{pr}_{TX}} & TX \\ & & \downarrow \text{pr}_{T_ZX} & & \downarrow \\ & & T_ZX & \longrightarrow & X, \end{array}$$

where the squares are cartesian.

Proposition 2.3.11. *Let $i : Z \hookrightarrow X$ be a closed immersion of smooth schemes. For all $\mathcal{F} \in D_{\text{ctf}}(X, \Lambda)$, the complex $\nu_{Z/X}(\mathcal{F}) \in D_{\text{ctf}}(T_ZX \times_k \eta_0, \Lambda)$ is a retract of*

$$(2.3.59) \quad \begin{aligned} & \alpha_!(\text{pr}_{TX}^* \nu \mathcal{H}om(\mathcal{F}, \mathcal{F}) \otimes \text{pr}_{T_ZX}^* \nu_{Z/X}(\mathcal{F})) \\ & \simeq \alpha_!(\text{pr}_{TX|Z}^*(\nu \mathcal{H}om(\mathcal{F}, \mathcal{F})|_{TX|Z}) \otimes \text{pr}_{T_ZX}^* \nu_{Z/X}(\mathcal{F})). \end{aligned}$$

The retraction maps are given by the following two compositions:

$$(2.3.60) \quad \begin{aligned} \nu_{Z/X}(\mathcal{F}) & \simeq \alpha_!(0_{TX}, \text{id})_!(0_{TX}, \text{id})^* \text{pr}_{T_ZX}^* \nu_{Z/X}(\mathcal{F}) \\ & \simeq \alpha_!((0_{TX}, \text{id})_! \Lambda \otimes \text{pr}_{T_ZX}^* \nu_{Z/X}(\mathcal{F})) \\ & \simeq \alpha_!(\text{pr}_{TX}^* 0_{TX!} \nu_{X/X}(\Lambda) \otimes \text{pr}_{T_ZX}^* \nu_{Z/X}(\mathcal{F})) \\ & \xrightarrow{(2.3.37)} \alpha_!(\text{pr}_{TX}^* \nu_{X/X \times X}(\delta_X! \Lambda) \otimes \text{pr}_{T_ZX}^* \nu_{Z/X}(\mathcal{F})) \\ & \xrightarrow{\text{coev}} \alpha_!(\text{pr}_{TX}^* \nu_{X/X \times X}(\mathcal{H}_{\mathcal{F}}) \otimes \text{pr}_{T_ZX}^* \nu_{Z/X}(\mathcal{F})) \\ & = \alpha_!(\text{pr}_{TX}^* \nu \mathcal{H}om(\mathcal{F}, \mathcal{F}) \otimes \text{pr}_{T_ZX}^* \nu_{Z/X}(\mathcal{F})), \end{aligned}$$

$$(2.3.61) \quad \begin{aligned} & \alpha_!(\text{pr}_{TX}^* \nu \mathcal{H}om(\mathcal{F}, \mathcal{F}) \otimes \text{pr}_{T_ZX}^* \nu_{Z/X}(\mathcal{F})) \\ & = \alpha_!(\text{pr}_{TX}^* \nu_{X/X \times X}(\mathcal{H}_{\mathcal{F}}) \otimes \text{pr}_{T_ZX}^* \nu_{Z/X}(\mathcal{F})) \\ & \xrightarrow{(2.3.35)} \alpha_!(\nu_{Z/X \times X}(\mathcal{H}_{\mathcal{F}}) \otimes \nu_{Z/X \times X}(\text{pr}_1^* \mathcal{F})) \\ & \xrightarrow{m} \alpha_! \nu_{Z/X \times X}(\mathcal{H}_{\mathcal{F}} \otimes \text{pr}_1^* \mathcal{F}) \\ & \xrightarrow{(2.3.37)} \nu_{Z/X}(\text{pr}_{2!}(\mathcal{H}_{\mathcal{F}} \otimes \text{pr}_1^* \mathcal{F})) \\ & \xrightarrow{\text{ev}} \nu_{Z/X}(\text{pr}_{2!} \text{pr}_2^! \mathcal{F}) \\ & \xrightarrow{\text{counit}} \nu_{Z/X}(\mathcal{F}), \end{aligned}$$

where m is induced by the canonical tensor map for the nearby cycles functor.

Proof. Consider the morphisms $X \xrightarrow{\delta_X} X \times X \xrightarrow{\text{pr}_2} X$. We first note that the morphisms (cf. (4.1.2))

$$(2.3.62) \quad \delta_X! \Lambda \otimes \text{pr}_1^* \mathcal{F} \xrightarrow{\text{coev} \otimes \text{id}} \mathcal{H}_{\mathcal{F}} \otimes \text{pr}_1^* \mathcal{F} \xrightarrow{\text{ev}} \text{pr}_2^! \mathcal{F}$$

induce a retraction

$$(2.3.63) \quad \mathcal{F} \simeq \mathrm{pr}_{2!}(\delta_{X!}\Lambda \otimes \mathrm{pr}_1^*\mathcal{F}) \rightarrow \mathrm{pr}_{2!}(\mathcal{H}_{\mathcal{F}} \otimes \mathrm{pr}_1^*\mathcal{F}) \rightarrow \mathcal{F}.$$

By naturality, we have a commutative diagram

$$(2.3.64) \quad \begin{array}{ccc} \alpha_!(\mathrm{pr}_{TX}^*\nu_{X/X \times X}(\delta_{X!}\Lambda) \otimes \mathrm{pr}_{TZ}^*\nu_{Z/X}(\mathcal{F})) & \xrightarrow{\mathrm{coev}} & \alpha_!(\mathrm{pr}_{TX}^*\nu_{X/X \times X}(\mathcal{H}_{\mathcal{F}}) \otimes \mathrm{pr}_{TZ}^*\nu_{Z/X}(\mathcal{F})) \\ \downarrow (2.3.35) & & \downarrow (2.3.35) \\ \alpha_!(\nu_{Z/X \times X}(\delta_{X!}\Lambda) \otimes \nu_{Z/X \times X}(\mathrm{pr}_1^*\mathcal{F})) & \xrightarrow{\mathrm{coev}} & \alpha_!(\nu_{Z/X \times X}(\mathcal{H}_{\mathcal{F}}) \otimes \nu_{Z/X \times X}(\mathrm{pr}_1^*\mathcal{F})) \\ \downarrow m & & \downarrow m \\ \alpha_!\nu_{Z/X \times X}(\delta_{X!}\Lambda \otimes \mathrm{pr}_1^*\mathcal{F}) & \xrightarrow{\mathrm{coev}} & \alpha_!\nu_{Z/X \times X}(\mathcal{H}_{\mathcal{F}} \otimes \mathrm{pr}_1^*\mathcal{F}) \\ \downarrow (2.3.37) & & \downarrow (2.3.37) \\ \nu_{Z/X}(\mathrm{pr}_{2!}(\delta_{X!}\Lambda \otimes \mathrm{pr}_1^*\mathcal{F})) & \xrightarrow{\mathrm{coev}} & \nu_{Z/X}(\mathrm{pr}_{2!}(\mathcal{H}_{\mathcal{F}} \otimes \mathrm{pr}_1^*\mathcal{F})). \end{array}$$

To prove Proposition 2.3.11, it suffices to show that the following diagram commutes:

$$(2.3.65) \quad \begin{array}{ccccc} \nu_{Z/X}(\mathcal{F}) & \xlongequal{\quad} & \nu_{Z/X}(\mathcal{F}) & & \\ \downarrow \simeq & & \downarrow \simeq & \searrow \text{diagonal} & \\ \alpha_!(\mathrm{pr}_{TX}^*0_{TX!}\Lambda \otimes \mathrm{pr}_{TZ}^*\nu_{Z/X}(\mathcal{F})) & \xrightarrow{(1)} & & & \\ \downarrow \simeq & & \downarrow & & \\ \alpha_!(\mathrm{pr}_{TX}^*\nu_{X/X \times X}(\delta_{X!}\Lambda) \otimes \mathrm{pr}_{TZ}^*\nu_{Z/X}(\mathcal{F})) & \xrightarrow{(2)} & \nu_{Z/X}(\mathrm{pr}_{2!}(\delta_{X!}\Lambda \otimes \mathrm{pr}_1^*\mathcal{F})) & & \\ \downarrow & & \downarrow & & \\ \alpha_!(\mathrm{pr}_{TX}^*\nu_{X/X \times X}(\mathcal{H}_{\mathcal{F}}) \otimes \mathrm{pr}_{TZ}^*\nu_{Z/X}(\mathcal{F})) & \xrightarrow{(3)} & \nu_{Z/X}(\mathrm{pr}_{2!}(\mathcal{H}_{\mathcal{F}} \otimes \mathrm{pr}_1^*\mathcal{F})) & \xrightarrow{\quad} & \nu_{Z/X}(\mathcal{F}), \end{array}$$

where (2) is given by the commutative diagram (2.3.64). Commutativity of (3) follows from (2.3.63). It remains to verify that the diagram (1) commutes.

Applying Lemma 2.3.1 to the morphism $A_{\mathrm{id}}(\delta_X) : A_Z(X) \rightarrow A_Z(X \times X)$ and $\Lambda \in D_{\mathrm{ctf}}(X, \Lambda)$, $\mathrm{pr}_1^*\mathcal{F} \in D_{\mathrm{ctf}}(X \times X, \Lambda)$, we obtain the following commutative diagram (note that $\nu_{X/X \times X}(\delta_{X!}\Lambda) \simeq 0_{TX!}\Lambda$ and $\nu_{Z/X}(\Lambda) \simeq \Lambda$)

$$(2.3.66) \quad \begin{array}{ccc} \mathrm{pr}_{TX}^*\nu_{X/X \times X}(\delta_{X!}\Lambda) \otimes \mathrm{pr}_{TZ}^*\nu_{Z/X}(\mathcal{F}) & & \\ \downarrow \simeq & & \\ (0_{TX}, \mathrm{id})_!\nu_{Z/X}(\Lambda) \otimes \nu_{Z/X \times X}(\mathrm{pr}_1^*\mathcal{F}) & \xrightarrow{(2.3.10)} & \nu_{Z/X \times X}(\delta_{X!}\Lambda \otimes \mathrm{pr}_1^*\mathcal{F}) \\ \downarrow \simeq & & \downarrow \simeq \\ (0_{TX}, \mathrm{id})_!(\nu_{Z/X}(\Lambda) \otimes (0_{TX}, \mathrm{id})^*\nu_{Z/X \times X}(\mathrm{pr}_1^*\mathcal{F})) & \xrightarrow{\quad} & \nu_{Z/X \times X}(\delta_{X!}(\Lambda \otimes \delta_X^*\mathrm{pr}_1^*\mathcal{F})) \\ \downarrow \simeq & & \downarrow \simeq \\ (0_{TX}, \mathrm{id})_!(0_{TX}, \mathrm{id})^*\nu_{Z/X \times X}(\mathrm{pr}_1^*\mathcal{F}) & \xrightarrow{\quad} & \nu_{Z/X \times X}(\delta_{X!}\delta_X^*\mathrm{pr}_1^*\mathcal{F}). \end{array}$$

We apply the natural transformation (2.3.37) associated with $A_Z(X \times X) \xrightarrow{A_{\text{id}}(\text{pr}_2)} A_Z(X)$ to the composite of the right-hand vertical arrows in (2.3.66). By naturality, we obtain a commutative diagram

$$(2.3.67) \quad \begin{array}{ccc} \alpha_! \nu_{Z/X \times X}(\delta_{X!} \Lambda \otimes \text{pr}_1^* \mathcal{F}) & \xrightarrow{(2.3.37)} & \nu_{Z/X}(\text{pr}_{2!}(\delta_{X!} \Lambda \otimes \text{pr}_1^* \mathcal{F})) \\ \simeq \downarrow & & \downarrow \simeq \\ \alpha_! \nu_{Z/X \times X}(\delta_{X!} \delta_X^* \text{pr}_1^* \mathcal{F}) & \xrightarrow{(2.3.37)} & \nu_{Z/X}(\text{pr}_{2!} \delta_{X!} \delta_X^* \text{pr}_1^* \mathcal{F}). \end{array}$$

Applying $\alpha_!$ to (2.3.66) and combining the resulting diagram with (2.3.67), we obtain the commutativity of the upper square in the following diagram:

$$(2.3.68) \quad \begin{array}{ccc} \alpha_!(\text{pr}_{TX}^* \nu_{X/X \times X}(\delta_{X!} \Lambda) \otimes \text{pr}_{TZ}^* \nu_{Z/X}(\mathcal{F})) & \xrightarrow{(2.3.64)} & \nu_{Z/X}(\text{pr}_{2!}(\delta_{X!} \Lambda \otimes \text{pr}_1^* \mathcal{F})) \\ \text{projection formula} \downarrow \simeq & & \simeq \downarrow \text{projection formula} \\ \alpha_!(0_{TX}, \text{id})_!(0_{TX}, \text{id})^* \text{pr}_{TZ}^* \nu_{Z/X}(\mathcal{F}) & \xrightarrow{\simeq} & \nu_{Z/X}(\text{pr}_{2!} \delta_{X!} \delta_X^* \text{pr}_1^* \mathcal{F}) \\ \simeq \downarrow & & \downarrow \simeq \\ \nu_{Z/X}(\mathcal{F}) & \xlongequal{\hspace{2cm}} & \nu_{Z/X}(\mathcal{F}). \end{array}$$

The lower square commutes by the transitivity of (2.3.37) for $\text{pr}_{2!} \circ \delta_{X!} = \text{id}_{X!}$ and of (2.3.35) for $\delta_X^* \circ \text{pr}_1^* = \text{id}_X^*$.

Now (2.3.68) implies that the diagram (1) in (2.3.65) commutes. This finishes the proof. $\square$

Let $f : X \rightarrow Y$ be a proper smooth morphism between smooth schemes over a field k . Consider the following cartesian diagrams:

$$(2.3.69) \quad \begin{array}{ccc} X \times_Y X & \xrightarrow{\text{pr}_{f,1}} & X \\ \downarrow \text{pr}_{f,2} & & \downarrow f \\ X & \xrightarrow{f} & Y, \end{array} \quad \begin{array}{ccc} X \times_Y X & \xleftarrow{i_f} & X \times X \\ \downarrow h_f & & \downarrow f \times f \\ Y & \xleftarrow{\delta_Y} & Y \times Y. \end{array}$$

By Lemma 2.2.5.(4), the following diagram is cartesian:

$$(2.3.70) \quad \begin{array}{ccc} A_{X \times_Y X}(X \times X) & \xrightarrow{A_{h_f}(f \times f)} & A_Y(Y \times Y) \\ \downarrow & & \downarrow \\ X \times X \times \mathbf{A}^1 & \xrightarrow{f \times f \times \text{id}} & Y \times Y \times \mathbf{A}^1. \end{array}$$

The morphism $A_{h_f}(f \times f)$ over the special fiber is the following morphism

$$(2.3.71) \quad T_{X \times_Y X}(X \times X) \xrightarrow{T_{h_f}(f \times f)} T_Y(Y \times Y).$$

Since f is a smooth morphism, we have an isomorphism

$$(2.3.72) \quad T_{X \times_Y X}(X \times X) \simeq X \times_Y X \times_Y TY,$$

together with a commutative diagram

$$(2.3.73) \quad \begin{array}{ccc} T_{X \times_Y X}(X \times X) & \xrightarrow{T_{h_f}(f \times f)} & T_Y(Y \times Y) \\ \simeq \downarrow & & \parallel \\ X \times_Y X \times_Y TY & \xrightarrow{h_f \times \text{id}_{TY} = \text{pr}_{TY}} & TY. \end{array}$$

Lemma 2.3.12. *Let $f : X \rightarrow Y$ be a proper smooth morphism between smooth schemes over a field k . Let $\mathcal{F}_1, \mathcal{F}_2 \in D_{\text{ctf}}(X, \Lambda)$. We put*

$$(2.3.74) \quad \mathcal{H}_X(\mathcal{F}_1, \mathcal{F}_2) = \mathcal{H}om_{X \times X}(\text{pr}_1^* \mathcal{F}_1, \text{pr}_2^! \mathcal{F}_2) \in D_{\text{ctf}}(X \times X, \Lambda),$$

and similarly for $\mathcal{H}_Y$. Then there are natural isomorphisms

$$(2.3.75) \quad (f \times f)_* \mathcal{H}_X(\mathcal{F}_1, \mathcal{F}_2) \simeq \mathcal{H}_Y(f_* \mathcal{F}_1, f_* \mathcal{F}_2),$$

$$(2.3.76) \quad (T_{h_f}(f \times f))_* \nu_{i_f} \mathcal{H}_X(\mathcal{F}_1, \mathcal{F}_2) \simeq \nu_{\delta_Y} \mathcal{H}_Y(f_* \mathcal{F}_1, f_* \mathcal{F}_2).$$

Proof. By [LZ22, Proposition 2.11] and the Künneth formula, we have

$$(2.3.77) \quad \begin{aligned} (f \times f)_* \mathcal{H}_X(\mathcal{F}_1, \mathcal{F}_2) &\simeq (f \times f)_*(D_X \mathcal{F}_1 \boxtimes \mathcal{F}_2) \\ &\simeq f_* D_X \mathcal{F}_1 \boxtimes f_* \mathcal{F}_2 \simeq D_Y(f_* \mathcal{F}_1) \boxtimes f_* \mathcal{F}_2 \simeq \mathcal{H}_Y(f_* \mathcal{F}_1, f_* \mathcal{F}_2). \end{aligned}$$

By (2.3.70), the morphism $A_{X \times_Y X}(X \times X) \rightarrow A_Y(Y \times Y)$ is proper over $\mathbf{A}^1$. By proper base change for nearby cycles (2.3.4), we obtain

$$(2.3.78) \quad (T_{h_f}(f \times f))_* \nu_{i_f} \mathcal{H}_X(\mathcal{F}_1, \mathcal{F}_2) \simeq \nu_{\delta_Y}(f \times f)_* \mathcal{H}_X(\mathcal{F}_1, \mathcal{F}_2) \simeq \nu_{\delta_Y} \mathcal{H}_Y(f_* \mathcal{F}_1, f_* \mathcal{F}_2).$$

□

Lemma 2.3.13. *Let $f : X \rightarrow Y$ be a proper smooth morphism between smooth schemes over k . Consider the diagrams (2.3.69), (2.3.70) and (2.3.73). Let $\mathcal{F} \in D_{\text{ctf}}(X, \Lambda)$ and $\mathcal{G} \in D_{\text{ctf}}(Y, \Lambda)$. Assume there is a retraction $\mathcal{G} \xrightarrow{i} f_* \mathcal{F} \xrightarrow{r} \mathcal{G}$ (i.e., $ri = \text{id}$). Then $\mathcal{H}_Y(\mathcal{G}, \mathcal{G})$ is a retract of $\mathcal{H}_Y(f_* \mathcal{F}, f_* \mathcal{F})$, and $\nu \mathcal{H}om_Y(\mathcal{G}, \mathcal{G}) := \nu_{\delta_Y} \mathcal{H}_Y(\mathcal{G}, \mathcal{G})$ is a retract of*

$$(2.3.79) \quad \nu_{\delta_Y} \mathcal{H}_Y(f_* \mathcal{F}, f_* \mathcal{F}) \stackrel{(2.3.76)}{\simeq} (T_{h_f}(f \times f))_* \nu_{i_f} \mathcal{H}_X(\mathcal{F}, \mathcal{F}).$$

The retraction $\mathcal{H}_Y(\mathcal{G}, \mathcal{G}) \rightarrow \mathcal{H}_Y(f_* \mathcal{F}, f_* \mathcal{F}) \rightarrow \mathcal{H}_Y(\mathcal{G}, \mathcal{G})$ is defined by (2.3.80) and (2.3.81) below.

Proof. Consider the following two morphisms

$$(2.3.80) \quad \mathcal{H}om_{Y \times Y}(\text{pr}_1^* \mathcal{G}, \text{pr}_2^! \mathcal{G}) \rightarrow \mathcal{H}om_{Y \times Y}(\text{pr}_1^* f_* \mathcal{F}, \text{pr}_2^! f_* \mathcal{F}),$$

$$(2.3.81) \quad \mathcal{H}om_{Y \times Y}(\text{pr}_1^* f_* \mathcal{F}, \text{pr}_2^! f_* \mathcal{F}) \rightarrow \mathcal{H}om_{Y \times Y}(\text{pr}_1^* \mathcal{G}, \text{pr}_2^! \mathcal{G}),$$

where (2.3.80) is defined by the functoriality for r on the first factor and i for the second factor, and the map (2.3.81) is defined analogously. By construction, the composition of (2.3.80) and (2.3.81) is the identity. Therefore, $\mathcal{H}_Y(\mathcal{G}, \mathcal{G})$ is a retract of $\mathcal{H}_Y(f_* \mathcal{F}, f_* \mathcal{F})$. Applying the functor ν_{δ_Y} , we get that $\nu \mathcal{H}om_Y(\mathcal{G}, \mathcal{G}) = \nu_{\delta_Y} \mathcal{H}_Y(\mathcal{G}, \mathcal{G})$ is a retract of

$$\nu_{\delta_Y} \mathcal{H}_Y(f_* \mathcal{F}, f_* \mathcal{F}) \stackrel{(2.3.76)}{\simeq} (T_{h_f}(f \times f))_* \nu_{i_f} \mathcal{H}_X(\mathcal{F}, \mathcal{F}).$$

This finishes the proof. □

We compute the nearby cycles complex for certain tamely ramified sheaves.

Lemma 2.3.14. *Let X be a scheme smooth over the semi-stable curve $\mathrm{Spec} k[s, t, u]/(st - u)$ over $\mathbf{A}^1 = \mathrm{Spec} k[u]$ and let $D_1, D_2 \subseteq X$ be the divisors defined by s and t respectively. Let $\mathcal{F}$ be a locally constant constructible sheaf of Λ -modules on $U = X - (D_1 \cup D_2)$ tamely ramified along $D_1 \cup D_2$ and let $\Psi = \Psi_{\mathcal{F}}$ be the nearby cycles complex on $X \times_{\mathbf{A}^1} 0 = D_1 \cup D_2$ with respect to the morphism $X \rightarrow \mathbf{A}^1 = \mathrm{Spec} k[u]$ at $u = 0$. Let $j_1: D_1^\circ = D_1 - (D_1 \cap D_2) \rightarrow D_1$ and $j_2: D_2^\circ = D_2 - (D_1 \cap D_2) \rightarrow D_2$ be the open immersions.*

1. *The restrictions $\Psi_1^\circ = \Psi|_{D_1^\circ}$ and $\Psi_2^\circ = \Psi|_{D_2^\circ}$ are locally constant and tamely ramified along $D_1 \cap D_2$.*

2. *The canonical morphisms $\Psi|_{D_1} \rightarrow j_{1*}\Psi_1^\circ$ and $\Psi|_{D_2} \rightarrow j_{2*}\Psi_2^\circ$ are isomorphisms.*

Proof. 1. Let $\mathbf{A}^{1t} = \mathrm{Spec} k[u^{1/n}; p \nmid n]$ be the maximal tamely ramified covering and $j^t: U \times_{\mathbf{A}^1} \mathbf{A}^{1t} \rightarrow X \times_{\mathbf{A}^1} \mathbf{A}^{1t}$ be the base change of the open immersion $j: U \rightarrow X$. Then, by the assumption that $\mathcal{F}$ is tamely ramified along $D_1 \cup D_2$, the extension $R^0 j_*^t \mathcal{F}$ is locally constant outside the inverse image of $D_1 \cap D_2$ and the restrictions $(R^0 j_*^t \mathcal{F})|_{D_1^\circ}$ and $(R^0 j_*^t \mathcal{F})|_{D_2^\circ}$ are tamely ramified along $D_1 \cap D_2$. Since $X - (D_1 \cap D_2)$ is smooth over $\mathbf{A}^1$, we have isomorphisms $(R^0 j_*^t \mathcal{F})|_{D_1^\circ} \rightarrow \Psi|_{D_1^\circ}$ and $(R^0 j_*^t \mathcal{F})|_{D_2^\circ} \rightarrow \Psi|_{D_2^\circ}$ of locally constant sheaves on D_1° and D_2° .

2. It suffices to show the isomorphisms at each geometric point x of $D_1 \cap D_2$. Let X_x be the strict localization and $U_x = X_x \times_{\mathrm{Spec} k[u]} \mathrm{Spec} k(u)$ be the generic fiber. Let $U_x^t = U_x \times_{\mathrm{Spec} k[s, t, u]/(st - u)} \mathrm{Spec} \varinjlim_{p \nmid n} k[s^{1/n}, t^{1/n}, u^{1/n}]/(s^{1/n}t^{1/n} - u^{1/n})$ be the universal tamely ramified covering. By cohomological purity, the canonical morphism $\Lambda \rightarrow \Gamma(U_x^t, \Lambda)$ is an isomorphism (cf. [Gro72, Théorème 3.3]). The canonical morphism $U \rightarrow \mathbf{A}^1$ is lifted to $U_x^t \rightarrow \mathbf{A}^{1t}$.

We identify the Galois group $\mathrm{Gal}(U_x^t/U_x)$ with the product $I_s \times I_t$ of $I_s = I_t = \varprojlim_{p \nmid n} \mu_n$ acting on $s^{1/n}, t^{1/n}$ by multiplication respectively. The morphism $U_x^t \rightarrow \mathbf{A}^{1t}$ induces $I_s \times I_t \rightarrow I_u = \mathrm{Gal}(\mathbf{A}^{1t}/\mathbf{A}^1)$. Let $I_0 = \mathrm{Ker}(I_s \times I_t \rightarrow I_u)$ be the kernel. The pull-back of $\mathcal{F}$ on U_x^t is a constant sheaf and the pull-back of $\mathcal{F}$ on U_x is the locally constant sheaf corresponding to the representation $M = H^0(U_x^t, \mathcal{F})$ of $I_s \times I_t$. The isomorphism $\Lambda \rightarrow \Gamma(U_x^t, \Lambda)$ induces an isomorphism $\Gamma(I_0, M) = \Gamma(I_0, \Gamma(U_x^t, \mathcal{F})) \rightarrow \Psi_x$.

For $i = 1, 2$, let $D_{ix} = D_i \times_X X_x$ and $D_{ix}^\circ = D_i^\circ \times_X X_x = D_{ix} - \{x\}$. Let $D_{ix}^{\circ t}$ be the maximal tamely ramified covering of D_{ix}° . We identify $\mathrm{Gal}(D_{1x}^{\circ t}/D_{1x}^\circ) = I_t$ and $\mathrm{Gal}(D_{2x}^{\circ t}/D_{2x}^\circ) = I_s$ with I_0 by the isomorphisms $I_0 \rightarrow I_t$ and $I_0 \rightarrow I_s$ induced by the projections. The canonical morphisms $\Lambda \rightarrow \Gamma(D_{ix}^{\circ t}, \Lambda)$ are also isomorphisms. The pull-backs of Ψ on $D_{1x}^{\circ t}$ and $D_{2x}^{\circ t}$ correspond to the representation M of I_0 . The isomorphisms $\Lambda \rightarrow \Gamma(D_{ix}^{\circ t}, \Lambda)$ induce isomorphisms $\Gamma(I_0, M) = \Gamma(I_0, \Gamma(D_{1x}^{\circ t}, \Psi_1^\circ)) \rightarrow (j_{1*}\Psi_1^\circ)_x$ and $\Gamma(I_0, M) = \Gamma(I_0, \Gamma(D_{2x}^{\circ t}, \Psi_2^\circ)) \rightarrow (j_{2*}\Psi_2^\circ)_x$. Hence we have canonical isomorphisms $\Psi_x \rightarrow (j_{1*}\Psi_1^\circ)_x$ and $\Psi_x \rightarrow (j_{2*}\Psi_2^\circ)_x$. $\square$

Let $X = \mathbf{A}^1 = \mathrm{Spec} k[x]$ and $Z = \{0\} \subseteq X$ be the reduced closed subscheme consisting of the origin. Let $(X \times \mathbf{A}^1)' \rightarrow X \times \mathbf{A}^1 = \mathrm{Spec} k[x, u]$ be the blow-up at $(0, 0)$ and $E = \mathbf{P}^1$ be the exceptional divisor. Let $(\{0\} \times \mathbf{A}^1)', (X \times \{0\})' \subseteq (X \times \mathbf{A}^1)'$ be the proper transforms of $\{0\} \times \mathbf{A}^1$ and $X \times \{0\}$ respectively and let $0, \infty \in E$ be the intersection points with $(\{0\} \times \mathbf{A}^1)', (X \times \{0\})'$. The complement $(X \times \mathbf{A}^1)' - (X \times \{0\})'$ is the deformation to normal bundle $A_Z X$ and $E - \{\infty\}$ is canonically identified with the normal bundle $T = T_Z X$.

Lemma 2.3.15. *Let $\mathcal{F}$ be a locally constant sheaf on $\mathbf{G}_m = \text{Spec } k[x^{\pm 1}]$ tamely ramified at 0 and let $j: \mathbf{G}_m \rightarrow X = \mathbf{A}^1 = \text{Spec } k[x]$ be the open immersion. Let $\text{pr}_1: X \times \mathbf{G}_m \rightarrow X$ be the projection and $\Psi = \Psi_{\text{pr}_1^* j!} \mathcal{F}$ be the nearby cycles complex on $(X \times \mathbf{A}^1)' \times_{\mathbf{A}^1} 0 = E \cup (X \times \{0\})'$ with respect to the morphism $(X \times \mathbf{A}^1)' \rightarrow \mathbf{A}^1 = \text{Spec } k[u]$ at $0 \in \mathbf{A}^1$. Let $\mathcal{M}$ be the restriction $\Psi|_{T^\times}$ on the $\mathbf{G}_m$ -torsor $T^\times = T_Z X - \{0\} \subseteq T_Z X = \text{Spec } k[v] \subseteq E$.*

1. *Let η_x be the generic point of the henselization at $0 \in X$ and let M be the representation of the tame decomposition group $G_x^{\text{tame}} = \text{Gal}(\eta_{x,\text{tame}}/\eta_x)$ corresponding to $\mathcal{F}|_{\eta_x}$. Let η_u be the generic point of the henselization of $\mathbf{A}^1 = \text{Spec } k[u]$ at 0 and define $\pi_1^{\text{tame}}(T^\times \times_k \eta_u)$ to be the fiber product $\pi_1^{\text{tame}}(T^\times) \times_{\text{Gal}(k_{\text{sep}}/k)} \text{Gal}(\eta_{u,\text{tame}}/\eta_u)$. Define a morphism $\pi_1^{\text{tame}}(T^\times \times_k \eta_u) \rightarrow G_x^{\text{tame}}$ induced by $T^\times \times (\mathbf{A}^1 - \{0\}) = \text{Spec } k[v^{\pm 1}, u^{\pm 1}] \rightarrow \mathbf{G}_m = \text{Spec } k[x^{\pm 1}]$ given by $x = uv$. Then the restriction $\mathcal{M}$ of Ψ on $T^\times \times_k \eta_u = \mathbf{G}_m \times_k \eta_u$ is a monodromic sheaf defined by the pull-back of M to $\pi_1^{\text{tame}}(T^\times \times_k \eta_u)$.*

2. *Let $j_0: T^\times \rightarrow T$ and $j_\infty: T \rightarrow E$ be the open immersions. Then, the restriction of Ψ on T is isomorphic to $\overline{\mathcal{M}} = j_{0!} \mathcal{M}$. The restriction of Ψ on the proper transform $(X \times \{0\})'$ is $j_* \mathcal{F}$ and the restriction of Ψ on E is isomorphic to $j_{\infty*} \overline{\mathcal{M}}$.*

Proof. 1. By Proposition 2.3.7.1, $\mathcal{M} = \Psi|_{T^\times}$ is monodromic. Let $\text{Spec } k[u, v] = A_Z X \subseteq (X \times \mathbf{A}^1)'$ be the deformation to the normal bundle where $v = x/u$. Then, $\text{pr}_1: X \times \mathbf{G}_m \rightarrow X$ is defined by $x = uv$. Hence $\mathcal{M}$ corresponds to the pull-back of M by $\pi_1^{\text{tame}}(T^\times \times_k \eta_u) \rightarrow G_x^{\text{tame}}$.

2. By Proposition 2.3.7.2, $\overline{\mathcal{M}} = j_{0!} \mathcal{M} \rightarrow \Psi|_T$ is an isomorphism. The isomorphisms $\Psi|_{(X \times \{0\})'} \rightarrow j_* \mathcal{F}$ and $\Psi|_E \rightarrow j_{\infty*} \overline{\mathcal{M}}$ follow from Lemma 2.3.14.2. $\square$

2.4 Radon transform

Definition 2.4.1. Let V be a fixed vector space over k of dimension $n + 1$ with $n \geq 1$, and put $P = \mathbf{P}(V)$ and $P^\vee = \mathbf{P}(V^\vee)$. Consider the incidence correspondence

$$(2.4.1) \quad Q = \{(x, H) \in P \times P^\vee : x \in H\} \subseteq P \times P^\vee.$$

Note that $P^\vee$ parametrizes hyperplanes $H \subseteq P$ and $Q \subseteq P \times P^\vee$ is the universal family of hyperplanes.

Let $p: Q \rightarrow P$ and $p^\vee: Q \rightarrow P^\vee$ be the projection maps. The Radon transform and the dual Radon transform are, respectively, the functors

$$(2.4.2) \quad \text{Rad} = p_* p^*[n-1]: D_{\text{ctf}}(P, \Lambda) \rightarrow D_{\text{ctf}}(P^\vee, \Lambda),$$

$$(2.4.3) \quad \text{Rad}^\vee = p_* p^{\vee*}[n-1]: D_{\text{ctf}}(P^\vee, \Lambda) \rightarrow D_{\text{ctf}}(P, \Lambda).$$

We define the anti-Radon transform

$$(2.4.4) \quad \overline{\text{Rad}} = \text{Rad}^\vee(n-1): D_{\text{ctf}}(P^\vee, \Lambda) \rightarrow D_{\text{ctf}}(P, \Lambda).$$

By Poincaré duality, $p^! = p^*[2n-2](n-1)$ and $p^{\vee!} = p^{\vee*}[2n-2](n-1)$. Consequently, $\overline{\text{Rad}}$ is both left and right adjoint to Rad .

It is well known that Rad and $\overline{\text{Rad}}$ do not define an adjoint equivalence for $n \geq 2$ in general. Nevertheless, the following retraction statement holds.

Proposition 2.4.2. *Assume $n \geq 1$. Let U, C denote the unit and counit of the adjunction $\text{Rad} \vdash \overline{\text{Rad}}$, and let $\overline{U}, \overline{C}$ denote the unit and counit of the adjunction $\overline{\text{Rad}} \vdash \text{Rad}$. For every $\mathcal{F} \in D_{\text{ctf}}(P, \Lambda)$, the following identity holds:*

$$(2.4.5) \quad \overline{C}_{\mathcal{F}} \circ U_{\mathcal{F}} \simeq (-1)^{n-1} \text{id}_{\mathcal{F}}.$$

In particular, $\mathcal{F}$ is a retract of $\overline{\text{Rad}}(\text{Rad}\mathcal{F})$.

Proof. Unwinding the definitions, the left-hand side of (2.4.5) is the composite

$$(2.4.6) \quad \begin{aligned} \mathcal{F} &\rightarrow p_* p^* \mathcal{F} \rightarrow p_* p^{\vee!} p_*^{\vee} p^* \mathcal{F} \\ &\simeq p_* p^{\vee*} p_*^{\vee} p^* \mathcal{F}[2n-2](n-1) \\ &\rightarrow p_* p^* \mathcal{F}[2n-2](n-1) \simeq p_* p^! \mathcal{F} \rightarrow \mathcal{F}. \end{aligned}$$

We first evaluate the two arrows involving $p^{\vee}$. By [ILO14, Exposé XVI, Définition 2.3.1 and Proposition 2.3.2], the composite

$$(2.4.7) \quad p^* \mathcal{F} \rightarrow p^{\vee!} p_*^{\vee} p^* \mathcal{F} \simeq p^{\vee*} p_*^{\vee} p^* \mathcal{F}[2n-2](n-1) \rightarrow p^* \mathcal{F}[2n-2](n-1)$$

is given by cup product with the Chern class $c_{n-1}(T(Q/P^{\vee}))$ of the relative tangent bundle $T(Q/P^{\vee})$. By the projection formula, we get

$$(2.4.8) \quad \overline{C}_{\mathcal{F}} U_{\mathcal{F}} \simeq p_* c_{n-1}(T(Q/P^{\vee})) \cup \text{id}_{\mathcal{F}}.$$

It remains to compute the cohomology class $p_* c_{n-1}(T(Q/P^{\vee})) \in H^0(P, \Lambda)$. The closed immersion $j: Q \hookrightarrow P \times P^{\vee}$ is defined by the identity section (cf. [Sai17a, 3.2])

$$(2.4.9) \quad \text{id} \in \text{End}(V) = V \otimes V^{\vee} = \Gamma(P \times P^{\vee}, \mathcal{O}(1, 1)).$$

Thus, for the ideal sheaf $\mathcal{I}_{Q/(P \times P^{\vee})}$ defining j , we have an isomorphism

$$(2.4.10) \quad \mathcal{I}_{Q/(P \times P^{\vee})} / \mathcal{I}_{Q/(P \times P^{\vee})}^2 \simeq \mathcal{O}_Q(-1, -1) = p^* \mathcal{O}_P(-1) \otimes p^{\vee*} \mathcal{O}_{P^{\vee}}(-1).$$

Now the morphisms $Q \xrightarrow{j} P \times P^{\vee} \xrightarrow{\text{pr}_2} P^{\vee}$ induce an exact sequence

$$(2.4.11) \quad 0 \rightarrow \mathcal{I}_{Q/(P \times P^{\vee})} / \mathcal{I}_{Q/(P \times P^{\vee})}^2 \rightarrow j^* \Omega_{P \times P^{\vee} / P^{\vee}}^1 \rightarrow \Omega_{Q/P^{\vee}}^1 \rightarrow 0.$$

Since $j^* \Omega_{P \times P^{\vee} / P^{\vee}}^1 \simeq p^* \Omega_{P/k}^1$, (2.4.11) induces the following exact sequence

$$(2.4.12) \quad 0 \rightarrow \mathcal{O}_Q(-1, -1) \rightarrow p^* \Omega_{P/k}^1 \rightarrow \Omega_{Q/P^{\vee}}^1 \rightarrow 0.$$

Since all three terms are locally free, after taking duality, we get an exact sequence of vector bundles

$$(2.4.13) \quad 0 \rightarrow T(Q/P^{\vee}) \rightarrow p^* TP \rightarrow \mathcal{O}_Q(1, 1) \rightarrow 0.$$

Now applying [Jou77, Lemme 8.4.1] to the above sequence, we get

$$(2.4.14) \quad p_* c_{n-1}(T(Q/P^{\vee})) = (-1)^{n-1} \cdot p_* c_{n-1}(T(Q/P^{\vee})^{\vee}) = (-1)^{n-1}.$$

This finishes the proof. □

Corollary 2.4.3. *Let V be a vector space over k of dimension $n + 1$ with $n \geq 1$, and put $P = \mathbf{P}(V)$ and $P^\vee = \mathbf{P}(V^\vee)$. Let $Q \subseteq P \times P^\vee$ be the incidence correspondence. Let $\mathcal{F} \in D_{\text{ctf}}(P, \Lambda)$, and set $\mathcal{G} = p^{\vee*} \text{Rad}(\mathcal{F})n-1 \in D_{\text{ctf}}(Q, \Lambda)$. Then $\mathcal{H}_P(\mathcal{F}, \mathcal{F})$ is a retract of $\mathcal{H}_P(p_*\mathcal{G}, p_*\mathcal{G})$, and $\nu \text{Hom}_P(\mathcal{F}, \mathcal{F}) := \nu_{\delta_P} \mathcal{H}_P(\mathcal{F}, \mathcal{F})$ is a retract of*

$$(2.4.15) \quad \nu_{\delta_P} \mathcal{H}_P(p_*\mathcal{G}, p_*\mathcal{G}) \stackrel{(2.3.76)}{\simeq} (h_p \times \text{id}_{TP})_* \nu_{i_p} \mathcal{H}_Q(\mathcal{G}, \mathcal{G}).$$

Proof. Note that

$$(2.4.16) \quad p_*\mathcal{G} = p_*p^{\vee*} \text{Rad}(\mathcal{F})n-1 = \overline{\text{Rad}} \text{Rad}(\mathcal{F}).$$

By Proposition 2.4.2, $\mathcal{F}$ is a retract of $p_*\mathcal{G}$. The retraction is defined by

$$(2.4.17) \quad \mathcal{F} \xrightarrow{i=U_{\mathcal{F}}} p_*\mathcal{G} \xrightarrow{r=(-1)^{n-1}\overline{C}_{\mathcal{F}}} \mathcal{F},$$

where $ri = \text{id}$. Now the results follow from Lemma 2.3.13 and (2.3.73). $\square$

Consider the following commutative diagram

$$(2.4.18) \quad \begin{array}{ccccc} Q \times_P Q & \xrightarrow{i_p} & Q \times Q & \xrightarrow{\delta} & (Q \times Q) \times_{P^\vee \times P^\vee} (Q \times Q) & \xrightarrow{\text{pr}_2} & Q \times Q \\ & & & \searrow & \downarrow \text{pr}_1 & & \downarrow p^\vee \times p^\vee \\ & & & & Q \times Q & \xrightarrow{p^\vee \times p^\vee} & P^\vee \times P^\vee. \end{array}$$

We apply Lemma 2.3.9 to the inclusion $i_p : Q \times_P Q \hookrightarrow Q \times Q$ and the morphism $p^\vee \times p^\vee : Q \times Q \rightarrow P^\vee \times P^\vee$. We now describe the action (2.2.28) in this setting. There are canonical identifications

$$(2.4.19) \quad T(Q \times Q / P^\vee \times P^\vee) = \text{pr}_{Q,1}^* T(Q / P^\vee) \oplus \text{pr}_{Q,2}^* T(Q / P^\vee) \quad \text{over } Q \times Q,$$

$$(2.4.20) \quad T_{Q \times_P Q}(Q \times Q) = Q \times_P Q \times_P TP \quad \text{over } Q \times_P Q.$$

Define a vector bundle W over $Q \times_P Q$ by

$$(2.4.21) \quad W = T(Q \times Q / P^\vee \times P^\vee)|_{Q \times_P Q} = \text{pr}_{p,1}^* T(Q / P^\vee) \oplus \text{pr}_{p,2}^* T(Q / P^\vee).$$

Under these identifications, the action (2.2.28) becomes an action of W on $Q \times_P Q \times_P TP$ over $Q \times_P Q$:

$$(2.4.22) \quad \alpha : W \times_P TP = W \times_{Q \times_P Q} (Q \times_P Q \times_P TP) \rightarrow Q \times_P Q \times_P TP.$$

More precisely, the action (2.4.22) is induced by the linear morphism

$$(2.4.23) \quad \beta : W \rightarrow Q \times_P Q \times_P TP, (w_1, w_2) \mapsto Tp(w_2) - Tp(w_1),$$

namely, the difference of the two morphisms

$$(2.4.24) \quad Tp \circ \text{pr}_2 : W \rightarrow \text{pr}_2^* T(Q / P^\vee) \rightarrow T(Q / P^\vee) \xrightarrow{Tp} TP,$$

$$(2.4.25) \quad Tp \circ \text{pr}_1 : W \rightarrow \text{pr}_1^* T(Q / P^\vee) \rightarrow T(Q / P^\vee) \xrightarrow{Tp} TP.$$

Here $Tp : T(Q/P^\vee) \rightarrow TP$ is the composition $T(Q/P^\vee) \rightarrow T(P \times P^\vee/P^\vee) \simeq TP \times P^\vee \xrightarrow{\text{pr}} TP$.

The natural morphism

$$(2.4.26) \quad \rho : T^*P \setminus 0_{T^*P} \rightarrow \mathbf{P}(T^*P) \simeq Q$$

defines a tautological line subbundle on $\delta_p(Q)$:

$$(2.4.27) \quad \mathcal{L} = \{(x, H, H, \xi) : x \in P, \xi \in \ell_{x,H}\} \subseteq \delta_p(Q) \times_P T^*P,$$

where $\ell_{x,H} \subseteq T_x^*P$ is the line represented by $(x, H) \in Q$. We have a canonical isomorphism

$$(2.4.28) \quad \mathcal{L} \simeq T_Q^*(P \times P^\vee).$$

Lemma 2.4.4. *Consider the diagram (2.4.18).*

1. *Let $\mathcal{F} \in D_{\text{ctf}}(P, \Lambda)$ and put $\mathcal{G} = p^{\vee*}\text{Rad}(\mathcal{F})n-1 \in D_{\text{ctf}}(Q, \Lambda)$. Then $\nu_{i_p}(\mathcal{H}_Q(\mathcal{G}, \mathcal{G}))$ on $Q \times_P Q \times_P TP$ is W -invariant.*

2. *Consider the dual morphism*

$$(2.4.29) \quad \beta^\vee : Q \times_P Q \times_P T^*P \rightarrow W^\vee = \text{pr}_{p,1}^*T^*(Q/P^\vee) \oplus \text{pr}_{p,2}^*T^*(Q/P^\vee),$$

and the projection morphism

$$(2.4.30) \quad h_p \times \text{id}_{T^*P} : Q \times_P Q \times_P T^*P \rightarrow T^*P.$$

The kernel $\ker(\beta^\vee)$ is the union of $Q \times_P Q \times_P 0_{T^*P}$ and $\mathcal{L}$ (cf. (2.4.27)).

Proof. We have isomorphisms

$$(2.4.31) \quad \begin{aligned} D_Q\mathcal{G} &= D_Q((p^\vee)^*\text{Rad}(\mathcal{F})n-1) \simeq (p^\vee)^!D_{P^\vee}\text{Rad}(\mathcal{F})1-n \\ &\simeq (p^\vee)^*D_{P^\vee}\text{Rad}(\mathcal{F})[n-1], \end{aligned}$$

$$(2.4.32) \quad \mathcal{H}_Q(\mathcal{G}, \mathcal{G}) \simeq D_Q\mathcal{G} \boxtimes \mathcal{G} \simeq (p^\vee \times p^\vee)^*(D_{P^\vee}(\text{Rad}(\mathcal{F})) \boxtimes \text{Rad}(\mathcal{F}))[2n-2](n-1).$$

Now (1) follows from Lemma 2.3.9. We prove (2). Considering $Q \subseteq P \times P^\vee \rightarrow P^\vee$, we have isomorphisms

$$(2.4.33) \quad \ker(\text{pr}_1 \circ \beta^\vee) \simeq (Q \times_P Q) \times_{\text{pr}_1, Q} T_Q^*(P \times P^\vee),$$

$$(2.4.34) \quad \ker(\text{pr}_2 \circ \beta^\vee) \simeq (Q \times_P Q) \times_{\text{pr}_2, Q} T_Q^*(P \times P^\vee).$$

Then $\ker(\beta^\vee) = \ker(\text{pr}_1 \circ \beta^\vee) \cap \ker(\text{pr}_2 \circ \beta^\vee) = (Q \times_P Q \times_P 0_{T^*P}) \cup \mathcal{L}$. $\square$

Let $f : X \rightarrow S$ be a morphism of noetherian schemes over k and $\mathcal{F} \in D_c^b(X, \Lambda)$. We denote by $E_f(\mathcal{F})$ the smallest closed subset of X such that $f : X \setminus E_f(\mathcal{F}) \rightarrow S$ is universally locally acyclic relatively to $\mathcal{F}$. We call $E_f(\mathcal{F})$ the non-ULA locus of $\mathcal{F}$ with respect to f .

For later convenience, we recall a result of Beilinson on the identification of the projectivization of singular support with the non-ULA locus. Recall the Legendre transform [Bei16]:

$$(2.4.35) \quad \mathbf{P}(T^*P) \simeq Q \simeq \mathbf{P}(T^*P^\vee).$$

Lemma 2.4.5 ([Bei16, Theorem 3.2]). *Let $\mathcal{F} \in D_{\text{ctf}}(P, \Lambda)$, and let $SS(\mathcal{F}) \subseteq T^*P$ be the singular support of $\mathcal{F}$. Under the Legendre transform $\mathbf{P}(T^*P) = Q$, we have*

$$(2.4.36) \quad \mathbf{P}(SS(\mathcal{F})) = E_p(p^{\vee*}\text{Rad}(\mathcal{F})).$$

3 Singular supports and microlocalization

In this section, let X be a separated smooth scheme of finite type over a field k of characteristic $p > 0$ and Λ be a finite local ring with residue characteristic $\ell \neq p$. We fix a non-trivial character $\psi: \mathbf{F}_p \rightarrow \Lambda^\times$.

3.1 Construction

We define the microlocalization of a constructible sheaf along a closed subscheme.

Definition 3.1.1. Let $i: Z \hookrightarrow X$ be a closed immersion of smooth schemes, and let η_0 be the generic point of the henselization of $\mathbf{A}^1$ at 0. Recall the specialization functor (2.3.27)

$$(3.1.1) \quad \nu_{Z/X}: D_{\text{ctf}}(X, \Lambda) \rightarrow D_{\text{ctf}}(T_Z X \times_k \eta_0, \Lambda).$$

For $\mathcal{F} \in D_{\text{ctf}}(X, \Lambda)$, we define its microlocalization along i to be the Fourier transform of its specialization:

$$(3.1.2) \quad \mu_{Z/X}(\mathcal{F}) = F_{\psi, T_Z X/Z}(\nu_{Z/X}(\mathcal{F})) \in D_{\text{ctf}}(T_Z^* X \times_k \eta_0, \Lambda).$$

In (2.3.51), we define the specialization $\nu\mathcal{H}om$ of the internal Hom object $\mathcal{H} = \mathcal{H}om$ (cf. (2.3.50)). Now we define its microlocal version. Let

$$(3.1.3) \quad F_\psi: D_{\text{ctf}}(TX \times_k \eta_0, \Lambda) \rightarrow D_{\text{ctf}}(T^* X \times_k \eta_0, \Lambda)$$

denote the Fourier transform associated with the tangent bundle (cf. Definition 2.1.2.2). We define a bifunctor

$$(3.1.4) \quad \mu\mathcal{H}om = \mu\mathcal{H}om_{\psi, \Lambda}: D_{\text{ctf}}(X, \Lambda)^{\text{op}} \times D_{\text{ctf}}(X, \Lambda) \rightarrow D_{\text{ctf}}(T^* X \times_k \eta_0, \Lambda),$$

by $\mu\mathcal{H}om = F_\psi \circ \nu\mathcal{H}om$.

In the following, we use the notation of Convention 2.3.5.

Proposition 3.1.2. Let $i: Z \hookrightarrow X$ be a closed immersion of smooth k -schemes, and let $q^\vee: T_Z^* X \rightarrow T^* X|_Z = T^* X \times_X Z$ denote the canonical closed immersion of the conormal bundle into the restricted cotangent bundle. For all $\mathcal{F} \in D_{\text{ctf}}(X, \Lambda)$, the complex

$$(3.1.5) \quad \mu_{Z/X}(\mathcal{F}) \in D_{\text{ctf}}(T_Z^* X \times_k \eta_0, \Lambda)$$

is a retract of

$$(3.1.6) \quad (q^\vee)^*(\mu\mathcal{H}om_X(\mathcal{F}, \mathcal{F})|_{T^* X|_Z}) \otimes \mu_{Z/X}(\mathcal{F}).$$

Note that this defines a canonical morphism

$$(3.1.7) \quad (q^\vee)^*(\mu\mathcal{H}om_X(\mathcal{F}, \mathcal{F})|_{T^* X|_Z}) \otimes \mu_{Z/X}(\mathcal{F}) \rightarrow \mu_{Z/X}(\mathcal{F}).$$

Proof. Applying Lemma 2.1.22 to $q: TX|_Z \rightarrow T_Z X$ and the following diagram

$$(3.1.8) \quad \begin{array}{ccc} & TX|_Z \times_Z T_Z X & \xrightarrow{\alpha} T_Z X \\ & \swarrow \text{pr}_{TX|_Z} & \downarrow \text{pr}_{T_Z X} \\ TX|_Z & & T_Z X, \end{array}$$

we see that (3.1.6) is the Fourier transform of (2.3.59). □

Lemma 3.1.3. *Let X be a smooth scheme over a field k of characteristic p , and let $\Lambda \rightarrow \Lambda'$ be a finite étale faithfully flat local extension of finite local rings with residue characteristic different from p . Suppose that Λ admits a non-trivial character $\psi : \mathbf{F}_p \rightarrow \Lambda^\times$, and let $\psi_{\Lambda'}$ be its extension to Λ' . For $\mathcal{F}, \mathcal{G} \in D_{\text{ctf}}(X, \Lambda)$, put $\mathcal{F}_{\Lambda'} = \mathcal{F} \otimes_{\Lambda} \Lambda'$ and $\mathcal{G}_{\Lambda'} = \mathcal{G} \otimes_{\Lambda} \Lambda'$. Let $\delta : X \hookrightarrow X \times X$ be the diagonal. We have canonical isomorphisms*

$$(3.1.9) \quad \mathcal{H}(\mathcal{F}, \mathcal{G}) \otimes_{\Lambda} \Lambda' \simeq \mathcal{H}(\mathcal{F}_{\Lambda'}, \mathcal{G}_{\Lambda'}) \quad \text{in } D_{\text{ctf}}(X \times_k X, \Lambda'),$$

$$(3.1.10) \quad \nu_{\delta}(\mathcal{H}(\mathcal{F}, \mathcal{G})) \otimes_{\Lambda} \Lambda' \simeq \nu_{\delta}(\mathcal{H}(\mathcal{F}_{\Lambda'}, \mathcal{G}_{\Lambda'})) \quad \text{in } D_{\text{ctf}}(TX \times_k \eta_0, \Lambda'),$$

$$(3.1.11) \quad \mu\text{Hom}_{\psi, \Lambda}(\mathcal{F}, \mathcal{G}) \otimes_{\Lambda} \Lambda' \simeq \mu\text{Hom}_{\psi_{\Lambda'}, \Lambda'}(\mathcal{F}_{\Lambda'}, \mathcal{G}_{\Lambda'}) \quad \text{in } D_{\text{ctf}}(T^*X \times_k \eta_0, \Lambda').$$

Proof. Since Λ' is finite free of positive rank over Λ , the functor $- \otimes_{\Lambda} \Lambda'$ is exact and conservative. It preserves constructibility and finite Tor-dimension and detects zero complexes, and is also compatible with the six operations, nearby cycles and specialization. Now (3.1.9) and (3.1.10) follow.

Let $E \rightarrow Y$ be a vector bundle, and let pr_1 and pr_2 be the projections from $E \times_Y E^\vee$. The formula

$$F_{\psi, E/Y, \Lambda}(\mathcal{N}) = \text{pr}_{2!}(\text{pr}_1^* \mathcal{N} \otimes_{\Lambda} \mathcal{L}_{\psi, \Lambda}(\langle \cdot, \cdot \rangle)),$$

together with $\mathcal{L}_{\psi, \Lambda} \otimes_{\Lambda} \Lambda' \simeq \mathcal{L}_{\psi_{\Lambda'}, \Lambda'}$, gives a canonical isomorphism

$$(3.1.12) \quad F_{\psi, E/Y, \Lambda}(\mathcal{N}) \otimes_{\Lambda} \Lambda' \simeq F_{\psi_{\Lambda'}, E/Y, \Lambda'}(\mathcal{N} \otimes_{\Lambda} \Lambda').$$

By the definition (3.1.4), the isomorphism (3.1.11) follows from (3.1.9), (3.1.10) and (3.1.12). $\square$

Let T be a k -scheme. An object $\mathcal{F} \in D_c^b(T \times_k \eta_0, \Lambda)$ has an underlying object in $D_c^b(T_{\bar{k}}, \Lambda)$, which we still denote by $\mathcal{F}$. This complex carries a continuous semilinear G_{η_0} -action, with the action on $T_{\bar{k}}$ induced by the quotient $G_{\eta_0} \rightarrow G_k$. The semilinear action, together with the surjectivity of $G_{\eta_0} \rightarrow G_k$, implies that $\text{supp}(\mathcal{F}) \subseteq T_{\bar{k}}$ is G_k -stable. Therefore by Galois descent for closed subsets, there is a unique closed subset $\text{supp}_k(\mathcal{F}) \subseteq T$ whose inverse image under $T_{\bar{k}} \rightarrow T$ is $\text{supp}(\mathcal{F})$:

$$(3.1.13) \quad (\text{supp}_k(\mathcal{F}))_{\bar{k}} = \text{supp}(\mathcal{F}).$$

Here $\text{supp}(\mathcal{F})$ is the closure of the set $\{t \in T_{\bar{k}} \mid \mathcal{F}_{\bar{t}} \neq 0\}$.

Definition 3.1.4. Let X be a smooth scheme over k , and let $\mathcal{F} \in D_{\text{ctf}}(X, \Lambda)$. Recall the object $\mu\text{Hom}(\mathcal{F}, \mathcal{F}) \in D_{\text{ctf}}(T^*X \times_k \eta_0, \Lambda)$ defined by (3.1.4). We define a closed subset $SS_{\mu}(\mathcal{F})$ of T^*X by

$$(3.1.14) \quad SS_{\mu}(\mathcal{F}) := \text{supp}_k(\mu\text{Hom}(\mathcal{F}, \mathcal{F})).$$

We say that $\mathcal{F} \in D_c^b(X, \Lambda)$ is locally constant if every cohomology sheaf $\mathcal{H}^a \mathcal{F}$ is locally constant.

Lemma 3.1.5. *Let $\mathcal{F} \in D_{\text{ctf}}(X, \Lambda)$.*

1. *The closed subset $SS_{\mu}(\mathcal{F}) \subseteq T^*X$ is conical.*
2. *Assume that $\mathcal{F}$ is locally constant and that X is irreducible. Then, we have $SS_{\mu}(\mathcal{F}) = T_X^*X$ if $\mathcal{F} \neq 0$ and $SS_{\mu}(\mathcal{F}) = \emptyset$ if $\mathcal{F} = 0$.*
3. *Assume $\dim X = 1$. Then, we have $SS_{\mu}(\mathcal{F}) \subseteq SS(\mathcal{F})$.*

Proof. 1. The specialization $\nu\mathcal{H}om(\mathcal{F}, \mathcal{F})$ and hence the microlocalization $\mu\mathcal{H}om(\mathcal{F}, \mathcal{F})$ are monodromic by Proposition 2.3.7.1 and [Zh24, Proposition 2.5(4)]. Thus its support $SS_\mu(\mathcal{F})$ is conical.

2. Since $\mathcal{H}om(\mathrm{pr}_1^*\mathcal{F}, \mathrm{pr}_2^!\mathcal{F})$ is locally constant, we have isomorphisms $e^*(\mathcal{H}om(\mathcal{F}, \mathcal{F}) \otimes K_X) \rightarrow \nu\mathcal{H}om(\mathcal{F}, \mathcal{F})$ and $0_*^\vee \mathcal{H}om(\mathcal{F}, \mathcal{F}) \rightarrow \mu\mathcal{H}om(\mathcal{F}, \mathcal{F})$ (2.1.8).

3. We may assume that X is irreducible. Let $U \subseteq X$ be the maximal open subset on which $\mathcal{F}|_U$ is locally constant and identify $T^*U = T^*X \times_X U$. Then by 2, we have $SS_\mu(\mathcal{F}) \cap T^*U = T_U^*U$ or $= \emptyset$ according to $\mathcal{F}|_U \neq 0$ or $= 0$. Since $SS(\mathcal{F})$ is the union of $SS_\mu(\mathcal{F}) \cap T^*U$ and fibers $T^*X - T^*U$, the assertion follows. $\square$

We show an equality for the direct image of singular support by closed immersions.

Lemma 3.1.6. *Let $i: X \rightarrow Y$ be a closed immersion of smooth schemes over k . Then, we have an equality*

$$(3.1.15) \quad SS_\mu(i_*\mathcal{F}) = i_* SS_\mu(\mathcal{F})$$

*of closed subsets of T^*Y .*

Proof. By Lemma 2.2.5.3, the morphism $A_X(X \times X) \rightarrow A_Y(Y \times Y)$ is a closed immersion. Hence the base change morphism $\nu\mathcal{H}om_Y(i_*\mathcal{F}, i_*\mathcal{F}) \rightarrow Ti_*\nu\mathcal{H}om_X(\mathcal{F}, \mathcal{F})$ is an isomorphism. Let

$$(3.1.16) \quad T^*X \xleftarrow{a=di} T^*Y \times_Y X \xrightarrow{b=\mathrm{pr}} T^*Y$$

be the canonical morphisms. Then the isomorphism $\nu\mathcal{H}om_Y(i_*\mathcal{F}, i_*\mathcal{F}) \rightarrow Ti_*\nu\mathcal{H}om_X(\mathcal{F}, \mathcal{F})$ induces an isomorphism $\mu\mathcal{H}om_Y(i_*\mathcal{F}, i_*\mathcal{F}) \rightarrow b_*a^*\mu\mathcal{H}om_X(\mathcal{F}, \mathcal{F})$ by Proposition 2.1.9. $\square$

Lemma 3.1.7. *Let $h: W \rightarrow X$ be a smooth morphism of smooth schemes over k . Then, we have an equality*

$$(3.1.17) \quad SS_\mu(h^*\mathcal{F}) = h^* SS_\mu(\mathcal{F})$$

*of closed subsets of T^*W .*

Proof. Let

$$(3.1.18) \quad \begin{array}{ccccc} T^*X & \xleftarrow{b=\mathrm{pr}} & T^*X \times_X W & \xrightarrow{a=dh} & T^*W \\ TX & \xleftarrow{b^\vee=\mathrm{pr}} & TX \times_X W & \xleftarrow{a^\vee=Th} & TW \end{array}$$

be the canonical morphisms. By Lemma 2.2.5.2, the morphism $A_W(W \times W) \rightarrow A_X(X \times X)$ is smooth. Hence the base change morphism $a^{\vee!}b^{\vee*}\nu\mathcal{H}om_X(\mathcal{F}, \mathcal{F}) \rightarrow \nu\mathcal{H}om_W(h^!\mathcal{F}, h^!\mathcal{F})$ is an isomorphism by the smooth base change theorem. This induces an isomorphism $a_!b^*\mu\mathcal{H}om_X(\mathcal{F}, \mathcal{F}) \rightarrow \mu\mathcal{H}om_W(h^!\mathcal{F}, h^!\mathcal{F}) = \mu\mathcal{H}om_W(h^*\mathcal{F}, h^*\mathcal{F})$ by Corollary 2.1.10.1. $\square$

If ψ does not exist, we can define SS_μ as follows.

Proposition 3.1.8. *Let X be a smooth k -scheme, and let $\mathcal{F} \in D_{\mathrm{ctf}}(X, \Lambda)$. Let $\Lambda \rightarrow \Lambda'$ be a finite étale faithfully flat local extension of finite local rings with residue characteristic*

different from p . Assume there is a non-trivial character $\psi : \mathbf{F}_p \rightarrow \Lambda'^\times$. The following formula defines a closed conical subset of T^*X :

$$(3.1.19) \quad SS_\mu(\mathcal{F}) := \text{supp}_k(\mu\mathcal{H}om_{\psi, \Lambda'}(\mathcal{F}_{\Lambda'}, \mathcal{F}_{\Lambda'})),$$

which is independent of the choice of Λ' and ψ .

If Λ admits a non-trivial character, this agrees with Definition 3.1.4.

Proof. Since Λ' is henselian and p is invertible in A , reduction induces a bijection between the p -th roots of unity in Λ' and those in its residue field. Every non-trivial character $\mathbf{F}_p \rightarrow \Lambda'^\times$ is therefore of the form $\psi_a(t) = \psi(at)$ for a unique $a \in \mathbf{F}_p^\times$. For a vector bundle $E \rightarrow Y$, let $m_a : E^\vee \rightarrow E^\vee$ be the multiplication by $a \in \mathbf{F}_p^\times$. We have the following isomorphism (cf. [AS11, the remark (i) after Definition 3.8])

$$(3.1.20) \quad F_{\psi_a, E/Y, \Lambda'}(\mathcal{N}) \simeq m_a^* F_{\psi, E/Y, \Lambda'}(\mathcal{N}).$$

By Lemma 3.1.5.1, $\text{supp}_k(\mu\mathcal{H}om_{\psi, \Lambda'}(\mathcal{F}_{\Lambda'}, \mathcal{F}_{\Lambda'}))$ is conical. Hence (3.1.20) shows that the right hand side of (3.1.19) is independent of the non-trivial character ψ .

Now consider two choices (Λ'_1, ψ_1) and (Λ'_2, ψ_2) . Choose a nonzero local factor B of $\Lambda'_1 \otimes_\Lambda \Lambda'_2$. It is finite étale and faithfully flat over each Λ'_i .

Since the functors $- \otimes_{\Lambda'_1} B$ and $- \otimes_{\Lambda'_2} B$ are conservative and detect zero complexes, we have

$$(3.1.21) \quad \begin{aligned} \text{supp}_k(\mu\mathcal{H}om_{\psi_1, \Lambda'_1}(\mathcal{F}_{\Lambda'_1}, \mathcal{F}_{\Lambda'_1})) &= \text{supp}_k(\mu\mathcal{H}om_{\psi_1 B, B}(\mathcal{F}_B, \mathcal{F}_B)) \\ &= \text{supp}_k(\mu\mathcal{H}om_{\psi_2 B, B}(\mathcal{F}_B, \mathcal{F}_B)) \\ &= \text{supp}_k(\mu\mathcal{H}om_{\psi_2, \Lambda'_2}(\mathcal{F}_{\Lambda'_2}, \mathcal{F}_{\Lambda'_2})). \end{aligned}$$

This proves independence of the choice of Λ' and ψ , as well as agreement with Definition 3.1.4 when a character exists over Λ . $\square$

Let k be any field, and let Λ be a finite local ring whose residue characteristic is invertible in k . Let $u : \mathbf{A}_k^1 \setminus \{1\} \hookrightarrow \mathbf{A}_k^1$ be the open immersion, and put $B = u_*\Lambda$. Let X be a noetherian scheme over k and let E be a vector bundle over X with dual $E^\vee$. Consider the projections and the canonical pairing

$$(3.1.22) \quad \begin{array}{ccc} E & \xleftarrow{\text{pr}_1} & E \times_X E^\vee \xrightarrow{\text{pr}_2} E^\vee \\ & & \downarrow \mu = \langle, \rangle \\ & & \mathbf{A}^1 \end{array}$$

Following [Wang, Section 2], but omitting the shift by $\text{rank } E$, we define

$$(3.1.23) \quad F_{B, E/X}(\mathcal{M}) = \text{pr}_{2!}(\text{pr}_1^* \mathcal{M} \otimes \langle, \rangle^* B).$$

Definition 3.1.9. If X is a smooth k -scheme X and $\mathcal{F} \in D_{\text{ctf}}(X, \Lambda)$, we define

$$(3.1.24) \quad \mu_B \mathcal{H}om(\mathcal{F}, \mathcal{F}) := F_{B, TX/X}(\nu \mathcal{H}om(\mathcal{F}, \mathcal{F})),$$

$$(3.1.25) \quad SS_{\mu, B}(\mathcal{F}) := \text{supp}_k(\mu_B \mathcal{H}om(\mathcal{F}, \mathcal{F})).$$

In positive characteristic, since $\nu \mathcal{H}om(\mathcal{F}, \mathcal{F})$ is monodromic, this construction agrees with (3.1.19). Consequently, whenever Λ admits a nontrivial character $\psi : \mathbf{F}_p \rightarrow \Lambda^\times$, it also agrees with Definition 3.1.4.

Proposition 3.1.10. *Let k be a field of characteristic $p > 0$, and let Λ be a finite local ring of residue characteristic $\ell \neq p$. Let X be a smooth scheme over k , and let $\mathcal{F} \in D_{\text{ctf}}(X, \Lambda)$. Then we have*

$$(3.1.26) \quad SS_\mu(\mathcal{F}) = SS_{\mu, B}(\mathcal{F}),$$

where the left hand side is defined by (3.1.19).

Proof. By Lemma 2.1.1, there exists a finite étale faithfully flat local Λ -algebra Λ' admitting a nontrivial character $\psi : \mathbf{F}_p \rightarrow (\Lambda')^\times$. Put $\mathcal{F}_{\Lambda'} = \mathcal{F} \otimes_\Lambda \Lambda'$. Since Λ' is finite free over Λ , the functor $- \otimes_\Lambda \Lambda'$ commutes with specialization and with F_B . Since $\Lambda' \rightarrow \Lambda$ is faithfully flat, hence $- \otimes_\Lambda \Lambda'$ preserves supports and we have

$$(3.1.27) \quad SS_{\mu, B}(\mathcal{F}) = SS_{\mu, B}(\mathcal{F}_{\Lambda'}).$$

By Proposition 2.3.7.1, the object $\nu\mathcal{H}om(\mathcal{F}_{\Lambda'}, \mathcal{F}_{\Lambda'})$ is monodromic. By [Wang, Corollary 4.5], the restrictions of F_B and F_ψ to geometric monodromic complexes over an algebraic closure of k are naturally isomorphic, although not canonically. Thus we have

$$(3.1.28) \quad \text{supp}_k(\mu_B\mathcal{H}om(\mathcal{F}_{\Lambda'}, \mathcal{F}_{\Lambda'})) = \text{supp}_k(\mu\mathcal{H}om_{\psi, \Lambda'}(\mathcal{F}_{\Lambda'}, \mathcal{F}_{\Lambda'})).$$

Together with (3.1.27) and (3.1.19), we get (3.1.26). $\square$

3.2 Inclusion of singular supports

In this subsection, we prove the inclusion $SS_\mu \subseteq SS$. We note the following fact. Let X be a smooth k -scheme, and let $\mathcal{F} \in D_{\text{ctf}}(X, \Lambda)$. Let $\Lambda \rightarrow \Lambda'$ be any finite étale faithfully flat local extension. Since the functor $- \otimes_\Lambda \Lambda'$ detects the vanishing of the vanishing cycles functor, by [Bei16, Theorem 1.5], we have

$$(3.2.1) \quad SS(\mathcal{F}) = SS(\mathcal{F} \otimes_\Lambda \Lambda').$$

In order to prove $SS_\mu \subseteq SS$, by Lemma 2.1.1 and Proposition 3.1.8, we may assume there exists a non-trivial character $\psi : \mathbf{F}_p \rightarrow \Lambda^\times$.

We first reduce the assertion to the case of projective spaces.

Definition 3.2.1. 1. Let X be a smooth k -scheme. We say that a complex $\mathcal{F} \in D_{\text{ctf}}(X, \Lambda)$ satisfies the singular support inclusion if

$$(3.2.2) \quad SS_\mu(\mathcal{F}) \subseteq SS(\mathcal{F}).$$

2. Let P be a projective space over k , and retain the notation of Definition 2.4.1. Set

$$(3.2.3) \quad \rho : (T^*P)^\times = T^*P \setminus T_P^*P \rightarrow \mathbf{P}(T^*P) \stackrel{(2.4.35)}{\simeq} Q,$$

which is the composite of the projectivization map and the Legendre transform. We say that a complex $\mathcal{F} \in D_{\text{ctf}}(P, \Lambda)$ satisfies the projective singular support inclusion if

$$(3.2.4) \quad SS_\mu(\mathcal{F})^\times = SS_\mu(\mathcal{F}) \setminus T_P^*P \subseteq \rho^{-1}(E_p(p^{\vee*}\text{Rad}(\mathcal{F}))) \stackrel{(2.4.36)}{=} \rho^{-1}(\mathbf{P}(SS(\mathcal{F}))).$$

Lemma 3.2.2. *Assume that, for every $n \geq 1$, every $(n + 1)$ -dimensional vector space V , and every $\mathcal{F} \in D_{\text{ctf}}(\mathbf{P}(V), \Lambda)$, the complex $\mathcal{F}$ satisfies the projective singular support inclusion.*

Then, for every smooth scheme X over k and every $\mathcal{F} \in D_{\text{ctf}}(X, \Lambda)$, the complex $\mathcal{F}$ satisfies the singular support inclusion. Equivalently, Theorem 3.2.5 holds.

Proof. When $\dim X \leq 1$, Theorem 3.2.5 follows from Lemma 3.1.5.2 and Lemma 3.1.5.3.

Now assume $\dim X > 1$. Combining Lemmas 3.1.6 and 3.1.7 with [Bei16, Lemmas 2.3 and 2.5], we may assume that $X = P = \mathbf{P}(V)$ is a projective space. Let $\mathcal{F} \in D_{\text{ctf}}(P, \Lambda)$. Since $\mathcal{F}$ satisfies the projective singular support inclusion, and both $SS_\mu(\mathcal{F})$ and $SS(\mathcal{F})$ are conical (cf. Lemma 3.1.5 and [Bei16, 1.3]), we have

$$(3.2.5) \quad SS_\mu(\mathcal{F})^\times \subseteq SS(\mathcal{F})^\times.$$

By [Bei16, Lemma 2.3], the base of $SS(\mathcal{F})$ is $\text{supp}(\mathcal{F})$. If $\mathcal{F}$ vanishes on an open neighborhood U of $x \in P$, then $\mathcal{H}_{\mathcal{F}}$ vanishes on $U \times U$. Consequently, $\nu\mathcal{H}om_X(\mathcal{F}, \mathcal{F})$ vanishes over TU and $\mu\mathcal{H}om_X(\mathcal{F}, \mathcal{F})$ vanishes over T^*U . It follows that

$$(3.2.6) \quad SS_\mu(\mathcal{F}) \cap T_P^*P \subseteq \text{supp}(\mathcal{F}) = SS(\mathcal{F}) \cap T_P^*P.$$

Now the result follows from (3.2.5) and (3.2.6). $\square$

For the remainder of this subsection, let P be a projective space over k and let $\mathcal{F} \in D_{\text{ctf}}(P, \Lambda)$. Retain the notation of Subsection 2.4, and set

$$(3.2.7) \quad \mathcal{G} = p^{\vee*}\text{Rad}(\mathcal{F})n-1 \in D_{\text{ctf}}(Q, \Lambda).$$

Universal local acyclicity is unaffected by cohomological shifts and Tate twists. Therefore, by Lemma 2.4.5, we have

$$(3.2.8) \quad E_p(p^{\vee*}\text{Rad}(\mathcal{F})) = E_p(\mathcal{G}).$$

Lemma 3.2.3. *Recall the closed immersion $i_p : Q \times_P Q \hookrightarrow Q \times Q$, and use the notation of Convention 2.3.5. Let $U \subseteq Q$ be an open subscheme. If $p|_U : U \rightarrow P$ is universally locally acyclic relatively to $\mathcal{G}|_U$, then the natural morphism*

$$(3.2.9) \quad i_p^*\mathcal{H}_{\mathcal{G}} \boxtimes \Lambda_{TP} \rightarrow \nu_{i_p}\mathcal{H}_{\mathcal{G}} \quad \text{in } D_{\text{ctf}}((Q \times_P Q \times_P TP) \times_k \eta_0, \Lambda)$$

restricts to an isomorphism over $U \times_P U \times_P TP$.

Proof. By [LZ22, Theorem 2.16], the morphism $p|_U \times p|_U : U \times U \rightarrow P \times P$ is universally locally acyclic relatively to the complex

$$(3.2.10) \quad \mathcal{H}_{\mathcal{G}}|_{U \times U} \simeq D_U(\mathcal{G}|_U) \boxtimes (\mathcal{G}|_U) \in D_{\text{ctf}}(U \times U, \Lambda).$$

Applying Lemma 2.3.6 to the cartesian diagram

$$(3.2.11) \quad \begin{array}{ccc} U \times_P U & \xleftarrow{i_p|_{U \times_P U}} & U \times U \\ h_{p|_{U \times_P U}} \downarrow & & \downarrow p|_{U \times_P U} \\ P & \xleftarrow{\delta_P} & P \times P, \end{array}$$

we obtain natural isomorphisms

$$(3.2.12) \quad \begin{aligned} \nu_{i_p} \mathcal{H}_{\mathcal{G}}|_{U \times_P U \times_P TP} &\simeq \nu_{i_p|_{U \times_P U}}(\mathcal{H}_{\mathcal{G}}|_{U \times U}) \\ &\simeq (\mathcal{H}_{\mathcal{G}}|_{U \times U})|_{U \times_P U \times_P TP} \simeq (i_p^* \mathcal{H}_{\mathcal{G}} \boxtimes \Lambda_{TP})|_{U \times_P U \times_P TP}. \end{aligned}$$

□

Recall the notation $h_p : Q \times_P Q \rightarrow P$ in (2.3.69) and that $h_p \times \text{id}_{TP}$ in (2.3.73) denotes the projection

$$(3.2.13) \quad h_p \times \text{id}_{TP} = \text{pr}_{TP} : Q \times_P Q \times_P TP \rightarrow TP.$$

Let $h_p \times \text{id}_{T^*P}$ be the projection

$$(3.2.14) \quad h_p \times \text{id}_{T^*P} = \text{pr}_{T^*P} : Q \times_P Q \times_P T^*P \rightarrow T^*P.$$

Lemma 3.2.4. *Let $U \subseteq Q$ be an open subscheme. Suppose that $p|_U$ is universally locally acyclic relatively to $\mathcal{G}|_U$. Then the complex*

$$(3.2.15) \quad F_{\psi, TP/P}((h_p \times \text{id}_{TP})_* \nu_{i_p} \mathcal{H}_{\mathcal{G}}) \simeq (h_p \times \text{id}_{T^*P})_* F_{\psi, Q \times_P Q \times_P TP/Q \times_P Q}(\nu_{i_p} \mathcal{H}_{\mathcal{G}})$$

vanishes over $\rho^{-1}(U)$.

Proof. There are canonical isomorphisms

$$(3.2.16) \quad \begin{aligned} D_Q \mathcal{G} &= D_Q((p^\vee)^* \text{Rad}(\mathcal{F})n-1) \simeq (p^\vee)^! D_{P^\vee} \text{Rad}(\mathcal{F})1-n \\ &\simeq (p^\vee)^* D_{P^\vee} \text{Rad}(\mathcal{F})[n-1], \\ \mathcal{H}_{\mathcal{G}} &\simeq D_Q \mathcal{G} \boxtimes \mathcal{G} \simeq (p^\vee \times p^\vee)^*(D_{P^\vee} \text{Rad}(\mathcal{F}) \boxtimes \text{Rad}(\mathcal{F}))[2n-2](n-1). \end{aligned}$$

By Lemma 2.4.4, the complex $\nu_{i_p} \mathcal{H}_{\mathcal{G}}$ is W -invariant under the action of W (cf. (2.4.22)). Let $\Sigma := \text{supp}_k(F_{\psi, Q \times_P Q \times_P TP/Q \times_P Q}(\nu_{i_p} \mathcal{H}_{\mathcal{G}})) \subseteq Q \times_P Q \times_P T^*P$ (cf. (3.1.13)). By Proposition 2.1.21 and Lemma 2.4.4, we have

$$(3.2.17) \quad \Sigma \subseteq \ker(\beta^\vee) = Q \times_P Q \times_P 0_{T^*P} \cup \mathcal{L}.$$

Let $J : U \times_P U \hookrightarrow Q \times_P Q$ be the inclusion, and let

$$(3.2.18) \quad J_{TP} = J \times \text{id}_{TP} : U \times_P U \times_P TP \hookrightarrow Q \times_P Q \times_P TP,$$

be its base change. By Lemma 3.2.3, the complex

$$(3.2.19) \quad J_{TP}^* \nu_{i_p} \mathcal{H}_{\mathcal{G}} = (i_p^* \mathcal{H}_{\mathcal{G}})|_{U \times_P U} \boxtimes \Lambda_{TP}$$

is the pull-back of $i_p^* \mathcal{H}_{\mathcal{G}}|_{U \times_P U}$ by the projection $U \times_P U \times_P TP \rightarrow U \times_P U$. Hence Lemma 2.1.3.3 gives

$$(3.2.20) \quad \Sigma|_{U \times_P U} = \text{supp}_k(F_{\psi, U \times_P U \times_P TP/U \times_P U}(J_{TP}^* \nu_{i_p} \mathcal{H}_{\mathcal{G}})) \subseteq U \times_P U \times_P 0_{T^*P}.$$

By (3.2.17) and (3.2.20), we see that

$$(3.2.21) \quad \Sigma \subseteq Q \times_P Q \times_P 0_{T^*P} \cup \mathcal{L}|_{\delta_p(Q \setminus U)}.$$

Hence Σ is disjoint from $Q \times_P Q \times_P \rho^{-1}(U)$. □

Theorem 3.2.5. *Let Λ be a finite local ring killed by a power of a prime $\ell \neq p$. Let X be a smooth k -scheme, and let $\mathcal{F} \in D_{\text{ctf}}(X, \Lambda)$. Then*

$$(3.2.22) \quad SS_\mu(\mathcal{F}) \subseteq SS(\mathcal{F}).$$

Proof. By Lemma 3.2.2, it suffices to prove that, for every projective space P and every $\mathcal{F} \in D_{\text{ctf}}(P, \Lambda)$, the complex $\mathcal{F}$ satisfies the projective singular support inclusion; that is,

$$(3.2.23) \quad SS_\mu(\mathcal{F})^\times = SS_\mu(\mathcal{F}) \setminus T_P^*P \subseteq \rho^{-1}(E_p(p^{\vee*}\text{Rad}(\mathcal{F}))).$$

Set $\mathcal{G} = p^{\vee*}\text{Rad}(\mathcal{F})n-1$. Let $U \subseteq Q$ be an open subscheme disjoint from $E_p(p^{\vee*}\text{Rad}(\mathcal{F}))$. Then $p|_U$ is universally locally acyclic relatively to $\mathcal{G}|_U$. By Lemma 3.2.4, the complex

$$(3.2.24) \quad F_{\psi, TP/P}((h_p \times \text{id}_{TP})_* \nu_{i_p} \mathcal{H}_{\mathcal{G}})$$

vanishes over $\rho^{-1}(U)$. By Corollary 2.4.3, the complex

$$(3.2.25) \quad \mu\mathcal{H}om_P(\mathcal{F}, \mathcal{F}) = F_{\psi, TP/P} \nu_{\delta_P} \mathcal{H}_{\mathcal{F}}$$

is a retract of (3.2.24), and therefore also vanishes over $\rho^{-1}(U)$. Therefore $SS_\mu(\mathcal{F})$ is disjoint from $\rho^{-1}(U)$. $\square$

Theorem 3.2.6. *Let $i : Z \hookrightarrow X$ be a closed immersion of smooth schemes and let $\mathcal{F} \in D_{\text{ctf}}(X, \Lambda)$. Then we have*

$$(3.2.26) \quad \text{supp}_k(\mu_{Z/X}(\mathcal{F})) \subseteq SS_\mu(\mathcal{F}) \cap T_Z^*X \subseteq SS(\mathcal{F}) \cap T_Z^*X,$$

where $\text{supp}_k(\mu_{Z/X}(\mathcal{F})) \subseteq T_Z^*X$ is the unique closed subset such that $(\text{supp}_k(\mu_{Z/X}(\mathcal{F})))_{\bar{k}} = \text{supp}(\mu_{Z/X}(\mathcal{F}))$ (cf. (3.1.13)).

Proof. Let $U \subseteq T^*X$ be an open subset where $\mu\mathcal{H}om_X(\mathcal{F}, \mathcal{F})$ vanishes. By Proposition 3.1.2, $\mu_{Z/X}(\mathcal{F})$ is a retract of

$$(3.2.27) \quad (q^\vee)^*(\mu\mathcal{H}om_X(\mathcal{F}, \mathcal{F})|_{T^*X|_Z}) \otimes \mu_{Z/X}(\mathcal{F})$$

which vanishes over $q^{\vee-1}(U) = U \cap T_Z^*X$ (note that $q^\vee : T_Z^*X \hookrightarrow T^*X|_Z$). It follows that

$$(3.2.28) \quad \begin{aligned} \text{supp}_k(\mu_{Z/X}(\mathcal{F})) &\subseteq \text{supp}_k(\mu\mathcal{H}om_X(\mathcal{F}, \mathcal{F})) \cap T_Z^*X \\ &= SS_\mu(\mathcal{F}) \cap T_Z^*X \subseteq SS(\mathcal{F}) \cap T_Z^*X, \end{aligned}$$

where the last inclusion follows from (3.2.22). $\square$

3.3 Equality of singular supports

In this subsection, we show that the inclusion $SS_\mu \subseteq SS$ in Theorem 3.2.5 is always an equality. As in the beginning of Subsection 3.2, we may assume there exists a non-trivial character $\psi : \mathbf{F}_p \rightarrow \Lambda^\times$.

Lemma 3.3.1. *Let X, Y be smooth k -schemes, $\mathcal{F}_1, \mathcal{F}_2 \in D_{\text{ctf}}(X, \Lambda)$ and $\mathcal{G}_1, \mathcal{G}_2 \in D_{\text{ctf}}(Y, \Lambda)$. We have canonical isomorphisms*

$$(3.3.1) \quad \mathcal{H}_{X \times Y}(\mathcal{F}_1 \boxtimes \mathcal{G}_1, \mathcal{F}_2 \boxtimes \mathcal{G}_2) \simeq \mathcal{H}_X(\mathcal{F}_1, \mathcal{F}_2) \boxtimes \mathcal{H}_Y(\mathcal{G}_1, \mathcal{G}_2),$$

$$(3.3.2) \quad \nu\mathcal{H}om_X(\mathcal{F}_1, \mathcal{F}_2) \boxtimes \nu\mathcal{H}om_Y(\mathcal{G}_1, \mathcal{G}_2) \simeq \nu\mathcal{H}om_{X \times Y}(\mathcal{F}_1 \boxtimes \mathcal{G}_1, \mathcal{F}_2 \boxtimes \mathcal{G}_2),$$

$$(3.3.3) \quad \mu\mathcal{H}om_X(\mathcal{F}_1, \mathcal{F}_2) \boxtimes \mu\mathcal{H}om_Y(\mathcal{G}_1, \mathcal{G}_2) \simeq \mu\mathcal{H}om_{X \times Y}(\mathcal{F}_1 \boxtimes \mathcal{G}_1, \mathcal{F}_2 \boxtimes \mathcal{G}_2).$$

Consequently, for any $\mathcal{F} \in D_{\text{ctf}}(X, \Lambda)$ and $\mathcal{G} \in D_{\text{ctf}}(Y, \Lambda)$, we have

$$(3.3.4) \quad SS_\mu(\mathcal{F} \boxtimes \mathcal{G}) = SS_\mu(\mathcal{F}) \times SS_\mu(\mathcal{G}).$$

Proof. The isomorphism (3.3.1) follows from [LZ22, Proposition 2.11]. Applying Proposition 2.2.7 to δ_X and δ_Y , we obtain an isomorphism

$$(3.3.5) \quad A_{X \times Y}(X \times X \times Y \times Y) \simeq A_X(X \times X) \times_{\mathbf{A}^1} A_Y(Y \times Y).$$

Applying the specialization functor to the isomorphism (3.3.1) along $\delta_{X \times Y} = \delta_X \times \delta_Y : X \times Y \rightarrow X \times X \times Y \times Y$ and then applying [LZ22, Proposition 3.1], we obtain the isomorphism (3.3.2). By Proposition 2.1.16, we obtain natural isomorphisms

$$\begin{aligned} & \mu\mathcal{H}om_{X \times Y}(\mathcal{F}_1 \boxtimes \mathcal{G}_1, \mathcal{F}_2 \boxtimes \mathcal{G}_2) \\ & \simeq F_{\psi, T(X \times Y)/X \times Y}(\nu\mathcal{H}om_{X \times Y}(\mathcal{F}_1 \boxtimes \mathcal{G}_1, \mathcal{F}_2 \boxtimes \mathcal{G}_2)) \\ (3.3.6) \quad & \stackrel{(3.3.2)}{\simeq} F_{\psi, TX \times TY/X \times Y}(\nu\mathcal{H}om_X(\mathcal{F}_1, \mathcal{F}_2) \boxtimes \nu\mathcal{H}om_Y(\mathcal{G}_1, \mathcal{G}_2)) \\ & \stackrel{(2.1.60)}{\simeq} F_{\psi, TX/X}(\nu\mathcal{H}om_X(\mathcal{F}_1, \mathcal{F}_2)) \boxtimes F_{\psi, TY/Y}(\nu\mathcal{H}om_Y(\mathcal{G}_1, \mathcal{G}_2)) \\ & \simeq \mu\mathcal{H}om_X(\mathcal{F}_1, \mathcal{F}_2) \boxtimes \mu\mathcal{H}om_Y(\mathcal{G}_1, \mathcal{G}_2). \end{aligned}$$

□

Lemma 3.3.2. *Let $i : Z \hookrightarrow X$ be a closed immersion of smooth k -schemes, and $\mathcal{F} \in D_{\text{ctf}}(X, \Lambda)$. If $\text{supp}_k(\mu_{Z/X}(\mathcal{F}))$ (cf. (3.2.26)) is contained in the image of the zero section $0^\vee : Z \hookrightarrow T_Z^*X$, then i is $\mathcal{F}$ -transversal.*

Proof. By Proposition 2.1.6, there exists a complex $\mathcal{G} \in D_{\text{ctf}}(Z \times_k \eta_0, \Lambda)$ such that

$$(3.3.7) \quad \nu_{Z/X}(\mathcal{F}) \simeq p^*\mathcal{G}.$$

Applying [Sai17a, Lemma 8.6.3] to $p : T_Z X \rightarrow Z$, we obtain that 0 is $\nu_{Z/X}(\mathcal{F})$ -transversal. We conclude by Proposition 2.3.8. □

Lemma 3.3.3. *The base of $SS_\mu(\mathcal{F})$ is equal to $\text{supp}(\mathcal{F})$.*

Proof. Outside the inverse image of $\text{supp}(\mathcal{F})$, $\mu\mathcal{H}om(\mathcal{F}, \mathcal{F})$ is zero. Thus the base of $SS_\mu(\mathcal{F})$ is contained in $\text{supp}(\mathcal{F})$. Conversely, let $U \subseteq X$ be an open subset outside of the base of $SS_\mu(\mathcal{F})$. Then $\mu\mathcal{H}om_X(\mathcal{F}, \mathcal{F})$ vanishes over T^*U , and hence

$$(3.3.8) \quad \nu\mathcal{H}om_U(\mathcal{F}|_U, \mathcal{F}|_U) = 0.$$

By Proposition 2.3.7,

$$\begin{aligned} (3.3.9) \quad 0 &= 0_{TU}^! \nu\mathcal{H}om_U(\mathcal{F}|_U, \mathcal{F}|_U) \simeq \delta_U^! \mathcal{H}om_{U \times U}(\text{pr}_{U,1}^* \mathcal{F}|_U, \text{pr}_{U,2}^* \mathcal{F}|_U) \\ &\simeq \mathcal{H}om_U(\mathcal{F}|_U, \mathcal{F}|_U). \end{aligned}$$

It follows that $\mathcal{F}|_U = 0$. □

Let X be a smooth k -scheme and $f : X \rightarrow \mathbf{A}^1$ be a function. Consider the cartesian diagram

$$(3.3.10) \quad \begin{array}{ccc} X \times_{\mathbf{A}^1} X & \xrightarrow{i_f} & X \times X \\ \downarrow h_f & & \downarrow f \times f \\ \mathbf{A}^1 & \xrightarrow{\delta_{\mathbf{A}^1}} & \mathbf{A}^1 \times \mathbf{A}^1. \end{array}$$

We denote by

$$(3.3.11) \quad \mathrm{pr}_{f,1}, \mathrm{pr}_{f,2} : X \times_{\mathbf{A}^1} X \rightarrow X$$

the projections over $\mathbf{A}^1$, and by

$$(3.3.12) \quad \mathrm{pr}_1, \mathrm{pr}_2 : X \times X \rightarrow X$$

the projections over k .

Lemma 3.3.4. *Let X be a smooth k -scheme and $f : X \rightarrow \mathbf{A}^1$ be a smooth morphism. Let $\mathcal{F} \in D_{\mathrm{ctf}}(X, \Lambda)$. Recall that $\mathcal{H}_{\mathcal{F}} = \mathcal{H}om_{X \times X}(\mathrm{pr}_1^* \mathcal{F}, \mathrm{pr}_2^! \mathcal{F}) = D_X(\mathcal{F}) \boxtimes \mathcal{F}$. Then f is universally locally acyclic with respect to $\mathcal{F}$ if and only if i_f is $\mathcal{H}_{\mathcal{F}}$ -transversal.*

Proof. We have a commutative diagram (cf. [YZ25, (3.54)])

$$(3.3.13) \quad \begin{array}{ccc} i_f^* \mathcal{H}om_{X \times X}(\mathrm{pr}_1^* \mathcal{F}, \mathrm{pr}_2^! \mathcal{F}) \otimes i_f^! \Lambda & \xrightarrow{c_{i_f, \mathcal{H}_{\mathcal{F}}}} & i_f^! \mathcal{H}om_{X \times X}(\mathrm{pr}_1^* \mathcal{F}, \mathrm{pr}_2^! \mathcal{F}) \\ \simeq \uparrow & & \downarrow \simeq \\ i_f^*(D_X(\mathcal{F}) \boxtimes \mathcal{F}) \otimes i_f^! \Lambda & & \\ \downarrow \simeq & & \\ D_{X/\mathbf{A}^1}(\mathcal{F}) \boxtimes_{\mathbf{A}^1} \mathcal{F} & \longrightarrow & \mathcal{H}om_{X \times_{\mathbf{A}^1} X}(\mathrm{pr}_{f,1}^* \mathcal{F}, \mathrm{pr}_{f,2}^! \mathcal{F}). \end{array}$$

We conclude using [LZ22, Theorem 2.16]. □

Lemma 3.3.5. *Let X be a smooth k -scheme and $\mathcal{F} \in D_{\mathrm{ctf}}(X, \Lambda)$. Let $f : X \rightarrow \mathbf{A}^1$ be a smooth function. Let u be the coordinate on $\mathbf{A}^1$. It induces a morphism*

$$(3.3.14) \quad du : \mathbf{A}^1 \rightarrow T^* \mathbf{A}^1 \simeq \mathbf{A}^1 \times \mathbf{A}^1.$$

Consider the composition

$$(3.3.15) \quad X \xrightarrow{du} X \times_{\mathbf{A}^1} T^* \mathbf{A}^1 \xrightarrow{df} T^* X,$$

which is also denoted by $df : X \rightarrow T^ X$. Assume that*

$$(3.3.16) \quad df(X) \cap SS_{\mu}(\mathcal{F}) = \emptyset.$$

Then f is universally locally acyclic relatively to $\mathcal{F}$.

Proof. By Lemma 3.3.1, we have

$$(3.3.17) \quad SS_\mu(\mathcal{H}_\mathcal{F}) = SS_\mu(D_X \mathcal{F}) \times SS_\mu(\mathcal{F}).$$

As f is smooth, $X \times_{\mathbf{A}^1} X$ is smooth and i_f is pure of codimension 1. Consider the conormal bundle

$$(3.3.18) \quad T_{X \times_{\mathbf{A}^1} X}^*(X \times X) \simeq X \times_{\mathbf{A}^1} X \times_{\mathbf{A}^1} T^* \mathbf{A}^1 \simeq X \times_{\mathbf{A}^1} X \times \mathbf{A}^1.$$

Since $SS_\mu(\mathcal{F})$ is conical, the condition (3.3.16) implies that

$$(3.3.19) \quad SS_\mu(\mathcal{H}_\mathcal{F}) \cap T_{X \times_{\mathbf{A}^1} X}^*(X \times X) \subseteq X \times_{\mathbf{A}^1} X \times_{\mathbf{A}^1} 0_{T^* \mathbf{A}^1}.$$

Then by Theorem 3.2.6, we obtain that

$$(3.3.20) \quad \text{supp}_k(\mu_{i_f}(\mathcal{H}_\mathcal{F})) \subseteq X \times_{\mathbf{A}^1} X \times_{\mathbf{A}^1} 0_{T^* \mathbf{A}^1}.$$

Applying Lemma 3.3.2 to i_f and $\mathcal{H}_\mathcal{F}$, we know that i_f is $\mathcal{H}_\mathcal{F}$ -transversal. Now the result follows from Lemma 3.3.4. $\square$

Theorem 3.3.6. *Let Λ be a finite local ring killed by a power of a prime $\ell \neq p$. Let X be a smooth k -scheme, and let $\mathcal{F} \in D_{\text{ctf}}(X, \Lambda)$. Then we have*

$$(3.3.21) \quad SS_\mu(\mathcal{F}) = SS(\mathcal{F}).$$

Proof. By Theorem 3.2.5, [Bei16, Theorem 1.5] and [Sai17a, Theorem 4.4], it suffices to show that

$$(3.3.22) \quad SS^w(\mathcal{F}) = SS(\mathcal{F}) \subseteq SS_\mu(\mathcal{F}),$$

where $SS^w(\mathcal{F})$ is defined in [Bei16, 1.5]. In the following, if necessary, we may take a finite extension of k by Lemma 3.1.7 and [Bei16, Theorem 1.4(iii)]. Consider a test function $f : U \rightarrow \mathbf{A}^1$ on X and a point $x \in U$, such that

$$(3.3.23) \quad (x, df(x)) \notin SS_\mu(\mathcal{F}|_U).$$

If $x \notin \text{supp}(\mathcal{F})$, then $\mathcal{F}$ vanishes near x and hence f is universally locally acyclic relatively to $\mathcal{F}$ at x . Otherwise, we may assume by Lemma 3.3.3 that $df(x) \neq 0$. After shrinking U , we may assume that $f : U \rightarrow \mathbf{A}^1$ is smooth and

$$(3.3.24) \quad df(U) \cap SS_\mu(\mathcal{F}|_U) = \emptyset.$$

By Lemma 3.3.5, we conclude that f is universally locally acyclic relatively to $\mathcal{F}$ at x . Thus by definition of $SS^w(\mathcal{F})$, we have $(x, df(x)) \notin SS^w(\mathcal{F})$. $\square$

4 Characteristic cycles and microlocalization

In this section, let X be a separated smooth scheme of finite type over a field k of characteristic $p > 0$ and Λ be a finite local ring with residue characteristic $\ell \neq p$. We fix a non-trivial character $\psi : \mathbf{F}_p \rightarrow \Lambda^\times$.

4.1 Construction

Let $\mathcal{F} \in D_{\text{ctf}}(X, \Lambda)$ and let $\mathcal{H}$ denote $\mathcal{H}om(\text{pr}_1^* \mathcal{F}, \text{pr}_2^! \mathcal{F}) \in D_{\text{ctf}}(X \times X, \Lambda)$. By [Gro77, (3.2.1)], we have a canonical isomorphism

$$(4.1.1) \quad \mathcal{H}om(\mathcal{F}, \mathcal{F}) \rightarrow \delta^! \mathcal{H}.$$

Hence the identity $1_{\mathcal{F}}$ defines a canonical morphism

$$(4.1.2) \quad \Lambda \rightarrow \delta^! \mathcal{H}.$$

Let $\boxtimes: D_{\text{ctf}}(X, \Lambda) \times D_{\text{ctf}}(X, \Lambda) \rightarrow D_{\text{ctf}}(X \times X, \Lambda)$ denote the bifunctor $\text{pr}_1^* - \otimes \text{pr}_2^* -$. The canonical isomorphism $D_X \mathcal{F} \boxtimes \mathcal{F} \rightarrow \mathcal{H}$ [Gro77, (3.1.1)] on $X \times X$ induces an isomorphism

$$(4.1.3) \quad D_X \mathcal{F} \otimes \mathcal{F} \rightarrow \delta^* \mathcal{H}.$$

Hence the evaluation morphism $D_X \mathcal{F} \otimes \mathcal{F} \rightarrow K_X$ defines a canonical morphism

$$(4.1.4) \quad \delta^* \mathcal{H} \rightarrow K_X.$$

The morphisms (4.1.2), (4.1.4) and the canonical morphism $\delta^! \rightarrow \delta^*$ define

$$(4.1.5) \quad \Lambda \rightarrow \delta^! \mathcal{H} \rightarrow \delta^* \mathcal{H} \rightarrow K_X.$$

Definition 4.1.1 ([AS07, Definition 2.1.1]). The canonical class $C_{X/k}(\mathcal{F}) \in H^0(X, K_X)$ is defined by the composition (4.1.5).

Under the identification $H^0(X, K_X) = H^0(T^*X, e^{\vee*} K_X)$, the (cohomological) characteristic class $C_{X/k}(\mathcal{F})$ is known to be the cycle class of $CC \mathcal{F}$ under the assumption that X is quasi-projective [YZ25].

The adjoint of (4.1.5) defines

$$(4.1.6) \quad \delta_! \Lambda \rightarrow \mathcal{H} \rightarrow \delta_* K_X.$$

Taking the specialization and the microlocalization, we obtain

$$(4.1.7) \quad 0_! \Lambda \rightarrow \nu \mathcal{H}om(\mathcal{F}, \mathcal{F}) \rightarrow 0_* K_X,$$

$$(4.1.8) \quad \Lambda \rightarrow \mu \mathcal{H}om(\mathcal{F}, \mathcal{F}) \rightarrow e^{\vee*} K_X.$$

Alternatively, by Proposition 2.3.7.2, (4.1.2) and (4.1.4) define the first and the last arrows in

$$(4.1.9) \quad \Lambda \rightarrow 0^! \nu_{X/X \times X} \mathcal{H} \rightarrow 0^* \nu_{X/X \times X} \mathcal{H} \rightarrow K_X.$$

The middle arrow is defined by the morphism $0^! \rightarrow 0^*$ of functors. Taking the adjoint, we obtain (4.1.7).

Let $i: SS_{\mu} \mathcal{F} \rightarrow T^*X$ be the closed immersion, and let $q: T^*X_{\bar{k}} \rightarrow T^*X$ be the projection. Since $(SS_{\mu} \mathcal{F})_{\bar{k}} \subseteq T^*X_{\bar{k}}$ is the support of $\mu \mathcal{H}om(\mathcal{F}, \mathcal{F})$, (4.1.8) defines

$$(4.1.10) \quad \Lambda \rightarrow i^* \mu \mathcal{H}om(\mathcal{F}, \mathcal{F}) \rightarrow i^! e^{\vee*} K_X$$

on $(SS_{\mu} \mathcal{F})_{\bar{k}}$ by adjunction. By the surjectivity of $G_{\eta_0} \rightarrow G_k$, the composition of (4.1.10) defines a G_k -invariant cohomology class in $H^0_{(SS_{\mu} \mathcal{F})_{\bar{k}}}(T^*X_{\bar{k}}, q^* e^{\vee*} K_X)$.

Let A and $A_{\bar{k}}$ denote the sets of irreducible components of dimension $\dim X$ of $SS_{\mu}(\mathcal{F})$ and of $SS_{\mu}(\mathcal{F})_{\bar{k}}$ respectively. Applying semi-purity [SGA 4 $\frac{1}{2}$ -Cycle, Rappel 2.2.8] to each connected component of X , we have canonical isomorphisms

$$(4.1.11) \quad \begin{aligned} H_{SS_{\mu}\mathcal{F}}^0(T^*X, e^{\vee*}K_X) &\simeq \bigoplus_{C_a \in A} \Lambda \cdot \text{cl}[C_a], \\ H_{(SS_{\mu}\mathcal{F})_{\bar{k}}}^0(T^*X_{\bar{k}}, q^*e^{\vee*}K_X) &\simeq \bigoplus_{C_b \in A_{\bar{k}}} \Lambda \cdot \text{cl}[C_b], \end{aligned}$$

Hence the canonical morphism

$$(4.1.12) \quad q^* : H_{SS_{\mu}\mathcal{F}}^0(T^*X, e^{\vee*}K_X) \xrightarrow{\sim} H_{(SS_{\mu}\mathcal{F})_{\bar{k}}}^0(T^*X_{\bar{k}}, q^*e^{\vee*}K_X)^{G_k}$$

is an isomorphism. This allows us to make the following definition.

Definition 4.1.2 (cf. [KS90, Definition 9.4.1]). Let $\mathcal{F} \in D_{\text{ctf}}(X, \Lambda)$. We define

$$(4.1.13) \quad CC_{\mu}(\mathcal{F}) = CC_{\mu, \psi, \Lambda}(\mathcal{F}) \in H^0(SS_{\mu}(\mathcal{F}), i^!e^{\vee*}K_X) = H_{SS_{\mu}(\mathcal{F})}^0(T^*X, e^{\vee*}K_X)$$

to be the unique class whose image by (4.1.12) equals the class of the composition of (4.1.10).

Remark 4.1.3. Let k be any field, and let Λ be a finite local ring whose residue characteristic is invertible in k . Let $u : \mathbf{A}_k^1 \setminus \{1\} \hookrightarrow \mathbf{A}_k^1$ be the open immersion, and put $B = u_*\Lambda$. Let X be a smooth scheme over k , and let $\mathcal{F} \in D_{\text{ctf}}(X, \Lambda)$. Since $0^*B \simeq \Lambda$, we have canonical isomorphisms $F_B(0_*A) \simeq e^{\vee*}A$. Applying F_B to (4.1.7), we obtain

$$(4.1.14) \quad \Lambda \longrightarrow \mu_B \mathcal{H}om(\mathcal{F}, \mathcal{F}) \longrightarrow e^{\vee*}K_X$$

on $T^*X \times_k \eta_0$. The composition defines a G_k -invariant class supported on $SS_{\mu, B}(\mathcal{F})_{\bar{k}}$, as in Definition 4.1.2. Thus by semi-purity and Galois descent as in (4.1.12), this class defines a unique class $CC_{\mu, B}(\mathcal{F}) \in H_{SS_{\mu, B}(\mathcal{F})}^0(T^*X, e^{\vee*}K_X)$ over k . In future work, we will investigate the functorial properties of $CC_{\mu, B}(\mathcal{F})$, and its relation to $CC_{\mu}(\mathcal{F})$ in the case where k has positive characteristic and Λ admits a non-trivial character $\psi : \mathbf{F}_p \rightarrow \Lambda^{\times}$.

Lemma 4.1.4. Assume that $\mathcal{F}$ is locally constant of rank $r = \text{rank } \mathcal{F}$ and that X is of dimension n . Then, we have

$$CC_{\mu}\mathcal{F} = (-1)^n \cdot \text{rank } \mathcal{F} \cdot \text{cl}(T_X^*X).$$

Proof. Since $\mathcal{H} = \mathcal{H}om(\text{pr}_1^*\mathcal{F}, \text{pr}_2^!\mathcal{F})$ is locally constant, we have an isomorphism $e^*(\mathcal{H}om(\mathcal{F}, \mathcal{F}) \otimes K_X) \rightarrow \nu\mathcal{H}om(\mathcal{F}, \mathcal{F})$. The morphisms (4.1.8) and (4.1.7) give a diagram

$$\begin{array}{ccccccc} F0_!\Lambda & \xrightarrow{F\text{adj}} & Fe^!\mathcal{H}om(\mathcal{F}, \mathcal{F}) & \xrightarrow{F(\text{res})} & F0_!(\mathcal{H}om(\mathcal{F}, \mathcal{F}) \otimes K_X) & \xrightarrow{F(\text{ev})} & F0_*K_X \\ (2.1.6) \downarrow & & \downarrow (2.1.7) & & \downarrow (2.1.6) & & \downarrow (2.1.6) \\ \Lambda & \xrightarrow{\text{res}} & 0_*^{\vee}\mathcal{H}om(\mathcal{F}, \mathcal{F}) & \xrightarrow{\text{adj}} & e^{\vee*}(\mathcal{H}om(\mathcal{F}, \mathcal{F}) \otimes K_X) & \xrightarrow{\text{Tr}} & e^{\vee*}K_X. \end{array}$$

The left square is commutative by Proposition 2.1.11.1 (2.1.36). The middle square is $(-1)^n$ -commutative by Proposition 2.1.11.2 (2.1.38). The right square is commutative by the functoriality of (2.1.6). Since the class of the composition of the lower line is $r \cdot \text{cl}(T_X^*X)$, the assertion follows. $\square$

Lemma 4.1.5. *The pull-back of $CC_\mu(\mathcal{F})$ by the 0-section*

$$0^{\vee*}: H_{SS_\mu(\mathcal{F})}^0(T^*X, e^{\vee*}K_X) \rightarrow H^0(X, K_X)$$

equals the canonical class $C_{X/k}(\mathcal{F})$.

Proof. The pull-back $0^{\vee*}: H_{SS_\mu(\mathcal{F})}^0(T^*X, e^{\vee*}K_X) \rightarrow H^0(X, K_X)$ is the same as the composition of the canonical morphism $H_{SS_\mu(\mathcal{F})}^0(T^*X, e^{\vee*}K_X) \rightarrow H^0(T^*X, e^{\vee*}K_X)$ and the inverse of the pull-back $e^{\vee*}: H^0(X, K_X) \rightarrow H^0(T^*X, e^{\vee*}K_X)$. Since the composition of (4.1.8) is the pull-back by $e^\vee$ of that of (4.1.5), the assertion follows. $\square$

Proposition 4.1.6. *Let F be a filtration on $\mathcal{F} \in D_{\text{ctf}}(X, \Lambda)$ such that $\text{Gr}^F \mathcal{F} = \bigoplus_i \text{Gr}_i^F \mathcal{F}$ is in $D_{\text{ctf}}(X, \Lambda)$. Then, we have $SS_\mu(\mathcal{F}) \subseteq S = SS_\mu(\text{Gr}^F \mathcal{F})$ and an equality*

$$(4.1.15) \quad CC_\mu(\mathcal{F}) = \sum_i CC_\mu(\text{Gr}_i^F \mathcal{F})$$

*in $H_S^0(T^*X, e^{\vee*}K_X)$.*

Proof. We may assume $k = \bar{k}$. The filtration F on $\mathcal{F}$ induces a filtration also denoted F on $\mu\mathcal{H}om(\mathcal{F}, \mathcal{F})$. Since the identity $1_{\mathcal{F}}$ is in $F^0 \mu\mathcal{H}om(\mathcal{F}, \mathcal{F})$, the class $CC_\mu(\mathcal{F})$ is the image of $1_{\mathcal{F}} \in H_S^0(T^*X, F^0 \mu\mathcal{H}om(\mathcal{F}, \mathcal{F}))$ by the evaluation morphism. Since the restriction of the evaluation morphism is the composition of $F^0 \mu\mathcal{H}om(\mathcal{F}, \mathcal{F}) \rightarrow \bigoplus_i \mu\mathcal{H}om(\text{Gr}_i^F \mathcal{F}, \text{Gr}_i^F \mathcal{F}) \rightarrow e^{\vee*}K_X$, the assertion follows. $\square$

Corollary 4.1.7. *Let $i: Z \rightarrow X$ be a closed immersion and $j: U = X - Z \rightarrow X$ be the open immersion of the complement. Then, we have*

$$(4.1.16) \quad CC_\mu(\mathcal{F}) = CC_\mu(j_! j^* \mathcal{F}) + CC_\mu(i_* i^* \mathcal{F})$$

in $H_{SS_\mu(j_! j^ \mathcal{F} \oplus i_* i^* \mathcal{F})}^0(T^*X, e^{\vee*}K_X)$.*

Proof. We define a filtration F on $\mathcal{F}$ by $F^0 \mathcal{F} = j_! j^* \mathcal{F} \subseteq \mathcal{F}$, $F^{-1} = 0$ and $F^1 = \mathcal{F}$. Then $\text{Gr}^F \mathcal{F} \in D_{\text{ctf}}(X, \Lambda)$ and we obtain (4.1.16). $\square$

We define a variant for a not necessarily smooth scheme admitting an immersion to a smooth scheme. Let $i: Z \rightarrow X$ be a closed immersion of schemes over k and assume that X is smooth over k and let $\gamma: Z \rightarrow Z \times X$ be the graph of i . Let $\mathcal{F} \in D_{\text{ctf}}(Z, \Lambda)$ and let $\mathcal{H}_i$ denote $\mathcal{H}om(\text{pr}_1^* \mathcal{F}, \text{pr}_2^! i_* \mathcal{F}) \in D_{\text{ctf}}(Z \times X, \Lambda)$. We have a canonical isomorphism

$$(4.1.17) \quad \mathcal{H}_i \simeq (1_Z \times i)_* \mathcal{H}om(\text{pr}_1^* \mathcal{F}, \text{pr}_2^! \mathcal{F}).$$

By [Gro77, (3.2.1)], we have a canonical isomorphism

$$(4.1.18) \quad \mathcal{H}om(\mathcal{F}, \mathcal{F}) \rightarrow \gamma^! \mathcal{H}_i.$$

Hence the identity $1_{\mathcal{F}}$ defines a canonical morphism

$$(4.1.19) \quad \Lambda \rightarrow \gamma^! \mathcal{H}_i.$$

Let $\boxtimes: D_{\text{ctf}}(Z, \Lambda) \times D_{\text{ctf}}(X, \Lambda) \rightarrow D_{\text{ctf}}(Z \times X, \Lambda)$ denote the bifunctor $\text{pr}_1^* - \otimes \text{pr}_2^*$. The canonical isomorphism $D_Z \mathcal{F} \boxtimes i_* \mathcal{F} \rightarrow \mathcal{H}_i$ [Gro77, (3.1.1)] on $Z \times X$ induces an isomorphism

$$(4.1.20) \quad D_Z \mathcal{F} \otimes \mathcal{F} \rightarrow \gamma^* \mathcal{H}_i.$$

Hence the evaluation morphism $D_Z \mathcal{F} \otimes \mathcal{F} \rightarrow K_Z$ defines a canonical morphism

$$(4.1.21) \quad \gamma^* \mathcal{H}_i \rightarrow K_Z.$$

The morphisms (4.1.19), (4.1.21) and the canonical morphism $\gamma^! \rightarrow \gamma^*$ define

$$(4.1.22) \quad \Lambda \rightarrow \gamma^! \mathcal{H}_i \rightarrow \gamma^* \mathcal{H}_i \rightarrow K_Z.$$

Since i is a closed immersion, the composition (4.1.22) defines the canonical class $C_{Z/k}(\mathcal{F}) \in H^0(Z, K_Z)$.

Let $e_i^\vee: T^*X \times_X Z \rightarrow Z$ be the projection. Then the adjoint of (4.1.22) defines

$$(4.1.23) \quad \gamma_! \Lambda \rightarrow \mathcal{H}_i \rightarrow \gamma_* K_Z.$$

Since γ is a regular immersion, the specialization functor to the normal bundle $TX \times_X Z$ is still well defined (cf. [Ver83, (8,2)]). Taking the specialization and the microlocalization, then using [Ver83, (8.5) and (8.6)], we obtain

$$(4.1.24) \quad 0_! \Lambda \rightarrow \nu_{Z/Z \times X} \mathcal{H}_i \rightarrow 0_* K_Z,$$

$$(4.1.25) \quad \Lambda \rightarrow \mu_{Z/Z \times X} \mathcal{H}_i \rightarrow e_i^{\vee*} K_Z.$$

Regard $SS_\mu(i_* \mathcal{F}) \rightarrow T^*X$ as a subset of $T^*X \times_X Z$ and let $s: SS_\mu(i_* \mathcal{F}) \rightarrow T^*X \times_X Z$ be the closed immersion. Let $q: (T^*X \times_X Z)_{\bar{k}} \rightarrow T^*X \times_X Z$ be the projection. Since $SS_\mu(i_* \mathcal{F})_{\bar{k}} \subseteq (T^*X \times_X Z)_{\bar{k}}$ equals the support of $\mu_{Z/Z \times X} \mathcal{H}_i$, (4.1.25) defines

$$(4.1.26) \quad \Lambda \rightarrow s^* \mu_{Z/Z \times X} \mathcal{H}_i \rightarrow s^! e_i^{\vee*} K_Z$$

on $SS_\mu(i_* \mathcal{F})_{\bar{k}}$ by adjunction.

Similarly to (4.1.12) we have a canonical isomorphism

$$(4.1.27) \quad q^*: H_{SS_\mu(i_* \mathcal{F})}^0(T^*X \times_X Z, e_i^{\vee*} K_Z) \xrightarrow{\sim} H_{SS_\mu(i_* \mathcal{F})_{\bar{k}}}^0((T^*X \times_X Z)_{\bar{k}}, q^* e_i^{\vee*} K_Z)^{G_k}.$$

Definition 4.1.8. Let $\mathcal{F} \in D_{\text{ctf}}(Z, \Lambda)$. Let $i: Z \rightarrow X$ be a closed immersion to a smooth scheme over k and $e_i^\vee: T^*X \times_X Z \rightarrow Z$ be the projection. We define

$$(4.1.28) \quad CC_{\mu,i}(\mathcal{F}) \in H^0(SS_\mu(i_* \mathcal{F}), s^! e_i^{\vee*} K_Z) = H_{SS_\mu(i_* \mathcal{F})}^0(T^*X \times_X Z, e_i^{\vee*} K_Z)$$

to be the class whose image by (4.1.27) equals the class of the composition of (4.1.26).

We prove a relation between $CC_{\mu,i}(\mathcal{F})$ and $CC_\mu(i_* \mathcal{F})$ in Proposition 4.3.2.2. By the same argument as in the proof of Lemma 4.1.5, using the isomorphism (4.1.17), we have the following.

Lemma 4.1.9. *The pull-back of $CC_{\mu,i}(\mathcal{F})$ by the 0-section*

$$0^*: H_{SS_\mu(i_* \mathcal{F})}^0(T^*X \times_X Z, e_i^{\vee*} K_Z) \rightarrow H^0(Z, K_Z)$$

equals the canonical class $C_{Z/k}(\mathcal{F})$.

4.2 Tamely ramified sheaves on curves

Let X be a smooth irreducible curve over k and let $x \in X$ be a k -rational point. Let $j: U = X - \{x\} \rightarrow X$ be the open immersion and $\mathcal{F}$ be a locally constant constructible sheaf of free Λ -modules of rank n on U . By Lemma 3.1.5, we know that $SS_\mu(j_!\mathcal{F}) \subseteq T^*X$ is a closed conical subset of the union of the 0-section T_X^*X and the fiber T_x^*X . By the semi-purity [SGA 4₂-Cycle, Rappel 2.2.8], we have $H_{T_X^*X \cup T_x^*X}^0(T^*X, e^{\vee*}K_X) = \Lambda[T_X^*X] \oplus \Lambda[T_x^*X]$. By Lemma 4.1.4, we have $CC_\mu(j_!\mathcal{F}) = -(n \cdot [T_X^*X] + a \cdot [T_x^*X])$ for some $a \in \Lambda$. We give a method to compute the coefficient $a \in \Lambda$ in Lemma 4.2.3.

Let $+: T_X^*X \cup T_x^*X \rightarrow T^*X$ be the closed immersion. Since $\mu\mathcal{H}om(j_!\mathcal{F}, j_!\mathcal{F})$ is supported on $SS_\mu(j_!\mathcal{F}) \subseteq T_X^*X \cup T_x^*X$, the adjoint of the morphism $\mu\mathcal{H}om(j_!\mathcal{F}, j_!\mathcal{F}) \rightarrow e^{\vee*}K_X$ (4.1.8) induced by ev defines $\mu\mathcal{H}om(j_!\mathcal{F}, j_!\mathcal{F}) \rightarrow +^!e^{\vee*}K_X$. Here and in the following, identify $\mu\mathcal{H}om(j_!\mathcal{F}, j_!\mathcal{F})$ with its restriction to $T_X^*X \cup T_x^*X$ by abuse of notation. By the isomorphism $F_\psi\mathcal{N}_\Lambda \rightarrow +^!e^{\vee*}K_X$ (2.1.54), this is induced by a morphism $\nu\mathcal{H}om(j_!\mathcal{F}, j_!\mathcal{F}) \rightarrow \mathcal{N}_\Lambda$.

We define a subcomplex $\mathcal{N}_\mathcal{F}$ of $e^*(j_*\mathcal{E}nd\mathcal{F} \otimes K_X)$ on TX similarly as $\mathcal{N}_\Lambda$ and construct a factorization

$$(4.2.1) \quad \nu\mathcal{H}om(j_!\mathcal{F}, j_!\mathcal{F}) \rightarrow \mathcal{N}_\mathcal{F} \rightarrow \mathcal{N}_\Lambda$$

such that the restriction to TU is $\nu\mathcal{H}om(\mathcal{F}, \mathcal{F}) \rightarrow e_U^*(\mathcal{E}nd\mathcal{F} \otimes K_U) \rightarrow e_U^*K_U$ where the second arrow is induced by $\text{Tr}: \mathcal{E}nd\mathcal{F} \rightarrow \Lambda_U$ and $e_U: TU \rightarrow U$ is the projection. The cohomology sheaves of $j_*\mathcal{E}nd\mathcal{F} \otimes K_X$ are $\mathcal{H}^{-2} = R^0j_*\mathcal{E}nd\mathcal{F}(1)$, $\mathcal{H}^{-1} = R^1j_*\mathcal{E}nd\mathcal{F}(1)$ and vanish otherwise. Let I_x denote the inertia group at $x \in X$. The stalk $(R^1j_*\mathcal{E}nd\mathcal{F}(1))|_x$ is canonically identified with the Galois cohomology $H^1(I_x, \text{End } F(1))$. Let $P \subseteq I_x$ denote the wild inertia subgroup. Since I_x/P is isomorphic to $\widehat{\mathbf{Z}}' = \varprojlim_{p|n} \mu_n$ and P is a pro- p group, the group $H^1(I_x, \text{End } F(1))$ is canonically identified with the coinvariants $(\text{End } F)_{I_x}$.

Let $T = TX \times_X x$ be the fiber at x and $g: T^\times = T - \{0\} \rightarrow T$ be the open immersion. Define a subcomplex $\mathcal{N}_\mathcal{F}$ of $e^*(j_*\mathcal{E}nd\mathcal{F} \otimes K_X)$ by replacing the cohomology sheaf $\mathcal{H}^{-1} = e^*R^1j_*\mathcal{E}nd\mathcal{F}(1) = (\text{End } F)_{I_x, T}$ at the top degree by the subsheaf $g_!(\text{End } F)_{I_x, T^\times}$. Here and in the following $(\text{End } F)_{I_x, T}$ denotes the geometrically constant sheaf on T with stalk $(\text{End } F)_{I_x}$. The trace morphism $e_U^*(\mathcal{E}nd\mathcal{F} \otimes K_U) \rightarrow e_U^*K_U$ induces $e^*(j_*\mathcal{E}nd\mathcal{F} \otimes K_X) \rightarrow e^*(j_*\Lambda \otimes K_X)$. Define a morphism $\text{Tr}: \mathcal{N}_\mathcal{F} \rightarrow \mathcal{N}_\Lambda$ to be the restriction.

Lemma 4.2.1. 1. The isomorphism $\nu\mathcal{H}om(j_!\mathcal{F}, j_!\mathcal{F})|_{TU} \rightarrow e_U^*(\mathcal{E}nd\mathcal{F} \otimes K_U)$ induces

$$(4.2.2) \quad \nu\mathcal{H}om(j_!\mathcal{F}, j_!\mathcal{F}) \rightarrow \mathcal{N}_\mathcal{F}.$$

2. The morphism (4.2.2) induces a factorization (4.2.1).

Proof. 1. The isomorphism $\nu\mathcal{H}om(j_!\mathcal{F}, j_!\mathcal{F})|_{TU} = \nu\mathcal{H}om(\mathcal{F}, \mathcal{F}) \rightarrow e_U^*(\mathcal{E}nd\mathcal{F} \otimes K_U)$ induces a morphism $\nu\mathcal{H}om(j_!\mathcal{F}, j_!\mathcal{F}) \rightarrow e^*(j_*\mathcal{E}nd\mathcal{F} \otimes K_X)$ by adjunction and the smooth base change theorem. Since $0^*\nu\mathcal{H}om(j_!\mathcal{F}, j_!\mathcal{F}) = \delta^*(D_X(j_!\mathcal{F}) \boxtimes j_!\mathcal{F}) = D_X(j_!\mathcal{F}) \otimes j_!\mathcal{F} = j_!\mathcal{E}nd\mathcal{F} \otimes K_X$ by Proposition 2.3.7.2, the stalk $\nu\mathcal{H}om(j_!\mathcal{F}, j_!\mathcal{F})_0$ at $0 \in T \subseteq TX$ is 0. Hence by adjunction, $\nu\mathcal{H}om(j_!\mathcal{F}, j_!\mathcal{F})|_{TX - \{0\}} \rightarrow e^*(j_*\mathcal{E}nd\mathcal{F} \otimes K_X)|_{TX - \{0\}} = \mathcal{N}_\mathcal{F}|_{TX - \{0\}}$ induces (4.2.2).

2. The restriction to TU of $\nu\mathcal{H}om(j_!\mathcal{F}, j_!\mathcal{F}) \rightarrow \mathcal{N}_\Lambda$ is induced by the trace $e_U^*(\mathcal{E}nd\mathcal{F} \otimes K_U) \rightarrow e_U^*K_U$. Hence the assertion follows by adjunction as in the proof of 1. $\square$

We have isomorphisms $j_*\mathcal{E}nd\mathcal{F} \rightarrow \mathcal{E}nd j_!\mathcal{F} \rightarrow 0^!\nu\mathcal{H}om(j_!\mathcal{F}, j_!\mathcal{F})$ by Proposition 2.3.7.2 and (4.1.1). This and the identity $1_{\mathcal{F}}$ define the first two arrows in

$$(4.2.3) \quad \Lambda \rightarrow j_*\mathcal{E}nd\mathcal{F} \rightarrow 0^!\nu\mathcal{H}om(j_!\mathcal{F}, j_!\mathcal{F}) \rightarrow 0^!\mathcal{N}_{\mathcal{F}} \rightarrow 0^!\mathcal{N}_{\Lambda}.$$

The last two arrows are induced by (4.2.1). Since the Fourier transform of its adjoint

$$0_!\Lambda \rightarrow 0_!\mathcal{E}nd j_!\mathcal{F} \rightarrow \nu\mathcal{H}om(j_!\mathcal{F}, j_!\mathcal{F}) \rightarrow \mathcal{N}_{\mathcal{F}} \rightarrow \mathcal{N}_{\Lambda}$$

defines $\Lambda \rightarrow \mu\mathcal{H}om(j_!\mathcal{F}, j_!\mathcal{F}) \rightarrow +^!e^{\vee*}K_X$ defining $CC_{\mu}(j_!\mathcal{F})$, the composition of (4.2.3) determines $CC_{\mu}(j_!\mathcal{F})$.

Computations similar to those in Lemma 2.1.13.1 and Corollary 2.1.15.1 work for $\mathcal{N}_{\mathcal{F}}$ and $0^!\mathcal{N}_{\mathcal{F}}$.

Lemma 4.2.2. *Let I_x denote the inertia group at $x \in X$ and I_0 denote the inertia group at $0 \in T$. Let F be the representation of I_x defined by $\mathcal{F}$ and consider the coinvariant quotient $(\text{End } F)_{I_x}$ as a trivial representation of I_0 .*

1. *The canonical morphism $\mathcal{N}_{\mathcal{F}} \rightarrow e^*(j_*\mathcal{E}nd\mathcal{F} \otimes K_X)$ induces a distinguished triangle*

$$(4.2.4) \quad \mathcal{N}_{\mathcal{F}} \rightarrow e^*(j_*\mathcal{E}nd\mathcal{F} \otimes K_X) \oplus g_!(\text{End } F)_{I_x, T^{\times}}[1] \rightarrow (\text{End } F)_{I_x, T}[1] \rightarrow .$$

2. *The distinguished triangle (4.2.4) induces a distinguished triangle*

$$(4.2.5) \quad 0^!\mathcal{N}_{\mathcal{F}} \rightarrow j_*\mathcal{E}nd\mathcal{F} \oplus \Gamma(I_0, (\text{End } F)_{I_x})_0 \rightarrow H^1(I_0, (\text{End } F)_{I_x})_0[-1] \rightarrow .$$

The last two terms are placed at $0 \in T \cap X \subseteq TX$.

Proof. 1. The distinguished triangle (4.2.4) is clear from the construction of $\mathcal{N}_{\mathcal{F}}$.

2. For the geometrically constant sheaf $\mathcal{R} = (\text{End } F)_{I_x, T}$ on T , we have canonical isomorphisms $0^!\mathcal{R} \rightarrow H^1(I_0, \mathcal{R})_0[-2]$ and $0^!g_!g^*\mathcal{R} \rightarrow \Gamma(I_0, \mathcal{R})_0[-1]$. Hence (4.2.4) induces (4.2.5). $\square$

Lemma 4.2.3. *Suppose that the composition*

$$(4.2.6) \quad \Lambda \rightarrow j_*\mathcal{E}nd\mathcal{F} \rightarrow 0^!\nu\mathcal{H}om(j_!\mathcal{F}, j_!\mathcal{F}) \rightarrow 0^!\mathcal{N}_{\mathcal{F}} \rightarrow \Gamma(I_0, (\text{End } F)_{I_x})_0$$

of (4.2.3) and the second component of the first arrow of (4.2.5) maps $1 \in H^0(X, \Lambda)$ to $a \in H^0(I_0, (\text{End } F)_{I_x}) = (\text{End } F)_{I_x}$ and let $\text{Tr}: (\text{End } F)_{I_x} \rightarrow \Lambda$ be the morphism induced by the trace. Then, we have

$$(4.2.7) \quad CC_{\mu}(j_!\mathcal{F}) = -(\text{rank } \mathcal{F} \cdot [T_X^*X] + \text{Tr } a \cdot [T_x^*X])$$

*in $H_{T_X^*X \cup T_x^*X}^0(T^*X, e^{\vee*}K_X)$.*

Proof. The trace morphism $\mathcal{E}nd\mathcal{F} \rightarrow \Lambda$ induces a commutative diagram

$$\begin{array}{ccc} 0^!\mathcal{N}_{\mathcal{F}} & \longrightarrow & j_*\mathcal{E}nd\mathcal{F} \oplus \Gamma(I_0, (\text{End } F)_{I_x})_0 \\ \text{Tr} \downarrow & & \downarrow \text{Tr} \oplus \text{Tr} \\ 0^!\mathcal{N}_{\Lambda} & \longrightarrow & j_*\Lambda \oplus \Gamma(I_0, \Lambda)_0. \end{array}$$

The composition of (4.2.3) and the first component of the first arrow of (4.2.5) maps $1 \in H^0(X, \Lambda)$ to $1_{\mathcal{F}}$. The equality (4.2.7) follows from this and Corollary 2.1.15.2. $\square$

In this section, we prove the following.

Proposition 4.2.4. *Let $j: \mathbf{G}_m \rightarrow X = \mathbf{A}^1$ be the open immersion and let $\mathcal{F}$ be a locally constant constructible sheaf of free Λ -modules of rank n on $\mathbf{G}_m$ tamely ramified at $0 \in X$. Let I_x denote the inertia group at $0 \in X$ and I_0 denote the inertia group at $0 \in T$. Then, the composition*

$$(4.2.8) \quad j_* \mathcal{E}nd \mathcal{F} \rightarrow 0^! \nu \mathcal{H}om(j_! \mathcal{F}, j_! \mathcal{F}) \rightarrow 0^! \mathcal{N}_{\mathcal{F}} \rightarrow \Gamma(I_0, (\text{End } F)_{I_x})_0$$

of (4.2.3) and the second component of the first arrow of (4.2.5) induces the canonical morphism $H^0(U, \mathcal{E}nd \mathcal{F}) \rightarrow H^0(I_0, (\text{End } F)_{I_x}) = (\text{End } F)_{I_x}$.

Proposition 4.2.4 implies the following equality by Lemma 4.2.3.

Corollary 4.2.5. *We have*

$$(4.2.9) \quad CC_\mu(j_! \mathcal{F}) = -n([T_X^* X] + [T_0^* X])$$

in $H_{T_X^ X \cup T_0^* X}^0(T^* X, e^{\vee*} K_X)$.*

As a preparation of the proof of Proposition 4.2.4, we first prove a lemma on the cohomology of a tame inertia group.

Lemma 4.2.6. *Let $\widehat{\mathbf{Z}}' = \varprojlim_{p \nmid n} \mathbf{Z}/n\mathbf{Z}$ and let I be a profinite group isomorphic to $\widehat{\mathbf{Z}}'^2$. Let $\sigma, \tau \in I$ be elements defining an isomorphism $\widehat{\mathbf{Z}}'^2 \rightarrow I$ and let $I_1 = \langle \sigma \rangle$, $I_2 = \langle \tau \rangle \subseteq I = I_1 \times I_2$ be the subgroups isomorphic to $\widehat{\mathbf{Z}}'$. Let M be a Λ -module with a continuous action of I such that the action of $\tau \in I$ is trivial.*

1. *The restriction morphisms define an isomorphism*

$$(4.2.10) \quad H^1(I, M) \rightarrow H^1(I_1, M) \oplus H^1(I_2, M)^{I_1}.$$

2. *Let $\rho = \sigma\tau$ and $I_3 = \langle \rho \rangle \subseteq I = I_3 \times I_2$. Then, we have a commutative diagram*

$$(4.2.11) \quad \begin{array}{ccc} H^1(I, M) & \xrightarrow{(4.2.10)} & H^1(I_1, M) \oplus H^1(I_2, M)^{I_1} \\ 1 \downarrow & & \downarrow \begin{pmatrix} i_{13}^* & i_{23}^* \\ 0 & 1 \end{pmatrix} \\ H^1(I, M) & \xrightarrow{(4.2.10)} & H^1(I_3, M) \oplus H^1(I_2, M)^{I_3} \end{array}$$

where $i_{13}^: H^1(I_1, M) \rightarrow H^1(I_3, M)$ and $i_{23}^*: H^1(I_2, M)^{I_1} = H^1(I_2, M^{I_1}) \rightarrow H^1(I_3, M)$ are induced by the projections $I_3 \rightarrow I_1$ and $I_3 \rightarrow I_2$.*

Proof. 1. Let $A = \Lambda[[I]]$ be the completed group algebra and regard M as an A -module. By the Koszul free resolution $A \xrightarrow{(\tau-1, -(\sigma-1))} A^2 \xrightarrow{(\sigma-1, \tau-1)} A$ of the A -module Λ , the complex $M \xrightarrow{(\sigma-1, \tau-1)} M^2 \xrightarrow{(\tau-1, -(\sigma-1))} M$ computes the cohomology $H^*(I, M)$. Since τ acts trivially on M , we obtain an isomorphism $H^1(I, M) = \text{Coker}(\sigma - 1: M \rightarrow M) \oplus \text{Ker}(\sigma - 1: M \rightarrow M) \rightarrow H^1(I_1, M) \oplus H^1(I_2, M)^{I_1}$.

2. We have a commutative diagram

$$\begin{array}{ccccc} M & \xrightarrow{(\sigma-1, \tau-1)} & M^2 & \xrightarrow{(\tau-1, -(\sigma-1))} & M \\ 1 \downarrow & & \begin{pmatrix} \tau & 1 \\ 0 & 1 \end{pmatrix} \downarrow & & \downarrow \tau \\ M & \xrightarrow{(\rho-1, \tau-1)} & M^2 & \xrightarrow{(\tau-1, -(\rho-1))} & M \end{array}$$

of complexes. The middle vertical arrow induces the right vertical arrow in (4.2.11). $\square$

To prove Proposition 4.2.4, we prepare a geometric construction. Let X denote $\mathbf{A}^1 = \operatorname{Spec} k[x]$. We define a commutative diagram

$$\begin{array}{ccc} (X \times X \times \mathbf{A}^1)^\sharp & \longrightarrow & (X \times X \times \mathbf{A}^1)^\natural \\ \downarrow & & \downarrow \\ (X \times X \times \mathbf{A}^1)' & \longrightarrow & X \times X \times \mathbf{A}^1 = \operatorname{Spec} k[x, y, u]. \end{array}$$

The right vertical arrow is the blow-up at the closed point $(0, 0, 0)$. The upper horizontal arrow is the blow-up at the inverse image of $\delta(X) \times \{0\}$, in other words, at the pull-back of the ideal $(y - x, u)$. The lower horizontal arrow is the blow-up at $\delta(X) \times \{0\}$. The left vertical arrow is defined by the universality of blow-up. The open subscheme $A_X(X \times X)^\sharp \subseteq (X \times X \times \mathbf{A}^1)^\sharp$ defined by the cartesian diagram

$$(4.2.12) \quad \begin{array}{ccc} A_X(X \times X)^\sharp & \longrightarrow & (X \times X \times \mathbf{A}^1)^\sharp \\ q \downarrow & & \downarrow \\ A_X(X \times X) & \longrightarrow & (X \times X \times \mathbf{A}^1)' \end{array}$$

is the complement of the proper transform of the divisor $(u) = X \times X \times \{0\} \subseteq X \times X \times \mathbf{A}^1$.

The diagonal immersion $\delta: X \times \mathbf{A}^1 \rightarrow X \times X \times \mathbf{A}^1$ is lifted to the immersion $0: (X \times \mathbf{A}^1)' \rightarrow A_X(X \times X)^\sharp$ from the blow-up $(X \times \mathbf{A}^1)' \rightarrow X \times \mathbf{A}^1$ at the closed point $(0, 0)$. The projections $X \times X \times \mathbf{A}^1 \rightarrow X \times \mathbf{A}^1$ are lifted to smooth morphisms $p_1, p_2: A_X(X \times X)^\sharp \rightarrow (X \times \mathbf{A}^1)'$.

On $A_X(X \times X)_0^\sharp = A_X(X \times X)^\sharp \times_{\mathbf{A}^1} 0$, we have a cartesian diagram

$$(4.2.13) \quad \begin{array}{ccccccc} \mathbf{A}^1 & \longrightarrow & E & \longrightarrow & (X \times \mathbf{A}^1)'_0 & \longleftarrow & X \\ \delta \downarrow & & \bar{\delta} \downarrow & & \downarrow 0 & & \downarrow 0 \\ A_0 = \mathbf{A}^2 & \longrightarrow & P & \longrightarrow & A_X(X \times X)_0^\sharp & \longleftarrow & TX \end{array}$$

defined as follows. The immersion $TX \rightarrow A_X(X \times X)$ is lifted to a closed immersion $TX \rightarrow A_X(X \times X)_0^\sharp$. The proper transform in $A_X(X \times X)^\sharp$ of the exceptional divisor of the second blow-up $(X \times X \times \mathbf{A}^1)^\sharp \rightarrow (X \times X \times \mathbf{A}^1)^\natural$ is the complement $P = \mathbf{P}^{2'} - L'$ in the blow-up $\mathbf{P}^{2'}$ of $\mathbf{P}^2$ at $(1, 1, 0) \in L = \mathbf{P}^2 - \mathbf{A}^2$ of the proper transform L' of the line L at infinity. In the middle cartesian square, $(X \times \mathbf{A}^1)'_0 \rightarrow A_X(X \times X)_0^\sharp$ is the base change of the lifting $0: (X \times \mathbf{A}^1)' \rightarrow A_X(X \times X)^\sharp$ of the morphism $\delta: X \times \mathbf{A}^1 \rightarrow X \times X \times \mathbf{A}^1$. The closed fiber $A_X(X \times X)_0^\sharp$ equals the union $P \cup TX$. The intersection $P \cap TX$ is the fiber $T = TX \times_X 0 \subseteq TX$ and is dense in the exceptional divisor of the blow-up $\mathbf{P}^{2'} \rightarrow \mathbf{P}^2$. The inverse image of P is the exceptional divisor E of the blow-up $(X \times \mathbf{A}^1)' \rightarrow X \times \mathbf{A}^1$. The immersion $\bar{\delta}: E \rightarrow P$ is an extension of the diagonal morphism $\delta: \mathbf{A}^1 \rightarrow \mathbf{A}^2$.

They are explicitly described as follows. The scheme $A_X(X \times X)^\sharp$ has an open covering by two affine spaces $A = \mathbf{A}^3 = \operatorname{Spec} k[x_1, y_1, u]$ and $B = \mathbf{A}^3 = \operatorname{Spec} k[x, w, v]$. The second morphism in

$$A_X(X \times X)^\sharp = A \cup B \xrightarrow{q} A_X(X \times X) = \operatorname{Spec} k[x, u, v] \rightarrow X \times X \times \mathbf{A}^1 = \operatorname{Spec} k[x, y, u]$$

is given by $y = x + uv$ and the first morphism is given by $x = x_1u, v = y_1 - x_1$ and $u = xw$. The closed subscheme $TX \subseteq A_X(X \times X)_0^\sharp$ is identified with $\operatorname{Spec} k[x, v] \subseteq B$. The closed

fiber $A_X(X \times X)_0^\# = A_X(X \times X)^\# \times_{\mathbf{A}^1} 0$ has an open covering by $A_0 = \mathbf{A}^2 = \text{Spec } k[x_1, y_1]$ and $B_0 = \text{Spec } k[x, w, v]/(xw)$.

The blow-up $(X \times \mathbf{A}^1)'$ is the union $A_1 \cup B_1$ of two affine planes $A_1 = \mathbf{A}^2 = \text{Spec } k[x_1, u]$ and $B_1 = \mathbf{A}^2 = \text{Spec } k[x, w]$. The morphism

$$(X \times \mathbf{A}^1)' = A_1 \cup B_1 \rightarrow X \times \mathbf{A}^1 = \text{Spec } k[x, u]$$

is given by $x = x_1 u$ and $u = xw$. The immersion $0: (X \times \mathbf{A}^1)' = A_1 \cup B_1 \rightarrow A_X(X \times X)^\# = A \cup B$ is given by $y_1 = x_1$ on $A_1 \subseteq A$ and by $v = 0$ on $B_1 \subseteq B$. The morphism $p_1: A_X(X \times X)^\# \rightarrow (X \times \mathbf{A}^1)'$ is given by the projections $A = \text{Spec } k[x_1, y_1, u] \rightarrow A_1 = \text{Spec } k[x_1, u]$ and $B = \text{Spec } k[x, w, v] \rightarrow B_1 = \text{Spec } k[x, w]$.

We fix notation for immersions of subschemes of $A_X(X \times X)_0^\#$ as in the diagram

$$\begin{array}{ccccccc} \mathbf{G}_m & \xrightarrow{j_A} & \mathbf{A}^1 & \xrightarrow{j_E} & E & \xrightarrow{T^\times} & X \xleftarrow{j} U = X - \{0\} \\ \delta \downarrow & & \bar{\delta} \downarrow & & g \downarrow & & \downarrow 0 \quad \downarrow 0 \\ \mathbf{A}^2 & \xrightarrow{j_P} & P & \xleftarrow{i_T} & T & \longrightarrow & TX \xleftarrow{j_{TX}} TU \end{array}$$

where $T^\times = T - \{0\}$ is the complement of the origin. The middle vertical arrow and the labeled horizontal arrows except i_T are open immersions. The other vertical arrows, the unlabeled horizontal arrows and i_T are closed immersions. We have $\mathbf{A}^2 = P - T$, $TU = TX - T$ and $T = P \cap TX$. Let $e: TX \rightarrow X$ be the canonical morphism.

Let $\mathcal{F}$ be a locally constant constructible sheaf of free Λ -modules on $\mathbf{G}_m$ tamely ramified at $0 \in X = \mathbf{A}^1$ as in Proposition 4.2.4. Define $\mathcal{H} = \mathcal{H}om(\text{pr}_1^* j_! \mathcal{F}, \text{pr}_2^! j_! \mathcal{F})$ on $X \times X$ and let $\mathcal{H}^\#$ be the pull-back on $A_X(X \times X)^\#$. Let $\Psi \mathcal{H}^\#$ on $A_X(X \times X)_0^\#$ be the nearby cycles complex with respect to $A_X(X \times X)^\# \rightarrow \mathbf{A}^1 = \text{Spec } k[u]$ at 0. Let $\mathcal{M} = \Psi \mathcal{F}|_{\mathbf{G}_m}$ be the restriction on the complement $\mathbf{G}_m = E - \{0, \infty\}$ of the nearby cycles complex $\Psi \mathcal{F}$ with respect to $(X \times \mathbf{A}^1)' \rightarrow \mathbf{A}^1 = \text{Spec } k[u]$ at 0 as in Lemma 2.3.15. Define $\mathcal{H}_\mathcal{M} = \mathcal{H}om(\text{pr}_1^* j_{A!} \mathcal{M}, \text{pr}_2^! j_{A!} \mathcal{M})$ on $\mathbf{A}^2$.

Proposition 4.2.7. *The distinguished triangle $\Psi \mathcal{H}^\# \rightarrow \Psi \mathcal{H}^\#|_{TX} \oplus \Psi \mathcal{H}^\#|_P \rightarrow \Psi \mathcal{H}^\#|_T \rightarrow$ defines a distinguished triangle*

$$(4.2.14) \quad \Psi \mathcal{H}^\# \rightarrow e^*(j_* \mathcal{E}nd \mathcal{F} \otimes K_X) \oplus j_{P*} \mathcal{H}_\mathcal{M} \rightarrow (j_{P*} \mathcal{H}_\mathcal{M})_T \rightarrow .$$

We have a canonical isomorphism

$$(4.2.15) \quad e^*(j_* \mathcal{E}nd \mathcal{F} \otimes K_X)|_T \rightarrow (j_{P*} \mathcal{H}_\mathcal{M})_T.$$

Proof. The tensor product $\Psi \mathcal{H}^\# \otimes$ with the exact sequence $0 \rightarrow \Lambda_{A_X(X \times X)_0^\#} \rightarrow \Lambda_{TX} \oplus \Lambda_P \rightarrow \Lambda_T \rightarrow 0$ defines a distinguished triangle $\Psi \mathcal{H}^\# \rightarrow \Psi \mathcal{H}^\#|_{TX} \oplus \Psi \mathcal{H}^\#|_P \rightarrow \Psi \mathcal{H}^\#|_T \rightarrow$. By identifying $A = \text{Spec } k[x_1, y_1, u]$ with the fiber product $\text{Spec } k[x_1, u] \times_{\text{Spec } k[u]} \text{Spec } k[y_1, u]$ and by applying [Ill94, Théorème 4.7], we obtain an isomorphism $\Psi \mathcal{H}^\#|_{\mathbf{A}^2} \rightarrow \mathcal{H}_\mathcal{M}$. Since $p_1, p_2: A_X(X \times X)^\# \rightarrow (X \times \mathbf{A}^1)'$ are smooth, we obtain isomorphisms $\Psi \mathcal{H}^\#|_P \rightarrow j_{P*} \mathcal{H}_\mathcal{M}$, $\Psi \mathcal{H}^\#|_{TX} \rightarrow j_{TX*} e^*(\mathcal{E}nd \mathcal{F} \otimes K_U) = e^*(j_* \mathcal{E}nd \mathcal{F} \otimes K_X)$ and

$$(4.2.16) \quad \Psi \mathcal{H}^\#|_T \rightarrow (j_{P*} \mathcal{H}_\mathcal{M})|_T \rightarrow e^*(j_* \mathcal{E}nd \mathcal{F} \otimes K_X)|_T$$

by Lemma 2.3.14. Hence we obtain (4.2.14) and (4.2.15). $\square$

The morphism $q: A_X(X \times X)^\sharp \rightarrow A_X(X \times X)$ induces a morphism $q_0: A_X(X \times X)_0^\sharp \rightarrow TX = \text{Spec } k[x, v]$ to the tangent bundle. On $A_0 = \text{Spec } k[x_1, y_1]$, it is given by $v = y_1 - x_1$, $x = 0$ and on $B_0 = \text{Spec } k[x, w, v]/(xw)$ by the injection $k[x, v] \rightarrow k[x, w, v]/(xw)$. The restriction to $TX \subseteq A_X(X \times X)_0^\sharp$ is the identity. The restriction to $\mathbf{A}^2$ induces the morphism $v: \mathbf{A}^2 = A_0 = \text{Spec } k[x_1, y_1] \rightarrow T = \text{Spec } k[v] \subseteq TX$ defined by $v = y_1 - x_1$. This is extended to a morphism $\bar{v}: P \rightarrow T$.

Corollary 4.2.8. 1. *The direct image of (4.2.14) by $q_0: A_X(X \times X)_0^\sharp \rightarrow TX$ defines a distinguished triangle*

$$(4.2.17) \quad \nu\mathcal{H}om(j_!\mathcal{F}, j_!\mathcal{F}) \rightarrow e^*(j_*\mathcal{E}nd\mathcal{F} \otimes K_X) \oplus v_*\mathcal{H}_\mathcal{M} \rightarrow (j_{P*}\mathcal{H}_\mathcal{M})_T \rightarrow .$$

The second component

$$(4.2.18) \quad v_*\mathcal{H}_\mathcal{M} \rightarrow (j_{P*}\mathcal{H}_\mathcal{M})_T$$

of (4.2.17) is given by the restriction $v_\mathcal{H}_\mathcal{M} = \bar{v}_*j_{P*}\mathcal{H}_\mathcal{M} \rightarrow (j_{P*}\mathcal{H}_\mathcal{M})_T$.*

2. *Taking $0^!$ of (4.2.17), we obtain a distinguished triangle*

$$(4.2.19) \quad 0^!\nu\mathcal{H}om(j_!\mathcal{F}, j_!\mathcal{F}) \rightarrow j_*\mathcal{E}nd\mathcal{F} \oplus \Gamma(\mathbf{G}_{m, \bar{k}}, \mathcal{E}nd\mathcal{M})_0 \rightarrow \Gamma(\mathbf{G}_{m, \bar{k}}, \mathcal{E}nd\mathcal{M})_0 \rightarrow .$$

The last two terms denote the geometrically constant complexes placed at $\infty = E \cap X \subseteq A_X(X \times X)_0^\sharp$ and at $0 = T \cap X \subseteq TX$.

Proof. 1. Since $q: A_X(X \times X)^\sharp \rightarrow A_X(X \times X)$ is proper, we have a canonical isomorphism $\nu\mathcal{H}om(j_!\mathcal{F}, j_!\mathcal{F}) \rightarrow q_{0*}\Psi\mathcal{H}^\sharp$. Since the restriction of q_0 to TX is the identity and that on $\mathbf{A}^2$ is $v = \bar{v} \circ j_P$, the distinguished triangle (4.2.14) induces (4.2.17).

The direct image $\bar{v}_*$ of the restriction morphism $j_{P*}\mathcal{H}_\mathcal{M} \rightarrow (j_{P*}\mathcal{H}_\mathcal{M})_T$ induces (4.2.18).

2. By the proper base change theorem and [Gro77, (3.2.1)], we have an isomorphism

$$0^!v_*\mathcal{H}_\mathcal{M} \rightarrow \Gamma(\mathbf{A}^1, 0^!\mathcal{H}_\mathcal{M}) \rightarrow \Gamma(\mathbf{A}_{\bar{k}}^1, \mathcal{E}nd j_{A!}\mathcal{M}) \rightarrow \Gamma(\mathbf{G}_{m, \bar{k}}, \mathcal{E}nd\mathcal{M}).$$

Hence (4.2.19) is induced by (4.2.17). $\square$

We construct a commutative diagram to compute (4.2.8).

Corollary 4.2.9. *Let I_E denote the tame inertia group at $\infty \in E$, and let I_T be the inertia group at $0 \in T$. Let M be the representation of I_E defined by $\mathcal{M}$ and consider the coinvariant quotient $(\text{End } M)_{I_E}$ as a trivial representation of I_T . We identify $(\text{End } F)_{I_x}$ in (4.2.8) and $(\text{End } M)_{I_E}$ in (4.2.20) by the isomorphism $e^*(j_*\mathcal{E}nd\mathcal{F} \otimes K_X)|_T \rightarrow (j_{P*}\mathcal{H}_\mathcal{M})_T$ (4.2.15). The morphism $\nu\mathcal{H}om(j_!\mathcal{F}, j_!\mathcal{F}) \rightarrow \mathcal{N}_\mathcal{F}$ (4.2.2) and the morphism $v_*\mathcal{H}_\mathcal{M} \rightarrow (j_{P*}\mathcal{H}_\mathcal{M})_T$ (4.2.18) induce a commutative diagram*

$$(4.2.20) \quad \begin{array}{ccc} 0^!\nu\mathcal{H}om(j_!\mathcal{F}, j_!\mathcal{F}) & \xrightarrow{(4.2.19)} & \Gamma(\mathbf{G}_{m, \bar{k}}, \mathcal{E}nd\mathcal{M})_0 \\ (4.2.2) \downarrow & & \downarrow (4.2.18) \\ 0^!\mathcal{N}_\mathcal{F} & \xrightarrow{(4.2.5)} & \Gamma(I_T, (\text{End } M)_{I_E})_0 \end{array}$$

where the terms in the right column are placed at $0 = T \cap X \subseteq TX$.

Proof. The morphism $\nu\mathcal{H}om(j_!\mathcal{F}, j_!\mathcal{F}) \rightarrow \mathcal{N}_{\mathcal{F}}$ (4.2.2) is induced by $\nu\mathcal{H}om(j_!\mathcal{F}, j_!\mathcal{F}) \rightarrow e^*(j_*\mathcal{E}nd\mathcal{F} \otimes K_X)$. We identify $e^*(j_*\mathcal{E}nd\mathcal{F} \otimes K_X)|_T$ with $(j_{P*}\mathcal{H}_{\mathcal{M}})_T$ by the isomorphism (4.2.15). Then by (4.2.19) and (4.2.5), we obtain (4.2.20). $\square$

To complete the proof of Proposition 4.2.4, we compute the morphism

$$(4.2.21) \quad H^0(\mathbf{G}_{m,\bar{k}}, \mathcal{E}nd \mathcal{M}) \rightarrow H^0(I_T, (\text{End } M)_{I_E})$$

induced by the right vertical arrow in the commutative diagram (4.2.20), in Lemma 4.2.10 below. By the canonical isomorphism $0^!\mathcal{H}_{\mathcal{M}} = j_{A*}\mathcal{E}nd \mathcal{M}$ [Gro77, (3.2.1)], we obtain a distinguished triangle $\Gamma(\mathbf{G}_{m,\bar{k}}, \mathcal{E}nd \mathcal{M}) \rightarrow \Gamma(\mathbf{A}_{\bar{k}}^2, \mathcal{H}_{\mathcal{M}}) \rightarrow \Gamma(\mathbf{A}_{\bar{k}}^2 - \delta(\mathbf{A}_{\bar{k}}^1), \mathcal{H}_{\mathcal{M}}) \rightarrow$. We have $\Gamma(\mathbf{A}_{\bar{k}}^2, \mathcal{H}_{\mathcal{M}}) = \Gamma(\mathbf{A}_{\bar{k}}^1, D(j_!\mathcal{M})) \otimes \Gamma(\mathbf{A}_{\bar{k}}^1, j_!\mathcal{M}) = 0$ by Lemma 2.3.4.2. Hence for the left hand side of (4.2.21), we obtain an isomorphism

$$(4.2.22) \quad H^{-1}(\mathbf{A}_{\bar{k}}^2 - \delta(\mathbf{A}_{\bar{k}}^1), \mathcal{H}_{\mathcal{M}}) \rightarrow H^0(\mathbf{G}_{m,\bar{k}}, \mathcal{E}nd \mathcal{M}).$$

Since the action of I_T is trivial, the right hand side of (4.2.21) is identified with

$$(4.2.23) \quad H^0(I_T, (\text{End } M)_{I_E}) = (\text{End } M)_{I_E} = H^1(I_E, \text{End } M(1)).$$

Let P_{∞} be the strict localization of P at $\infty = E \cap T$ and let $V = P_{\infty} \times_P (\mathbf{A}^2 - \delta(\mathbf{A}^1)) \subseteq P_{\infty}$. We identify the tame fundamental group $\pi_1(V)^{\text{tame}}$ with $I_E \times I_T$ and $\Gamma(V, \mathcal{H}_{\mathcal{M}})$ with $\Gamma(I_E \times I_T, \text{End } M(1))[2]$. We define

$$(4.2.24) \quad H^{-1}(V, \mathcal{H}_{\mathcal{M}}) = H^1(I_E \times I_T, \text{End } M(1)) \rightarrow H^1(I_E, \text{End } M(1))$$

to be the restriction morphism for $I_E \rightarrow I_E \times I_T$. Since the second component of the right vertical arrow in (4.2.20) is induced by $v_*\mathcal{H}_{\mathcal{M}} \rightarrow (j_{P*}\mathcal{H}_{\mathcal{M}})|_T$, we have a commutative diagram

$$(4.2.25) \quad \begin{array}{ccc} H^{-1}(\mathbf{A}_{\bar{k}}^2 - \delta(\mathbf{A}_{\bar{k}}^1), \mathcal{H}_{\mathcal{M}}) & \longrightarrow & H^{-1}(V, \mathcal{H}_{\mathcal{M}}) \\ \textcolor{red}{(4.2.22)} \downarrow & & \downarrow \textcolor{red}{(4.2.24)} \\ H^0(\mathbf{G}_{m,\bar{k}}, \mathcal{E}nd \mathcal{M}) & \xrightarrow{\textcolor{red}{(4.2.21)}} & H^1(I_E, \text{End } M(1)). \end{array}$$

The upper horizontal arrow is the pull-back by $V \rightarrow \mathbf{A}_{\bar{k}}^2 - \delta(\mathbf{A}_{\bar{k}}^1)$.

To compute the morphism (4.2.21), we introduce a geometric construction. Let $\mathbf{P}_1^2$ be the strict localization of $\mathbf{P}^2$ at $(1, 1, 0) \in L = \mathbf{P}^2 - \mathbf{A}^2$ and let $U = \mathbf{P}_1^2 \times_{\mathbf{P}^2} (\mathbf{A}^2 - \delta(\mathbf{A}^1)) \subseteq \mathbf{P}_1^2$. Let I_L be the tame inertia group at $(1, 1, 0) \in L$ and let $I_{\mathbf{P}^1}$ be the tame inertia group at $(1, 1, 0) \in \mathbf{P}^1$ in the closure $\mathbf{P}^1$ of $\delta(\mathbf{A}^1)$. We identify the tame fundamental group $\pi_1(U)^{\text{tame}}$ with $I_{\mathbf{P}^1} \times I_L$. We regard $\text{End } M(1)$ as a representation of $I_{\mathbf{P}^1} \times I_L$ with the canonical action of $I_{\mathbf{P}^1}$ and the trivial action of I_L . Then we have an identification

$$(4.2.26) \quad \Gamma(U, \mathcal{H}_{\mathcal{M}}) = \Gamma(I_{\mathbf{P}^1} \times I_L, \text{End } M(1))[2].$$

The restriction $P \rightarrow \mathbf{P}^2$ of the blow-up at $(1, 1, 0)$ induces a morphism $V \rightarrow U$. We have a commutative diagram

$$(4.2.27) \quad \begin{array}{ccc} \pi_1(U)^{\text{tame}} & \longleftarrow & I_{\mathbf{P}^1} \times I_L \\ \uparrow & & \uparrow \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \\ \pi_1(V)^{\text{tame}} & \longleftarrow & I_E \times I_T \end{array}$$

of isomorphisms for the tame fundamental groups.

We consider the diagram

$$(4.2.28) \quad \begin{array}{ccccc} H^{-1}(\mathbf{A}_k^2 - \delta(\mathbf{A}_k^1), \mathcal{H}_{\mathcal{M}}) & \longrightarrow & H^{-1}(U, \mathcal{H}_{\mathcal{M}}) & \longrightarrow & H^{-1}(V, \mathcal{H}_{\mathcal{M}}) \\ \downarrow & & \downarrow (4.2.26) & & \downarrow = (4.2.24) \\ H^0(\mathbf{G}_{m,\bar{k}}, \mathcal{E}nd \mathcal{M}) & \longrightarrow & H^1(I_{\mathbf{P}^1} \times I_L, \text{End } M(1)) & \longrightarrow & H^1(I_E \times I_T, \text{End } M(1)). \end{array}$$

The upper horizontal arrows are induced by $V \rightarrow U \rightarrow \mathbf{A}^2 - \delta(\mathbf{A}^1)$. The lower right horizontal arrow is induced by the left vertical arrow of (4.2.27) and the right square is commutative. To complete the proof of Proposition 4.2.4, it suffices to show the following lemma.

Lemma 4.2.10. *We identify*

$$H^0(\mathbf{G}_{m,\bar{k}}, \mathcal{E}nd \mathcal{M}) = H^0(I_{\mathbf{P}^1}, H^1(I_L, \text{End } M(1))) = H^0(I_E, H^1(I_T, \text{End } M(1))).$$

1. Define $H^0(\mathbf{G}_{m,\bar{k}}, \mathcal{E}nd \mathcal{M}) \rightarrow H^1(I_{\mathbf{P}^1} \times I_L, \text{End } M(1))$ to be the inclusion of the direct summand $H^0(I_{\mathbf{P}^1}, H^1(I_L, \text{End } M(1))) \subseteq H^1(I_{\mathbf{P}^1} \times I_L, \text{End } M(1))$ in Lemma 4.2.6.1. Then, the left square in the diagram (4.2.28) is commutative.

2. The morphism (4.2.21) equals the morphism

$$(4.2.29) \quad H^0(I_E, H^1(I_T, \text{End } M(1))) = H^1(I_T, H^0(I_E, \text{End } M(1))) \rightarrow H^1(I_E, \text{End } M(1))$$

induced by the canonical isomorphism $I_E \rightarrow I_T$ compatible with the inclusion $H^0(I_E, \text{End } M(1)) \rightarrow \text{End } M(1)$.

Proof. 1. Similarly as (4.2.25), we have a commutative diagram

$$(4.2.30) \quad \begin{array}{ccc} \Gamma(\mathbf{A}_k^2 - \delta(\mathbf{A}_k^1), \mathcal{H}_{\mathcal{M}}) & \longrightarrow & \Gamma(U, \mathcal{H}_{\mathcal{M}}) \\ \downarrow & & \downarrow \\ \Gamma(\mathbf{G}_{m,\bar{k}}, \mathcal{E}nd \mathcal{M})[1] & \longrightarrow & \Gamma(I_{\mathbf{P}^1}, H^1(I_L, \text{End } M(1)))[1]. \end{array}$$

By Lemma 4.2.6.1, it suffices to show that the composition

$$H^{-1}(\mathbf{A}_k^2 - \delta(\mathbf{A}_k^1), \mathcal{H}_{\mathcal{M}}) \rightarrow H^{-1}(U, \mathcal{H}_{\mathcal{M}}) \rightarrow H^1(I_L \times I_{\mathbf{P}^1}, \text{End } M(1)) \rightarrow H^1(I_{\mathbf{P}^1}, \text{End } M(1))$$

is the 0-mapping.

Let $j_{\mathbf{P}^2}: \mathbf{A}^2 \rightarrow \mathbf{P}^2$ be the open immersion and set $\mathcal{G} = (j_{\mathbf{P}^2*} \mathcal{H}_{\mathcal{M}})|_L$. Let $0, 1, \infty \in L$ be the intersections of the closures of $0 \times \mathbf{A}^1$, $\delta(\mathbf{A}^1)$, $\mathbf{A}^1 \times 0$ with L in $\mathbf{P}^2$. Let L_1 be the strict localization of L at a geometric point above $1 \in L$ and set $U_1 = L_1 \times_L (L - \{1\})$. Let $j_0: L - \{0\} \rightarrow L$ be the open immersion. Since $\mathcal{G} \rightarrow j_{0*} j_0^* \mathcal{G}$ is an isomorphism, we have a commutative diagram

$$(4.2.31) \quad \begin{array}{ccccc} H^{-1}(\mathbf{A}_k^2 - \delta(\mathbf{A}_k^1), \mathcal{H}_{\mathcal{M}}) & \longrightarrow & H^{-1}(U, \mathcal{H}_{\mathcal{M}}) & \longrightarrow & H^1(I_{\mathbf{P}^1}, \text{End } M(1)) \\ \downarrow & & \downarrow & & \downarrow \\ H^{-1}(L - \{0, 1\}, \mathcal{G}) & \longrightarrow & H^{-1}(U_1, \mathcal{G}) & \longrightarrow & H^0(U_1, \mathcal{H}^{-1} \mathcal{G}). \end{array}$$

To show that the composition of the lower line is 0, we consider the commutative diagram

$$\begin{array}{ccc} H^{-1}(L - \{0, 1\}, \mathcal{G}) & \longrightarrow & H^{-1}(U_1, \mathcal{G}) \\ \downarrow & & \downarrow \\ H^0(L - \{0, 1\}, \mathcal{H}^{-1}\mathcal{G}) & \longrightarrow & H^0(U_1, \mathcal{H}^{-1}\mathcal{G}) \end{array}$$

Let $j_\infty: L - \{\infty\} \rightarrow L$ be the open immersion. Since $j_{\infty!}j_\infty^*\mathcal{G} \rightarrow \mathcal{G}$ is an isomorphism, we have $H^0(L - \{0, 1\}, \mathcal{H}^{-1}\mathcal{G}) = 0$ and the composition of the lower line of (4.2.31) is 0. Since the right vertical arrow in (4.2.31) is an isomorphism, the composition of the upper line is also 0.

2. Define $H^0(\mathbf{G}_{m, \bar{k}}, \mathcal{E}nd \mathcal{M}) \rightarrow H^1(I_E \times I_T, \mathcal{E}nd M(1))$ to be the direct sum

$$\begin{aligned} H^0(I_E, H^1(I_T, \mathcal{E}nd M(1))) &\rightarrow H^1(I_E, \mathcal{E}nd M(1)) \oplus H^0(I_E, H^1(I_T, \mathcal{E}nd M(1))) \\ &= H^1(I_E \times I_T, \mathcal{E}nd M(1)) \end{aligned}$$

of (4.2.29) and the identity for the direct sum decomposition in Lemma 4.2.6.1. Then, the big rectangle in the diagram (4.2.28) is commutative by the commutative diagram

$$\begin{array}{ccc} I_{\mathbf{P}^1} \times I_L & \xleftarrow{(4.2.27)} & I_E \times I_T \\ 1 \times 1 \uparrow & & \uparrow (1, 0) \\ I_E \times I_T & \xleftarrow{\text{diag}} & I_E, \end{array}$$

Lemma 4.2.6.2 and 1. Hence (4.2.29) at the position of (4.2.21) in (4.2.25) makes the diagram (4.2.25) commutative. Since (4.2.22) is an isomorphism, such a morphism is unique and (4.2.21) equals (4.2.29). $\square$

4.3 Proper push-forward

Lemma 4.3.1. *Let*

$$(4.3.1) \quad \begin{array}{ccc} C & \xrightarrow{g} & D \\ c \downarrow & & \downarrow d \\ X & \xrightarrow{f} & Y \end{array}$$

be a commutative diagram of separated schemes of finite type over k .

1. Define $f_!c_!c^! \rightarrow d_!d^!f_!$ to be the adjoint of the composition of the canonical isomorphism $f_!c_!c^!f^! \rightarrow d_!g_!g^!d^!$ and the morphism $d_!g_!g^!d^! \rightarrow d_!d^!$ induced by adj_g . Then the diagram

$$(4.3.2) \quad \begin{array}{ccc} f_!c_!c^! & \longrightarrow & d_!d^!f_! \\ \text{adj}_c \downarrow & & \downarrow \text{adj}_d \\ f_! & \equiv & f_! \end{array}$$

is commutative.

2. Define $f_*c_*c^* \leftarrow d_*d^*f_*$ to be the adjoint of the composition of the canonical isomorphism $f_*c_*c^*f^* \rightarrow d_*g_*g^*d^*$ and the morphism $d_*g_*g^*d^* \rightarrow d_*d^*$ induced by adj_g . Then the diagram

$$(4.3.3) \quad \begin{array}{ccc} f_* & \xlongequal{\quad} & f_* \\ \text{adj}_c \downarrow & & \downarrow \text{adj}_d \\ f_*c_*c^* & \longleftarrow & d_*d^*f_* \end{array}$$

is commutative.

Proof. 1. By the commutative diagram (4.3.1), we have a commutative diagram

$$\begin{array}{ccc} f_!c_!c^!f^! & \xrightarrow{\text{adj}_g} & d_!d^! \\ \text{adj}_c \downarrow & & \downarrow \text{adj}_d \\ f_!f^! & \xrightarrow{\text{adj}_f} & 1_Y. \end{array}$$

By taking the adjoint, we obtain (4.3.2).

2. Part 2 is proved similarly as 1. $\square$

Let $f: X \rightarrow Y$ be a morphism of smooth schemes over k and $i: Z \hookrightarrow X$ be a closed immersion such that $g = fi: Z \rightarrow Y$ is proper. Consider the following commutative diagram

$$(4.3.4) \quad \begin{array}{ccccccc} T^*X & \xleftarrow{c^\vee = \text{pr}_{T^*X}} & T^*X \times_X Z & \xleftarrow{a^\vee = df} & T^*Y \times_Y Z & \xrightarrow{b^\vee = \text{pr}_{T^*Y}} & T^*Y \\ e_X^\vee \downarrow & & e_i^\vee \downarrow & & \downarrow e_g^\vee & & \downarrow e_Y^\vee \\ X & \xleftarrow{i} & Z & \xleftarrow{1_Z} & Z & \xrightarrow{g} & Y. \end{array}$$

Let $\mathcal{F} \in D_{\text{ctf}}(Z, \Lambda)$. By [Bei16, Lemma 2.2 (ii)] and Theorem 3.3.6, we have

$$(4.3.5) \quad SS_\mu(g_*\mathcal{F}) = SS(g_*\mathcal{F}) \subseteq f_*SS(i_*\mathcal{F}) = f_*SS_\mu(i_*\mathcal{F}) = b^\vee(a^{\vee-1}(SS_\mu(i_*\mathcal{F}))).$$

The base of $SS_\mu(i_*\mathcal{F})$ is contained in Z and we have $SS_\mu(i_*\mathcal{F}) \subseteq T^*X \times_X Z$. We define

$$(4.3.6) \quad f_! : H_{SS_\mu(i_*\mathcal{F})}^0(T^*X, e_X^{\vee*}K_X) \rightarrow H_{f_*SS_\mu(i_*\mathcal{F})}^0(T^*Y, e_Y^{\vee*}K_Y)$$

to be the following composition

$$(4.3.7) \quad \begin{aligned} H_{SS_\mu(i_*\mathcal{F})}^0(T^*X, e_X^{\vee*}K_X) &\simeq H_{SS_\mu(i_*\mathcal{F})}^0(T^*X \times_X Z, c^{\vee!}e_X^{\vee*}K_X) \\ &\simeq H_{SS_\mu(i_*\mathcal{F})}^0(T^*X \times_X Z, e_i^{\vee*}i^!K_X) \\ &\simeq H_{SS_\mu(i_*\mathcal{F})}^0(T^*X \times_X Z, e_i^{\vee*}K_Z) \\ &\xrightarrow{g_! = b_*^\vee a^{\vee*}} H_{f_*SS_\mu(i_*\mathcal{F})}^0(T^*Y, e_Y^{\vee*}K_Y). \end{aligned}$$

Proposition 4.3.2. *Let $f: X \rightarrow Y$ be a morphism of smooth schemes over k and $i: Z \hookrightarrow X$ be a closed immersion such that $g = fi: Z \rightarrow Y$ is proper. Let $\mathcal{F} \in D_{\text{ctf}}(Z, \Lambda)$.*

1. ([AS07, Proposition 2.1.6], [Gro77, Théorème 4.4, Corollaire 4.8]) *We have*

$$(4.3.8) \quad C_{Y/k}(g_*\mathcal{F}) = g_*C_{Z/k}(\mathcal{F})$$

in $H^0(Y, K_Y)$.

2. The canonical isomorphism $H_{SS_\mu(i_*\mathcal{F})}^0(T^*X \times_X Z, e_i^{\vee*}K_Z) \rightarrow H_{SS_\mu(i_*\mathcal{F})}^0(T^*X, e_X^{\vee*}K_X)$ induced by the isomorphism $K_Z \rightarrow i^!K_X$ maps $CC_{\mu,i}(\mathcal{F})$ to $CC_\mu(i_*\mathcal{F})$.

3. We have

$$(4.3.9) \quad CC_\mu(g_*\mathcal{F}) = f_!CC_\mu(i_*\mathcal{F})$$

$$\text{in } H_{f_*SS_\mu(i_*\mathcal{F}) \cup SS_\mu(g_*\mathcal{F})}^0(T^*Y, e_Y^{\vee*}K_Y) = H_{f_*SS_\mu(i_*\mathcal{F})}^0(T^*Y, e_Y^{\vee*}K_Y).$$

Proof. We apply Lemma 4.3.1 to the commutative diagram

$$(4.3.10) \quad \begin{array}{ccccccc} X & \xleftarrow{i} & Z & \xrightarrow{1_Z} & Z & \xrightarrow{g} & Y \\ \delta_X \downarrow & & \gamma_i \downarrow & & \downarrow \gamma_g & & \downarrow \delta_Y \\ X \times X & \xleftarrow{i \times 1_X} & Z \times X & \xrightarrow{1_Z \times f} & Z \times Y & \xrightarrow{g \times 1_Y} & Y \times Y \end{array}$$

as follows. Set

$$(4.3.11) \quad \begin{aligned} \mathcal{H}_X &= \mathcal{H}om_{X \times X}(\text{pr}_1^*i_*\mathcal{F}, \text{pr}_2^!i_*\mathcal{F}), & \mathcal{H}_i &= \mathcal{H}om_{Z \times X}(\text{pr}_1^*\mathcal{F}, \text{pr}_2^!i_*\mathcal{F}), \\ \mathcal{H}_Y &= \mathcal{H}om_{Y \times Y}(\text{pr}_1^*g_*\mathcal{F}, \text{pr}_2^!g_*\mathcal{F}), & \mathcal{H}_g &= \mathcal{H}om_{Z \times Y}(\text{pr}_1^*\mathcal{F}, \text{pr}_2^!g_*\mathcal{F}), \end{aligned}$$

and identify them with

$$(4.3.12) \quad \begin{aligned} \boxtimes_X &= D_X(i_*\mathcal{F}) \boxtimes i_*\mathcal{F}, & \boxtimes_i &= D_Z(\mathcal{F}) \boxtimes i_*\mathcal{F}, \\ \boxtimes_Y &= D_Y(g_*\mathcal{F}) \boxtimes g_*\mathcal{F}, & \boxtimes_g &= D_Z(\mathcal{F}) \boxtimes g_*\mathcal{F}, \end{aligned}$$

by the canonical isomorphisms [Gro77, (3.1.1)]. The canonical morphism $\boxtimes_X \rightarrow (i \times 1_X)_*\boxtimes_i$ is an isomorphism.

By applying Lemma 4.3.1.1 and by [Gro77, (3.2.1)], we obtain commutative diagrams

$$(4.3.13) \quad \begin{array}{ccc} \delta_{X!}\mathcal{H}om_X(i_*\mathcal{F}, i_*\mathcal{F}) & \xleftarrow{\simeq} & (i \times 1_X)_!\gamma_{i!}\mathcal{H}om_Z(\mathcal{F}, \mathcal{F}) \\ \downarrow & & \downarrow \\ \mathcal{H}_X & \xleftarrow{\simeq} & (i \times 1_X)_!\mathcal{H}_i, \end{array}$$

$$(4.3.14) \quad \begin{array}{ccc} (1_Z \times f)_!\gamma_{i!}\mathcal{H}om_Z(\mathcal{F}, \mathcal{F}) & \longrightarrow & \gamma_{g!}\mathcal{H}om_Z(\mathcal{F}, g^!g_!\mathcal{F}) \\ \downarrow & & \downarrow \\ (1_Z \times f)_!\mathcal{H}_i & \xrightarrow{\simeq} & \mathcal{H}_g, \end{array}$$

$$(4.3.15) \quad \begin{array}{ccc} (g \times 1_Y)_!\gamma_{g!}\mathcal{H}om_Z(\mathcal{F}, g^!g_!\mathcal{F}) & \xrightarrow{\simeq} & \delta_{Y!}\mathcal{H}om_Y(g_!\mathcal{F}, g_!\mathcal{F}) \\ \downarrow & & \downarrow \\ (g \times 1_Y)_!\mathcal{H}_g & \xrightarrow{\simeq} & \mathcal{H}_Y. \end{array}$$

Since i and g are proper, we identify $i_! = i_*$ and $g_! = g_*$. Defining vertical arrows by $1_{i_*\mathcal{F}}$, $1_{\mathcal{F}}$, $\text{adj}: \mathcal{F} \rightarrow g^!g_!\mathcal{F}$ and $1_{g_!\mathcal{F}}$, we obtain commutative diagrams

$$(4.3.16) \quad \begin{array}{ccc} \delta_{X!}\Lambda & \xrightarrow{\text{res}} & (i \times 1_X)_!\gamma_{i!}\Lambda \\ 1_{i_*\mathcal{F}} \downarrow & & \downarrow 1_{\mathcal{F}} \\ \delta_{X!}\mathcal{H}om_X(i_*\mathcal{F}, i_*\mathcal{F}) & \xleftarrow{\simeq} & (i \times 1_X)_!\gamma_{i!}\mathcal{H}om_Z(\mathcal{F}, \mathcal{F}), \end{array}$$

$$\begin{array}{ccc}
(1_Z \times f)_! \gamma_{i!} \Lambda & \xrightarrow{\cong} & \gamma_{g!} \Lambda \\
1_{\mathcal{F}} \downarrow & & \downarrow \text{adj} \\
(1_Z \times f)_! \gamma_{i!} \mathcal{H}om_Z(\mathcal{F}, \mathcal{F}) & \xrightarrow{\text{adj}} & \gamma_{g!} \mathcal{H}om_Z(\mathcal{F}, g^! g_! \mathcal{F}),
\end{array}
\tag{4.3.17}$$

$$\begin{array}{ccc}
(g \times 1_Y)_! \gamma_{g!} \Lambda & \xleftarrow{\quad} & \delta_{Y!} \Lambda \\
\text{adj} \downarrow & & \downarrow 1_{g_! \mathcal{F}} \\
(g \times 1_Y)_! \gamma_{g!} \mathcal{H}om_Z(\mathcal{F}, g^! g_! \mathcal{F}) & \xrightarrow{\cong} & \delta_{Y!} \mathcal{H}om_Y(g_! \mathcal{F}, g_! \mathcal{F}).
\end{array}
\tag{4.3.18}$$

By applying Lemma 4.3.1.2, we obtain commutative diagrams

$$\begin{array}{ccc}
\boxtimes_X & \xrightarrow{\cong} & (i \times 1_X)_* \boxtimes_i \\
\downarrow & & \downarrow \\
\delta_{X*}(D_X i_* \mathcal{F} \otimes i_* \mathcal{F}) & \xrightarrow{\cong} & (i \times 1_X)_* \gamma_{i*}(D_Z \mathcal{F} \otimes \mathcal{F}),
\end{array}
\tag{4.3.19}$$

$$\begin{array}{ccc}
(1_Z \times f)_! \boxtimes_i & \xrightarrow{\cong} & \boxtimes_g \\
\downarrow & & \downarrow \\
(1_Z \times f)_! \gamma_{i*}(D_Z \mathcal{F} \otimes \mathcal{F}) & \xleftarrow{\text{adj}} & \gamma_{g*}(D_Z \mathcal{F} \otimes g^* g_* \mathcal{F}),
\end{array}
\tag{4.3.20}$$

$$\begin{array}{ccc}
(g \times 1_Y)_* \boxtimes_g & \xrightarrow{\cong} & \boxtimes_Y \\
\downarrow & & \downarrow \\
(g \times 1_Y)_* \gamma_{g*}(D_Z \mathcal{F} \otimes g^* g_* \mathcal{F}) & \xleftarrow{\cong} & \delta_{Y*}(D_Y g_* \mathcal{F} \otimes g_* \mathcal{F}).
\end{array}
\tag{4.3.21}$$

The lower line in (4.3.21) is induced by the isomorphism $g_*(D_Z \mathcal{F} \otimes g^* g_* \mathcal{F}) \rightarrow g_* D_Z \mathcal{F} \otimes g_* \mathcal{F}$ given by the projection formula and the canonical isomorphism $g_* D_Z \mathcal{F} \rightarrow D_Y g_! \mathcal{F} \rightarrow D_Y g_* \mathcal{F}$.

Defining vertical arrows by $\text{ev}_X: D_X i_* \mathcal{F} \otimes i_* \mathcal{F} \rightarrow K_X$, $\text{ev}_Z: D_Z \mathcal{F} \otimes \mathcal{F} \rightarrow K_Z$, $\text{ev}_Z \circ (1 \otimes \text{adj}): D_Z \mathcal{F} \otimes g^* g_* \mathcal{F} \rightarrow D_Z \mathcal{F} \otimes \mathcal{F} \rightarrow K_Z$ and $\text{ev}_Y: D_Y g_* \mathcal{F} \otimes g_* \mathcal{F} \rightarrow K_Y$, we obtain commutative diagrams

$$\begin{array}{ccc}
\delta_{X*}(D_X i_* \mathcal{F} \otimes i_* \mathcal{F}) & \xrightarrow{\cong} & (i \times 1_X)_* \gamma_{i*}(D_Z \mathcal{F} \otimes \mathcal{F}), \\
\text{ev}_X \downarrow & & \downarrow \text{ev}_Z \\
\delta_{X*} K_X & \xleftarrow{\text{adj}} & (i \times 1_X)_* \gamma_{i*} K_Z,
\end{array}
\tag{4.3.22}$$

$$\begin{array}{ccc}
(1_Z \times f)_! \gamma_{i*}(D_Z \mathcal{F} \otimes \mathcal{F}) & \xleftarrow{\text{adj}} & \gamma_{g*}(D_Z \mathcal{F} \otimes g^* g_* \mathcal{F}) \\
\text{ev}_Z \downarrow & & \downarrow \text{ev}_Z \circ (1 \otimes \text{adj}) \\
(1_Z \times f)_! \gamma_{i*} K_Z & \xrightarrow{\cong} & \gamma_{g*} K_Z,
\end{array}
\tag{4.3.23}$$

$$\begin{array}{ccc}
(g \times 1_Y)_* \gamma_{g*} (D_Z \mathcal{F} \otimes g^* g_* \mathcal{F}) & \longrightarrow & \delta_{Y*} (D_Y g_* \mathcal{F} \otimes g_* \mathcal{F}) \\
\text{ev}_Z \circ (1 \otimes \text{adj}) \downarrow & & \downarrow \text{ev}_Y \\
(g \times 1_Y)_* \gamma_{g*} K_Z & \xrightarrow{\text{adj}} & \delta_{Y*} K_Y.
\end{array}
\tag{4.3.24}$$

The lower line in (4.3.24) is induced by the adjunction $g_* K_Z = g_! g^! K_Y \rightarrow K_Y$.

Commutative diagrams (4.3.13), (4.3.16), (4.3.19) and (4.3.22) define a commutative diagram

$$\begin{array}{ccc}
\delta_{X!} \Lambda & \xrightarrow{\text{res}} & (i \times 1_X)_* \gamma_{i!} \Lambda \\
1_{i_* \mathcal{F}} \downarrow & & \downarrow 1_{\mathcal{F}} \\
\mathcal{H}_X & \xrightarrow{\simeq} & (i \times 1_X)_* \mathcal{H}_i \\
\text{ev}_X \downarrow & & \downarrow \text{ev}_Z \\
\delta_{X*} K_X & \xleftarrow{\text{adj}} & (i \times 1_X)_* \gamma_{i*} K_Z,
\end{array}
\tag{4.3.25}$$

Similarly, commutative diagrams (4.3.14), (4.3.17), (4.3.20) and (4.3.23) define a commutative diagram

$$\begin{array}{ccc}
(1_Z \times f)_! \gamma_{i!} \Lambda & \xrightarrow{\simeq} & \gamma_{g!} \Lambda \\
1_{\mathcal{F}} \downarrow & & \downarrow \text{adj} \\
(1_Z \times f)_! \mathcal{H}_i & \xrightarrow{\simeq} & \mathcal{H}_g \\
\text{ev}_Z \downarrow & & \downarrow \text{ev}_g \circ \text{adj} \\
(1_Z \times f)_! \gamma_{i*} K_Z & \xrightarrow{\simeq} & \gamma_{g*} K_Z.
\end{array}
\tag{4.3.26}$$

Similarly, commutative diagrams (4.3.15), (4.3.18), (4.3.21) and (4.3.24) define a commutative diagram

$$\begin{array}{ccc}
(g \times 1_Y)_* \gamma_{g!} \Lambda & \longleftarrow & \delta_{Y!} \Lambda \\
\text{adj} \downarrow & & \downarrow 1_{g_* \mathcal{F}} \\
(g \times 1_Y)_* \mathcal{H}_g & \xrightarrow{\simeq} & \mathcal{H}_Y \\
\text{ev} \circ \text{adj} \downarrow & & \downarrow \text{ev} \\
(g \times 1_Y)_* \gamma_{g*} K_Z & \longrightarrow & \delta_{Y*} K_Y.
\end{array}
\tag{4.3.27}$$

Since i is a closed immersion, the diagrams (4.3.26) and (4.3.27) imply (4.3.8).

The commutative diagram (4.3.10) induces a commutative diagram

$$\begin{array}{ccccccc}
X & \xleftarrow{i} & Z & \xrightarrow{1_Z} & Z & \xrightarrow{g} & Y \\
0_X \downarrow & & 0_i \downarrow & & \downarrow 0_g & & \downarrow 0_Y \\
TX & \xleftarrow{c} & TX \times_X Z & \xrightarrow{a} & TY \times_Y Z & \xrightarrow{b} & TY.
\end{array}
\tag{4.3.28}$$

The vertical arrows are the 0-sections.

Since the left square in (4.3.10) is cartesian with proper morphisms, the base change morphism $\nu_{X/X \times X}(i \times 1_X)_* \rightarrow c_*\nu_{Z/Z \times X}$ (2.3.36) is an isomorphism. Hence applying the functor $\nu_{X/X \times X}$ to (4.3.25) defines a commutative diagram

$$(4.3.29) \quad \begin{array}{ccc} 0_{X!}\Lambda & \longrightarrow & c_*0_{i!}\Lambda \\ \downarrow & & \downarrow \\ \nu_{X/X \times X}\mathcal{H}_X & \xrightarrow{\simeq} & c_*\nu_{Z/Z \times X}\mathcal{H}_i \\ \downarrow & & \downarrow \\ 0_{X*}K_X & \longleftarrow & c_*0_{i*}K_Z. \end{array}$$

We apply $\nu_{Z/Z \times Y}$ to (4.3.26) and apply the morphism $a_!\nu_{Z/Z \times X} \rightarrow \nu_{Z/Z \times Y}(1_Z \times f)_!$ of functors (2.3.37) to the left column. Then, we obtain a commutative diagram

$$(4.3.30) \quad \begin{array}{ccc} a_!0_{i!}\Lambda & \xrightarrow{\simeq} & 0_{g!}\Lambda \\ \downarrow & & \downarrow \\ a_!\nu_{Z/Z \times X}\mathcal{H}_i & \longrightarrow & \nu_{Z/Z \times Y}\mathcal{H}_g \\ \downarrow & & \downarrow \\ a_!0_{i*}K_Z & \xrightarrow{\simeq} & 0_{g*}K_Z. \end{array}$$

Since the right square in (4.3.10) is cartesian with proper morphisms, the base change morphism $\nu_{Y/Y \times Y}(g \times 1_Y)_* \rightarrow b_*\nu_{Z/Z \times Y}$ (2.3.36) is an isomorphism. Hence applying the functor $\nu_{Y/Y \times Y}$ to (4.3.27) defines a commutative diagram

$$(4.3.31) \quad \begin{array}{ccc} b_*0_{g!}\Lambda & \longleftarrow & 0_{Y!}\Lambda \\ \downarrow & & \downarrow \\ b_*\nu_{Z/Z \times Y}\mathcal{H}_g & \xrightarrow{\simeq} & \nu_{Y/Y \times Y}\mathcal{H}_Y \\ \downarrow & & \downarrow \\ b_*0_{g*}K_Z & \longrightarrow & 0_{Y*}K_Y. \end{array}$$

The lower line of the commutative diagram (4.3.28) induces the upper line of the commutative diagram

$$(4.3.32) \quad \begin{array}{ccccccc} T^*X & \xleftarrow{c^\vee = \text{pr}_{T^*X}} & T^*X \times_X Z & \xleftarrow{a^\vee = df} & T^*Y \times_Y Z & \xrightarrow{b^\vee = \text{pr}_{T^*Y}} & T^*Y \\ e_X^\vee \downarrow & & e_i^\vee \downarrow & & \downarrow e_g^\vee & & \downarrow e_Y^\vee \\ X & \xleftarrow{i} & Z & \xleftarrow{1_Z} & Z & \xrightarrow{g} & Y, \end{array}$$

where $a^\vee$ is the dual of a . The vertical arrows are the projections. By applying the Fourier transform to (4.3.29) and the isomorphism (2.1.49) to the left column and using

the isomorphisms (2.1.6), we obtain a commutative diagram

$$(4.3.33) \quad \begin{array}{ccc} \Lambda & \longrightarrow & c_*^\vee \Lambda \\ \downarrow & & \downarrow \\ \mu_{X/X \times X} \mathcal{H}_X & \longrightarrow & c_*^\vee \mu_{Z/Z \times X} \mathcal{H}_i \\ \downarrow & & \downarrow \\ e_X^{\vee*} K_X & \longleftarrow & c_*^\vee e_i^{\vee*} K_Z. \end{array}$$

By the canonical isomorphism $K_Z \rightarrow i^! K_X$, the bottom arrow induces an isomorphism $H_{SS(i_*\mathcal{F})}^0(T^*X, e_X^{\vee*} K_X) \leftarrow H_{SS(i_*\mathcal{F})}^0(T^*X \times_X Z, e_i^{\vee*} K_Z)$. By (4.3.33), the canonical isomorphism $H_{SS\mu(i_*\mathcal{F})}^0(T^*X \times_X Z, e_i^{\vee*} K_Z) \rightarrow H_{SS\mu(i_*\mathcal{F})}^0(T^*X, e_X^{\vee*} K_X)$ maps $CC_{\mu,i}\mathcal{F}$ to $CC_{\mu}i_*\mathcal{F}$.

Similarly by applying the Fourier transform to (4.3.30) and the isomorphism (2.1.24) to the left column and using the isomorphisms (2.1.6), we obtain a commutative diagram

$$(4.3.34) \quad \begin{array}{ccc} a^{\vee*} \Lambda & \xleftarrow{\simeq} & \Lambda \\ \downarrow & & \downarrow \\ a^{\vee*} \mu_{Z/Z \times X} \mathcal{H}_i & \longrightarrow & \mu_{Z/Z \times Y} \mathcal{H}_g \\ \downarrow & & \downarrow \\ a^{\vee*} e_i^{\vee*} K_Z & \xleftarrow{\simeq} & e_g^{\vee*} K_Z. \end{array}$$

Similarly by applying the Fourier transform to (4.3.31) and the isomorphism (2.1.49) to the left column and using the isomorphisms (2.1.6), we obtain a commutative diagram

$$(4.3.35) \quad \begin{array}{ccc} b_*^\vee \Lambda & \longleftarrow & \Lambda \\ \downarrow & & \downarrow \\ b_*^\vee \mu_{Z/Z \times Y} \mathcal{H}_g & \longrightarrow & \mu_{Y/Y \times Y} \mathcal{H}_Y \\ \downarrow & & \downarrow \\ b_*^\vee e_g^{\vee*} K_Z & \longrightarrow & e_Y^{\vee*} K_Y. \end{array}$$

Combining these three diagrams, we obtain the equality (4.3.9). $\square$

Corollary 4.3.3. *Let Z be a proper scheme over k and let $i: Z \rightarrow X$ be a closed immersion to a smooth scheme over k . Let $\mathcal{F} \in D_{\text{ctf}}(Z, \Lambda)$. Then, we have*

$$(4.3.36) \quad \chi(Z_{\bar{k}}, \mathcal{F}) = (CC_{\mu,i}(\mathcal{F}), T_X^* X \times_X Z)_{T^* X \times_X Z}$$

in Λ . The right hand side denotes the image of $CC_{\mu,i}(\mathcal{F})$ by the composition

$$(4.3.37) \quad H_{SS\mu(i_*\mathcal{F})}^0(T^*X \times_X Z, e_i^{\vee*} K_Z) \xrightarrow{0_Z^*} H^0(Z, K_Z) \xrightarrow{\text{Tr}_{Z/k}} \Lambda.$$

Proof. Let $f: X \rightarrow Y = \text{Spec } k$ be the canonical morphism and $g = fi$. Then, we have $\chi(Z_{\bar{k}}, \mathcal{F}) = CC_{\mu}(g_*\mathcal{F})$. Since $(CC_{\mu,i}(\mathcal{F}), T_X^* X \times_X Z)_{T^* X \times_X Z} = f_! CC_{\mu} i_* \mathcal{F}$, the equality (4.3.36) follows from Proposition 4.3.2.3. $\square$

4.4 Smooth pull-back

Lemma 4.4.1. *Let*

$$(4.4.1) \quad \begin{array}{ccc} C & \xleftarrow{g} & D \\ c \downarrow & & \downarrow d \\ X & \xleftarrow{h} & W \end{array}$$

be a commutative diagram of separated schemes of finite type over k .

1. *Define $h^!c_!c^! \leftarrow d_!d^!h^!$ to be the adjoint of the composition of the canonical isomorphism $h_!d_!d^!h^! \rightarrow c_!g_!g^!c^!$ and the morphism $c_!g_!g^!c^! \rightarrow c_!c^!$ induced by adj_g . Then the diagram*

$$(4.4.2) \quad \begin{array}{ccc} h^!c_!c^! & \xleftarrow{\quad} & d_!d^!h^! \\ \text{adj}_c \downarrow & & \downarrow \text{adj}_d \\ h^! & \xlongequal{\quad} & h^!. \end{array}$$

is commutative.

2. *Define $h^*c_*c^* \rightarrow d_*d^*h^*$ to be the adjoint of the composition of the canonical isomorphism $c_*g_*g^*c^* \rightarrow h_*d_*d^*h^*$ and the morphism $c_*c^* \rightarrow c_*g_*g^*c^*$ induced by adj_g . Then the diagram*

$$(4.4.3) \quad \begin{array}{ccc} h^* & \xlongequal{\quad} & h^* \\ \text{adj}_c \downarrow & & \downarrow \text{adj}_d \\ h^*c_*c^* & \longrightarrow & d_*d^*h^* \end{array}$$

is commutative.

Proof. Similar to Lemma 4.3.1. □

Let $h: W \rightarrow X$ be a smooth morphism of relative dimension $r = \dim W - \dim X$ of smooth schemes over k . Let (3.1.18) be the canonical morphisms of cotangent bundles and let $e_X^\vee: T^*X \rightarrow X$, $e_h^\vee: T^*X \times_X W \rightarrow W$ and $e_W^\vee: T^*W \rightarrow W$ be the projections. For a closed conical subset $S \subseteq T^*X$, define a morphism

$$(4.4.4) \quad h^!: H_S^0(T^*X, e_X^{\vee*}K_X) \rightarrow H_{h^*S}^0(T^*W, e_W^{\vee*}K_W)$$

to be $(-1)^r$ -times the composition of $b^*: H_S^0(T^*X, e_X^{\vee*}K_X) \rightarrow H_{b^{-1}S}^0(T^*X \times_X W, b^*e_X^{\vee*}K_X)$ and $a_*: H_{b^{-1}S}^0(T^*X \times_X W, b^*e_X^{\vee*}K_X) \rightarrow H_{a(b^{-1}S)}^0(T^*W, e_W^{\vee*}K_W)$. Note that the definition of $h^!$ involves the sign $(-1)^{\dim X - \dim W}$.

Lemma 4.4.2. *Let $h: W \rightarrow X$ be a smooth morphism of pure relative dimension $r = \dim W - \dim X$ of smooth schemes over k . Define a morphism*

$$(4.4.5) \quad h^!: H^0(X, K_X) \rightarrow H^0(W, K_W)$$

to be the composition of $h^: H^0(X, K_X) \rightarrow H^0(W, h^*K_X)$ and the cup-product with the top Chern class $c_r(TW/X) = (-1)^r c_r(T^*W/X) \in H^0(W, h^!\Lambda)$ of the relative tangent bundle.*

Then, the diagram

$$(4.4.6) \quad \begin{array}{ccc} H_S^0(T^*X, e_X^{\vee*}K_X) & \xrightarrow{h^!} & H_{h^0S}^0(T^*W, e_W^{\vee*}K_W) \\ \downarrow & & \downarrow \\ H^0(X, K_X) & \xrightarrow{h^!} & H^0(W, K_W) \end{array}$$

is commutative. The vertical arrows are defined by the pull-back by the 0-sections.

Proof. It suffices to show that the diagram

$$(4.4.7) \quad \begin{array}{ccccc} H_S^0(T^*X, e_X^{\vee*}K_X) & \xrightarrow{b^*} & H_{b^{-1}S}^0(T^*X \times_X W, e_h^{\vee*}h^*K_X) & \xrightarrow{(-1)^ra_*} & H_{h^0S}^0(T^*W, e_W^{\vee*}K_W) \\ \downarrow & & \downarrow & & \downarrow \\ H^0(X, K_X) & \xrightarrow{h^*} & H^0(W, h^*K_X) & \xrightarrow{c_r(TW/X)} & H^0(W, K_W) \end{array}$$

is commutative. The commutativity of the left square is clear from the functoriality. By the exact sequence $0 \rightarrow h^*\Omega_X^1 \rightarrow \Omega_W^1 \rightarrow \Omega_{W/X}^1 \rightarrow 0$, the morphism $a_*: H_{b^{-1}S}^0(T^*X \times_X W, e_h^{\vee*}K_X) \rightarrow H_{h^0S}^0(T^*W, e_W^{\vee*}K_W)$ is compatible with the cup-product

$$(4.4.8) \quad \cup_{c_r(T^*W/X)}: H^0(W, h^*K_X) \rightarrow H^0(W, K_W).$$

Hence the assertion follows. $\square$

Proposition 4.4.3. *Let $h: W \rightarrow X$ be a smooth morphism of smooth schemes over k and $\mathcal{F} \in D_{\text{ctf}}(X, \Lambda)$. Then, we have*

$$(4.4.9) \quad CC_\mu h^*(\mathcal{F}) = h^! CC_\mu(\mathcal{F})$$

in $H_{h^0(SS_\mu\mathcal{F})}^0(T^*W, e_W^{\vee*}K_W)$ and

$$(4.4.10) \quad C_{W/k}(h^*\mathcal{F}) = h^! C_{X/k}(\mathcal{F})$$

in $H^0(W, K_W)$.

Proof. We apply Lemma 4.4.1 to the commutative diagram

$$(4.4.11) \quad \begin{array}{ccccc} X & \xleftarrow{h} & W & \xleftarrow{1_W} & W \\ \delta_X \downarrow & & \gamma_h \downarrow & & \downarrow \delta_W \\ X \times X & \xleftarrow{h \times 1_X} & W \times X & \xleftarrow{1_W \times h} & W \times W \end{array}$$

as follows. Set

$$(4.4.12) \quad \begin{aligned} \mathcal{H}_X &= \mathcal{H}om_{X \times X}(\text{pr}_1^*\mathcal{F}, \text{pr}_2^!\mathcal{F}), & \mathcal{H}_h &= \mathcal{H}om_{W \times X}(\text{pr}_1^*h^!\mathcal{F}, \text{pr}_2^!\mathcal{F}), \\ \mathcal{H}_W &= \mathcal{H}om_{W \times W}(\text{pr}_1^*h^!\mathcal{F}, \text{pr}_2^!h^!\mathcal{F}) \end{aligned}$$

and identify them with

$$(4.4.13) \quad \boxtimes_X = D_X(\mathcal{F}) \boxtimes \mathcal{F}, \quad \boxtimes_h = D_W(h^!\mathcal{F}) \boxtimes \mathcal{F}, \quad \boxtimes_W = D_W(h^!\mathcal{F}) \boxtimes h^!\mathcal{F}$$

by the canonical isomorphisms [Gro77, (3.1.1)]. We have canonical isomorphisms $(h \times 1_X)^* \mathcal{H}_X \rightarrow \mathcal{H}_h$ and $(1_W \times h)^! \mathcal{H}_h \rightarrow \mathcal{H}_W$. We consider the commutative diagram

$$(4.4.14) \quad \begin{array}{ccc} (1_W \times h)^! \mathcal{H}_h & \longrightarrow & \mathcal{H}_W \\ \uparrow & & \uparrow \\ (1_W \times h)^* \boxtimes_h \otimes \text{pr}_2^* h^! \Lambda & \longrightarrow & \boxtimes_W \end{array}$$

where the upper horizontal arrow is an isomorphism.

By applying Lemma 4.4.1.1 and by [Gro77, (3.2.1)], we obtain commutative diagrams

$$(4.4.15) \quad \begin{array}{ccc} (h \times 1_X)^! \delta_{X!} \mathcal{H}om_X(\mathcal{F}, \mathcal{F}) & \longleftarrow & \gamma_{h!}(\mathcal{H}om_W(h^! \mathcal{F}, h^! \mathcal{F}) \otimes h^! \Lambda) \\ \downarrow & & \downarrow \\ (h \times 1_X)^! \mathcal{H}_X & \longleftarrow & \mathcal{H}_h \otimes \text{pr}_1^* h^! \Lambda, \end{array}$$

$$(4.4.16) \quad \begin{array}{ccc} (1_W \times h)^! \gamma_{h!} \mathcal{H}om_W(h^! \mathcal{F}, h^! \mathcal{F}) & \longleftarrow & \delta_{W!} \mathcal{H}om_W(h^! \mathcal{F}, h^! \mathcal{F}) \\ \downarrow & & \downarrow \\ (1_W \times h)^! \mathcal{H}_h & \xrightarrow{\simeq} & \mathcal{H}_W. \end{array}$$

Since h is smooth, the canonical morphism $(h \times 1_X)^*(-) \otimes \text{pr}_1^* h^! \Lambda \rightarrow (h \times 1_X)^!$ of functors is an isomorphism and (4.4.15) defines a commutative diagram

$$(4.4.17) \quad \begin{array}{ccc} (h \times 1_X)^* \delta_{X!} \mathcal{H}om_X(\mathcal{F}, \mathcal{F}) & \longleftarrow & \gamma_{h!} \mathcal{H}om_W(h^! \mathcal{F}, h^! \mathcal{F}) \\ \downarrow & & \downarrow \\ (h \times 1_X)^* \mathcal{H}_X & \longleftarrow & \mathcal{H}_h \end{array}$$

Since h is smooth, the canonical morphism $(1_W \times h)^*(-) \otimes \text{pr}_2^* h^! \Lambda \rightarrow (1_W \times h)^!$ is an isomorphism and (4.4.16) defines a commutative diagram

$$(4.4.18) \quad \begin{array}{ccc} (1_W \times h)^* \gamma_{h!} \mathcal{H}om_W(h^! \mathcal{F}, h^! \mathcal{F}) \otimes \text{pr}_2^* h^! \Lambda & \longleftarrow & \delta_{W!} \mathcal{H}om_W(h^! \mathcal{F}, h^! \mathcal{F}) \\ \downarrow & & \downarrow \\ (1_W \times h)^* \mathcal{H}_h \otimes \text{pr}_2^* h^! \Lambda & \xrightarrow{\simeq} & \mathcal{H}_W. \end{array}$$

Defining vertical arrows by $1_{\mathcal{F}}$ and $1_{h^! \mathcal{F}}$, we obtain commutative diagrams

$$(4.4.19) \quad \begin{array}{ccc} (h \times 1_X)^* \delta_{X!} \Lambda & \xrightarrow{\simeq} & \gamma_{h!} \Lambda \\ 1_{\mathcal{F}} \downarrow & & \downarrow 1_{h^! \mathcal{F}} \\ (h \times 1_X)^* \delta_{X!} \mathcal{H}om_X(\mathcal{F}, \mathcal{F}) & \xrightarrow{\simeq} & \gamma_{h!} \mathcal{H}om_W(h^! \mathcal{F}, h^! \mathcal{F}), \end{array}$$

$$(4.4.20) \quad \begin{array}{ccc} (1_W \times h)^! \gamma_{h!} \Lambda & \xleftarrow{\text{adj}} & \delta_{W!} \Lambda \\ 1_{h^! \mathcal{F}} \downarrow & & \downarrow 1_{h^! \mathcal{F}} \\ (1_W \times h)^! \gamma_{h!} \mathcal{H}om_W(h^! \mathcal{F}, h^! \mathcal{F}) & \longleftarrow & \delta_{W!} \mathcal{H}om_W(h^! \mathcal{F}, h^! \mathcal{F}). \end{array}$$

The upper horizontal arrow in (4.4.20) is the adjoint of the isomorphism $\gamma_{h!} \leftarrow (1_W \times h)_! \delta_{W!}$.

By applying Lemma 4.4.1.2, we obtain commutative diagrams

$$(4.4.21) \quad \begin{array}{ccc} (h \times 1_X)^* \boxtimes_X & \xrightarrow{\cong} & \boxtimes_h \\ \downarrow & & \downarrow \\ (h \times 1_X)^* \delta_{X*} (D_X \mathcal{F} \otimes \mathcal{F}) & \xrightarrow{\cong} & \gamma_{h*} (D_W h^! \mathcal{F} \otimes h^* \mathcal{F}), \end{array}$$

$$(4.4.22) \quad \begin{array}{ccc} (1_W \times h)^* \boxtimes_h \otimes \text{pr}_2^* h^! \Lambda & \xrightarrow{\cong} & \boxtimes_W \\ \downarrow & & \downarrow \\ (1_W \times h)^* \gamma_{h*} (D_W h^! \mathcal{F} \otimes h^* \mathcal{F} \otimes h^! \Lambda) & \xrightarrow{\text{adj}} & \delta_{W*} (D_W h^! \mathcal{F} \otimes h^! \mathcal{F}). \end{array}$$

The lower line is induced by the canonical morphism $h^* \otimes h^! \Lambda \rightarrow h^!$ and the adjoint of the isomorphism $\gamma_{h*} \rightarrow (1_W \times h)_* \delta_{W*}$.

Defining vertical arrows by $\text{ev}_X: D_X \mathcal{F} \otimes \mathcal{F} \rightarrow K_X$, $\text{ev}_X \circ (1 \otimes \text{adj}): D_X \mathcal{F} \otimes h^* h_* \mathcal{F} \rightarrow D_X \mathcal{F} \otimes \mathcal{F} \rightarrow K_X$ and $\text{ev}_W: D_W h^* \mathcal{F} \otimes h^* \mathcal{F} \rightarrow K_W$, we obtain commutative diagrams

$$(4.4.23) \quad \begin{array}{ccc} (h \times 1_X)^* \delta_{X*} (D_X \mathcal{F} \otimes \mathcal{F}) & \xleftarrow{\cong} & \gamma_{h*} (D_W h^! \mathcal{F} \otimes h^* \mathcal{F}) \\ \text{ev} \downarrow & & \downarrow \text{ev} \\ (h \times 1_X)^* \delta_{X*} K_X & \xrightarrow{\cong} & \gamma_{h*} h^* K_X, \end{array}$$

$$(4.4.24) \quad \begin{array}{ccc} (1_W \times h)^* \gamma_{h*} (D_W h^! \mathcal{F} \otimes h^* \mathcal{F} \otimes h^! \Lambda) & \xrightarrow{\text{adj}} & \delta_{W*} (D_W h^! \mathcal{F} \otimes h^! \mathcal{F}) \\ \text{ev} \downarrow & & \downarrow \text{ev} \\ (1_W \times h)^* \gamma_{h*} (h^* K_X \otimes h^! \Lambda) & \xrightarrow{\text{adj}} & \delta_{W*} K_W. \end{array}$$

The horizontal lines in (4.4.24) are induced by the morphism $h^* \otimes h^! \Lambda \rightarrow h^!$. The lower line is further induced by the adjoint of the isomorphism $\gamma_{h*} \rightarrow (1_W \times h)_* \delta_{W*}$ and $h^* K_X \otimes h^! \Lambda \rightarrow K_W$. Commutative diagrams (4.4.17), (4.4.19), (4.4.21) and (4.4.23) define a commutative diagram

$$(4.4.25) \quad \begin{array}{ccc} (h \times 1_X)^* \delta_{X!} \Lambda & \xrightarrow{\cong} & \gamma_{h!} \Lambda \\ 1_{\mathcal{F}} \downarrow & & \downarrow 1_{h^! \mathcal{F}} \\ (h \times 1_X)^* \mathcal{H}_X & \xrightarrow{\cong} & \mathcal{H}_h \\ \text{ev} \downarrow & & \downarrow \text{ev} \\ (h \times 1_X)^* \delta_{X*} K_X & \xrightarrow{\cong} & \gamma_{h*} h^* K_X. \end{array}$$

Similarly, commutative diagrams (4.4.16), (4.4.20), (4.4.22) and (4.4.24) define a commu-

tative diagram

$$\begin{array}{ccc}
(1_W \times h)^* \gamma_{h!} h^! \Lambda & \xleftarrow{\text{adj}} & \delta_{W!} \Lambda \\
1_{h^! \mathcal{F}} \downarrow & & \downarrow 1_{h^! \mathcal{F}} \\
(1_W \times h)^* \mathcal{H}_h \otimes \text{pr}_2^* h^! \Lambda & \xrightarrow{\simeq} & \mathcal{H}_W \\
\text{ev} \downarrow & & \downarrow \text{ev} \\
(1_W \times h)^* \gamma_{h*} (h^* K_X \otimes h^! \Lambda) & \longrightarrow & \delta_{W*} K_W.
\end{array}
\tag{4.4.26}$$

The commutative diagram (4.4.11) induces a commutative diagram

$$\begin{array}{ccccc}
X & \xleftarrow{h} & W & \xleftarrow{1_W} & W \\
0_X \downarrow & & \downarrow 0_h & & \downarrow 0_W \\
TX & \xleftarrow{b} & TX \times_X W & \xleftarrow{a} & TW.
\end{array}
\tag{4.4.27}$$

The vertical arrows are the 0-sections. Applying the functor $\nu_{W/W \times X}$ to (4.4.25) and the base change morphism $b^* \nu_{X/X \times X} \rightarrow \nu_{W/W \times X} (h \times 1_X)^*$ (2.3.35) to the left column, we obtain a commutative diagram

$$\begin{array}{ccc}
b^* 0_{X!} \Lambda & \xrightarrow{\simeq} & 0_{h!} \Lambda \\
\downarrow & & \downarrow \\
b^* \nu_{X/X \times X} \mathcal{H}_X & \longrightarrow & \nu_{W/W \times X} \mathcal{H}_h \\
\downarrow & & \downarrow \\
b^* 0_{X*} K_X & \xrightarrow{\simeq} & 0_{h*} h^* K_X.
\end{array}
\tag{4.4.28}$$

Applying the functor $\nu_{W/W \times W}$ to (4.4.26) and the base change morphism $a^* \nu_{W/W \times W} \rightarrow \nu_{W/W \times X} (1_W \times h)^*$ (2.3.38) to the left column, we obtain a commutative diagram

$$\begin{array}{ccc}
a^! 0_{h!} \Lambda & \xleftarrow{\text{adj}_a} & 0_{W!} \Lambda \\
\downarrow & & \downarrow \\
a^! \nu_{W/W \times X} \mathcal{H}_h & \longleftarrow & \nu_{W/W \times W} \mathcal{H}_W \\
\downarrow & & \downarrow \\
a^! 0_{h*} h^* K_X & \xrightarrow{\text{res}_a} & 0_{W*} K_W.
\end{array}
\tag{4.4.29}$$

Let $b^\vee: T^*X \times_X W \rightarrow T^*X$ be the canonical morphism and $a^\vee: T^*X \times_X W \rightarrow T^*W$ be the dual of a . Let $e_X^\vee: T^*X \rightarrow X$, $e_h^\vee: T^*X \times_X W \rightarrow W$ and $e_W^\vee: T^*W \rightarrow W$ be the projections. By applying the Fourier transforms to (4.4.28) and the isomorphism (2.1.50) to the left column and using the isomorphisms (2.1.6), we obtain a commutative diagram

$$\begin{array}{ccc}
b^{\vee*} \Lambda & \xrightarrow{\simeq} & \Lambda \\
\downarrow & & \downarrow \\
b^{\vee*} \mu_{X/X \times X} \mathcal{H}_X & \longrightarrow & \mu_{W/W \times X} \mathcal{H}_h \\
\downarrow & & \downarrow \\
b^{\vee*} e_X^{\vee*} K_X & \xrightarrow{\simeq} & e_h^{\vee*} h^* K_X.
\end{array}
\tag{4.4.30}$$

By applying the Fourier transforms to (4.4.29) and the isomorphism (2.1.27) to the left column and using the isomorphism (2.1.6), we obtain a commutative diagram

$$(4.4.31) \quad \begin{array}{ccc} a_!^\vee \Lambda & \longleftarrow & \Lambda \\ \downarrow & & \downarrow \\ a_!^\vee \mu_{W/W \times X} \mathcal{H}_h & \longleftarrow & \mu_{W/W \times W} \mathcal{H}_W \\ \downarrow & & \downarrow \\ a_!^\vee e_h^{\vee*} h^* K_X & \longrightarrow & e_W^{\vee*} K_W. \end{array}$$

By Proposition 2.1.12.2 and 1., the top and the bottom horizontal arrows in (4.4.31) are $\text{res}_{a^\vee}$ and $(-1)^r \text{adj}_{a^\vee}$ respectively. Hence (4.4.30) and (4.4.31) imply (4.4.9). $\square$

4.5 External products

Let X and Y be smooth k -schemes. Let $\mathcal{F} \in D_{\text{ctf}}(X, \Lambda)$ and $\mathcal{G} \in D_{\text{ctf}}(Y, \Lambda)$. We identify

$$(4.5.1) \quad T^*(X \times Y) = T^*X \times T^*Y$$

by the canonical decomposition $\Omega_{X \times Y/k}^1 = \text{pr}_X^* \Omega_{X/k}^1 \oplus \text{pr}_Y^* \Omega_{Y/k}^1$.

For cohomology classes with support, $\alpha \boxtimes \beta$ denotes the cohomological external product followed by the canonical isomorphism (cf. [YZ25, (3.2)])

$$(4.5.2) \quad K_{X/k} \boxtimes K_{Y/k} \xrightarrow{\sim} K_{X \times Y/k}.$$

Lemma 4.5.1. *Let $\mathcal{H}_{\mathcal{F}} = \mathcal{H}om(p_{1,X}^* \mathcal{F}, p_{2,X}^! \mathcal{F})$ and $\mathcal{H}_{\mathcal{G}} = \mathcal{H}om(p_{1,Y}^* \mathcal{G}, p_{2,Y}^! \mathcal{G})$. There are canonical isomorphisms*

$$(4.5.3) \quad \mathcal{H}_{\mathcal{F} \boxtimes \mathcal{G}} \simeq \mathcal{H}_{\mathcal{F}} \boxtimes \mathcal{H}_{\mathcal{G}},$$

$$(4.5.4) \quad \nu_{\delta_{X \times Y}}(\mathcal{H}_{\mathcal{F} \boxtimes \mathcal{G}}) \simeq \nu_{\delta_X}(\mathcal{H}_{\mathcal{F}}) \boxtimes_{\eta_0} \nu_{\delta_Y}(\mathcal{H}_{\mathcal{G}}).$$

Proof. This follows from Lemma 3.3.1. $\square$

Theorem 4.5.2 (Microlocal Künneth formula). *After the identification (4.5.1), there is a canonical isomorphism*

$$(4.5.5) \quad \mu \mathcal{H}om(\mathcal{F} \boxtimes \mathcal{G}, \mathcal{F} \boxtimes \mathcal{G}) \xrightarrow{\sim} \mu \mathcal{H}om(\mathcal{F}, \mathcal{F}) \boxtimes \mu \mathcal{H}om(\mathcal{G}, \mathcal{G}).$$

Consequently, we have

$$(4.5.6) \quad SS_\mu(\mathcal{F} \boxtimes \mathcal{G}) = SS_\mu(\mathcal{F}) \times SS_\mu(\mathcal{G}),$$

$$(4.5.7) \quad CC_\mu(\mathcal{F} \boxtimes \mathcal{G}) = CC_\mu(\mathcal{F}) \boxtimes CC_\mu(\mathcal{G}).$$

Proof. (4.5.5) and (4.5.6) have been proved in Lemma 3.3.1. The equality (4.5.7) follows from Theorem 3.3.6, Theorem 4.6.2 and [Sai17b, Theorem 2.2]. Here, we provide an independent proof of (4.5.7).

Consider the kernel $\mathcal{H}_{\mathcal{F}}$. The cohomological characteristic class of $\mathcal{F}$ is represented by the composition

$$(4.5.8) \quad \Lambda_X \xrightarrow{1_{\mathcal{F}}} \delta_X^! \mathcal{H}_{\mathcal{F}} \longrightarrow \delta_X^* \mathcal{H}_{\mathcal{F}} \xrightarrow{\text{ev}_{\mathcal{F}}} K_{X/k}.$$

We have the analogous composites for $\mathcal{G}$ and $\mathcal{F} \boxtimes \mathcal{G}$. The compatibility of (4.5.3) with the identity morphisms gives a commutative diagram

$$(4.5.9) \quad \begin{array}{ccc} \Lambda_X \boxtimes \Lambda_Y & \xrightarrow{1_{\mathcal{F}} \boxtimes 1_{\mathcal{G}}} & \delta_X^! \mathcal{H}_{\mathcal{F}} \boxtimes \delta_Y^! \mathcal{H}_{\mathcal{G}} \\ \simeq \downarrow & & \downarrow \simeq \\ \Lambda_{X \times Y} & \xrightarrow{1_{\mathcal{F} \boxtimes \mathcal{G}}} & \delta_{X \times Y}^! \mathcal{H}_{\mathcal{F} \boxtimes \mathcal{G}} \end{array}$$

The canonical morphisms $\delta^! \rightarrow \delta^*$ give a commutative diagram

$$(4.5.10) \quad \begin{array}{ccc} \delta_X^! \mathcal{H}_{\mathcal{F}} \boxtimes \delta_Y^! \mathcal{H}_{\mathcal{G}} & \longrightarrow & \delta_X^* \mathcal{H}_{\mathcal{F}} \boxtimes \delta_Y^* \mathcal{H}_{\mathcal{G}} \\ \simeq \downarrow & & \downarrow \simeq \\ \delta_{X \times Y}^! \mathcal{H}_{\mathcal{F} \boxtimes \mathcal{G}} & \longrightarrow & \delta_{X \times Y}^* \mathcal{H}_{\mathcal{F} \boxtimes \mathcal{G}}. \end{array}$$

Its commutativity follows from the naturality of $\delta^! \rightarrow \delta^*$ and its compatibility with external products. The evaluation morphisms give a commutative diagram

$$(4.5.11) \quad \begin{array}{ccc} \delta_X^* \mathcal{H}_{\mathcal{F}} \boxtimes \delta_Y^* \mathcal{H}_{\mathcal{G}} & \xrightarrow{\text{ev}_{\mathcal{F}} \boxtimes \text{ev}_{\mathcal{G}}} & K_{X/k} \boxtimes K_{Y/k} \\ \simeq \downarrow & & \downarrow \simeq \\ \delta_{X \times Y}^* \mathcal{H}_{\mathcal{F} \boxtimes \mathcal{G}} & \xrightarrow{\text{ev}_{\mathcal{F} \boxtimes \mathcal{G}}} & K_{X \times Y/k}. \end{array}$$

The vertical morphisms are the external-product isomorphisms and (4.5.2). Diagram (4.5.9) follows from $\text{id}_{\mathcal{F} \boxtimes \mathcal{G}} = \text{id}_{\mathcal{F}} \boxtimes \text{id}_{\mathcal{G}}$. To verify (4.5.11), use $\mathcal{H}_{\mathcal{F}} \simeq D_X \mathcal{F} \boxtimes \mathcal{F}$ and $\mathcal{H}_{\mathcal{G}} \simeq D_Y \mathcal{G} \boxtimes \mathcal{G}$. Under $D_{X \times Y}(\mathcal{F} \boxtimes \mathcal{G}) = D_X \mathcal{F} \boxtimes D_Y \mathcal{G}$, the evaluation $\text{ev}_{\mathcal{F} \boxtimes \mathcal{G}}$ is the composite of the symmetry isomorphism

$$(4.5.12) \quad (D_X \mathcal{F} \boxtimes D_Y \mathcal{G}) \otimes (\mathcal{F} \boxtimes \mathcal{G}) \xrightarrow{\sim} (D_X \mathcal{F} \otimes \mathcal{F}) \boxtimes (D_Y \mathcal{G} \otimes \mathcal{G})$$

and $\text{ev}_{\mathcal{F}} \boxtimes \text{ev}_{\mathcal{G}}$, followed by (4.5.2). Hence (4.5.11) is commutative.

By adjunction, (4.5.8) defines a correspondence

$$(4.5.13) \quad \delta_{X!} \Lambda_X \longrightarrow \mathcal{H}_{\mathcal{F}} \longrightarrow \delta_{X*} K_{X/k}.$$

Let $0_X : X \hookrightarrow TX$ be the zero section. Applying Verdier specialization to (4.5.13), we obtain

$$(4.5.14) \quad 0_{X!} \Lambda_X \longrightarrow \nu_{\delta_X}(\mathcal{H}_{\mathcal{F}}) \longrightarrow 0_{X*} K_{X/k}.$$

Here we use the canonical isomorphisms $\nu_{\delta_X}(\delta_{X!} \Lambda_X) = 0_{X!} \Lambda_X$ and $\nu_{\delta_X}(\delta_{X*} K_{X/k}) = 0_{X*} K_{X/k}$. Applying the Fourier transformation and the canonical isomorphisms

$$(4.5.15) \quad F_{\psi, TX/X}(0_{X!} \Lambda_X) \simeq \Lambda_{T^*X}, \quad F_{\psi, TX/X}(0_{X*} K_{X/k}) \simeq e_X^{\vee*} K_{X/k}$$

we obtain morphisms

$$(4.5.16) \quad \Lambda_{T^*X} \xrightarrow{u_{\mathcal{F}}} \mu \mathcal{H}om(\mathcal{F}, \mathcal{F}) \xrightarrow{v_{\mathcal{F}}} e_X^{\vee*} K_{X/k}.$$

These are the two structural morphisms defining $CC_\mu(\mathcal{F})$.

By [LZ22, Proposition 3.1] and Proposition 2.1.16, the three commutative diagrams (4.5.9), (4.5.10), and (4.5.11) can be specialized simultaneously. We obtain a commutative diagram

$$(4.5.17) \quad \begin{array}{ccccc} \Lambda_{T^*X} \boxtimes \Lambda_{T^*Y} & \xrightarrow{u_{\mathcal{F}} \boxtimes u_{\mathcal{G}}} & \mu\mathcal{H}om(\mathcal{F}, \mathcal{F}) \boxtimes \mu\mathcal{H}om(\mathcal{G}, \mathcal{G}) & \xrightarrow{v_{\mathcal{F}} \boxtimes v_{\mathcal{G}}} & e_X^{\vee*} K_{X/k} \boxtimes e_Y^{\vee*} K_{Y/k} \\ \simeq \downarrow & & \simeq \downarrow & & \downarrow \simeq \\ \Lambda_{T^*(X \times Y)} & \xrightarrow{u_{\mathcal{F} \boxtimes \mathcal{G}}} & \mu\mathcal{H}om(\mathcal{F} \boxtimes \mathcal{G}, \mathcal{F} \boxtimes \mathcal{G}) & \xrightarrow{v_{\mathcal{F} \boxtimes \mathcal{G}}} & e_{X \times Y}^{\vee*} K_{X \times Y/k}. \end{array}$$

The left vertical morphism is induced by the external product of constant sheaves, and the right one is induced by (4.5.2). Thus (4.5.17) is compatible with both structural morphisms (4.5.16) defining CC_μ .

We now pass to cohomology with support. Let $i_{\mathcal{F}} : SS_\mu(\mathcal{F}) \hookrightarrow T^*X$ and $i_{\mathcal{G}} : SS_\mu(\mathcal{G}) \hookrightarrow T^*Y$ be the closed immersions. By (4.5.6), we have $SS_\mu(\mathcal{F} \boxtimes \mathcal{G}) = SS_\mu(\mathcal{F}) \times SS_\mu(\mathcal{G})$. By definition, $CC_\mu(\mathcal{F})$ is represented on $SS_\mu(\mathcal{F})$ by the composite

$$(4.5.18) \quad \Lambda \longrightarrow i_{\mathcal{F}}^* \mu\mathcal{H}om(\mathcal{F}, \mathcal{F}) \longrightarrow i_{\mathcal{F}}^! e_X^{\vee*} K_{X/k},$$

obtained from (4.5.16) by adjunction (cf. (4.1.10)). By (4.5.17), the external product of (4.5.18) and its analogue for $\mathcal{G}$ is the corresponding composite on $SS_\mu(\mathcal{F}) \times SS_\mu(\mathcal{G})$. Here we use the canonical isomorphism

$$(4.5.19) \quad (i_{\mathcal{F}}^! e_X^{\vee*} K_{X/k}) \boxtimes (i_{\mathcal{G}}^! e_Y^{\vee*} K_{Y/k}) \xrightarrow{\sim} (i_{\mathcal{F}} \times i_{\mathcal{G}})^! e_{X \times Y}^{\vee*} K_{X \times Y/k}.$$

Taking H^0 , the resulting class is $CC_\mu(\mathcal{F}) \boxtimes CC_\mu(\mathcal{G})$. By the lower row of (4.5.17), it is also $CC_\mu(\mathcal{F} \boxtimes \mathcal{G})$. This proves (4.5.7). $\square$

4.6 Characteristic cycles

In this subsection, we prove that $CC_\mu = \text{cl}(CC)$. We first prove the curve case.

Proposition 4.6.1. *Let X be a smooth curve over a perfect field k and $\mathcal{F} \in D_{\text{ctf}}(X, \Lambda)$. Then $CC_\mu(\mathcal{F})$ is the cycle class of $CC(\mathcal{F})$.*

Proof. Let $j : U \rightarrow X$ be a dense open immersion such that $j^*\mathcal{F}$ is locally constant. Since the assertion is local on X and since we may replace k by an algebraic closure, we may assume that $U = X - \{x\}$ is the complement of a k -rational point $x \in X$. It suffices to show that the coefficient a_x of the fiber T_x^*X in $CC_\mu(\mathcal{F})$ equals $-a_x(\mathcal{F})$.

Let $i : x \rightarrow X$ be the closed immersion. Then, by Corollary 4.1.7, we have $CC_\mu(\mathcal{F}) = CC_\mu(j_! j^* \mathcal{F}) + CC_\mu(i_* i^* \mathcal{F})$. Since $CC_\mu(i_* i^* \mathcal{F}) = \text{rank } i^* \mathcal{F} \cdot [T_x^* X]$ by Lemma 4.1.4 and Proposition 4.3.2.2, we may assume that $j_! j^* \mathcal{F} \rightarrow \mathcal{F}$ is an isomorphism.

Since $j^* \mathcal{F} \in D_{\text{ctf}}(U, \Lambda)$, by [SGA 4₂¹-Rapp, Lemme 4.5.1], there exist a finite complex $\mathcal{L}_\bullet$ of locally constant sheaves of free Λ -modules of finite rank and a quasi-isomorphism $\mathcal{L}_\bullet \rightarrow j^* \mathcal{F}$. Hence by Proposition 4.1.6, we may further assume that $j^* \mathcal{F}$ is a locally constant sheaf of free Λ -modules of finite rank.

In the case where $X = \mathbf{A}^1$, $x = 0$ and $\mathcal{F}$ is a locally constant constructible sheaf of free Λ -modules of rank n on $U = \mathbf{G}_m$ tamely ramified at x , we have $CC_\mu(\mathcal{F}) = -n([T_x^* X] + [T_x^* X])$ by Corollary 4.2.5 and hence $CC_\mu(\mathcal{F}) = \text{cl } CC(\mathcal{F})$.

We show the general case by reducing to this case. Since the assertion is étale local on X , by [Kat86, Theorem 4.1], we may assume that $X = \mathbf{P}^1 - \{0\}$, $x = \infty$ and $U = \mathbf{G}_m$ and that $j^*\mathcal{F}$ is a locally constant constructible sheaf of free Λ -modules of rank n tamely ramified at 0. Let $\bar{j}: X = \mathbf{P}^1 - \{0\} \rightarrow \mathbf{P}^1$ be the open immersion. We have $SS_\mu(\bar{j}_!\mathcal{F}) \subseteq T_{\mathbf{P}^1}^*\mathbf{P}^1 \cup T_0^*\mathbf{P}^1 \cup T_\infty^*\mathbf{P}^1$. By the tamely ramified case proved above and by the semi-purity [SGA 4 $\frac{1}{2}$ -Cycle, Rappel 2.2.8], we have $CC_\mu(\bar{j}_!\mathcal{F}) = -n([T_{\mathbf{P}^1}^*\mathbf{P}^1] + [T_0^*\mathbf{P}^1]) - a_\infty[T_\infty^*\mathbf{P}^1]$ for some $a_\infty \in \Lambda$. We have $\chi(\mathbf{P}_{\bar{k}}^1, \bar{j}_!\mathcal{F}) = \chi_c(X_{\bar{k}}, \mathcal{F}) = n - a_\infty$ by Corollary 4.3.3. Since $\chi_c(X_{\bar{k}}, \mathcal{F}) = n - a_\infty$ by the Grothendieck–Ogg–Shafarevich formula, we obtain $a_\infty = a_\infty \mathcal{F}$ and $CC_\mu(\mathcal{F}) = \text{cl } CC(\mathcal{F})$. $\square$

Theorem 4.6.2. *Let X be a smooth connected scheme over a perfect field k and $\mathcal{F} \in D_{\text{ctf}}(X, \Lambda)$. We have*

$$(4.6.1) \quad CC_\mu(\mathcal{F}) = \text{cl}(CC(\mathcal{F})),$$

where $\text{cl}: Z_{\dim(X)}(SS(\mathcal{F})) = \bigoplus_{i \in I} \mathbf{Z} \cdot C_i \rightarrow H_{SS(\mathcal{F})}^0(T^*X, e_X^{\vee*}K_{X/k})$ is the cycle class map and $C_i (i \in I)$ are irreducible components of $SS(\mathcal{F})$.

Proof. The proof is similar to that of the functorial characterization of characteristic cycles [Sai18, Proposition 8]. Let $C = SS_\mu(\mathcal{F}) = SS(\mathcal{F}) \subseteq T^*X$. By [Bei16, 1.3 Theorem (ii)], we have $\dim C \leq \dim X$. Hence by the semi-purity [SGA 4 $\frac{1}{2}$ -Cycle, Rappel 2.2.8], we have $H_C^0(T^*X, e_X^{\vee*}K_X) = \bigoplus_a \Lambda[C_a]$ where C_a runs through irreducible components of C of dimension $\dim X$ and hence $CC_\mu(\mathcal{F}) = \sum_a \mu_a[C_a]$ for some $\mu_a \in \Lambda$. Let $CC(\mathcal{F}) = \sum_a m_a[C_a]$. It suffices to show the equality $\mu_a = m_a$ in Λ for each irreducible component C_a .

Since the question is local on X , we may assume that X is affine. By applying Proposition 4.3.2 to an immersion $X \rightarrow \mathbf{A}^n$, we may assume $X = \mathbf{A}^n$. Further, since the question is local on X , we may assume that X is projective. We fix a closed immersion $X \rightarrow \mathbf{P}^N$.

Let C_a be an irreducible component of C of dimension $\dim X$. Then, there exists a linear subvariety $A \subseteq \mathbf{P}^N$ of codimension 2 and $x \in X - X \cap A$ satisfying the following conditions. The intersection $X \cap A$ is transversal. Let $X' \rightarrow X$ be the blow-up at $X \cap A$ and let $f: X' \rightarrow Y = \mathbf{P}^1$ be the morphism to the projective line parametrizing hyperplanes containing A . The fibers of f are the intersections with hyperplanes. The point x is an isolated characteristic point of $f: X' \rightarrow Y$ and x is the unique characteristic point in the fiber $f^{-1}(y)$ for $y = f(x)$. The component $C_a \subseteq T^*X$ is the unique irreducible component of C meeting the section df at x . The intersection number $(C_a, df)_x$ equals 1, if necessary replacing X by $X \times \mathbf{P}^1$ in the case $p = 2$ [Sai17a, Proposition 5.19].

Let m_a be the coefficient of C_a in $CC(\mathcal{F})$ and $\mu_a \in \Lambda$ be the coefficient of $\text{cl } C_a$ in $CC_\mu(\mathcal{F})$. Then, we have $(CC(\mathcal{F}), df)_x = m_a(C_a, df)_x \in \mathbf{Z}$ and $(CC_\mu(\mathcal{F}), df)_x = \mu_a(C_a, df)_x \in \Lambda$. By the Milnor formula, for the complex of vanishing cycles $\Phi_x(\mathcal{F}, f)$, we have $\dim \text{tot } \Phi_x(\mathcal{F}, f) = -(CC(\mathcal{F}), df)_x = -m_a(C_a, df)_x$. Since $f_!CC_\mu(\mathcal{F}) = CC_\mu(f_*\mathcal{F})$ by Proposition 4.3.2, the intersection product $(CC_\mu(\mathcal{F}), df)_x = \mu_a(C_a, df)_x$ equals the coefficient $-a_y f_*\mathcal{F}$ of $\text{cl } T_y^*Y$ in $f_!CC_\mu(\mathcal{F}) = CC_\mu(f_*\mathcal{F})$. By the distinguished triangle $\Gamma(X_{\bar{y}}, \mathcal{F}) \rightarrow \Gamma(X_{\bar{\eta}}, \mathcal{F}) \rightarrow \Phi_x(\mathcal{F}, f) \rightarrow$ for a geometric generic point $\bar{\eta}$ of the strict localization at $\bar{y}$, we have $\dim \text{tot } \Phi_x(\mathcal{F}, f) = a_y f_*\mathcal{F}$. Thus we have $\dim \text{tot } \Phi_x(\mathcal{F}, f) = -m_a(C_a, df)_x = -\mu_a(C_a, df)_x$. Since $(C_a, df)_x = 1$, we obtain $m_a = \mu_a$ in Λ and hence $CC_\mu(\mathcal{F}) = \text{cl } CC(\mathcal{F})$. $\square$

Corollary 4.6.3. *Let $f: X \rightarrow Y$ be a morphism of smooth schemes over a perfect field k and $i: Z \hookrightarrow X$ be a closed immersion such that $g = fi: Z \rightarrow Y$ is proper. Let $\mathcal{F} \in D_{\text{ctf}}(Z, \Lambda)$. We have*

$$(4.6.2) \quad \text{cl}CC(g_*\mathcal{F}) = f_!\text{cl}CC(i_*\mathcal{F}) \quad \text{in} \quad H_{f_*SS(i_*\mathcal{F})}^0(T^*Y, e_Y^{\vee*}K_Y).$$

In particular, if $Y = \text{Spec } k$, then we have

$$(4.6.3) \quad \chi(Z_k, \mathcal{F}) = (CC(i_*\mathcal{F}), T_X^*X \times_X Z)_{T^*X \times_X Z} \quad \text{in} \quad \Lambda.$$

Proof. This follows from Proposition 4.3.2.3 and Theorem 4.6.2. $\square$

Remark 4.6.4. In [Sai18, Conjecture 1], the first named author conjectures that we have equality $CC(g_*\mathcal{F}) = f_!CC(i_*\mathcal{F})$ in $\text{CH}_{\dim Y}(f_*SS(i_*\mathcal{F}))$. If $\dim f_*SS(i_*\mathcal{F}) \leq \dim Y$, then $H_{f_*SS(i_*\mathcal{F})}^0(T^*Y, e_Y^{\vee*}K_Y) \simeq \text{CH}_{\dim Y}(f_*SS(i_*\mathcal{F})) \otimes \Lambda \simeq Z_{\dim Y}(f_*SS(i_*\mathcal{F})) \otimes \Lambda$. When f is quasi-projective and Y is proper, (4.6.2) is proved in [Sai21, Theorem 2.2.5]. For (4.6.3), we show in Corollary 4.9.5 that the equality holds in $\mathbf{Z}$.

Let $f: X \rightarrow Y$ be a morphism from a smooth scheme X to a smooth connected curve Y over a perfect field k , and let $i: Z \hookrightarrow X$ be a closed immersion such that $g = fi: Z \rightarrow Y$ is proper. Let $\mathcal{F} \in D_{\text{ctf}}(Z, \Lambda)$. Assume that f is properly $SS(i_*\mathcal{F})$ -transversal over a dense open subset $V \subseteq Y$. Let $y \in Y \setminus V$ be a closed point. We define an intersection number $(CC(i_*\mathcal{F}), df)_{T^*X, Z_y}$ as follows.

Let t be a local parameter at y . After shrinking Y around y , we may assume that $Y \setminus \{y\} \subseteq V$ and that dt is nowhere zero. Thus dt defines a section $dt: Y \rightarrow T^*Y$. Let $\theta_{f,t}: X \rightarrow T^*X$ be the composition

$$(4.6.4) \quad X \xrightarrow{dt} X \times_Y T^*Y \xrightarrow{df} T^*X.$$

Since f is properly $SS(i_*\mathcal{F})$ -transversal over $Y \setminus \{y\}$, we have $\theta_{f,t}^{-1}SS(i_*\mathcal{F}) \subseteq Z_y$. Hence

$$(4.6.5) \quad \theta_{f,t}^*\text{cl}CC(i_*\mathcal{F}) \in H_{Z_y}^0(X, K_{X/k}) \quad \text{and} \quad \theta_{f,t}^!CC(i_*\mathcal{F}) \in \text{CH}_0(Z_y),$$

where $\theta_{f,t}^!$ is the refined Gysin pull-back along $\theta_{f,t}$. The compatibility between the cycle class map and refined Gysin pull-back gives the commutative diagram (cf. [AS07, (1.11)])

$$(4.6.6) \quad \begin{array}{ccc} \text{CH}_{\dim X}(SS(i_*\mathcal{F})) & \xrightarrow{\text{cl}} & H_{SS(i_*\mathcal{F})}^0(T^*X, e_X^{\vee*}K_{X/k}) \\ \theta_{f,t}^! \downarrow & & \downarrow \theta_{f,t}^* \\ \text{CH}_0(Z_y) & \xrightarrow{\text{cl}} & H_{Z_y}^0(X, K_{X/k}). \end{array}$$

It follows that

$$(4.6.7) \quad \theta_{f,t}^*\text{cl}CC(i_*\mathcal{F}) = \text{cl}(\theta_{f,t}^!CC(i_*\mathcal{F})) \quad \text{in} \quad H_{Z_y}^0(X, K_{X/k}).$$

We define (cf. [Sai21, (2.5)])

$$(4.6.8) \quad (CC(i_*\mathcal{F}), df)_{T^*X, Z_y} := \deg_{k(y)}(\theta_{f,t}^!CC(i_*\mathcal{F})).$$

Here $\deg_{k(y)}$ denotes the degree map $\text{CH}_0(Z_y) \rightarrow \text{CH}_0(y) = \mathbf{Z}$, which is defined since Z_y is proper over y . The resulting class and its degree are independent of the choice of the local parameter t , since $CC(i_*\mathcal{F})$ is conical.

Proposition 4.6.5. *Let $f: X \rightarrow Y$ be a morphism from a smooth scheme X to a smooth connected curve Y over a perfect field k , and let $i: Z \hookrightarrow X$ be a closed immersion such that $g = fi: Z \rightarrow Y$ is proper. Let $\mathcal{F} \in D_{\text{ctf}}(Z, \Lambda)$. Let $y \in Y$ be a closed point. Assume that f is properly $SS(i_*\mathcal{F})$ -transversal over a dense open subset $V \subseteq Y$. Then we have*

$$(4.6.9) \quad -a_y(g_*\mathcal{F}) = (CC(i_*\mathcal{F}), df)_{T^*X, Z_y} \quad \text{in} \quad H_y^0(Y, K_{Y/k}) \simeq \Lambda,$$

where a_y is the Artin conductor.

Assuming an additional condition, we prove in Corollary 4.9.6 that the equality (4.6.9) holds in $\mathbf{Z}$.

Proof. Let t be a local parameter at y . After shrinking Y around y , we may assume that $Y \setminus \{y\} \subseteq V$ and that dt is nowhere zero. Consider the correspondence

$$(4.6.10) \quad T^*X \xleftarrow{df} X \times_Y T^*Y \xrightarrow{\text{pr}} T^*Y.$$

By the assumption, we have $dt^{-1}(f_*SS(i_*\mathcal{F})) \subseteq \{y\}$ and a commutative diagram

$$(4.6.11) \quad \begin{array}{ccc} H_{SS(i_*\mathcal{F})}^0(T^*X, e_X^{\vee*}K_{X/k}) & \xrightarrow{f_!} & H_{f_*SS(i_*\mathcal{F})}^0(T^*Y, e_Y^{\vee*}K_{Y/k}) \\ \theta_{f,t}^* \downarrow & & \downarrow \theta_{\text{id},t}^* \\ H_{Z_y}^0(X, K_{X/k}) & \xrightarrow{g_*} & H_y^0(Y, K_{Y/k}). \end{array}$$

By Corollary 4.6.3, we have

$$(4.6.12) \quad f_! \text{cl}CC(i_*\mathcal{F}) = \text{cl}CC(g_*\mathcal{F}) \quad \text{in} \quad H_{f_*SS(i_*\mathcal{F})}^0(T^*Y, e_Y^{\vee*}K_{Y/k}).$$

On the curve Y , the coefficient of T_y^*Y in $CC(g_*\mathcal{F})$ is $-a_y(g_*\mathcal{F})$. Since dt is disjoint from the zero section and meets T_y^*Y transversally with multiplicity one, it follows that

$$(4.6.13) \quad g_*\theta_{f,t}^*(\text{cl}CC(i_*\mathcal{F})) = -a_y(g_*\mathcal{F}) \cdot \text{cl}[y].$$

The equality (4.6.9) now follows from (4.6.7), (4.6.8) and (4.6.13). $\square$

4.7 An Artin–Schreier sheaf

In this section, we compute CC_μ for an Artin–Schreier sheaf with wild ramification, directly without using Proposition 4.6.1.

Lemma 4.7.1. *Let*

$$\begin{array}{ccc} C & \xrightarrow{g} & D \\ c \downarrow & & \downarrow d \\ X & \xrightarrow{q} & A \end{array}$$

be a cartesian diagram of separated schemes of finite type over k .

1. Assume that the vertical arrows are closed immersions and define a morphism $q^*d_!d^! \rightarrow c_!c^!q^*$ by $q^*\mathcal{H}om(d_*\Lambda, -) \rightarrow \mathcal{H}om(c_*\Lambda, q^*-)$. Then, the diagram

$$(4.7.1) \quad \begin{array}{ccc} c_!c^!q^* & \longleftarrow & q^*d_!d^! \\ \text{adj}_c \downarrow & & \downarrow \text{adj}_d \\ q^* & \xlongequal{\quad} & q^* \end{array}$$

is commutative.

2. The diagram

$$(4.7.2) \quad \begin{array}{ccc} q^* & \xrightarrow{\text{adj}_d} & q^*d_*d^* \\ \text{adj}_c \downarrow & & \downarrow \text{bc} \\ c_*c^*q^* & \xleftarrow{\text{can}} & c_*g^*d^* \end{array}$$

is commutative.

Proof. 1. This follows from the commutative diagram

$$\begin{array}{ccc} q^*\mathcal{H}om(d_*\Lambda, -) & \longrightarrow & \mathcal{H}om(c_*\Lambda, q^*-) \\ \downarrow & & \downarrow \\ q^*\mathcal{H}om(\Lambda, -) & \xlongequal{\quad} & \mathcal{H}om(\Lambda, q^*-). \end{array}$$

2. By adjunction and by the definition of $q^*d_* \rightarrow c_*g^*$, this follows from the commutative diagram

$$\begin{array}{ccc} 1 & \xrightarrow{\text{adj}_d} & d_*d^* \\ \text{adj}_{qc} \downarrow & & \downarrow \text{adj}_g \\ q_*c_*c^*q^* & \xleftarrow{\text{can}} & d_*g_*g^*d^*. \end{array}$$

□

Proposition 4.7.2. Let $j_X: X - \{0\} = \text{Spec } k[x^{\pm 1}] \rightarrow X = \mathbf{A}^1 = \text{Spec } k[x]$ be the open immersion and let $\mathcal{F} = j_{X!}\mathcal{L}_\psi(1/x)$. Then, we have

$$(4.7.3) \quad CC_\mu(\mathcal{F}) = -\text{cl}([T_X^*X] + 2 \cdot [T_0^*X])$$

in $H_{T_X^*X \cup T_0^*X}^0(T^*X, e^{\vee*}K_X)$.

Proof. Set $X \times X = \text{Spec } k[x, y]$ and let

$$A_X(X \times X) = \mathbf{A}^3 = \text{Spec } k[x, u, v] \rightarrow \mathbf{A}^1 = \text{Spec } k[u]$$

be the deformation to the normal bundle $T = TX = \text{Spec } k[x, v]$ for $v = (y - x)/u$. Let $q: X \rightarrow W = \text{Spec } k[w]$ be the morphism defined by $k[w] \rightarrow k[x]$ sending w to $-x^2$. We construct a commutative diagram

$$(4.7.4) \quad \begin{array}{ccccc} X \times \mathbf{A}^1 & \longrightarrow & W \times \mathbf{A}^1 = \text{Spec } k[w, u] & \xleftarrow{1} & W \times \mathbf{A}^1 \\ \delta \downarrow & & \downarrow 0_A & & \downarrow 0_B \\ A_X(X \times X) & \xrightarrow{q_A} & A = \text{Spec } k[w, u, v] & \xleftarrow{\bar{g}} & \bar{B} \supset B = \text{Spec } k[w, u, s]. \end{array}$$

The left vertical arrow $\delta: X \times \mathbf{A}^1 = \operatorname{Spec} k[x, u] \rightarrow A_X(X \times X) = \operatorname{Spec} k[x, u, v]$ is a lifting of the diagonal $X \times \mathbf{A}^1 \rightarrow X \times X \times \mathbf{A}^1$ and is defined by the surjection $k[x, u, v] \rightarrow k[x, u]$ sending v to 0. The morphism $q_A: A_X(X \times X) = \operatorname{Spec} k[x, u, v] \rightarrow A = \operatorname{Spec} k[w, u, v]$ is defined by $w = -x(x + uv)$ and the left square is cartesian. The morphism $\bar{g}: \bar{B} \rightarrow A$ is the blow-up at the ideal (w, v) and $B \subseteq \bar{B}$ is the complement of the proper transform of the divisor $w = 0$. The restriction $g: B \rightarrow A$ of $\bar{g}$ is a morphism of line bundles over $W \times \mathbf{A}^1$ defined by $v = ws$ and the vertical arrows in the left square are the 0-sections.

The base change of the lower line to the origin $0 \rightarrow \mathbf{A}^1 = \operatorname{Spec} k[u]$ gives the upper line of the diagram

$$(4.7.5) \quad \begin{array}{ccccc} TX & \xrightarrow{q_T} & E = \operatorname{Spec} k[w, v] & \xleftarrow{\bar{g}_F} & \bar{F} \supset F = \operatorname{Spec} k[w, s] \\ e \downarrow & & \downarrow e_E & & \downarrow e_F \\ X & \xrightarrow{q} & W & \xleftarrow{1} & W \end{array}$$

of line bundles. The right square in (4.7.4) is the base change of that in (4.7.5).

We have an isomorphism

$$(j_X \times j_X)_! \mathcal{L}(1/y - 1/x) \otimes \operatorname{pr}_1^* K_X \rightarrow \mathcal{H}om(\operatorname{pr}_1^* \mathcal{F}, \operatorname{pr}_2^! \mathcal{F})$$

on $X \times X$. Let $\mathcal{H}_X$ on $X \times X \times \mathbf{G}_m$ be the pull-back of $\mathcal{H}om(\operatorname{pr}_1^* \mathcal{F}, \operatorname{pr}_2^! \mathcal{F})$. Let $e_A: A \rightarrow W$ be the projection and $j_A: \operatorname{Spec} k[w^{\pm 1}, u, v] \rightarrow A$ be the open immersion. Define $\mathcal{H}_A$ on A by

$$\mathcal{H}_A = j_{A!} \mathcal{L}_\psi(uv/w) \otimes e_A^* K_W$$

and let $\mathcal{H}_{A^\circ}$ be the restriction on $A^\circ = A - E$. We extend the canonical isomorphism $(q^* K_W)|_{X - \{0\}} \rightarrow K_X|_{X - \{0\}}$ uniquely to an isomorphism $q^* K_W \rightarrow K_X$ and identify them and let $q_{A^\circ}: X \times X \times \mathbf{G}_m \rightarrow A^\circ$ be the restriction of q_A . Then, since

$$1/y - 1/x = -(y - x)/xy = -(y - x)/(x(x + uv)) = uv/w,$$

we have a canonical isomorphism

$$(4.7.6) \quad \mathcal{H}_X \leftarrow q_{A^\circ}^* \mathcal{H}_{A^\circ}.$$

By Lemma 4.7.1 applied to $q_{A^\circ}: X \times X \times \mathbf{G}_m \rightarrow A^\circ$, the isomorphism (4.7.6) defines a commutative diagram

$$(4.7.7) \quad \begin{array}{ccc} \delta_!^\circ \delta^{\circ!} \mathcal{H}_X & \longleftarrow & q_{A^\circ}^* 0_{A^\circ!} 0_{A^\circ}^! \mathcal{H}_{A^\circ} \\ \downarrow & & \downarrow \\ \mathcal{H}_X & \longleftarrow & q_{A^\circ}^* \mathcal{H}_{A^\circ} \\ \downarrow & & \downarrow \\ \delta_*^\circ \delta^{\circ*} \mathcal{H}_X & \longleftarrow & q_{A^\circ}^* 0_{A^\circ*} 0_{A^\circ}^* \mathcal{H}_{A^\circ}. \end{array}$$

Here and in the following, the superscript $^\circ$ indicates the base change by $\mathbf{G}_m \rightarrow \mathbf{A}^1$. By the isomorphisms [Gro77, (3.2.1) and (3.1.1)], the identity of $\mathcal{F}$ and the evaluation define isomorphisms

$$(4.7.8) \quad j_{X*} \Lambda \rightarrow \mathcal{H}om(\mathcal{F}, \mathcal{F}) \rightarrow \delta^{\circ!} \mathcal{H}_X, \quad \delta^{\circ*} \mathcal{H}_X \rightarrow D_X \mathcal{F} \otimes \mathcal{F} \rightarrow j_{X!} \Lambda.$$

Let $e_B: B \rightarrow W$ be the projection and let $j_B: B \rightarrow \overline{B}$ be the open immersion. Define $\mathcal{H}_B$ on B by

$$\mathcal{H}_B = \mathcal{L}_\psi(us) \otimes e_B^* K_W$$

and set $\mathcal{H}_{\overline{B}} = j_{B!} \mathcal{H}_B$. Let $g^\circ: B^\circ = B - F \rightarrow A^\circ = A - E$ be the restriction of g and $\overline{g}^\circ: \overline{B}^\circ = \overline{B} - \overline{F} \rightarrow A^\circ$ be the restriction of $\overline{g}$. Let $\mathcal{H}_{B^\circ}$ and $\mathcal{H}_{\overline{B}^\circ}$ be the restrictions of $\mathcal{H}_B$ and $\mathcal{H}_{\overline{B}}$. Then, since u is invertible on B° , we have isomorphisms

$$(4.7.9) \quad \overline{g}_*^\circ j_{B!}(\mathcal{L}(us)|_{B^\circ}) = g_!^\circ(\mathcal{L}(us)|_{B^\circ}) = \mathcal{L}(uv/w)|_{A^\circ}.$$

This induces an isomorphism

$$(4.7.10) \quad \mathcal{H}_{A^\circ} \rightarrow \overline{g}_*^\circ \mathcal{H}_{\overline{B}^\circ} = g_!^\circ \mathcal{H}_{B^\circ}.$$

We consider the cartesian diagram

$$\begin{array}{ccccc} A & \xleftarrow{\overline{g}} & \overline{B} & \xleftarrow{j_B} & B \\ 0_A \uparrow & & \uparrow +_{\overline{B}} & & \uparrow +_B \\ W \times \mathbf{A}^1 & \xleftarrow{\quad} & \overline{g}^{-1}(W \times \mathbf{A}^1) & \xleftarrow{\quad} & g^{-1}(W \times \mathbf{A}^1). \end{array}$$

The inverse image $\overline{g}^{-1}(W \times \mathbf{A}^1)$ is the union of the 0-section $0_B(W \times \mathbf{A}^1)$ and the fiber $\overline{B}_0 = \overline{B} \times_W 0$. Similarly as (4.7.7), by Lemma 4.3.1 applied to $\overline{g}^\circ: \overline{B}^\circ \rightarrow A^\circ$, the isomorphism (4.7.10) defines a commutative diagram

$$(4.7.11) \quad \begin{array}{ccc} 0_{A^\circ}! 0_{A^\circ}^! \mathcal{H}_{A^\circ} & \longrightarrow & \overline{g}_*^\circ +_{\overline{B}!}^\circ +_{\overline{B}^\circ}^! \mathcal{H}_{\overline{B}^\circ} \\ \downarrow & & \downarrow \\ \mathcal{H}_{A^\circ} & \xleftarrow{\simeq} & \overline{g}_*^\circ \mathcal{H}_{\overline{B}^\circ} \\ \downarrow & & \downarrow \\ 0_{A^\circ*} 0_{A^\circ}^* \mathcal{H}_{A^\circ} & \xleftarrow{\quad} & g_*^\circ +_{\overline{B}*}^\circ +_{\overline{B}^\circ}^* \mathcal{H}_{\overline{B}^\circ} \end{array}$$

on A° . By proper base change theorem, the horizontal arrows are isomorphisms.

We compute the top and the bottom of the right column of (4.7.11). Let $j_W: \mathbf{G}_m \rightarrow W$ and $j_{W \times \mathbf{A}^1}: \mathbf{G}_m \times \mathbf{A}^1 \rightarrow W \times \mathbf{A}^1$ be the open immersions. The canonical morphism $+_{\overline{B}}^* \mathcal{H}_{\overline{B}^\circ} \otimes +_{\overline{B}}^! \Lambda \rightarrow +_{\overline{B}}^! \mathcal{H}_{\overline{B}^\circ}$ is an isomorphism and we have a distinguished triangle $\Lambda_{\overline{B}_0} \rightarrow +_{\overline{B}}^! e_{\overline{B}}^* K_W \rightarrow 0_{\overline{B}*} j_{W \times \mathbf{A}^1} \Lambda \rightarrow$. Since $\overline{g}_*^\circ(\mathcal{L}(us)|_{\overline{B}^\circ}) = 0$ (4.7.9), we have canonical isomorphisms

$$(4.7.12) \quad \overline{g}_*^\circ +_{\overline{B}!}^\circ +_{\overline{B}^\circ}^! \mathcal{H}_{\overline{B}^\circ} \rightarrow 0_{A^\circ}! j_{W \times \mathbf{A}^1*} \Lambda = \overline{g}_*^\circ 0_{B^\circ}! j_{W \times \mathbf{A}^1*} \Lambda,$$

$$(4.7.13) \quad \overline{g}_*^\circ +_{\overline{B}*}^\circ +_{\overline{B}^\circ}^* \mathcal{H}_{\overline{B}^\circ} \leftarrow 0_{A^\circ*} j_{W \times \mathbf{A}^1!} K_W = \overline{g}_*^\circ 0_{B^\circ}! j_{W \times \mathbf{A}^1!} K_W.$$

Let η_0 be the generic point of the henselization of $\mathbf{A}^1 = \text{Spec } k[u]$ at $0 \in \mathbf{A}^1$ and let $\nu_A: D_{\text{ctf}}(A^\circ, \Lambda) \rightarrow D_{\text{ctf}}(E \times_k \eta_0, \Lambda)$ be the specialization functor. Let $j_F: F \rightarrow \overline{F}$ be the open immersion. By Proposition 2.1.4.1, the projection $\overline{B} \rightarrow W \times \mathbf{A}^1$ is locally acyclic relatively to $\mathcal{H}_{\overline{B}}$. Hence by [Ill77, Corollaire 2.7], the composition $\overline{B} \rightarrow \mathbf{A}^1 = \text{Spec } k[u]$ is locally acyclic relatively to $\mathcal{H}_{\overline{B}}$ and we have $\Psi \mathcal{H}_{\overline{B}} = \mathcal{H}_{\overline{B}}|_{\overline{F}} = j_{F!} e_F^* K_W$. Here Ψ denotes the nearby cycles functor for $\overline{B} \rightarrow \mathbf{A}^1 = \text{Spec } k[u]$ at $0 \in \mathbf{A}^1$. Let $g_F: F \rightarrow E$ be the

restriction of $g: B \rightarrow A$. Then, by proper base change theorem, the isomorphisms (4.7.12) and (4.7.13) induce isomorphisms

$$(4.7.14) \quad \nu_A \bar{g}_*^{\circ} +_{\bar{B}!}^{\circ} +_{\bar{B}}^{\circ!} \mathcal{H}_{\bar{B}^{\circ}} \rightarrow g_{F!} 0_{F!} j_{W*} \Lambda, \quad \nu_A \bar{g}_*^{\circ} +_{\bar{B}*}^{\circ} +_{\bar{B}}^{\circ*} \mathcal{H}_{\bar{B}^{\circ}} \rightarrow g_{F!} 0_{F!} j_{W!} K_W.$$

Hence the specialization of (4.7.11) defines a commutative diagram

$$(4.7.15) \quad \begin{array}{ccc} \nu_A 0_{A^{\circ}!} 0_{A^{\circ}}^! \mathcal{H}_{A^{\circ}} & \longrightarrow & g_{F!} 0_{F!} j_{W*} \Lambda \\ \downarrow & & \downarrow \\ \nu_A \mathcal{H}_{A^{\circ}} & \xleftarrow{\simeq} & g_{F!} e_F^* K_W \\ \downarrow & & \downarrow \\ \nu_A 0_{A^{\circ}*} 0_{A^{\circ}}^* \mathcal{H}_{A^{\circ}} & \longleftarrow & g_{F!} 0_{F*} j_{W!} K_W \end{array}$$

on E where the horizontal arrows are isomorphisms.

Let $0_{\bar{B}^{\circ}}: W \times \mathbf{G}_m \rightarrow \bar{B}^{\circ}$ be the 0-section and consider the canonical morphisms

$$(4.7.16) \quad 0_{\bar{B}^{\circ}!} 0_{\bar{B}^{\circ}}^! \mathcal{H}_{\bar{B}^{\circ}} \rightarrow +_{\bar{B}!}^{\circ} +_{\bar{B}}^{\circ!} \mathcal{H}_{\bar{B}^{\circ}} \rightarrow \mathcal{H}_{\bar{B}^{\circ}} \rightarrow +_{\bar{B}*}^{\circ} +_{\bar{B}}^{\circ*} \mathcal{H}_{\bar{B}^{\circ}} \rightarrow 0_{\bar{B}^{\circ}*} 0_{\bar{B}^{\circ}}^* \mathcal{H}_{\bar{B}^{\circ}},$$

where the middle arrows induce the right columns of (4.7.11) and (4.7.15). Then, the specialization of compositions in (4.7.16) defines

$$(4.7.17) \quad 0_{F!} \Lambda \rightarrow e_F^* K_W \rightarrow 0_{F*} K_W$$

where the first arrow is the adjoint of the canonical isomorphism $\Lambda \rightarrow 0_F^! e_F^* K_W$ and the second arrow is the restriction.

Let $\mu_A: D_{\text{ctf}}(A^{\circ}, \Lambda) \rightarrow D_{\text{ctf}}(E^{\vee} \times_k \eta_0, \Lambda)$ denote the composition of ν_A with the Fourier transform F_{ψ} . Let $g_F^{\vee}: E^{\vee} \rightarrow F^{\vee}$ be the dual of $g_F: F \rightarrow E$ and define a cartesian diagram

$$\begin{array}{ccc} g_F^{\vee-1}(W) & \longrightarrow & W \\ +_{E^{\vee}} \downarrow & & \downarrow 0_{F^{\vee}} \\ E^{\vee} & \xrightarrow{g_F^{\vee}} & F^{\vee}. \end{array}$$

Let $e_{E^{\vee}}: E^{\vee} \rightarrow W$ and $e_{F^{\vee}}: F^{\vee} \rightarrow W$ be the canonical morphisms. Since $F_{\psi} g_{F!} e_F^* K_W = g_F^{\vee*} 0_{F^{\vee}*} \Lambda = +_{E^{\vee}*} \Lambda$ by Proposition 2.1.9, the Fourier transform of (4.7.15) gives

$$(4.7.18) \quad \begin{array}{ccc} \mu_A 0_{A^{\circ}!} 0_{A^{\circ}}^! \mathcal{H}_{A^{\circ}} & \longrightarrow & e_{E^{\vee}}^* j_{W*} \Lambda \\ \downarrow & & \downarrow \\ \mu_A \mathcal{H}_{A^{\circ}} & \xleftarrow{\simeq} & +_{E^{\vee}*} \Lambda \\ \downarrow & & \downarrow \\ \mu_A 0_{A^{\circ}*} 0_{A^{\circ}}^* \mathcal{H}_{A^{\circ}} & \longleftarrow & e_{E^{\vee}}^* j_{W!} K_W \end{array}$$

on the dual $E^{\vee}$. To compute the Fourier transform of the right column, we apply the isomorphism $F g_{F!} \rightarrow g_F^{\vee*} F$ (2.1.24) and the isomorphisms (2.1.6) and (2.1.8). We consider

the commutative diagram

$$\begin{array}{ccccc}
e_{E^\vee}^* \Lambda & \longleftarrow & F_\psi 0_{E!} \Lambda & \longleftarrow & g_F^{\vee*} \Lambda \\
\downarrow & & \downarrow & & \downarrow \\
+_{E^\vee*} \Lambda & \longleftarrow & F_\psi g_{F!} e_F^* K_W & \longleftarrow & g_F^{\vee*} 0_{F^\vee!} \Lambda \\
\downarrow & & \downarrow & & \downarrow \\
e_{E^\vee}^* K_W & \longleftarrow & F_\psi 0_{E*} K_W & \longleftarrow & g_F^{\vee*} e_{F^\vee}^* K_W.
\end{array}
\tag{4.7.19}$$

The left column is obtained from the right column of (4.7.18) by taking the compositions with $\Lambda \rightarrow j_{W*} \Lambda$ and $j_{W!} K_W \rightarrow K_W$. The right half is obtained by applying the isomorphism $F_\psi g_{F!} \leftarrow g_F^{\vee*} F_\psi$ (2.1.24) to (4.7.17). By the description above of (4.7.17) and Proposition 2.1.11, the composition of the right column is $g_F^{\vee*}$ of -1 -times the cycle class of the 0-section. By the commutativity of the diagram, the left column is -1 -times the cycle class of $g_F^{\vee-1}(W)$.

Let $\nu_X: D_{\text{ctf}}(X \times X \times \mathbf{G}_m, \Lambda) \rightarrow D_{\text{ctf}}(TX \times_k \eta_0, \Lambda)$ be the specialization functor and let μ_X be the composition with the Fourier transform. Let $0_X: X \rightarrow TX$ be the 0-section. The specialization of the commutative diagram (4.7.7) defines a commutative diagram

$$\begin{array}{ccc}
0_{X!} 0_X^! \nu_X \mathcal{H}_X & \longleftarrow & q_T^* \nu_A 0_{A!} 0_A^! \mathcal{H}_A \\
\downarrow & & \downarrow \\
\nu_X \mathcal{H}_X & \longleftarrow & q_T^* \nu_A \mathcal{H}_A \\
\downarrow & & \downarrow \\
0_{X*} 0_X^* \nu_X \mathcal{H}_X & \longleftarrow & q_T^* \nu_A 0_{A*} 0_A^* \mathcal{H}_A.
\end{array}
\tag{4.7.20}$$

By Proposition 2.3.7.2, the isomorphisms (4.7.8) induce isomorphisms

$$j_{X*} \Lambda \rightarrow 0_X^! \nu_X \mathcal{H}_X, \quad 0_X^* \nu_X \mathcal{H}_X \rightarrow j_{X!} \Lambda.
\tag{4.7.21}$$

Let $q_{T*}: T^*X \rightarrow E^\vee$ denote the morphism induced by q_T on the dual. Then the Fourier transform of (4.7.20) gives

$$\begin{array}{ccc}
e^{\vee*} j_{X*} \Lambda & \longleftarrow & q_{T*}^* \mu_A 0_{A!} 0_A^! \mathcal{H}_A \\
\downarrow & & \downarrow \\
\mu_X \mathcal{H}_X & \longleftarrow & q_{T*}^* \mu_A \mathcal{H}_A \\
\downarrow & & \downarrow \\
e^{\vee*} j_{X!} K_{X-0} & \longleftarrow & q_{T*}^* \mu_A 0_{A*} 0_A^* \mathcal{H}_A
\end{array}
\tag{4.7.22}$$

on T^*X . Since the ramification index of $q: X \rightarrow W$ is 2, the pull-back $q_{T*}^*: H_+^0(E^\vee, e_E^{\vee*} K_W) \rightarrow H_+^0(T^*X, e_X^{\vee*} K_X)$ maps the class of the composition of the left column in (4.7.19) to the class $-([T_X^* X] + 2 \cdot [T_0^* X])$. Thus, combining (4.7.18) and (4.7.22), we obtain (4.7.3). $\square$

4.8 Characteristic classes and characteristic cycles

In this subsection, we show that the cohomological characteristic class $C_{X/k}$ is the cycle class of the characteristic cycle CC . This proves the conjecture [Sai17a, Conjecture 6.8] formulated by the first named author.

Let X be a smooth scheme over a perfect field k . Let $e_X : TX \rightarrow X$ and $e_X^\vee : T^*X \rightarrow X$ be the projections, and let $0 : X \hookrightarrow TX$, $0^\vee : X \hookrightarrow T^*X$ be the zero sections. Let $\delta : X \hookrightarrow X \times X$ be the diagonal morphism.

Let $i : Z \hookrightarrow X$ be a closed immersion. By transitivity of extraordinary pull-back, we have a canonical isomorphism $K_{Z/k} = i^! K_{X/k}$. Excision gives an isomorphism

$$(4.8.1) \quad \iota_Z : H^0(Z, K_{Z/k}) \xrightarrow{\sim} H_Z^0(X, K_{X/k}).$$

Let $0_X : X \hookrightarrow T^*X$ be the zero section. Let $\mathcal{F} \in D_{\text{ctf}}(Z, \Lambda)$. Since both $SS_\mu(i_*\mathcal{F})$ and $SS(i_*\mathcal{F})$ lie over Z , pull-back by the zero-section defines a morphism

$$(4.8.2) \quad 0_{X,Z}^* : H_{SS(i_*\mathcal{F})}^0(T^*X, e_X^{\vee*} K_{X/k}) \longrightarrow H_Z^0(X, K_{X/k}).$$

By Lemma 4.1.9 and Proposition 4.3.2.2, we have

$$(4.8.3) \quad 0_{X,Z}^* CC_\mu(i_*\mathcal{F}) = \iota_Z(CC_{Z/k}(\mathcal{F})).$$

Lemma 4.8.1. *Let $i : Z \hookrightarrow X$ be a closed immersion to a smooth scheme X of dimension d over k . Let $\text{cl}_X : \text{CH}_d(SS(i_*\mathcal{F})) \rightarrow H_{SS(i_*\mathcal{F})}^0(T^*X, e_X^{\vee*} K_{X/k})$ be the cycle-class map with support. Then we have*

$$(4.8.4) \quad 0_{X,Z}^* \text{cl}_X(CC(i_*\mathcal{F})) = \iota_Z(\text{cl}_Z(cc_{Z,0}(\mathcal{F}))),$$

where $cc_{Z,0}(\mathcal{F})$ denotes the characteristic class associated with the cycle $CC(i_*\mathcal{F})$ (see [Sai17a, Definition 6.7]).

Proof. The cycle $CC(i_*\mathcal{F})$ is a cycle on T^*X whose base is contained in Z . Since the zero section is a regular immersion, the compatibility of the cycle-class map with refined Gysin pull-back gives a commutative diagram (cf. [AS07, (1.11)])

$$(4.8.5) \quad \begin{array}{ccc} \text{CH}_d(SS(i_*\mathcal{F})) & \xrightarrow{\text{cl}_X} & H_{SS(i_*\mathcal{F})}^0(T^*X, e_X^{\vee*} K_{X/k}) \\ 0_X^! \downarrow & & \downarrow 0_{X,Z}^* \\ \text{CH}_0(Z) & \xrightarrow{\iota_Z \circ \text{cl}_Z} & H_Z^0(X, K_{X/k}). \end{array}$$

The lower horizontal morphism is the cycle-class map followed by the excision isomorphism, since the intersection is supported on Z . Applying the diagram to $CC(i_*\mathcal{F})$, we obtain

$$(4.8.6) \quad 0_{X,Z}^* \text{cl}_X(CC(i_*\mathcal{F})) = \iota_Z \text{cl}_Z(0_X^! CC(i_*\mathcal{F})) = \iota_Z \text{cl}_Z(cc_{Z,0}(\mathcal{F})),$$

where the second equality follows from [Sai17a, Definition 6.7.2 and Lemma 6.9.1]. $\square$

Theorem 4.8.2 ([Sai17a, Conjecture 6.8]). *Let Z be a k -scheme admitting a closed immersion into a smooth k -scheme. For every $\mathcal{F} \in D_{\text{ctf}}(Z, \Lambda)$, we have*

$$(4.8.7) \quad C_{Z/k}(\mathcal{F}) = \text{cl}_Z(cc_{Z,0}(\mathcal{F})) \quad \text{in } H^0(Z, K_{Z/k}).$$

Proof. Take a closed immersion $i : Z \hookrightarrow X$ with X smooth. We first assume that there exists a non-trivial character $\psi : \mathbf{F}_p \rightarrow \Lambda^\times$. Applying Theorem 4.6.2, we have

$$(4.8.8) \quad CC_\mu(i_*\mathcal{F}) = \text{cl}_X(CC(i_*\mathcal{F})) \quad \text{in } H_{SS(i_*\mathcal{F})}^0(T^*X, e_X^{\vee*} K_{X/k}).$$

Apply $0_{X,Z}^*$ to (4.8.8). By (4.8.3) and Lemma 4.8.1, the two sides are, respectively,

$$(4.8.9) \quad \iota_Z(C_{Z/k}(\mathcal{F})) \quad \text{and} \quad \iota_Z(\text{cl}_Z(cc_{Z,0}(\mathcal{F}))).$$

Since ι_Z is an isomorphism, we obtain (4.8.7).

In general, let (Λ', ψ) be as in Lemma 2.1.1. By [Sai17a, (5.15)], we have

$$(4.8.10) \quad CC(i_*\mathcal{F}) = CC(i_*\mathcal{F}_{\Lambda'}).$$

The characteristic class $C_{Z/k}(\mathcal{F})$ is defined using the identity endomorphism, adjunction, and evaluation. These constructions commute with coefficient extension. Thus, the image of $C_{Z/k}(\mathcal{F})$ under the canonical map

$$(4.8.11) \quad \tau : H^0(Z, \mathcal{K}_{Z/k}) \rightarrow H^0(Z, \mathcal{K}_{Z/k}) \otimes_{\Lambda} \Lambda' \simeq H^0(Z, \mathcal{K}_{Z/k, \Lambda'})$$

is $C_{Z/k}(\mathcal{F}_{\Lambda'})$. By the previous case, we have equality in $H^0(Z, \mathcal{K}_{Z/k, \Lambda'})$:

$$(4.8.12) \quad \tau(C_{Z/k}(\mathcal{F})) = C_{Z/k}(\mathcal{F}_{\Lambda'}) = \text{cl}_Z(cc_{Z,0}(\mathcal{F}_{\Lambda'})) \stackrel{(4.8.10)}{=} \tau(\text{cl}_Z(cc_{Z,0}(\mathcal{F}))).$$

Since $\Lambda \rightarrow \Lambda'$ is faithfully flat, the map τ is injective. From (4.8.12), we get the equality (4.8.7). $\square$

4.9 Microlocalization for ℓ -adic coefficients

Let $E/\mathbf{Q}_{\ell}$ be a finite extension, let $\mathcal{O}_E$ be the ring of integers in E , and let π be a uniformizer. After replacing E by a finite extension, we assume that there exists a non-trivial character $\psi : \mathbf{F}_p \rightarrow \mathcal{O}_E^{\times}$.

Let Y be a scheme of finite type over k . Let $D_{\text{cons}}(Y_{\text{proét}}, \mathcal{O}_E)$ be the stable ∞ -category of constructible pro-étale sheaves on Y with $\mathcal{O}_E$ -coefficients. We use analogous notation for $D_{\text{cons}}(Y_{\text{proét}}, E)$ and $D_{\text{cons}}(Y_{\text{proét}}, \mathcal{O}_E/\pi^m)$. By [HRS23, Theorem 7.7], there are natural equivalences

$$(4.9.1) \quad \begin{aligned} D_{\text{cons}}(Y_{\text{proét}}, \mathcal{O}_E/\pi^m) &\simeq D_{\text{ctf}}(Y, \mathcal{O}_E/\pi^m), \\ D_{\text{cons}}(Y_{\text{proét}}, \mathcal{O}_E) &\simeq D_{\text{c}}^b(Y, \mathcal{O}_E), \\ D_{\text{cons}}(Y_{\text{proét}}, E) &\simeq D_{\text{c}}^b(Y, E). \end{aligned}$$

By [HRS23, Theorem 1.6 (1) and (2)], there is an equivalence of stable ∞ -categories

$$(4.9.2) \quad D_{\text{cons}}(Y_{\text{proét}}, \mathcal{O}_E) \simeq \varprojlim_m D_{\text{cons}}(Y_{\text{proét}}, \mathcal{O}_E/\pi^m).$$

By [HS23, Corollary 2.4], every constructible E -sheaf has an integral model. By [Bar24, Lemma 2.1], reduction modulo π is conservative on derived complete constructible objects.

Assume Y is smooth. The operations used to form the diagonal endomorphism kernel, Verdier specialization, and Fourier transform preserve constructibility and are compatible with reduction. Hence, for an integral model $\mathcal{F}_0$ of $\mathcal{F} \in D_{\text{cons}}(Y_{\text{proét}}, E)$ and $\mathcal{F}_m = \mathcal{F}_0 \otimes_{\mathcal{O}_E}^L \mathcal{O}_E/\pi^m$, there are canonical isomorphisms

$$(4.9.3) \quad \mu\mathcal{H}om_{\mathcal{O}_E}(\mathcal{F}_0, \mathcal{F}_0) \otimes_{\mathcal{O}_E}^L \mathcal{O}_E/\pi^m \xrightarrow{\sim} \mu\mathcal{H}om_{\mathcal{O}_E/\pi^m}(\mathcal{F}_m, \mathcal{F}_m),$$

$$(4.9.4) \quad \mu\mathcal{H}om_E(\mathcal{F}, \mathcal{F}) \xrightarrow{\sim} \mu\mathcal{H}om_{\mathcal{O}_E}(\mathcal{F}_0, \mathcal{F}_0) \otimes_{\mathcal{O}_E} E.$$

By [Bar24, Proposition 3.6], universal local acyclicity on the pro-étale site is equivalent to dualizability in the relative correspondence category. It is preserved by smooth pull-back and proper push-forward. Moreover, Beilinson's singular support and its projective Radon description hold for constructible E -sheaves (cf. [Bar24]). Following [UYZ20], the characteristic cycle of an ℓ -adic sheaf $\mathcal{F}$ is defined by

$$(4.9.5) \quad CC(\mathcal{F}) := CC(\mathcal{F}_1),$$

where $\mathcal{F}_1 = \mathcal{F}_0 \otimes^L \mathcal{O}_E/\pi$ for any integral model $\mathcal{F}_0$ of $\mathcal{F}$.

Theorem 4.9.1. *Let X be a smooth k -scheme, and let $\mathcal{F} \in D_{\text{cons}}(X_{\text{proét}}, E)$. Then*

$$(4.9.6) \quad SS_\mu(\mathcal{F}) \subseteq SS(\mathcal{F}).$$

Proof. The assertion is clear if $\mathcal{F} = 0$. By [Bar24, Theorem 1.5 (ii) and (v)], there exists an integral model $\mathcal{F}_0 \in D_{\text{cons}}(X_{\text{proét}}, \mathcal{O}_E)$ of $\mathcal{F}$ such that

$$(4.9.7) \quad \mathcal{F}_0 \otimes_{\mathcal{O}_E} E \simeq \mathcal{F}, \quad SS(\mathcal{F}) = SS(\mathcal{F}_0) = SS(\mathcal{F}_1),$$

where $\mathcal{F}_1 = \mathcal{F}_0 \otimes_{\mathcal{O}_E}^L \mathcal{O}_E/\pi$. We next compare the microlocal Hom complexes. The isomorphisms (4.9.3) and (4.9.4) give

$$(4.9.8) \quad \begin{aligned} \mu\mathcal{H}om_{\mathcal{O}_E}(\mathcal{F}_0, \mathcal{F}_0) \otimes_{\mathcal{O}_E}^L \mathcal{O}_E/\pi &\xrightarrow{\sim} \mu\mathcal{H}om_{\mathcal{O}_E/\pi}(\mathcal{F}_1, \mathcal{F}_1), \\ \mu\mathcal{H}om_{\mathcal{O}_E}(\mathcal{F}_0, \mathcal{F}_0) \otimes_{\mathcal{O}_E} E &\xrightarrow{\sim} \mu\mathcal{H}om_E(\mathcal{F}, \mathcal{F}). \end{aligned}$$

Let us recall why these compatibilities hold. The diagonal endomorphism kernel can be written as

$$(4.9.9) \quad \mathcal{H}_{\mathcal{O}_E}(\mathcal{F}_0, \mathcal{F}_0) \simeq D_{\mathcal{O}_E}(\mathcal{F}_0) \boxtimes \mathcal{F}_0.$$

For constructible complexes, Verdier duality and external products commute with reduction modulo π and with rationalization. Specialization along δ_X has the same coefficient compatibility. Finally, the Fourier transform is constructed from pull-back, tensor product with the Artin–Schreier sheaf, and compactly supported push-forward, all of which commute with these changes of coefficients. This proves (4.9.8).

We claim that

$$(4.9.10) \quad SS_\mu(\mathcal{F}) \subseteq SS_\mu(\mathcal{F}_1).$$

Let $U = T^*X \setminus \text{supp}_k(\mu\mathcal{H}om_{\mathcal{O}_E/\pi}(\mathcal{F}_1, \mathcal{F}_1))$. By definition, $\mu\mathcal{H}om_{\mathcal{O}_E/\pi}(\mathcal{F}_1, \mathcal{F}_1)|_U = 0$. Using the first isomorphism in (4.9.8), we obtain $\mu\mathcal{H}om_{\mathcal{O}_E}(\mathcal{F}_0, \mathcal{F}_0)|_U \otimes_{\mathcal{O}_E}^L \mathcal{O}_E/\pi = 0$. The complex

$$(4.9.11) \quad \mathcal{N} = \mu\mathcal{H}om_{\mathcal{O}_E}(\mathcal{F}_0, \mathcal{F}_0)|_U$$

is constructible and derived π -complete. Derived Nakayama therefore gives $\mathcal{N} = 0$. More explicitly, the distinguished triangle

$$(4.9.12) \quad \mathcal{N} \xrightarrow{\pi} \mathcal{N} \longrightarrow \mathcal{N} \otimes_{\mathcal{O}_E}^L \mathcal{O}_E/\pi \longrightarrow$$

shows that multiplication by π on $\mathcal{N}$ is an isomorphism. Hence $\mathcal{N} \otimes_{\mathcal{O}_E}^L \mathcal{O}_E/\pi^m = 0$ for every $m \geq 1$. Since $\mathcal{N}$ is derived π -complete, we have $\mathcal{N} \simeq R\varprojlim_m (\mathcal{N} \otimes_{\mathcal{O}_E}^L \mathcal{O}_E/\pi^m) = 0$.

Rationalizing and using the second isomorphism in (4.9.8), we obtain

$$(4.9.13) \quad \mu\mathcal{H}om_E(\mathcal{F}, \mathcal{F})|_U \simeq \mu\mathcal{H}om_{\mathcal{O}_E}(\mathcal{F}_0, \mathcal{F}_0)|_U \otimes_{\mathcal{O}_E} E = 0.$$

It follows that

$$(4.9.14) \quad \text{supp}_k(\mu\mathcal{H}om_E(\mathcal{F}, \mathcal{F})) \subseteq \text{supp}_k(\mu\mathcal{H}om_{\mathcal{O}_E/\pi}(\mathcal{F}_1, \mathcal{F}_1)),$$

which proves (4.9.10).

Finally, $\mathcal{F}_1$ is a constructible complex with coefficients in the finite field $\mathcal{O}_E/\pi$. Under the equivalence (4.9.1) between constructible torsion sheaves on the pro-étale and the étale sites, Theorem 3.3.6 applies to $\mathcal{F}_1$ and gives $SS_\mu(\mathcal{F}_1) = SS(\mathcal{F}_1)$. Together with (4.9.10) and (4.9.7), we obtain

$$(4.9.15) \quad SS_\mu(\mathcal{F}) \stackrel{(4.9.10)}{\subseteq} SS_\mu(\mathcal{F}_1) \stackrel{(3.3.21)}{=} SS(\mathcal{F}_1) \stackrel{(4.9.7)}{=} SS(\mathcal{F}),$$

as required. $\square$

Within the six-functor formalism, the microlocal characteristic cycle CC_μ for an object $\mathcal{F} \in D_{\text{cons}}(X_{\text{proét}}, E)$ is also well-defined.

Theorem 4.9.2. *Let X be a smooth connected scheme over a perfect field k , and let $\mathcal{F} \in D_{\text{cons}}(X_{\text{proét}}, E)$. Then we have*

$$(4.9.16) \quad CC_\mu(\mathcal{F}) = \text{cl}(CC(\mathcal{F})),$$

where $\text{cl} : Z_{\dim(X)}(SS(\mathcal{F})) \rightarrow H_{SS(\mathcal{F})}^0(T^*X, e_X^{\vee*} K_{X/k})$ is the cycle class map.

In particular, if X is a smooth curve and $\mathcal{F}$ is smooth outside a finite set $Z \subseteq X$, then we have (cf. [Sai17a, Lemma 5.11.3])

$$(4.9.17) \quad CC_\mu(\mathcal{F}) = -\text{rank}\mathcal{F} \cdot \text{cl}[T_X^*X] - \sum_{x \in Z} a_x(\mathcal{F}) \cdot \text{cl}[T_x^*X],$$

where $a_x(\mathcal{F})$ is the Artin conductor of $\mathcal{F}$ at x .

Proof. By Proposition 4.1.6 and [Sai17a, Lemma 5.13.1], both CC_μ and CC are additive for distinguished triangles. By [Bar24, Theorem 1.5(viii)], we have

$$(4.9.18) \quad SS(\mathcal{F}) = \bigcup_i SS({}^p\mathcal{H}^i \mathcal{F}).$$

By additivity and (4.9.18), it suffices to prove the assertion when $\mathcal{F}$ is a nonzero perverse E -sheaf.

Take a torsion-free perverse integral model $\mathcal{F}_0$ of $\mathcal{F}$. Put $\mathcal{F}_m = \mathcal{F}_0 \otimes^L \mathcal{O}_E/\pi^m$. By [Bar24, Theorem 1.5(v)] and [Sai17a, Corollary 4.5.4], we have

$$(4.9.19) \quad SS(\mathcal{F}) = SS(\mathcal{F}_0) = SS(\mathcal{F}_m).$$

By definition,

$$(4.9.20) \quad CC(\mathcal{F}) = CC(\mathcal{F}_0) = CC(\mathcal{F}_m).$$

By [Bei16, Theorem 1.3], we have $\dim SS(\mathcal{F}) = \dim X$. By semi-purity, we have

$$(4.9.21) \quad \begin{aligned} H_{SS(\mathcal{F})}^r(T^*X, e_X^{\vee*} K_{X/k, \mathcal{O}_E/\pi^m}) &= 0 \quad (r < 0), \\ H_{SS(\mathcal{F})}^0(T^*X, e_X^{\vee*} K_{X/k, \mathcal{O}_E/\pi^m}) &\simeq \bigoplus_a \mathcal{O}_E/\pi^m \cdot \text{cl}_{\mathcal{O}_E/\pi^m}[C_a], \\ H_{SS(\mathcal{F})}^0(T^*X, e_X^{\vee*} K_{X,E}) &= \bigoplus_a E \cdot \text{cl}_E[C_a], \end{aligned}$$

where C_a runs through the irreducible components of $SS(\mathcal{F})$ (of dimension $\dim X$). These identifications are compatible with reduction. By (4.9.21), we have an isomorphism

$$(4.9.22) \quad H_{SS(\mathcal{F})}^0(T^*X, e_X^{\vee*} K_{X/k, \mathcal{O}_E}) \xrightarrow{\sim} \varprojlim_m H_{SS(\mathcal{F})}^0(T^*X, e_X^{\vee*} K_{X/k, \mathcal{O}_E/\pi^m}).$$

Indeed, the relevant supported complexes lie in $D^{\geq 0}$, so the possible $\varprojlim^1 H^{-1}$ -term vanishes.

Since the evaluation morphism and identity section are compatible with reduction, by the definition of CC_μ (cf. (4.1.10)) and the isomorphism (4.9.3), we have compatibility

$$(4.9.23) \quad \begin{aligned} H_{SS(\mathcal{F})}^0(T^*X, e_X^{\vee*} K_{X/k, \mathcal{O}_E}) &\rightarrow H_{SS(\mathcal{F})}^0(T^*X, e_X^{\vee*} K_{X/k, \mathcal{O}_E/\pi^m}) \rightarrow H_{SS(\mathcal{F})}^0(T^*X, e_X^{\vee*} K_{X/k, \mathcal{O}_E/\pi^{m-1}}) \\ CC_{\mu, \mathcal{O}_E}(\mathcal{F}_0) &\mapsto CC_{\mu, \mathcal{O}_E/\pi^m}(\mathcal{F}_m) \mapsto CC_{\mu, \mathcal{O}_E/\pi^{m-1}}(\mathcal{F}_{m-1}). \end{aligned}$$

Similarly, for rationalization, we have

$$(4.9.24) \quad \begin{aligned} H_{SS(\mathcal{F})}^0(T^*X, e_X^{\vee*} K_{X/k, \mathcal{O}_E}) \otimes_{\mathcal{O}_E} E &\xrightarrow{\sim} H_{SS(\mathcal{F})}^0(T^*X, e_X^{\vee*} K_{X/k, E}) \\ CC_{\mu, \mathcal{O}_E}(\mathcal{F}_0) &\mapsto CC_{\mu, E}(\mathcal{F}) \end{aligned}$$

By Theorem 4.6.2, we have

$$(4.9.25) \quad CC_\mu(\mathcal{F}_m) = \text{cl}_{\mathcal{O}_E/\pi^m}[CC(\mathcal{F}_m)] \quad \text{in} \quad H_{SS(\mathcal{F})}^0(T^*X, e_X^{\vee*} K_{X/k, \mathcal{O}_E/\pi^m}).$$

Passing to the inverse limit and then inverting π , we get $CC_\mu(\mathcal{F}) = \text{cl}(CC(\mathcal{F}))$. $\square$

Corollary 4.9.3. *Let $f: X \rightarrow Y$ be a morphism of smooth schemes over a perfect field k and $i: Z \hookrightarrow X$ be a closed immersion such that $g = fi: Z \rightarrow Y$ is proper. Let $\mathcal{F} \in D_{\text{cons}}(Z_{\text{proét}}, E)$. We have*

$$(4.9.26) \quad \text{cl}CC(g_*\mathcal{F}) = f_!\text{cl}CC(i_*\mathcal{F}) \quad \text{in} \quad H_{f_*SS(i_*\mathcal{F})}^0(T^*Y, e_Y^{\vee*} K_Y).$$

Proof. By Theorem 4.9.2, we only need to show the following equality:

$$(4.9.27) \quad CC_\mu(g_*\mathcal{F}) = f_!CC_\mu(i_*\mathcal{F}) \quad \text{in} \quad H_{f_*SS(i_*\mathcal{F})}^0(T^*Y, e_Y^{\vee*} K_Y).$$

Note that the proof of Proposition 4.3.2.3 is within the six-functor formalism. It still works in the ℓ -adic case, and the same argument proves (4.9.27). $\square$

Corollary 4.9.4. *Let $f: X \rightarrow Y$ be a morphism from a smooth scheme X to a smooth connected curve Y over a perfect field k . Let $i: Z \hookrightarrow X$ be a closed immersion such that $g = fi: Z \rightarrow Y$ is proper. Let $\mathcal{F} \in D_{\text{cons}}(Z_{\text{proét}}, E)$ and let $y \in Y$ be a closed point. Assume that f is properly $SS(i_*\mathcal{F})$ -transversal over a dense open subset $V \subseteq Y$. Then we have*

$$(4.9.28) \quad -a_y(g_*\mathcal{F}) = (CC(i_*\mathcal{F}), df)_{T^*X, Z_y} \quad \text{in} \quad E.$$

In particular, this extends [Sai21, Theorem 2.2.3] by removing its quasi-projectivity assumption on f .

Proof. This follows from Corollary 4.9.3 (cf. proof of [Sai18, Lemma 3]). For more details, see the proof of Proposition 4.6.5. $\square$

We apply the preceding results to establish two results for finite coefficients.

Corollary 4.9.5. *Let X be a smooth scheme over a perfect field k of characteristic $p > 0$, and let Λ be a finite local ring over $\mathbf{Z}/\ell^n\mathbf{Z}$ for a prime number $\ell \neq p$. Let $i: Z \hookrightarrow X$ be a closed immersion and $\mathcal{F} \in D_{\text{ctf}}(Z, \Lambda)$. Let $cc_{Z,0}(\mathcal{F}) = 0_X^! CC(i_*\mathcal{F}) \in \text{CH}_0(Z)$ be the refined Gysin pull-back of $CC(i_*\mathcal{F})$ by the zero section $0_X: X \rightarrow T^*X$. Assume Z is proper over k . Then we have an equality in $\mathbf{Z}$*

$$(4.9.29) \quad \chi(Z_{\bar{k}}, \mathcal{F}) = \deg cc_{Z,0}(\mathcal{F}).$$

When Z is projective and smooth, this is proved in [Sai17a, Theorem 7.13].

Proof. By compatibility of characteristic cycles with extension of the base field, we may assume that k is algebraically closed. Let Λ_0 be the residue field of Λ . Since $\chi(Z_{\bar{k}}, \mathcal{F}) = \chi(Z_{\bar{k}}, \mathcal{F} \otimes_{\Lambda}^L \Lambda_0)$ and $cc_{Z,0}(\mathcal{F}) = cc_{Z,0}(\mathcal{F} \otimes_{\Lambda}^L \Lambda_0)$, we may assume that Λ is a finite field of characteristic $\ell \neq p$. Let $K(Z, \overline{\mathbf{Q}}_{\ell})$ be the Grothendieck group of the category of constructible $\overline{\mathbf{Q}}_{\ell}$ -sheaves (cf. [UYZ20, 5.2.2]), and similarly for $K(Z, \overline{\mathbf{F}}_{\ell})$. Note that the decomposition homomorphism $d_Z: K(Z, \overline{\mathbf{Q}}_{\ell}) \rightarrow K(Z, \overline{\mathbf{F}}_{\ell})$ defined in [UYZ20, 5.2.6] is surjective. Let $\mathcal{G} \in K(Z, \overline{\mathbf{Q}}_{\ell})$ such that $d_Z(\mathcal{G}) = \mathcal{F} \otimes_{\Lambda} \overline{\mathbf{F}}_{\ell}$. We have $\chi(Z_{\bar{k}}, \mathcal{F}) = \chi(Z_{\bar{k}}, \mathcal{G})$ and $cc_{Z,0}(\mathcal{F}) = cc_{Z,0}(\mathcal{G})$. By Theorem 4.9.2 and Corollary 4.9.3, we have an equality in $\overline{\mathbf{Q}}_{\ell}$:

$$(4.9.30) \quad \chi(Z_{\bar{k}}, \mathcal{G}) = \deg cc_{Z,0}(\mathcal{G}).$$

From this, we obtain (4.9.29). $\square$

Corollary 4.9.6. *Let $f: X \rightarrow Y$ be a morphism from a smooth scheme X to a smooth connected curve Y over a perfect field k of characteristic $p > 0$, and let Λ be a finite local ring over $\mathbf{Z}/\ell^m\mathbf{Z}$ for a prime number $\ell \neq p$. Let $i: Z \hookrightarrow X$ be a closed immersion such that $g = fi: Z \rightarrow Y$ is proper. Let $\mathcal{F} \in D_{\text{ctf}}(Z, \Lambda)$ and let $y \in Y$ be a closed point. Assume that there is a dense open subset $V \subseteq Y$ such that the restriction $g_V: Z_V = Z \times_Y V \rightarrow V$ is a smooth morphism and $\mathcal{F}|_{Z_V}$ is locally constant. Then we have*

$$(4.9.31) \quad -a_y(g_*\mathcal{F}) = (CC(i_*\mathcal{F}), df)_{T^*X, Z_y} \quad \text{in} \quad \mathbf{Z}.$$

Proof. By compatibility of characteristic cycles and Artin conductors with extension of the base field, we may assume that k is algebraically closed. Let Λ_0 be the residue field of Λ . Since $a_y(g_*\mathcal{F}) = a_y(g_*\mathcal{F} \otimes_{\Lambda}^L \Lambda_0)$ and $CC(i_*\mathcal{F}) = CC(i_*\mathcal{F} \otimes_{\Lambda}^L \Lambda_0)$, we may assume that Λ is a finite field of characteristic $\ell \neq p$.

Let $A(\overline{\mathbf{F}}_{\ell})$ (resp. $A(\overline{\mathbf{Q}}_{\ell})$) be the subgroups of $K(Z, \overline{\mathbf{F}}_{\ell})$ (resp. $K(Z, \overline{\mathbf{Q}}_{\ell})$) generated by the classes of constructible sheaves $\mathcal{G}$ such that $\mathcal{G}|_{Z_V}$ are smooth. Let $L(Z_V, \overline{\mathbf{F}}_{\ell})$ (resp. $L(Z_V, \overline{\mathbf{Q}}_{\ell})$) be the subgroups of $K(Z_V, \overline{\mathbf{F}}_{\ell})$ (resp. $K(Z_V, \overline{\mathbf{Q}}_{\ell})$) generated by the classes of smooth sheaves $\mathcal{G}$ on Z_V . Then, we have

$$(4.9.32) \quad A(\overline{\mathbf{F}}_{\ell}) = K(Z \setminus Z_V, \overline{\mathbf{F}}_{\ell}) \oplus L(Z_V, \overline{\mathbf{F}}_{\ell}),$$

$$(4.9.33) \quad A(\overline{\mathbf{Q}}_{\ell}) = K(Z \setminus Z_V, \overline{\mathbf{Q}}_{\ell}) \oplus L(Z_V, \overline{\mathbf{Q}}_{\ell}).$$

The decomposition homomorphism d defined in [UYZ20, 5.2.6] induces a surjective map

$$(4.9.34) \quad d: A(\overline{\mathbf{Q}}_\ell) \rightarrow A(\overline{\mathbf{F}}_\ell).$$

Let $\mathcal{G} \in A(\overline{\mathbf{Q}}_\ell)$ such that $d(\mathcal{G}) = \mathcal{F} \otimes_\Lambda \overline{\mathbf{F}}_\ell$. By Corollary 4.9.4, we have

$$(4.9.35) \quad -a_y(g_*\mathcal{G}) = (CC(i_*\mathcal{G}), df)_{T^*X, Z_y} \quad \text{in} \quad \overline{\mathbf{Q}}_\ell.$$

Since $a_y(g_*\mathcal{G}) = a_y(g_*\mathcal{F})$ and $CC(i_*\mathcal{G}) = CC(i_*\mathcal{F})$, we obtain (4.9.31). $\square$

Remark 4.9.7. Retain the hypotheses of Corollary 4.9.6, except for the local constancy of $\mathcal{F}|_{Z_V}$. Suppose that $g_V : Z_V \rightarrow V$ is smooth and that $D \subset Z_V$ is a relative divisor with simple normal crossings. Let $j : Z_V \setminus D \hookrightarrow Z_V$ be the open immersion. Assume that $\mathcal{F}|_{Z_V} \simeq j_*\mathcal{L}$, where $\mathcal{L} \in D_{\text{ctf}}(Z_V \setminus D, \Lambda)$ is smooth and tamely ramified along D . Then the same argument shows (4.9.31).

Corollary 4.9.8 (cf. [Blo87, Conjecture]). *Let $f : X \rightarrow Y$ be a proper flat morphism of smooth schemes over a perfect field k of characteristic $p > 0$, and let Λ be a finite local ring over $\mathbf{Z}/\ell^m\mathbf{Z}$ for a prime number $\ell \neq p$. Let $y \in Y$ be a closed point. Assume that $\dim X = n$, $\dim Y = 1$ and that there is a dense open subset $V \subseteq Y$ such that the base change $f_V : X \times_Y V \rightarrow V$ is smooth. Then we have*

$$(4.9.36) \quad -a_y(f_*\Lambda) = (-1)^n \deg_{k(y)} \left(c_{nX_y}^X(\Omega_{X/Y}^1) \cap [X] \right) \quad \text{in} \quad \mathbf{Z},$$

where $c_{nX_y}^X(\Omega_{X/Y}^1)$ is the localized Chern class defined in [Blo87, Section 1].

When f is projective, this is proved in [Sai21, Corollary 2.2.4].

Proof. By Corollary 4.9.6, we have $-a_y(f_*\Lambda) = (-1)^n (T_X^*X, df)_{T^*X, X_y}$. By [Sai21, Lemma 2.1.4], we have $(T_X^*X, df)_{T^*X, X_y} = \deg \left(c_{nX_y}^X(\Omega_{X/Y}^1) \cap [X] \right)$, which proves (4.9.36). $\square$

Proposition 4.9.9. *For every perverse E -sheaf $\mathcal{P}$ on a smooth k -scheme X , we have*

$$(4.9.37) \quad SS_\mu(\mathcal{P}) = SS(\mathcal{P}).$$

Proof. By definition of SS_μ , we have $SS_\mu(\mathcal{P})_{\bar{k}} = SS_\mu(\mathcal{P}|_{X_{\bar{k}}})$. By [Bar24, Theorem 1.5(xi)], we have $SS(\mathcal{P})_{\bar{k}} = SS(\mathcal{P}|_{X_{\bar{k}}})$. Therefore, we may assume $k = \bar{k}$ is perfect. The assertion may be checked on the connected components of X . We may therefore assume that X is connected, and put $n = \dim X$. The assertion is clear if $\mathcal{P} = 0$. By Theorem 4.9.1, we have

$$(4.9.38) \quad SS_\mu(\mathcal{P}) \subseteq SS(\mathcal{P}).$$

We first establish the positivity property of $CC(\mathcal{P})$. By [Bar24, Proposition 2.6 and the proof of Theorem 1.5(v)], we may choose an integral model $\mathcal{P}_0 \in \text{Perv}(X, \mathcal{O}_E)$ of $\mathcal{P}$ which is torsion-free as a perverse sheaf. Put $\mathcal{P}_1 = \mathcal{P}_0 \otimes_{\mathcal{O}_E}^L \mathcal{O}_E/\pi$. The distinguished triangle

$$(4.9.39) \quad \mathcal{P}_0 \xrightarrow{\pi} \mathcal{P}_0 \longrightarrow \mathcal{P}_1 \longrightarrow$$

shows that $\mathcal{P}_1$ is perverse. Indeed, ${}^p\mathrm{H}^{-1}(\mathcal{P}_1) = \ker(\pi : \mathcal{P}_0 \rightarrow \mathcal{P}_0) = 0$ because $\mathcal{P}_0$ is torsion-free, while ${}^p\mathrm{H}^0(\mathcal{P}_1) = \operatorname{coker}(\pi : \mathcal{P}_0 \rightarrow \mathcal{P}_0)$. By [Bar24, Theorem 1.5(ii),(v)], we have

$$(4.9.40) \quad SS(\mathcal{P}_1) = SS(\mathcal{P}_0) = SS(\mathcal{P}).$$

By [Sai17a, Proposition 5.14], the characteristic cycle $CC(\mathcal{P}_1)$ is effective and

$$(4.9.41) \quad \operatorname{supp} CC(\mathcal{P}_1) = SS(\mathcal{P}_1).$$

Since the characteristic cycle with E -coefficients is defined by reduction of an integral model, if $SS(\mathcal{P}) = \bigcup_{a \in A} C_a$ is the decomposition into irreducible components, then

$$(4.9.42) \quad CC(\mathcal{P}) = CC(\mathcal{P}_1) = \sum_{a \in A} m_a [C_a], \quad m_a \in \mathbf{Z}_{>0}.$$

In particular, $SS(\mathcal{P})$ is purely n -dimensional, and every m_a has nonzero image in the characteristic-zero field E .

We now recall the consequence of semi-purity needed below. If $W \subseteq T^*X$ is a closed subset of dimension at most n , then the cycle-class map induces an isomorphism

$$(4.9.43) \quad \operatorname{cl}_{W,E} : Z_n(W) \otimes_{\mathbf{Z}} E \xrightarrow{\sim} H_W^0(T^*X, e_X^{\vee*} K_{X/k}),$$

where $Z_n(W)$ denotes the group of n -dimensional cycles supported on W .

Indeed, since X is smooth of dimension n , we have $e_X^{\vee*} K_{X/k} \simeq E(n)[2n]$. A closed subset of W of dimension strictly smaller than n has codimension at least $n+1$ in the smooth scheme T^*X of dimension $2n$. Semi-purity therefore shows that such a subset contributes neither to degree 0 nor to degree 1 cohomology with coefficients in $e_X^{\vee*} K_{X/k}$. After removing a closed subset of dimension less than n , the n -dimensional irreducible components of W become pairwise disjoint regular closed subschemes of codimension n . Absolute purity then gives one copy of E for each such component, proving (4.9.43). Moreover, since these identifications are induced by cycle classes, they are compatible with extension of supports.

Let $C = SS(\mathcal{P})$, $C_\mu = SS_\mu(\mathcal{P})$ and

$$\rho_{C_\mu, C} : H_{C_\mu}^0(T^*X, e_X^{\vee*} K_{X/k}) \longrightarrow H_C^0(T^*X, e_X^{\vee*} K_{X/k})$$

be the extension-of-supports morphism. Since $C_\mu \subseteq C$ (cf. (4.9.6)) and C is purely n -dimensional, every n -dimensional irreducible component of C_μ is one of the C_a . Consequently, (4.9.43) gives

$$(4.9.44) \quad \operatorname{Im}(\rho_{C_\mu, C}) \simeq \bigoplus_{\{a \in A : C_a \subseteq C_\mu\}} E \cdot \operatorname{cl}_{C,E}([C_a]) \subseteq H_C^0(T^*X, e_X^{\vee*} K_{X/k}).$$

By Theorem 4.9.2, interpreted after extension of supports along $C_\mu \subseteq C$, we have

$$(4.9.45) \quad \begin{aligned} \rho_{C_\mu, C}(CC_\mu(\mathcal{P})) &= \operatorname{cl}_{C,E}(CC(\mathcal{P})) \\ &= \sum_{a \in A} m_a \operatorname{cl}_{C,E}([C_a]) \quad \text{in} \quad H_C^0(T^*X, e_X^{\vee*} K_{X/k}). \end{aligned}$$

The left-hand side of (4.9.45) belongs to the subspace described in (4.9.44). On the right-hand side, however, every coefficient m_a is nonzero in E . It follows that $C_a \subseteq C_\mu$ for every $a \in A$. Therefore $C \subseteq C_\mu$. Together with (4.9.38), this proves $SS_\mu(\mathcal{P}) = SS(\mathcal{P})$. $\square$

Lemma 4.9.10. *Let X be a smooth k -scheme of pure dimension n . The functor*

$$(4.9.46) \quad \mu_{\delta_X}[n] : D_{\text{cons}}((X \times X)_{\text{proét}}, E) \rightarrow D_{\text{cons}}(T^*X_{\text{proét}}, E)$$

is perverse t -exact.

Proof. It suffices to show that both ν_{δ_X} and $F_{\psi, TX/X}[n]$ are perverse t -exact. The first assertion follows from [Ill94, Corollaire 4.5] and the second assertion follows from [Lau87, Theorem 1.3.2.3]. $\square$

Lemma 4.9.11. *Let X be a smooth k -scheme of pure dimension n , and let $\mathcal{F}, \mathcal{G} \in D_{\text{cons}}(X_{\text{proét}}, E)$. For every integer $r \in \mathbf{Z}$, we have a canonical isomorphism*

$$(4.9.47) \quad {}^p\text{H}^r(\mu\mathcal{H}om_X(\mathcal{F}, \mathcal{G})[n]) \simeq \bigoplus_{j-i=r} \mu\mathcal{H}om_X({}^p\text{H}^i\mathcal{F}, {}^p\text{H}^j\mathcal{G})[n].$$

In particular, we have $\bigcup_i SS_\mu({}^p\text{H}^i\mathcal{F}) \subseteq SS_\mu(\mathcal{F})$.

Proof. By [BBDG, Propositions 1.3.21 and 4.2.8], we have isomorphisms

$$(4.9.48) \quad \begin{aligned} {}^p\text{H}^r(\mathcal{H}_X(\mathcal{F}, \mathcal{G})) &\simeq {}^p\text{H}^r(D_X\mathcal{F} \boxtimes \mathcal{G}) \\ &\simeq \bigoplus_{j-i=r} D_X({}^p\text{H}^i\mathcal{F}) \boxtimes {}^p\text{H}^j\mathcal{G} \\ &\simeq \bigoplus_{j-i=r} \mathcal{H}_X({}^p\text{H}^i\mathcal{F}, {}^p\text{H}^j\mathcal{G}). \end{aligned}$$

By Lemma 4.9.10, the functor $\mu_{\delta_X}[n]$ is perverse t -exact. Applying it to (4.9.48), we obtain the isomorphism (4.9.47).

Now take $\mathcal{G} = \mathcal{F}$ and $r = 0$. Each $\mu\mathcal{H}om_X({}^p\text{H}^i\mathcal{F}, {}^p\text{H}^i\mathcal{F})[n]$ is a direct summand of ${}^p\text{H}^0(\mu\mathcal{H}om_X(\mathcal{F}, \mathcal{F})[n])$. Thus $\bigcup_i SS_\mu({}^p\text{H}^i\mathcal{F}) \subseteq SS_\mu(\mathcal{F})$. $\square$

Theorem 4.9.12. *Let X be a smooth k -scheme and $\mathcal{F} \in D_{\text{cons}}(X_{\text{proét}}, E)$. We have*

$$(4.9.49) \quad SS_\mu(\mathcal{F}) = SS(\mathcal{F}).$$

Proof. By [Bar24, Theorem 1.5.(viii)] and Lemma 4.9.11, we have

$$(4.9.50) \quad SS(\mathcal{F}) = \bigcup_i SS({}^p\text{H}^i\mathcal{F}) \stackrel{(4.9.37)}{=} \bigcup_i SS_\mu({}^p\text{H}^i\mathcal{F}) \subseteq SS_\mu(\mathcal{F}).$$

Combining (4.9.6) and (4.9.50), we obtain (4.9.49). $\square$

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